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1. Title page

A randomized, double-blind, placebo-controlled Phase III study of ODM-201 versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer

ODM-201 in addition to standard androgen deprivation therapy and docetaxel in metastatic hormone-sensitive prostate cancer

Acronym: **ARASENS**

Test drug: BAY 1841788 / ODM-201

Study purpose: Efficacy and Safety

Clinical study phase: III Date: 25 MAY 2016

EudraCT: no: 2015-002590-38 Version no.: 1.0

Sponsor's study no.: 17777

Sponsor:

- a) Sponsor (Non-US): Bayer AG, [REDACTED]
- b) Sponsor (US territory): Bayer HealthCare Pharmaceuticals Inc., [REDACTED]

Sponsor's medical expert: [REDACTED]

The study will be conducted in compliance with the protocol, ICH-GCP and any applicable regulatory requirements.

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Signature of the sponsor's medically responsible person

The signatory agrees to the content of the final clinical study protocol as presented.

Name: [REDACTED]

Role: [REDACTED]

Date:

June 9, 2016

Signature:

[Handwritten Signature]

Signature of principal investigator

The signatory agrees to the content of the final clinical study protocol as presented.

Name:

Affiliation:

Date:

Signature:

Signed copies of this signature page are stored in the sponsor's study file and in the respective center's Investigator site file.

2. Synopsis

Title	A randomized, double-blind, placebo-controlled Phase III study of ODM-201 versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer
Short title	ODM-201 in addition to standard androgen deprivation therapy (ADT) and docetaxel in metastatic hormone-sensitive prostate cancer
Acronym	ARASENS
Clinical study phase	III
Study objective(s)	The primary objective of this study is: - To demonstrate the superiority in overall survival (OS) of ODM-201 in addition to standard ADT and docetaxel over placebo in addition to standard ADT and docetaxel The secondary objectives of this study are to evaluate: - Time to castration-resistant prostate cancer - Time to initiation of subsequent antineoplastic therapy - Symptomatic skeletal event free survival (SSE-FS) - Time to first symptomatic skeletal event (SSE) - Time to initiation of opioid use for ≥ 7 consecutive days - Time to pain progression - Time to worsening of physical symptoms of disease based on functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire (NCCN-FACT FPSI-17) - Safety The exploratory objectives of this study include: - Quality of life - Medical resource use - Prostate-specific antigen (PSA) assessments - Pharmacokinetics and exposure-response analysis - Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease
Test drug(s)	ODM-201 (BAY 1841788)

Name of active ingredient	ODM-201 (BAY 1841788)
Dose(s)	600 mg (2 x 300-mg tablets) twice daily (bid) with food to a daily dose of 1200 mg
Route of administration	Oral
Duration of treatment	Until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.
Reference drug(s)	Matching placebo
Name of active ingredient	Not applicable
Dose(s)	The dosing of placebo is the same as for ODM-201 (BAY 1841788), please see above.
Route of administration	Oral
Duration of treatment	Until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.
Background treatment	ADT (luteinizing hormone-releasing hormone [LHRH] agonist/antagonist or orchiectomy) and 6 cycles of docetaxel
Indication	Metastatic hormone-sensitive prostate cancer (mHSPC)
Diagnosis and main criteria for inclusion /exclusion	Inclusion criteria • Written informed consent • Males ≥ 18 years of age • Histologically or cytologically confirmed adenocarcinoma of prostate • Metastatic disease documented either by a positive bone scan, or for soft tissue or visceral metastases, either by contrast-enhanced abdominal/pelvic/chest computed tomography (CT) or magnetic resonance imaging (MRI) scan assessed by Investigator and confirmed by central radiology review. Metastatic disease is defined as either malignant lesions in bone scan or measurable lymph nodes above the aortic bifurcation or soft tissue/visceral lesions according to Response Evaluation Criteria In Solid Tumors (RECIST) 1.1. Lymph nodes are measurable if the short axis diameter is ≥ 15 mm, soft tissue/visceral lesions are measurable if the long axis diameter is ≥ 10 mm. Subjects with regional lymph node metastases only (N1, below the aortic bifurcation) will not be eligible for the study. Only subjects with non-regional lymph node metastases (M1a) and/or bone metastases (M1b) and/or other sites of metastases with or without bone disease

(M1c) will be eligible.

- Subjects must be candidates for ADT and docetaxel therapy per Investigator's judgment
- Started ADT (LHRH agonist/antagonist or orchiectomy) with or without first generation anti-androgen, but no longer than 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks, prior to randomization is recommended. First generation anti-androgen has to be stopped prior to randomization.
- An Eastern Cooperative Oncology Group performance status of 0 or 1
- Blood counts at Screening: hemoglobin ≥ 9.0 g/dL, absolute neutrophil count $\geq 1.5 \times 10^9/L$, platelet count $\geq 100 \times 10^9/L$ (subject must not have received any growth factor within 4 weeks or a blood transfusion within 7 days of the hematology laboratory sample obtained at Screening)
- Screening values of serum alanine aminotransferase and/or aspartate transaminase $\leq 1.5 \times$ upper limit of normal (ULN), total bilirubin $\leq$ ULN, creatinine $\leq 2.0 \times$ ULN
- Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the treatment with ODM-201/placebo and for 3 months after the end of the treatment with ODM-201/placebo and 6 months after treatment with docetaxel.

Exclusion criteria

- Prior treatment with:
 - LHRH agonist/antagonists started more than 12 weeks before randomization
 - Second-generation androgen receptor (AR) inhibitors such as enzalutamide, ARN-509, ODM-201, other investigational AR inhibitors
 - Cytochrome P 17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer
 - Chemotherapy or immunotherapy for prostate cancer prior to randomization
- Treatment with radiotherapy (external beam radiation therapy, brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization
- Known hypersensitivity to any of the study drugs, study drug classes, or excipients in the formulation of the study drugs
- Contraindication to both CT and MRI contrast agent
- Had any of the following within 6 months before randomization: stroke, myocardial infarction, severe/unstable angina pectoris,

	coronary/peripheral artery bypass graft, congestive heart failure (New York Heart Association Class III or IV) • Uncontrolled hypertension as indicated by a resting systolic blood pressure (BP) ≥ 160 mmHg or diastolic BP ≥ 100 mmHg despite medical management • Had a prior malignancy. Adequately treated basal cell or squamous cell carcinoma of skin or superficial bladder cancer that has not spread behind the connective tissue layer (i.e., pTis, pTa, and pT1) is allowed, as well as any other cancer for which treatment has been completed ≥ 5 years before randomization and from which the subject has been disease-free • A gastrointestinal disorder or procedure which is expected to interfere significantly with absorption of study drug • An active viral hepatitis, known human immunodeficiency virus infection with detectable viral load, or chronic liver disease with a need for treatment • Previous (within 28 days before the start of study drug or 5 half-lives of the investigational treatment of the previous study, whichever is longer) or concomitant participation in another clinical study with investigational medicinal product(s) • Any other serious or unstable illness, or medical, social, or psychological condition, that could jeopardize the safety of the subject and/or his/her compliance with study procedures, or may interfere with the subject's participation in the study or evaluation of the study results • Inability to swallow oral medications • Close affiliation with the investigational site (e.g., a close relative of the Investigator, dependent person [e.g., employee or student of the investigational site]) • Previous assignment to treatment in this study
Study design	Randomized, double-blind, placebo-controlled, multicenter phase III study. Approximately 1,300 subjects will be randomized (1:1 ratio) to receive one of the following study drugs: • ODM-201, 600 mg (2 x 300-mg tablets) bid orally with food • Placebo matching ODM-201 tablets in appearance, bid orally with food All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended. Six cycles of docetaxel will be administered after randomization. Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator.

	Subjects will be stratified at randomization as follows: • Extent of disease ○ Non-regional lymph nodes metastases only ○ Bone metastases with or without lymph node metastases ○ Visceral metastases with or without lymph node metastases or with or without bone metastases • Alkaline Phosphatase (ALP) ○ ALP<ULN ○ ALP≥ULN Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.
Methodology	The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up. Screening period: All trial-related procedures and evaluations will only be performed after the subject has agreed to participate and has signed the informed consent form. The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met. Once eligibility is confirmed and documented, eligible subjects will be randomized in a ratio of 1:1 to treatment with ODM-201 or placebo. Treatment period: Study treatment will be provided for all subjects bid until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, and non-compliance. Subjects will be evaluated every 12 weeks for castration-resistant prostate cancer, initiation of subsequent antineoplastic therapy, SSEs, opioid use for ≥7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, adverse events (AEs) and serious adverse events (SAEs), Quality of life (QoL) and PSA assessment. Active Follow-up period: After treatment discontinuation, subjects will enter the Active Follow-up period, which includes: • End of Treatment Visit An End of Treatment (EOT) Visit will be conducted 30 (+7) days after the last dose of study drug. The following assessments will be performed at the EOT Visit: QoL, pain assessment, analgesic consumption, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical progression, radiological progression, toxicity, or other), SSEs, and all AEs

	and SAEs regardless of causality. Active Follow-up visits During the Active Follow-up period, the following assessments will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical progression, radiological progression, toxicity, or other), SSEs, and study drug-related SAEs. After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data. Long-term (survival) Follow-up period: After Active Follow-up, subjects will continue to be contacted approximately every 12 weeks (by phone) to capture all antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs and survival status. Additional contacts may be required during this period, for example, for the collection of survival information for survival sweep data collection. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study.
Type of control	Placebo
Data Monitoring Committee	Yes
Number of subjects	Approximately 1,300 subjects in total are to be randomized
Primary variable(s)	Overall survival, defined as the time (in days) from date of randomization until death from any cause.
Time point/frame of measurement for primary variable(s)	The primary analysis will be performed when approximately 509 deaths occur in the 2 treatment arms combined. The expected study duration for 509 deaths is approximately 70 months.
Plan for statistical analysis	All randomized subjects will be included in the analysis of overall survival (OS), the primary efficacy endpoint. The primary efficacy endpoint of OS will be analyzed using a one-sided stratified log-rank test with an overall alpha of 0.025. Analysis will be according to treatment groups as randomized, with stratification factors extent of disease and alkaline phosphatase level. The hazard ratio (HR) (ODM-201/placebo) and its 95% confidence interval will be calculated using the Cox model, stratified by the same randomization factors. Kaplan-Meier estimates and plots will be presented for each treatment group. The secondary efficacy time-to-event endpoints will be analyzed using stratified log-rank test with randomization stratification factors. HR and



	95% confidence intervals will be provided using Cox model. Interim analyses for futility at approximately 30% of total planned events and for futility and efficacy at approximately 70% of total planned events will be conducted.
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List of abbreviations

5-HT	5-hydroxytryptamine
ADT	Androgen deprivation therapy
AE	Adverse event
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
AR	Androgen receptor
AST	Aspartate aminotransferase
AUC	The area under the concentration-time curve
AUC(0-t _{last})	The area under the concentration-time curve from time zero to the last sample with the quantifiable concentration
AUC(0-8)	The area under the concentration-time curve from time zero to the 8 hour blood sample calculated with linear trapezoidal rule
AUC(0-12)	The area under the concentration-time curve from time zero to the 12 hour blood sample calculated with linear trapezoidal rule
AUC(0-24)	The area under the concentration-time curve from time zero to the 24 hour blood sample calculated with linear trapezoidal rule
AV	Atrioventricular
bid	Twice daily
BCRP	Breast cancer resistance protein
BP	Blood pressure
BPI-SF	Brief Pain Inventory – Short Form
CAB	Complete androgen blockade
CI	Confidence interval
C _{max}	Maximum concentration
CNS	Central nervous system
CRF	Case report form
CRO	Contract research organization
CRPC	Castration-resistant prostate cancer
CT	Computed tomography
CTC	Circulating tumor cell
ctDNA	Circulating tumor DNA
CV	Coefficient of variation
CYP	Cytochrome P
dL	Deciliter
DMC	Data Monitoring Committee
DNA	Deoxyribonucleic acid
EBRT	External beam radiation therapy
ECG	Electrocardiogram

ECOG PS	Eastern Cooperative Oncology Group Performance Status
eCRF	Electronic case report form
EDC	Electronic data capture
EOT	End of treatment
ePRO	Electronic patient reported outcome
ESMO	European Society of Medical Oncology
EU	European Union
FAS	Full analysis set
FPSI-DRS-P	FACT/NCCN FPSI-17 disease-related symptoms - physical
g	Gram
GABA _A	Type-A γ -aminobutyric acid
GCP	Good Clinical Practice
GI	Gastrointestinal
GMP	Good Manufacturing Practice
HDPE	High density polyethylene
HR	Hazard ratio, heart rate
IB	Investigator's brochure
IC	Informed consent
IC ₅₀	Half maximal inhibitory concentration
ICF	Informed consent form
IEC	Independent Ethics Committee
IRB	Institutional Review Board
IV	Intravenous
IXRS	Interactive Voice/Web Response System
KM	Kaplan-Meier
LC-MS/MS	Liquid chromatography-tandem mass spectrometry
LHRH	Luteinizing hormone releasing hormone
LLOQ	Lower limit of quantification
M&S	Modeling & Simulation
mCRPC	Metastatic castration-resistant prostate cancer
mCSPC	Metastatic castration-sensitive prostate cancer
mHSPC	Metastatic hormone-sensitive prostate cancer
MedDRA	Medical Dictionary for Regulatory Activities
mg	Milligram
mL	Milliliter
mmHg	Millimeter of mercury
MRI	Magnetic resonance imaging
NCCN	National Comprehensive Cancer Network
NCCN-FACT FPSI-17	Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire

NCI-CTCAE v 4.03	National Cancer Institute-Common Terminology Criteria for Adverse Events; version 4.03
ng	Nanogram
nM	Nanomolar
OS	Overall survival
P-gp	P-glycoprotein
PCWG3	Prostate Cancer Clinical Trials Working Group
PD	Pharmacodynamic
PI	Principal investigator
PI3K	Phosphatidylinositol 3-kinase
PIK3CA	Phosphatidylinositol-4,5-bisphosphate 3-kinase, catalytic subunit alpha
PK	Pharmacokinetics
PKS	Pharmacokinetic Analysis Set
PRO	Subject-reported outcomes
PSA	Prostate-specific antigen
PTEN	Phosphatase and tensin homolog
QoL	Quality of life
RECIST	Response Evaluation Criteria In Solid Tumors
RNA	Ribonucleic acid
SAE	Serious adverse event
SAP	Statistical analysis plan
SAS	Statistical Analysis Software
SD	Standard deviation
SHBG	Sex hormone-binding globulin
SNP	Single gene polymorphism
SSE	Symptomatic skeletal event
SSE-FS	Symptomatic skeletal event free survival
SUSAR	Suspected unexpected serious adverse reaction
t_{last}	Time to the last quantifiable concentration
t_{max}	Time to the maximum plasma concentration
ULN	Upper limit of normal
UVB	Ultraviolet B
VCaP	Vertebral cancer of the prostate
WPS	Worst pain subscale

3. Introduction

3.1 Background

Prostate cancer is the second most common cancer and the fifth leading cause of cancer death among men worldwide (1). It is the most common non-cutaneous cancer and the second leading cause of cancer-related deaths in men in Europe (2) and the United States (3).

At the time of initial diagnosis, most men are found to have localized disease and can be treated with curative surgery or radiotherapy. However, a proportion of subjects will recur either locally or in distant sites. In addition, approximately 5% of new diagnoses present with de novo hormone-sensitive metastatic disease. The most common site of metastases is bone. However, soft tissue and visceral metastases can also occur (4). Upon development of metastatic disease, treatment is palliative, with the goal of prolonging quantity and maintaining quality of life (QoL) (5; 6; 7).

The mainstay of therapy for newly diagnosed metastatic hormone-sensitive prostate cancer (mHSPC), also known as metastatic castration-sensitive prostate cancer (mCSPC), remains androgen deprivation therapy (ADT), as recommended by the prostate cancer guidelines from the American Society of Clinical Oncology, the National Comprehensive Cancer Network (NCCN), and the European Association of Urology (8; 9; 10). Androgen deprivation therapy can be accomplished either by surgical orchiectomy (castration) or medical castration (using either a luteinizing hormone releasing hormone [LHRH] agonist or an LHRH antagonist). When LHRH therapy is initiated, a transient rise in luteinizing hormone can cause a surge in serum testosterone, which may stimulate prostate cancer growth. The flare phenomenon can be effectively prevented with the addition of an anti-androgen therapy to ADT, a complete androgen blockade (CAB), which blocks the effect of the increased serum testosterone (8; 9; 10).

Over the years, efforts have been made to improve upon ADT by modified timing (early versus deferred) and schedule (intermittent versus continuous) or by adding adjunctive agents, but the optimal management of this disease state is still not defined. Combined androgen blockade using both an LHRH agonist and an anti-androgen has been extensively studied as upfront therapy in mHSPC (6; 7; 11). Several meta-analyses have shown a modest 2% to 5% improvement in 5-year survival with CAB over castration, although with increased toxicity. One of the largest meta-analyses included the individual level data of 8,275 subjects from 27 trials. Although there was an overall trend toward improved 5-year survival with CAB over castration, this did not reach statistical significance ($P=0.11$) (12; 13; 14). According to NCCN prostate cancer guidelines 2014, CAB provides modest to no benefit over castration alone in metastatic subjects.

While ADT demonstrates antitumor activity in mHSPC with prolonged disease control, unfortunately resistance ultimately occurs and subjects die of castration-resistant prostate cancer (CRPC). Approximately 10% to 20% of prostate cancer subjects develop CRPC within 5 years, with a median time of 12-24 months (15), and have a poor median survival expectancy of 9 to 30 months (16). In the setting of metastatic castration-resistant prostate cancer (mCRPC), new therapeutic agents with different mechanisms of action have recently

become available and have significantly improved clinical outcomes, including chemotherapeutic agents (cabazitaxel), androgen-receptor targeted agents (abiraterone and enzalutamide), immunotherapies (sipuleucel-T), and radiopharmaceuticals (radium-223) (17; 18; 19; 20; 21; 22; 23). Conversely, the overall treatment paradigm of mHSPC has remained substantially stable and has not been strongly challenged until more recently, when the randomized controlled CHAARTED trial examined the addition of chemotherapy to standard ADT for metastatic hormone-sensitive disease, thus bringing chemotherapy to an arena historically dominated by hormone therapy only. The CHAARTED study demonstrated that combining 6 cycles of docetaxel chemotherapy with standard ADT was associated with a statistically significant overall survival (OS) benefit in metastatic hormone-sensitive disease compared with ADT alone, particularly in subjects with high volume disease (24; 25). Recent data from another ongoing study in Europe, the Medical Research Council STAMPEDE trial, confirm a significant OS benefit for docetaxel combined with ADT compared with ADT alone in subjects with high risk locally advanced or newly diagnosed metastatic disease (26; 27).

Chemo-hormonal therapy is currently recommended by the NCCN as first-line therapy for high volume mHSPC (11) and is recommended by ESMO Clinical guidelines as first-line treatment of metastatic, hormone-naïve disease in men fit enough for chemotherapy (33).

3.2 Overview of ODM-201 (BAY 1841788)

3.2.1 Pharmacology

ODM-201 is a novel non-steroidal androgen receptor (AR) inhibitor which binds with high selectivity and binding affinity (9 nM) to AR when compared to known second-generation anti-androgens. ODM-201 is a 1:1 mixture of 2 pharmacologically active diastereomers: ORM-16497 and ORM-16555. Diastereomers as well as the main circulating metabolite from ODM-201, ORM-15341, show no major differences in pharmacological activity and function as antagonists in AR-overexpressing cells.

In the vertebral cancer of the prostate (VCaP) cell line (derived from a bone metastasis of a subject with CRPC) with endogenous AR gene amplification and AR overexpression, ODM-201 suppressed dose dependently androgen-induced VCaP cell proliferation more efficiently than bicalutamide.

W741L (tryptophan to leucine) mutation of AR has been shown to have a role in the mechanism of bicalutamide resistance. In this mutation, ODM-201 functioned as an antagonist, whereas bicalutamide was a pure agonist, as previously shown.

In addition, ODM-201 inhibits testosterone-mediated nuclear translocation of AR in AR overexpressing cells. ODM-201 and ORM-15341 were found to decrease 0.3 nM testosterone-induced nuclear translocation at concentrations ≥ 100 nM and ≥ 300 nM, respectively; whereas bicalutamide failed to block testosterone-induced AR nuclear translocation at all tested concentrations.

In the immature rat assay, ODM-201 showed anti-androgenic activity, by dose dependently and significantly antagonizing testosterone-induced growth of androgen-sensitive tissues in

ventral prostate and seminal vesicles after 6-day oral dosing, with maximal inhibition at 100 mg/kg/day.

In the castration-resistant VCaP xenograft model, ODM-201 showed significant anti-tumor activity. The VCaP tumors grew significantly more slowly in mice treated with 50 mg/kg of ODM-201 orally once daily for 37 days compared with untreated castrated mice, whereas tumor growth was fully inhibited after orally twice daily (bid) dosing at 50 mg/kg.

In the orthotopic prostate cancer model, where VCaP cells were inoculated into prostate of nude intact male mice, inhibition of tumor growth and reduced increase in serum prostate-specific antigen (PSA) level were observed in mice treated with 50 mg/kg of ODM-201 orally twice daily for 3 weeks compared with vehicle controls. ODM-201 did not have any effect on serum testosterone levels.

3.2.2 Nonclinical pharmacokinetics and metabolism

ODM-201 is rapidly absorbed in mice and rats from preclinical formulations. Dog studies show that fasted and fed states differ in ODM-201 absorption with fed states showing higher total exposure than fasted states. Upon repeated dosing, no accumulation was observed. Systemic exposure to ODM-201 increased less than dose proportionally, suggesting solubility-limited absorption at the high dose levels.

ODM-201 consists of 1:1 ratio of diastereomers ORM-16497 and ORM-16555, which are able to interconvert and exhibit different ratios in animals. In rat plasma the diastereomer ratio remains about 1:1, but mice and dogs show higher exposure to ORM-16497 (i.e. ORM-16555 is eliminated more rapidly than ORM-16497 in these species). ORM-16555 is the main human diastereomer (about 85%).

The main circulating metabolite ORM-15341 is formed by oxidation of a secondary alcohol to form a ketone. The diastereomers ORM-16497 and ORM-16555 are able to interconvert via ORM-15341. In rats and dogs, the metabolite/parent ODM-201 ratio is close to unity, while in mice and humans the ratio is around 2 to 3. No human-specific disproportionate metabolites have been found so far.

Systemic clearance of ODM-201 in nonclinical species is low (0.2 L/h/kg in dogs) which is in agreement with the results obtained *in vitro* in hepatocyte incubations. Terminal volume of distribution of ODM-201 in dogs is 1.8 L/kg.

Based on *in vitro* data, ODM-201 is a substrate and inhibitor of P-glycoprotein (P-gp) and Breast Cancer Resistance Protein (BCRP). According to additional *in vitro* data, ODM-201 is also an inhibitor of the uptake transporters OATP1B1 and OATP1B3. The clinical relevance is currently not known.

In general, the potential for ODM-201 to inhibit CYP reaction pathways *in vitro* in human liver microsomes is low. The lowest IC₅₀ of 30 µM was observed for CYP2C9. For its pharmacologically active metabolite ORM-15341, a K_i value of 27 µM was determined. Applying a mechanistic static model, no inhibition potential towards CYP2C9 *in vivo* is expected.

The CYP450 induction potential of ODM-201 and ORM-15341 was evaluated *in vitro* in primary cultures of human hepatocytes from three individual donors. Based on these data, a risk to observe weak to moderate CYP3A4 induction by ODM-201 in the clinic cannot be ruled out. The clinical significance of these potential interactions is not known.

In vitro ODM-201 has a free fraction of 4.5% to 8% in the plasma of human, rat, and mouse. The free fraction in dog plasma is dependent on ODM-201 concentration (range: 0.3% to 11%). The major binding protein is serum albumin. There are no differences in plasma protein binding between the diastereomers. The main circulating metabolite ORM-15341 has a free fraction <1% in most species.

Tissue distribution studies in rodents suggest that ODM-201 and its metabolite ORM-15341 have very limited access to brain (brain/plasma ratio <0.05). Prostate and heart have clearly higher tissue/plasma ratios of ODM-201. The kidney/plasma ratio of ODM-201 is around 1 and the liver/plasma ratio is around 3. The results from a whole body autoradiography study in rats confirm that the ¹⁴C-ODM-201-related radioactivity concentrations in tissues (except in the liver) are lower than those in plasma.

In vivo mass balance study in rats suggests that about two-thirds of ODM-201-related radioactivity is excreted to feces and the rest to urine. Maximally, 45% of ODM-201 is excreted unchanged. The similar excretion pattern after intravenous (IV) and oral administration supports complete oral absorption at the studied low dose (11 mg/kg). Excretion is almost complete by 48 hours post-dose.

3.2.3 Safety pharmacology and toxicology

The repeat-dose toxicity studies with ODM-201 have been conducted up to 26 weeks in the rat and up to 39 weeks in the dog. ODM-201 has been well tolerated up to the highest dose levels in both species. The no observable adverse effect level was >2 x 500 mg/kg/day in the rat and >2 x 200 mg/kg/day in the dog. All the observed effects were directly related to the anti-androgenic mechanism of action of ODM-201 and no off-target toxicity could be identified in either species. The effects were also shown to be reversible after discontinuation of administration of ODM-201. Increase of exposure to ODM-201 was less than dose-proportional both in rats and dogs, and no accumulation was observed. The exposure multiples of ODM-201 at the highest dose level in the rat and dog were approximately up to 3-fold and 1.8-fold, respectively, compared with AUC(0-24h) at the 700-mg bid dose level in subjects with mCRPC in the clinical phase I/II ARADES study (17829 / 3104001). The corresponding exposure multiples of the metabolite ORM-15341 in the rat and dog were approximately up to 1.5-fold and 0.7-fold, respectively.

No stand-alone developmental and reproductive toxicity studies have been conducted with ODM-201. Based on the known pharmacologic effects of anti-androgens (11; 12; 13; 14), decreased fertility in males and developmental toxicity would be expected. Also it has to be taken into account that the subjects in this clinical study use gonadotropin releasing hormone agonist or antagonist treatment or have bilateral orchiectomy, which also affects fertility.

Based on the results from the completed *in vitro* and *in vivo* genotoxicity studies, ODM-201 is not considered to represent any significant genotoxic risk for man. Although both ODM-201

and its metabolite, ORM-15341, absorb light in the UVB wavelength range of 290 to 320 nm with the absorption maximum at 290 nm, an *in vitro* 3T3 neutral red uptake phototoxicity study demonstrated that ODM-201 is not phototoxic.

Cardiovascular safety evaluation of ODM-201 investigated within the frame of systemic toxicity studies in dogs, did not reveal any significant cardiovascular effects up to the highest dose levels tested after oral dosing. The mean maximum concentrations ($C_{\max}$) of ODM-201 in dog plasma at the highest dose levels in the 28-day, 13-week and 39-week toxicity studies were 9,990, 6,030 to 8,350 and 12070-12440 ng/mL, respectively.

In an anesthetized dog model using bolus IV administration, ODM-201 and its main human diastereomer ORM-16555, at observed mean $C_{\max}$ values of >9,300 ng/mL and >4,000 mg/mL in plasma, respectively, induced vasodilatation, especially at the peripheral level, leading to decreased arterial blood pressure (BP). This is not regarded to be of biological relevance as this was observed at free peak plasma concentrations of ≥ 3 -fold higher than the plasma free peak concentrations observed at 700 mg bid dose level in the Phase I/II clinical study in mCRPC subjects.

In the study with ORM-16555, 1 dog out of 4 displayed a reversible complete atrioventricular (AV) block at the highest bolus dose. At this dose, the mean $C_{\max}$ of ORM-16555 in dog plasma was 27,200 ng/mL, indicating that ORM-16555 may affect AV conduction at very high systemic exposure levels. The mechanism and etiology of vasodilatation and AV block are not known, but they might be multifactorial and at least partly related to a synergistic effect of the anesthetic agents used and secondary pharmacodynamic (PD) effects of ODM-201 and its main human diastereomer ORM-16555. Based on *in vitro* human ether-a-go-go-related gene and calcium channel inhibition and *in vivo* dog studies, the risk for QT interval prolongation appears low.

Functional *in vitro* studies evaluating secondary PD effects showed that ODM-201 inhibits 5-hydroxytryptamine (5-HT) uptake and type-A γ -aminobutyric acid (GABA_A) receptors. However, due to the low brain/plasma ratio, central nervous system (CNS) effects via 5-HT uptake and GABA_A inhibition are not expected. *In vivo* safety pharmacology evaluations confirmed that ODM-201 has no remarkable effects on CNS or respiratory function. At high local doses/concentrations in the gastrointestinal (GI) tract, ODM-201 may affect gastric emptying and intestinal transit.

3.2.4 Drug-drug interactions

ODM-201 has a low potential for drug-drug interaction with other substances based on *in vitro* studies. No inhibitory effect of ODM-201 is expected on the PK of any substrate primarily metabolized by CYP enzymes. A weak to moderate inducing effect by ODM-201 on the exposure of substrates which are mainly metabolized by CYP3A4 cannot be ruled out.

ODM-201 is a substrate for P-gp and BCRP. Therefore administration of strong inhibitors of P-gp and BCRP may increase the plasma concentrations of ODM-201. ODM-201 also inhibits BCRP and P-gp as well as OATP1B1 and OATP1B3 with half maximal inhibitory concentration (IC_{50}) values of 1.3, 16, 17 and 39 μ M, respectively. With the exception of the

IC₅₀ value for BCRP, these values are above the mean maximum concentration expected during steady state conditions of ODM-201 treatment.

Docetaxel will be administered after randomization according to standard treatment practice. Docetaxel is mainly metabolized by CYP3A4. Thus, inducers or inhibitors of CYP3A4 may alter the PK of docetaxel. ODM-201 has no inhibitory but may have some inducing effect on CYP3A4 substrates. Docetaxel is also a substrate of P-gp and OATB1P3. An effect of ODM-201 on the PK of docetaxel can, therefore, not be excluded. However, based on the *in vitro* transporter inhibition results with ODM-201, if any, only a minor increase of docetaxel exposure is expected. Furthermore, from the known safety profiles of ODM-201 and docetaxel, the major toxicities (bone marrow suppression and neutropenic infections) are not overlapping. In order to address the remaining risk of a potential pharmacokinetic interaction, it is planned to specifically investigate the potential impact of ODM-201 on docetaxel PK and the respective impact on safety on the day of the first docetaxel administration in the first 20 randomized subjects.

3.2.5 Clinical studies with ODM-201

As of 02 OCT 2015, 173 subjects with mCRPC have been treated with ODM-201.

ODM-201 has been studied in 6 clinical trials: a first-in-man study with cohort expansion (Study 17829 / 3104001, ARADES), a long-term safety extension study for the first-in-man study (Study 18035 / 3104002, ARADES-EXT), and a bioavailability study (Study 17830 / 3104003, ARAFOR). Additionally, there is a phase III efficacy and safety study of ODM-201 in men with high risk non-metastatic ongoing CRPC (Study 17712 / 3104007, ARAMIS), a Phase I mass balance and bioavailability study in healthy male subjects, as well as a phase I study in Japanese subjects.

ODM-201 treatment has been well tolerated up to the highest pre-specified dose of 900 mg bid. The maximum tolerated dose was not reached, and no dose-limiting toxicities were observed in the phase I dose-escalation. Escalation of doses in the phase I study was discontinued because a plateau in bioavailability of ODM-201 was observed.

Following oral administration of ODM-201 at doses of 100, 200, 300, 500, 700, and 900 mg bid, both diastereomers, ORM-16497 and ORM-16555, were slowly absorbed after administration with a light breakfast, the median t_{max} was 2-5 h.

AUC and C_{max} of ODM-201, ORM-16497, and ORM-16555 increased in a nearly dose-proportional manner at doses 100-700 mg bid. At the 900 mg bid dose level, there was no further increase of C_{max} or AUC observed compared to 700 mg dose bid. The concentration of the active metabolite ORM-15341 was 1.6-2.4 times higher than of the parent compound ODM-201. This ratio was dose-independent. After repeated oral administration of ODM-201, steady-state conditions seemed to be reached after 7 days at the latest. The diastereomer ratio of ORM-16497 to ORM-16555 was about 1:6 under steady-state conditions. The terminal half-life of ODM-201 was in the range of 11.5-16.1 hours after single oral administration of a tablet formulation.

A food effect was observed when ODM-201 was administered after a standardized high-fat, high-calorie breakfast. The exposure and C_{max} were about 50% lower when ODM-201 was administered under fasting conditions.

Considering the 164 mCRPC patients as of 02 OCT 2015 treated with ODM-201 in the Studies 17829 / 3104001, ARADES, 17830 / 3104003, ARAFOR, and 18035 / 3104002, ARADES-EXT, the integrated safety data showed 93.3% of subjects developed at least 1 adverse event (AE) and Grade 3 or higher AEs were reported in 28% of subjects. The most common AEs have been fatigue in 26.8% of subjects, back pain (22%). The most common AEs considered related to study treatment by the Investigator have been fatigue (9.8%) and hot flush (4.3%). The majority of AEs (91%) were classified as Grade 1 to 2. No dose-related trends were noted for the occurrence of any AEs.

Forty-five subjects (27.4%) have experienced a total of 125 serious adverse events (SAEs) from the start of study treatment and 7 subjects (4%) have died. None of these events was considered related to the study treatment by the Investigator or by the Sponsor.

Eleven subjects (6.7%) have withdrawn from the studies due to an AE. Only 1 of these AEs (fatigue, Grade 2) was considered related to the study treatment by the Investigator.

No events of seizure have been reported during ODM-201 treatment period.

Seven patients (7/164) experienced falls or accidental falls. Two out of those seven patients who had a fall or accidental fall experienced SAEs. All SAEs were resolved/improved after treatment. Nine patients (9/164) experienced non-pathological fractures. Four patients experienced SAEs. All serious events of fractures were secondary to a fall, an accidental wound, or syncope. Among the other five patients who experienced non-serious fractures, only one patient had a fall. None of those events were assessed as related to ODM-201 or resulted in permanent discontinuation from the study due to the events.

No safety concerns were raised from the ongoing Studies. In Japanese Phase I study, one SAE of grade 3 nausea was reported as related to ODM-201 by investigator, however, due to preexisting confounding factors, the sponsor assessed the event as not related with ODM-201 treatment.

Efficacy results show that ODM-201 has substantial antitumor activity in subjects with metastatic CRPC for time to PSA progression and for 12-week changes from baseline in serum PSA, soft tissue and bone lesions, and circulating tumor cell (CTC) count. Anti-tumor effects were observed at all dose levels. Anti-tumor activity was better in chemotherapy-naïve CYP17i-naïve subjects compared to post-chemotherapy CYP17i-naïve subjects and post-CYP17i subjects. The higher doses seemed to have a better response than the lower doses, particularly in chemotherapy-naïve CYP17i-naïve subjects.

Further details can be found in the latest available version of the Investigator's Brochure (IB), which contains comprehensive information on the study drug.

3.3 Rationale of the study

Management of advanced prostate cancer has evolved considerably in recent years and new therapeutic agents with diverse mechanisms of action have dramatically changed the

paradigm of treatment. However, the vast majority of advances have been made in subjects with metastatic castration-resistant disease. Fewer advances have been made in the earlier, hormone-responsive stage of metastatic disease.

ADT remains the mainstay of treatment in metastatic hormone-sensitive prostate cancer. However, despite the demonstrated antitumor activity of ADT in mHSPC, subjects ultimately progress and die of CRPC (5; 6; 7).

The variations upon ADT timing and schedule have not resulted in substantial changes in clinical outcome. Similarly, combined androgen blockade using both an LHRH agonist and an anti-androgen has shown a modest survival benefit over castration, with higher toxicity and costs (12; 13; 14).

The treatment scenario of mHSPC has only recently been challenged, when the randomized controlled CHAARTED trial showed improved survival with upfront chemotherapy (docetaxel) combined with ADT compared to standard ADT in men with metastatic hormone-sensitive prostate cancer, with a 17-month median OS benefit in those with high volume disease (25). Recent data from the STAMPEDE trial confirm the role of upfront chemo-hormonal therapy in mHSPC, showing a significant improvement in OS for subjects with metastatic disease treated with docetaxel in addition to ADT (26; 27).

Docetaxel combined with ADT is, therefore, anticipated to become a new standard for subjects with mHSPC.

It is clear that while ADT is the backbone of treatment, and upfront chemotherapy combined with ADT considerably improves survival, hormone-sensitive disease remains an area of active investigations. Novel treatment approaches are needed to improve disease control and survival and delay castration resistance.

ODM-201 is a novel non-steroidal AR inhibitor lacking a significant agonist action on AR, unlike first generation anti-androgens. It has high binding affinity to AR and a distinctive ability to decrease testosterone-mediated nuclear localization of AR in AR overexpressing cells. In phase I-II clinical studies, ODM-201 has demonstrated a very favorable safety profile, with no dose-limiting toxicities observed, and substantial antitumor activity in mCRPC, with effect on both serum PSA and soft tissue and bone lesions. A phase III trial is ongoing to compare ODM-201 versus placebo in men with non-metastatic CRPC.

The excellent tolerability and the clinical activity demonstrated so far in phase I/II in mCRPC provide the rationale to explore the efficacy of ODM-201 in the earlier stages of metastatic hormone-sensitive prostate cancer, where novel therapeutic strategies are needed to improve the outcome of subjects before castration resistance occurs.

This randomized phase III study aims to demonstrate that the addition of ODM-201 to ADT and docetaxel chemotherapy significantly prolongs OS over placebo in mHSPC subjects.

Based on *in vitro* transporter inhibition and *in vitro* CYP3A4 induction results with ODM-201, a small effect on docetaxel exposure during concomitant treatment with ODM-201 cannot be excluded. It is, therefore, planned to investigate the potential impact of ODM-201 on docetaxel PK and the respective impact on safety on the day of the first docetaxel administration in the first 20 randomized subjects (see Section 13.1.1). Based on the

mechanisms described above as well as known safety profiles from ODM-201 and docetaxel, the toxicities from the combination therapy are expected to be well managed. Therefore, the study will continue subject accrual during this PK and safety review period.

3.4 Benefit-risk assessment

ODM-201 treatment has been well tolerated in clinical studies in mCRPC subjects. There is no important risk identified. Fall and non-pathological fractures are currently considered important potential risks. No dose-limiting toxicities were observed in the phase I dose escalation. The AE profile observed reflects a profile that is expected in a population of subjects with advanced prostate cancer. Most AEs were Grade 1 to 2 and assessed by the investigators as not related to the study treatment. No dose-related trends were noted in the AE profile. Antitumor activity was seen at all dose levels as evaluated by PSA, CTC counts, and soft and bone lesion imaging. Subjects who were naïve to treatment with chemotherapy or CYP17i have responded best to ODM-201 treatment.

With its potent anti-androgenic profile, there is a strong rationale to study ODM-201 in subjects with mHSPC to delay disease progression and death. The available safety and efficacy data suggest that subjects who participate in this trial are not placed at undue risk.

As presented in the previous section, ADT and docetaxel have demonstrated significant survival benefits with the tolerable toxicities and is anticipated to become a new standard for subjects with mHSPC.

Overall, the benefit/risk assessment for ODM-201 in addition to standard ADT and docetaxel therapy for mHSPC subjects is positive.

4. Study objectives

The primary objective of this study is:

- To demonstrate the superiority in OS of ODM-201 in addition to standard ADT and docetaxel over placebo in addition to standard ADT and docetaxel

The secondary objectives of this study are to evaluate:

- Time to castration-resistant prostate cancer
- Time to initiation of subsequent antineoplastic therapy
- Symptomatic skeletal event free survival (SSE-FS)
- Time to first symptomatic skeletal event (SSE)
- Time to initiation of opioid use for ≥ 7 consecutive days
- Time to pain progression
- Time to worsening of physical symptoms of disease based on NCCN-FACT FPSI-17
- Safety

The exploratory objectives of this study include:

- Quality of life
- Medical resource use
- Prostate-specific antigen (PSA) assessments
- Pharmacokinetics and exposure-response analysis
- Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease

5. Study design

Design overview

This is an international, randomized, double-blind, placebo-controlled phase III study of ODM-201 in subjects with mHSPC.

The estimated sample size is approximately 1,300 subjects (see Section 10.4).

The start of the study period is defined by signing of the informed consent form (ICF). After an up to 28-day Screening period, subjects meeting the eligibility criteria (i.e. all of the inclusion criteria and none of the exclusion criteria [see Sections 6.1 and 6.2]) will be randomly assigned in a 1:1 ratio to treatment with one of the study drugs (ODM-201 or placebo):

- ODM-201 600 mg (2 x 300-mg tablets) bid orally with food to a total dose of 1200 mg
- Placebo matching ODM-201 tablets in appearance, bid orally with food

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended.

Six cycles of docetaxel will be administered after randomization.

Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator.

For the first 20 subjects, in which a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

Subjects will be stratified at randomization as follows:

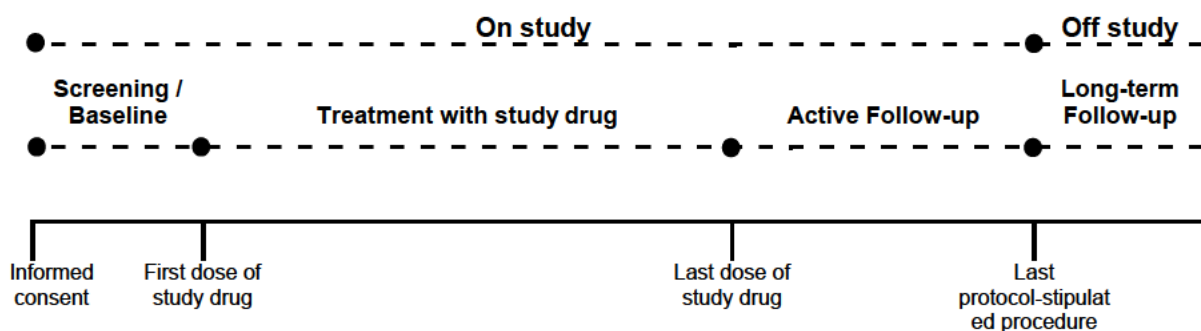
- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases

- Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase (ALP)
 - $ALP < \text{upper limit of normal (ULN)}$
 - $ALP \geq \text{ULN}$

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

A schematic of the study periods is presented in [Figure 5–1](#).

Figure 5–1 Study periods schematic



The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up.

Screening period:

All trial-related procedures and evaluations will only be performed after the subject has agreed to participate and has signed the ICF. The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met.

Once eligibility is confirmed and documented, eligible subjects will be randomized in a ratio of 1:1 to treatment with ODM-201 or placebo.

Treatment period:

The start of the treatment period is defined by the first administration of study drug. Treatment will be provided for all subjects twice daily until disease progression (symptomatic progressive disease, change of antineoplastic therapy), unacceptable toxicity, consent withdrawal or withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.

Subjects will be evaluated every 12 weeks for CRPC, initiation of subsequent antineoplastic therapy, SSEs, opioid use for ≥ 7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, AEs and SAEs, QoL and PSA assessment.

Active Follow-up period:

The Active Follow-up period is the interval from the end-of-study drug intake to the end of all protocol-specified post-treatment interventions.

It includes:

- End of Treatment Visit (EOT)

An EOT Visit will be conducted 30 (+7) days after the last dose of study drug. The following assessment will be performed: QoL, pain assessment, analgesic consumption, subsequent antineoplastic treatments for prostate cancer, which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, other), SSEs, and all AEs and SAEs regardless of causality.

- Active Follow-up visits

During the Active Follow-up period, the following assessment will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent antineoplastic treatments for prostate cancer, which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, other), SSEs, and study drug-related SAEs with concomitant medications received. After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data.

Long-term Follow-up period:

After Active Follow-up, subjects will continue to be contacted approximately every 12 weeks (by phone) to capture all antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs with concomitant medications received, and survival status. Additional contacts may be required during this period, for example, for the collection of survival information for survival sweep data collection. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study.

End-of-study

The end of the study as a whole will be reached as soon as the last visit of the last subject has been reached in all centers in all participating countries (European Union [EU] and non-EU).

Primary completion

The first analysis of OS will be performed as an interim futility analysis when approximately 153 deaths have occurred. In case the study continues after the futility analysis, an interim efficacy analysis of OS will be performed when approximately 357 deaths have occurred and a final efficacy analysis of OS will be performed when approximately 509 deaths have occurred. Should the Data Monitoring Committee (DMC) stop the study after the interim efficacy analysis, then the related database cut-off date will be the date of primary

completion. Should the study continue after the interim efficacy analysis, then the database cut-off date for the final analysis of OS will be the date of primary completion.

6. Study population

6.1 Inclusion criteria

Subjects must meet the following criteria at the time of Screening:

1. Written informed consent.
2. Males ≥ 18 years of age.
3. Histologically or cytologically confirmed adenocarcinoma of prostate.
4. Metastatic disease documented either by a positive bone scan, or for soft tissue or visceral metastases, either by contrast-enhanced abdominal/pelvic/chest computed tomography (CT) or magnetic resonance imaging (MRI) scan assessed by Investigator and confirmed by central radiology review. Metastatic disease is defined as either malignant lesions in bone scan or measurable lymph nodes above the aortic bifurcation or soft tissue/visceral lesions according to Response Evaluation Criteria In Solid Tumors (RECIST) 1.1 (31). Lymph nodes are measurable if the short axis diameter is ≥ 15 mm, soft tissue/visceral lesions are measurable if the long axis diameter is ≥ 10 mm.
 - Subjects with regional lymph node metastases only (N1, below the aortic bifurcation) will not be eligible for the study. Only subjects with non-regional lymph node metastases (M1a) and/or bone metastases (M1b) and/or other sites of metastases with or without bone disease (M1c) will be eligible.
5. Subjects must be candidates for ADT and docetaxel therapy, per Investigator's judgment.
6. Started ADT (LHRH agonist/antagonist or orchiectomy) with or without first generation anti-androgen, but no longer than 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended. First generation anti-androgen has to be stopped prior to randomization.
7. An Eastern Cooperative Oncology Group performance status (ECOG PS) of 0 or 1.
8. Blood counts at Screening: hemoglobin ≥ 9.0 g/dL, absolute neutrophil count $\geq 1.5 \times 10^9$ /L, platelet count $\geq 100 \times 10^9$ /L (subject must not have received any growth factor within 4 weeks or a blood transfusion within 7 days of the hematology laboratory sample obtained at Screening).
9. Screening values of serum alanine aminotransferase (ALT) and/or aspartate transaminase (AST) $\leq 1.5 \times \text{ULN}$, total bilirubin $\leq \text{ULN}$, creatinine $\leq 2.0 \times \text{ULN}$.

10. Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the treatment with ODM-201/placebo and for 3 months after treatment with ODM-201/placebo and 6 months after treatment with docetaxel.

6.2 Exclusion criteria

Subjects who meet any of the following criteria at the time of Screening will be excluded:

1. Prior treatment with:
 - LHRH agonist/antagonists started more than 12 weeks before randomization
 - Second-generation AR inhibitors such as enzalutamide, ARN-509, ODM-201, other investigational AR inhibitors
 - CYP17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer
 - Chemotherapy or immunotherapy for prostate cancer prior to randomization
2. Treatment with radiotherapy (external beam radiation therapy [EBRT], brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization.
3. Known hypersensitivity to any of the study drugs, study drug classes, or excipients in the formulation of the study drugs.
4. Contraindication to both CT and MRI contrast agent.
5. Had any of the following within 6 months before randomization: stroke, myocardial infarction, severe/unstable angina pectoris, coronary/peripheral artery bypass graft, congestive heart failure (New York Heart Association Class III or IV).
6. Uncontrolled hypertension as indicated by a resting systolic BP ≥ 160 mmHg or diastolic BP ≥ 100 mmHg despite medical management.
7. Had a prior malignancy. Adequately treated basal cell or squamous cell carcinoma of skin or superficial bladder cancer that has not spread behind the connective tissue layer (i.e., pTis, pTa, and pT1) is allowed, as well as any other cancer for which treatment has been completed ≥ 5 years before randomization and from which the subject has been disease-free.
8. A GI disorder or procedure which is expected to interfere significantly with absorption of study drug.
9. An active viral hepatitis, known human immunodeficiency virus infection with detectable viral load, or chronic liver disease with a need of treatment.
10. Previous (within 28 days before the start of study drug or 5 half-lives of the investigational treatment of the previous study, whichever is longer) or concomitant participation in another clinical study with investigational medicinal product(s).

11. Any other serious or unstable illness, or medical, social, or psychological condition, that could jeopardize the safety of the subject and/or his/her compliance with study procedures, or may interfere with the subject's participation in the study or evaluation of the study results.
12. Inability to swallow oral medications.
13. Close affiliation with the investigational site (e.g., a close relative of the Investigator, dependent person [e.g., employee or student of the investigational site]).
14. Previous assignment to treatment in this study.

For prohibited concomitant therapy, please refer to Section [8.1.3](#).

6.3 Justification of selection criteria

The inclusion/exclusion criteria used in this study, including concomitant conditions and medications as outlined above and in Section [8.1](#) are based upon knowledge of the known toxicity profile of the agent under investigation and the understanding of the concomitant conditions associated with prostate cancer. To ensure that subjects are in the earliest stages of mHSPC, before castration resistance occurs, ADT initiation within 12 weeks before randomization was selected as an inclusion criterion.

The selection criteria were chosen to ensure, to the best of the Sponsor's ability, the safety of the subjects who may participate in this study.

6.4 Withdrawal of subjects from study

6.4.1 Withdrawal

Withdrawal criteria

General procedures

In all cases, the reason for withdrawal must be recorded in the case report form (CRF) and in the subject's medical records.

The subject may object to the generation and processing of post-withdrawal data as specified in Section [13.4](#).

Details for the premature termination of the study as a whole (or components thereof) are provided in Section [12](#).

Withdrawal from the treatment with study drug

Subjects must be withdrawn from the treatment with study drug if any of the following occurs:

- If, in the Investigator's opinion, continuation of the study drug would be harmful to the subject's well-being
- Disease progression (symptomatic progressive disease, change of antineoplastic therapy)
- Unacceptable toxicity
- Any subject requiring study drug interruption >28 consecutive days
- Any subject requiring dosing of the study drug below 300 mg bid
- Occurrence of Grade 3 or higher treatment-related AE while the subject is on 300 mg bid

Note: Study drug discontinuation (i.e., discontinuation during the treatment period) does not constitute withdrawal from the study. Every effort should be made to retain subjects who discontinue the treatment period for any reason. These subjects are to be encouraged to remain on the study for follow-up of primary, secondary, and exploratory endpoints (i.e., continue in the Active Follow-up period or Long-term Follow-up period). Subjects are expected to participate in follow-up unless they explicitly object. Withdrawal of consent should be documented in the subject's medical file. For guidance on docetaxel background treatment, see Section 7.4.3.

Withdrawal from study

Subjects must be withdrawn from the study if any of the following occurs:

- At their own request or at the request of their legally acceptable representative. At any time during the study and without giving reasons, a subject may decline to participate further. The subject will not suffer any disadvantage as a result.
- At the specific request of the Sponsor and in liaison with the Investigator (e.g. obvious non-compliance, safety concerns)
- Sponsor terminates the study

Depending on the time point of withdrawal, a withdrawn subject is referred to as either "screening failure" or "dropout" as specified below:

Screening failure

A subject who, for any reason (e.g. failure to satisfy the selection criteria), terminates the study before the time point used for the definition of "dropout" (see below) is regarded a "screening failure". The reason will be documented in the source documentation and the electronic case report form (eCRF) and the subject will discontinue participation in study with no further follow-up required.

Re-screening of a subject who did not fulfill the eligibility criteria is allowed within 2 weeks from the prior screen failure date. If a subject is re-screened, a new screening number will be allocated and a link will be made between the screening numbers of the same subject.

Re-screening of screen failed subjects may only be allowed once after discussion with the Bayer-designated medical representative or Sponsor and after approval by the Sponsor. Sponsor approval of re-screening for a screen failed subject must be documented.

The subjects who need to repeat the screening procedures will be re-consented.

In any case, the Investigator has to ensure that the repeated screening procedures do not expose the subject to an unjustifiable health risk.

Dropout

A subject who discontinues study participation prematurely for any reason is defined as a “dropout” if the subject has already been randomized.

6.4.2 Replacement

A subject withdrawn from the treatment with study drug will not be replaced.

6.5 Subject identification

The subject number is a 9-digit number consisting of:

Digits 1 to 2 = Country code

Digits 3 to 5 = Center number within the country

(Digits 1 to 5 = Trial unit)

Digits 6 to 9 = Current subject number within the center

7. Treatment(s)

7.1 Treatments to be administered

Study drugs

The following treatment groups are defined for this study:

- ODM-201 600 mg (2 x 300-mg tablets) bid with food to a daily dose of 1200 mg
- Placebo matching ODM-201 tablets in appearance, bid orally with food

Background treatment

All subjects must receive ADT of Investigator’s choice (LHRH agonist/antagonists or orchiectomy) as standard therapy.

Six cycles of docetaxel will be administered after randomization. Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the

Investigator. The first cycle of docetaxel should be administered within 6 weeks after randomization. For the first 20 subjects, in which a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug). Docetaxel should be prepared at the study centers in accordance with the product information and/or local standards and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy following within one hour after the administration of study drug. Administration of study drug in these 20 subjects should come before docetaxel administration.

7.2 Identity of study treatment

7.2.1 ODM-201 and placebo

All study drugs will be labeled according to the requirements of local law and legislation. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request.

For all study drugs, a system of numbering in accordance with all requirements of Good Manufacturing Practice (GMP) will be used, ensuring that each dose of study drug can be traced back to the respective bulk batch of the ingredients. Lists linking all numbering levels will be maintained by the Sponsor's clinical supplies Quality Assurance group.

A complete record of batch numbers and expiry dates of all study drugs as well as the labels will be maintained in the Sponsor's study file.

ODM-201 and matching placebo tablets will be provided by Bayer AG.

The items in [Table 7–1](#) describe the elements used to identify each study drug.

Table 7–1 Study drug identifiers

Name	ODM-201	Placebo
Formulation	Film coated tablets for oral administration	Film coated tablets for oral administration
Appearance	Blue oval shaped tablets embossed with 'OR-300'	Blue oval shaped tablets embossed with 'OR-300'
Galencial form	Film coated tablet	Film coated tablet
Composition	Drug substance: 300 mg BAY1841788 Excipients: Lactose monohydrate Calcium hydrogen phosphate anhydrous Croscarmellose sodium Povidone Magnesium stearate Laquer blue	Excipients: Lactose monohydrate Cellulose microcrystalline Calcium hydrogen phosphate anhydrous Croscarmellose sodium Povidone Magnesium stearate Laquer blue
Strength	300 mg	Not applicable
Bayer number	BAY 1841788	Not applicable
Packaging	White plastic bottles of HDPE each containing 140 tablets	White plastic bottles of HDPE each containing 140 tablets

Abbreviations: HDPE = high density polyethylene.

7.2.2 Background docetaxel treatment

Docetaxel is commercially available. The participating investigators are required to consult the product information for docetaxel.

All treatment provided by the Sponsor will be labeled according to the requirements of local law and legislation. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request. For all study treatment provided by the Sponsor, a system of numbering in accordance with all requirements of GMP will be used, ensuring that each dose of study treatment can be traced back to the respective bulk ware of the ingredients. Lists linking all numbering levels will be maintained by the Sponsor's clinical supplies Quality Assurance group. A complete record of batch numbers and expiry dates as well as the labels of all study treatments provided by the Sponsor will be maintained in the Sponsor study file.

7.3 Treatment assignment

At the end of the screening period, all subjects meeting all of the inclusion criteria and none of the exclusion criteria will be randomly assigned on a 1:1 basis in a blinded fashion to treatment with ODM-201 or matching placebo plus ADT and docetaxel. In addition, randomization will be stratified by:

- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases
 - Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase
 - ALP<ULN
 - ALP≥ULN

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

A computer-generated randomization list will be prepared by the Bayer Randomization Manager. The randomization number will be used to link the subject to a treatment group and to a medication package.

The Investigator (or designated associate) will obtain the randomization number and the medication package numbers assigned to a subject by accessing the Interactive Voice/Web Response System (IXRS) after confirming that the subject meets all the inclusion criteria and does not meet any of the exclusion criteria.

An IXRS description and handbook will be provided to each investigational site.

7.4 Dosage and administration of study treatment

7.4.1 Dose modifications of study drug

The study drug will be administered as oral 300 mg tablets. The dose of study drug to be administered is 600 mg (2 x 300-mg tablets) bid to a daily dose of 1200 mg. Placebo matching ODM-201 tablets in appearance will be administered bid orally. It is recommended that ODM-201 or placebo be taken with food.

The dose or dosing schedule of the study drug may be modified following the occurrence of clinically significant AEs. Doses of study drug may be delayed or reduced in case of clinically significant toxicities that are considered by the Investigator to be related to study drug. Toxicities will be graded using the National Cancer Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) v 4.03. All dose modifications regardless of relatedness should be recorded on the eCRF. If a subject experiences several study drug-related toxicities with different grading, the recommendation of the worst grading should be used.

7.4.1.1 Dose interruption

The maximum time for a dose interruption period is 28 consecutive days. Any subject requiring treatment interruption >28 consecutive days must be withdrawn from treatment with study drug.

7.4.1.2 Dose reduction

If considered necessary for subject's safety, the dose of study drug may be reduced to 300 mg bid. The medical monitor must be notified of any ODM-201 dose reduction.

Dosing of the study drug below 300 mg bid is not allowed. If a Grade 3 or higher treatment-related AE occurs while the subject is on 300 mg bid, the subject must be withdrawn from the treatment with study drug.

7.4.1.3 General recommendations for dose modifications of study drug

A subject who experiences a treatment-related Grade 3 or 4 AE should interrupt study drug until the AE improves to Grade 2 or less. Treatment with study drug is then to be restarted at 300 mg bid.

Additional details are provided in [Table 7–2](#).

Table 7–2 Study drug dose modifications

Severity grade (NCI-CTCAE v 4.03)	Dose modifications	Study treatment withdrawal
Grade 0-2	Treat on time. Per Investigator's decision to interrupt or reduce study drug ^{a,b}	---
Grade 3 or 4	Delay until ≤ Grade 2 ^a When the severity is Grade ≤2, restart at a reduced dose of 300 mg bid ^{b,c}	If the dosing of the study drug is temporally or permanently reduced to 300 mg bid and a Grade 3 or higher treatment-related AE occurs while the subject is on a dose of 300 mg bid, the subject must be withdrawn from treatment with study drug

Excludes clinically nonsignificant and asymptomatic laboratory abnormalities.

Abbreviations: AE = adverse event; bid = twice daily; NCI-CTCAE v 4.03 = National Cancer Institute-Common Terminology Criteria for Adverse Events version 4.03.

^a If there is no recovery after 28 consecutive days, treatment with study drug should be permanently discontinued.

^b When AE returns to baseline or is resolved, dose escalation to 600 mg bid may be considered at the discretion of the Investigator.

^c If, following a dose reduction, dose is escalated to 600 mg bid and a second treatment-related AE with a severity of Grade 3 or higher occurs, a permanent dose reduction to 300 mg bid is required. A third occurrence of a Grade 3 or higher treatment-related AE requires permanent discontinuation of treatment with study drug.

7.4.2 Dosage and administration for background treatment

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy.

The docetaxel recommended dose to be administered is 75 mg/m² Day 1 as 1 hour IV infusion every 21 days. The cycle should be repeated every 3 weeks for 6 cycles. Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator. The recommended pre-medication regimen is oral dexamethasone 8 mg, 12 hours, 3 hours and 1 hour before the docetaxel infusion. However other schedules are also acceptable if summary of product characteristics regimen is different to local practice. Anti-emetic regimens are recommended as per local clinical practice.

7.4.3 Dose modifications of docetaxel background treatment

Dose adjustments of docetaxel should be made based on the specific types of toxicities observed, graded using NCI-CTCAE v4.03.

Docetaxel should be administered when the neutrophil count is $\geq 1,500$ cells/mm³. In order to monitor the occurrence of neutropenia, which may be severe and result in infection, frequent blood cell counts should be performed on all subjects receiving docetaxel.

Docetaxel should not be given to subjects with bilirubin >ULN, or to subjects with AST and/or ALT >1.5 x ULN. Bilirubin, AST or ALT values should be obtained prior to each cycle of docetaxel. In subjects who experienced either febrile neutropenia, neutrophil count <500 cells/mm³ for more than one week, severe or cumulative cutaneous reactions or severe peripheral neuropathy during docetaxel therapy, the dose of docetaxel should be reduced from 75 to 60 mg/m². If the subject continues to experience these reactions at 60 mg/m², the docetaxel treatment should be discontinued. Should treatment with docetaxel be discontinued for any reason before completion of 6 cycles, the treatment with study drug may be continued.

Please refer to the local country/region full prescribing information for further guidance.

7.5 Blinding

Subjects will be randomized to receive ODM-201 or matching placebo in a double-blind fashion such that neither the Investigator, nor the Sponsor, nor the subject will know which agent is being administered. The randomization number will be assigned through the IXRS based on information supplied by the Investigator at the time of randomization.

ODM-201 and placebo will be identical in appearance in order to preserve blinding. To maintain the blind, study drugs (either ODM-201 or matching placebo) will be packaged in a drug pack labeled with a unique drug pack number. The study drug pack number will be assigned to the subject through the IXRS.

The DMC will review the safety data and PK of the first 20 subjects after at least 1 cycle of docetaxel in an unblinded manner. However, the investigators, subjects and the Sponsor will remain blinded.

Bioanalytic analysis of ODM-201 and docetaxel PK samples will be performed by an unblinded bioanalyst at an external laboratory.

SUSAR unblinding

In the event of a Suspected Unexpected Serious Adverse Reaction (SUSAR) (see Section 9.6.1.4) related to the blinded treatment, the subject's treatment code will usually be broken before reporting to health authorities and central ethics committees in an unblinded fashion. In compliance with the European Commission's Detailed Guidance on the collection and verification of presentation of AE/reaction reports arising from clinical trials on medicinal products for human use, investigators and persons responsible for the ongoing conduct of the study will not usually receive unblinded copies of SUSAR reports, unless unblinded information is judged necessary for safety reasons.

Emergency unblinding by the Investigator

Investigators may only unblind subjects under emergency unblinding rules. If a subject is unblinded by the Investigator, they must discontinue study drugs.

Investigators should note that emergency unblinding is reserved for emergency situations where lack of knowledge about the actual study drug treatment interferes with appropriate emergency management. The occurrence of an SAE or progressive disease should not routinely precipitate the immediate unblinding of the label.

If emergency unblinding is necessary for the treatment of a subject for an SAE, the treatment with study drug can be unblinded via the IXRS system (refer to the IXRS manual for instructions). The participating site has unrestricted and immediate access to break the treatment code in IXRS. Should the blind code be broken for a subject, the medical monitor or designee should be contacted by the principal investigator (PI) within 1 working day of unblinding to discuss the rationale for the premature unblinding.

7.6 Drug logistics and accountability

All study drugs will be stored at the investigational site in accordance with Good Clinical Practice (GCP) and GMP requirements and the instructions given by the clinical supplies department of the Sponsor (or its affiliate/contract research organization [CRO]), and will be inaccessible to unauthorized personnel. Special storage conditions and a complete record of batch numbers and expiry dates can be found in the Sponsor's study file; the site-relevant elements of this information will be available in the Investigator site file. On the day of receipt, the responsible site personnel will confirm receipt of study drug via IXRS. The personnel will use the study drug only within the framework of this clinical study and in accordance with this protocol. Receipt, distribution, return, and destruction (if any) of the study drug must be properly documented according to the Sponsor's agreed and specified procedures.

Written instructions on medication destruction will be made available to affected parties as applicable.

The Investigator or designated associate(s) are responsible for dispensing the study drugs (ODM-201 or placebo).

The Investigator is responsible for control of the distribution and use of the study drug. Bottles must be returned to the Investigator with all unused medication. Throughout the study,

all unused study drug must be accounted for. This information will be recorded in the drug dispensing log.

Study subjects should be instructed to return all unused tablets to the study site at their next study visit for drug accountability and proper disposal. Study subjects must be instructed not to dispose of unused tablets on their own. Drug accountability should be done at the time bottles are returned by the subject. Any discrepancies should be reported immediately upon discovery.

7.7 Treatment compliance

Any discrepancies between actual and expected amount of returned study medication must be discussed with the subject at the time of the visit, and any explanation must be documented in the source records.

8. Non-study therapy

8.1 Prior and concomitant therapy

All subjects must receive an ADT of the Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy started ≤ 12 weeks before randomization (if combined with a first-generation anti-androgen, such as bicalutamide, flutamide, nilutamide, or cyproterone acetate, it must be stopped before randomization). For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended.

Six cycles of docetaxel will be administered after randomization.

Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator.

8.1.1 Prior therapy

All medications taken in the first 28 days prior to randomization (including start/stop dates, dose frequency, route of administration, and indication) must be recorded in the subject's source documentation, as well as entered in the appropriate pages of the eCRF. Prior local treatment for prostate cancer (surgery and/or radiation) before the time of informed consent (IC) is allowed and should be recorded in the medical record and in the appropriate eCRF. Prior treatment with LHRH agonist/antagonists started more than 12 weeks before randomization, second-generation AR inhibitors, other investigational AR inhibitors, CYP17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer is not allowed. Prior chemotherapy or immunotherapy for prostate cancer is not allowed. Treatment with radiotherapy (EBRT, brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization is not allowed. Blood transfusions are not allowed within 7 days and growth factors are not allowed within 4 weeks of the hematology laboratory sample obtained at Screening. For a detailed list of prohibited medications and treatments, see Sections [6.2](#) and [8.1.3](#).

8.1.2 Permissible concomitant medications and treatments

All concomitant treatments from the time of IC until the end-of-study treatment visit must be recorded on the CRFs. Once the subject has been withdrawn from the treatment with study drug, follow-up treatments will be recorded if used to treat new study drug-related SAEs or unresolved related AEs or if being used as a systemic antineoplastic therapy for prostate cancer.

Analgesic use will be captured via eCRF. Physicians are to record which opioid and non-opioid medications were used since the last visit, and physicians are to record exact daily doses of each analgesic consumed in the last 24 hours prior to the visit (Analgesic 24 hour consumption log –eCRF). Note that analgesic medication should be recorded via eCRF for each visit that pain is assessed via the Brief Pain Inventory short form (BPI-SF).

Palliative radiation therapy or surgical intervention as needed are allowed during study treatment. Treatment with bisphosphonates and denosumab is allowed.

For docetaxel pre-medication please refer to Section 7.4.2. Supportive care in case of toxicity related to docetaxel including use of biologic response modifiers such as granulocyte colony-stimulating factor or granulocyte-macrophage colony-stimulating factor, should be applied according to standard practice.

The administration of strong inhibitors of P-gp (e.g. verapamil, dronedarone) and BCRP (e.g. pantoprazole, eltrombopag) may increase the plasma concentrations of ODM-201 and should be used with caution.

Plasma concentration of drugs that are sensitive P-gp or BCRP substrates might be increased by ODM-201. Medicinal products that are sensitive substrates for P-gp (e.g. digoxin) or BCRP (e.g. methotrexate) should be used with caution when co-administered with ODM-201.

ODM-201 may decrease the plasma concentrations of concomitantly given drugs which are mainly metabolized via CYP3A4 enzyme (e.g. simvastatin, verapamil, dronedarone). The clinical significance of these potential interaction is not known. Subjects receiving such drugs should be monitored for signs of decreased efficacy.

8.1.3 Prohibited concomitant medications and treatments

Concomitant treatment with another systemic antineoplastic therapy or another investigational medicinal product is prohibited with the exception of ADT throughout the study and 6 cycles of docetaxel after randomization.

Initiation of the following medications during the study treatment period is prohibited:

- Any investigational medicinal product
- Radiopharmaceuticals
- Immunotherapy (e.g. sipuleucel-T)
- Cytotoxic chemotherapy other than docetaxel for 6 cycles after randomization
- Enzalutamide, ARN-509, bicalutamide, flutamide, nilutamide

- Abiraterone acetate, TAK-700, or other CYP17 inhibitors
- Systemic ketoconazole as antineoplastic treatment for prostate cancer
- Another systemic antineoplastic therapy may be initiated no sooner than 7 days after the last dose of study drug

For prohibited prior therapy, please refer to Section [6.2](#).

8.2 Post-study therapy

After treatment discontinuation, subjects will enter the Active Follow-up period and then the Long-term (Survival) Follow-up period. Therapy after discontinuation of study treatment will be collected as follows:

During Active Follow-up (includes End of Treatment Visit and Active Follow-up visits):

- End of Treatment Visit: collection of medications related to pain and QoL, and collection of medications for study drug-related SAEs.
- Active Follow-up Visit: collection of all subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change PSA progression, clinical progression, radiological progression, toxicity, or other); collection of medications related to pain and QoL, and collection of medications for study drug-related SAEs.

During Long-term (Survival) Follow-up period:

- Long-term Follow-up visits: collection of subsequent antineoplastic treatments for prostate cancer which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, or other), and collection of medications for study drug-related SAEs.

9. Procedures and variables

9.1 Tabular schedule of evaluations

All procedures, including efficacy and safety measurements obtained during the course of the study, are summarized in the schedule of assessments ([Table 9–1](#)).

Table 9–1 Schedule of assessments

Procedures	Screening Period		Treatment Period ^a		Follow-up Period		
			Visit 1	Visit 2 and subsequent visits	End of Treatment Visit	Active Follow-up Visits	Long-term Follow-up Period
	Within 28 days before randomization	Within 7 days before randomization	Day 1 (±3 days)	Week 12 (±7 days) and every 12 weeks	30 days (+7 days) after last dose of study drug	Approximately every 12 weeks from discontinuation of treatment up to 1 year	Approximately every 12 weeks after Active Follow-up, until death, lost to follow-up, consent withdrawal, or end-of-study
Informed consent	X						
Pharmacogenetic consent (whole blood only)	X						
Demography	X						
Medical history	X						
Prostate cancer history ^b	X						
Eligibility criteria	X	X					
Medical resource use			X	X	X	X	
Symptom/QoL questionnaire NCCN-FACT-FPSI-17 ^c		X	X ^d	X	X	X	
Pain questionnaire BPI-SF ^{c,e}		X	X ^d	X	X	X	
Analgesic 24 hour consumption log – eCRF (physician recording analgesic pain meds consumed over last 24 hours) ^{c,e}		X	X ^d	X	X	X	
Randomization			X				
Physical examination including weight and height ^f	X		X ^g	X	X		
12-lead ECG		X	X ^g	X	X		
Vital signs (BP, HR)		X	X ^g	X	X		
Laboratory safety assessments (hematology, general chemistry and urinalysis) ^h		X		X	X		
Serum PSA ⁱ	X			X	X		
Testosterone ^h	X			X	X		

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Procedures	Screening Period		Treatment Period ^a		Follow-up Period		
			Visit 1	Visit 2 and subsequent visits	Active Follow-up Period		Long-term Follow-up Period
					End of Treatment Visit	Active Follow-up Visits	
	Within 28 days before randomization	Within 7 days before randomization	Day 1 (±3 days)	Week 12 (±7 days) and every 12 weeks	30 days (+7 days) after last dose of study drug	Approximately every 12 weeks from discontinuation of treatment up to 1 year	Approximately every 12 weeks after Active Follow-up, until death, lost to follow-up, consent withdrawal, or end-of-study
Alkaline Phosphatase for screening ^h	X						
PK samples ⁱ			X	X			
Plasma collection for genetic biomarkers		X	X	X	X		
Plasma collection for non-genetic biomarkers		X	X	X	X		
Whole blood for CTC analyses			X	X ^j	X		
Whole blood collection for pharmacogenetic analysis – only if pharmacogenetic consent is signed			X				
Collection of tumor tissue (e.g. biopsy)	X ^k				X ^k		
ECOG PS	X		X	X	X		
CT/MRI ^l	X ^m			(X) ⁿ			
Bone scan ^l	X ^o			(X) ^p			
First SSE				X	X	X	
SSE-FS				X	X	X	
Post-study antineoplastic therapy ^q						X	X
Survival status						X	X
AEs/SAEs	X	X	X	X	X	X ^r	X ^r
Concomitant treatments	X	X	X	X	X	X ^r	X ^r
Study drug dispensing			X	X			

AE=adverse events, BP=blood pressure, BPI-SF=Brief Pain Inventory – Short Form; CT=computed tomography, CTC=circulating tumor cells, ECG=electrocardiogram, ECOG PS=Eastern Cooperative Oncology Group Performance Status, eCRF=electronic case report form, HR=heart rate, MRI=magnetic resonance imaging, NCCN-FACT FFSI-17=Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17-item questionnaire; PK=pharmacokinetics, PSA=prostate-specific antigen, QoL=quality of life, SAE=serious adverse event, SSE=symptomatic skeletal event, SSE-FS=symptomatic skeletal event free survival

^a Subjects will receive 6 cycles of docetaxel every 21 days, as per local practice. The first cycle of docetaxel should be administered within 6 weeks after randomization. For the first 20 subjects, in which a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

^b Including treatments and procedures.

^c Questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

^d If not collected at the Screening visit.

^e Analgesic consumption and pain assessment questionnaires must be collected on the same day.

^f Height to be measured at Screening only.

^g If not previously done.

^h PSA and testosterone values are to be assessed by a central laboratory during the study period. In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), using a liquid chromatography-tandem mass spectrometry method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT treatment. In addition, sex hormone-binding globulin and a bumin concentrations will be determined to calculate the concentration of non-protein-bound (free) testosterone in these subjects. Blood samples to test alkaline phosphatase for stratification purpose will be analyzed in a central laboratory and thereafter by a local laboratory. Other laboratory assessments are to be performed by local laboratories. Specific laboratory assessments are detailed in Section 9.6.2.1.

ⁱ In the first 20 subjects; several blood samples will be collected on the day of first docetaxel administration, which should be at least 14 days after start of study drug. On this day, blood samples will be collected at the following time points: Pre-Dose (prior to the administration of ODM-201 and docetaxel), 20±10 minutes after start of docetaxel infusion, 1 hour (within 5 minutes before the end of the infusion), 90 minutes, 2, 3, 4, 6, and 8 hours following the start of docetaxel infusion. For planned time points between 90 minutes and 8 hours, blood samples should be collected within ±15 minutes of the planned time. For all other randomized subjects, 2 blood samples for PK will be collected at least one hour apart on the first day of study drug administration independent of docetaxel administration. The first sample should be taken at least 30 minutes after the first study drug dose. Additional blood samples will be collected at 2 visits (i.e., one sample per visit), at any time relative to study drug and docetaxel administration, from Visit 2 onwards in all randomized subjects.

^j To be performed at Visit 2 only.

^k Collection of tumor tissue at Screening is only if available. Biopsy at disease progression is optional and if available.

^l Chest, abdomen and pelvic CT or MRI and bone scan should be performed at the end of docetaxel treatment (within 30 days from last cycle of docetaxel) and yearly thereafter.

^m Contrast-enhanced chest, abdomen, and pelvic CT or MRIs performed within 42 days prior to start of treatment with study drug are acceptable as screening scans if the standard acquisition procedure according to the Imaging Manual was followed.

ⁿ Chest, abdomen and pelvic CT or MRI can be performed at any time in case of PSA progression, symptomatic progressive disease, or change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment; (X) indicates optional assessment.

^o Bone scan performed within 42 days prior to start of treatment with study drug are acceptable as screening scan if the standard acquisition procedure according to the Imaging Manual was followed.

^p Bone scan can be performed at any time in case of PSA progression, symptomatic progressive disease, change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment; (X) indicates optional assessment.

^q Please note that documentation of post-study antineoplastic therapy must include the start and stop dates of each treatment and reason for treatment changes (PSA progression, clinical progression, radiological progression, toxicity, other).

^r Only study drug-related SAEs with concomitant medication received.

9.2 Visit description

Subject study visits are to occur every 12 weeks (± 7 days) starting from Day 1. The EOT Visit should occur within a minimum of 30 days (+7 days) after last dose of study drug.

The details of these visits are listed in [Table 9–1](#).

9.2.1 Screening period

A prospective study subject will receive both written and verbal information about the study, and will have an opportunity to ask questions and should have sufficient time to decide whether or not to participate in the study. The original signed and dated ICF must be retained in the Investigator's study file and a copy must be provided to the subject.

All subjects for whom the IC procedure is completed successfully will be assigned a subject number.

Subject screening must occur within 28 days prior to randomization at Visit 1 (Day 1). Screening procedures can be performed on separate occasions within the allowed 28 day timeframe. Re-testing is allowed during the 28-day window only if there is a medical or logistical reason. In case screening procedures need to be repeated outside the allowed 28 day timeframe of the screening period, subject should be considered a screen failure. It is estimated that it will take 7 to 14 days to obtain results from the safety laboratory assessments, testosterone, and PSA.

Each subject's eligibility will be confirmed by the Bayer-designated medical representative's written approval. The Investigator will complete a paper eligibility form confirming the inclusion and exclusion criteria, this will be forwarded to the Bayer-designated medical representative for review and eligibility confirmation which will take into account the results of the central image review for eligibility. The subject will be identified only by their study-specific subject number.

A subject who, for any reason (e.g. failure to satisfy the selection criteria), terminates the study at any time prior to randomization is regarded a "screening failure".

Please refer to [Section 6.4.1](#) for guidance on re-screening procedures.

The following procedures will be performed during the Screening period:

Within **28** days before randomization:

- Signed and dated ICF
- Informed consent for pharmacogenetic consent (whole blood only)
- Demographic data (date of birth, age, racial group)
- Medical history and current medical condition
- Prostate cancer history including prior treatments and procedures

- Assess eligibility criteria
 - Eligibility must be confirmed by written approval of the Bayer-designated medical representative
- ECOG PS
- Physical examination including weight and height
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), using a liquid chromatography-tandem mass spectrometry (LC-MS/MS) method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT treatment. In addition, sex hormone-binding globulin (SHBG) and albumin concentrations will be determined to calculate the concentration of non-protein-bound (free) testosterone in these subjects
- Alkaline phosphatase for screening (stratification)
 - Note for alkaline phosphatase test during Screening: blood samples to test alkaline phosphatase for stratification purpose will be analyzed in central laboratory and during treatment period by local laboratory
- Contrast-enhanced chest, abdomen, and pelvic CT or MRI. CT/MRI performed within 42 days prior to start of treatment with study drug are acceptable as screening scans if fulfilling the requirements of the Imaging Manual.

Radiologic tumor assessments will include, chest, abdomen and pelvis, and will be evaluated locally at the study site and by the central review. The first radiological assessment must be comprised of an IV (and oral, if indicated, per Imaging Manual) contrast enhanced CT/MRI scans of chest, abdomen and pelvis. Scans done up to 42 days prior to first dose can be used as baseline scans fulfilling the requirements in the Imaging Manual. Contrast-enhanced MRI shall be performed instead of CT when local regulations or allergy to CT contrast media do not permit the use of CT as requested per protocol schedule. Copies of scans should be forwarded to the central imaging facility for independent assessment. Subjects with a contraindication to both, CT and MRI contrast agent, should not be enrolled in this study (see exclusion criterion 4).

- Bone scan. Bone scan performed within 42 days prior to start of treatment with study drug is acceptable as screening scan if performed according to the Imaging Manual. Copies of scans should be forwarded to the central imaging facility for independent assessment
- AEs/SAEs and concomitant treatments, including opioid use for pain

- Optional collection of archival tissue – if available- for tissue biomarker analyses unless precluded by local guidelines, e.g. Independent Ethics Committee (IEC)/ Institutional Review Board (IRB) or regulatory authorities

Within 7 days before randomization:

- Functional assessment of cancer therapy/National Comprehensive Cancer Network prostate cancer symptom index (NCCN-FACT FPSI-17) questionnaire (collected via electronic patient reported outcome [ePRO] device)
- Pain questionnaire - BPI-SF (collected via ePRO device)
- Analgesic 24-hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject). Should be collected on the same day as pain assessment questionnaires
- A 12-lead ECG
- Vital signs (BP and HR)
- Laboratory safety assessments (hematology, chemistry and urinalysis) (see Section 9.6.2.1)
- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities if eligibility has been confirmed (see Section 13)
- AEs/SAEs
- Concomitant treatments, including opioid use for pain
- Confirmation of eligibility

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

9.2.2 Study treatment period

9.2.2.1 Day 1 (Visit 1)

The subject will arrive at the study center on the morning of Day 1 ($\pm$ 3 days). The first dose of study drug will be administered with breakfast at the study center. The following procedures will be performed:

- Randomization
- Medical resource use
- Blood draw for 2 PK samples, which are collected at least 1 hour apart, and the first sample should be taken at least 30 minutes after the first ODM-201 dose

In the first 20 subjects, detailed mandatory blood sampling for PK analysis of ODM-201, ORM-16555, ORM-16497, ORM-15341, and docetaxel will be collected on the day of the first docetaxel administration which should be at least 14 days after randomization (i.e. start of study drug; see Section 9.5).

- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities (see Section 13)
- Whole blood draw for CTC
- Whole blood draw for pharmacogenomics analysis if PI/ICF Pharmacogenetic consent is signed unless precluded by local guidelines, e.g., IEC/IRB or regulatory authorities (see Section 13)
- ECOG PS
- Current medical condition, AEs, SAEs, and concomitant treatments
- Study drug will be dispensed for 12 weeks

If not performed at Screening:

- Functional assessment of cancer therapy/National Comprehensive Cancer Network prostate cancer symptom index (NCCN-FACT- FPSI-17) questionnaire (collected via ePRO device)
- Pain questionnaire - BPI-SF (collected via ePRO device)
- Analgesic 24-hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Physical examination including weight
- A 12-lead ECG
- Vital signs (BP and HR)

Subjects will continue taking study drug at home throughout the study treatment period.

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

Note for docetaxel administration: Subjects will receive 6 cycles of docetaxel every 21 days, as per local practice. The first cycle of docetaxel should be administered within 6 weeks after randomization. For the first 20 subjects, in which a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

9.2.2.2 Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days)

The following procedures will be performed:

- Medical resource use
- Symptom / QoL questionnaire NCCN-FACT-FPSI-17
- Pain questionnaire - BPI-SF
- Analgesic 24-hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Physical examination including weight
- A 12-lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry and urinalysis) (see Section 9.6.2.1)
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), using a LC-MS/MS method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT treatment. In addition, SHBG and albumin concentrations will be determined to calculate the concentration of non-protein-bound (free) testosterone in these subjects
- At 2 visits, an additional blood draw for PK analysis (see Section 9.5)
- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities (see Section 13)
- Collect whole blood sample for CTC (at Visit 2 only)
- ECOG PS

- Chest, abdomen and pelvic CT/MRI: (please refer to Section 9.2.2.3)
- Bone scan: (please refer to Section 9.2.2.3)
- Date of first SSE (if applicable)
- Current medical condition, AEs, SAEs, and concomitant treatments
- Study drug will be dispensed for 12 weeks

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

9.2.2.3 Requirement for chest, abdomen and pelvic CT/MRI and bone scan during the study treatment period

Contrast-enhanced chest, abdomen and pelvic CT or MRI for the assessment of soft tissue lesions and bone scan for the assessment of bone lesions should be performed at the end of docetaxel treatment (within 30 days from last cycle of docetaxel) and yearly thereafter.

In addition, contrast-enhanced chest, abdomen and pelvic CT or MRI and bone scan can be performed at any time in case of PSA progression, symptomatic progressive disease, change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment. (Section 9.2.5).

If a subject develops contraindication to CT contrast media during the course of the study after the baseline imaging, contrast enhanced MRI shall be performed. In rare cases where the subject develops contraindication to both CT and MRI contrast media, the post-baseline imaging examination could continue without contrast media.

The method chosen at the baseline should be the same throughout the study. The same technique (e.g., slice thickness, field of view) should be used for all scans during the study treatment period. Preferably all scans should be interpreted by the same investigator/radiologist during the study whenever possible.

Radiological progression by soft tissue lesions is determined according to RECIST criteria, version 1.1 (31), based on MRI / CT scans of the chest, abdomen, and pelvis performed by the Investigator, as recommended by Prostate Cancer Clinical Trials Working Group (PCWG3) (32).

Bone lesions will be recorded separately from soft tissue lesions. Radiological progression by bone lesions is determined according to PCWG3 criteria based on whole body ^{99m}Tc methylene diphosphonate bone scans performed by the Investigator.

Further guidance for acquisition of images and assessment rules will be provided in the Imaging Manual.

9.2.3 Active Follow-up period

End-of-study treatment visit

Subjects will have an EOT Visit 30 days (+7 days) after the last dose of study drug. Another systemic antineoplastic therapy may be initiated no sooner than 7 days after the last dose of study drug.

The following procedures will be performed during the end-of-study treatment visit:

- Medical resource use
- Symptom / QoL questionnaire NCCN-FACT-FPSI-17
- Pain questionnaire - BPI-SF
- Analgesic consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
- Physical examination including weight
- A 12-lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry, and urinalysis) (see Section 9.6.2.1)
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at screening and repeated every 6 months (Visit 3, Visit 5, etc.), using a LC-MS/MS method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT treatment. In addition, SHBG and albumin concentrations will be determined to calculate the concentration of non-protein-bound (free) testosterone in these subjects
- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities
- Whole blood draw sample for CTC analysis
- Optional collection of biopsy tissue taken at disease progression, if available, for tissue biomarker analyses unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities
- ECOG PS

- Date of first SSE (if applicable)
- Current medical condition, AEs, SAEs and concomitant treatments

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

Active Follow-up Visits

After treatment discontinuation, subjects will begin the Active Follow-up period. Active Follow-up visits occur as standard of care clinic visits approximately every 12 weeks for up to 1 year to collect the following:

- Resource utilization
- Review handheld device or tablet and ensure subjects complete the following:
 - Symptom / QoL questionnaire NCCN-FACT-FPSI-17
 - Pain questionnaire - BPI-SF
 - Analgesic consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Record SSEs if applicable
- Subsequent antineoplastic treatments for prostate cancer which should be provided with start and stop dates and reason for treatment change (PSA progression, clinical progression, radiological progression, toxicity, other)
- Record survival status
- Study drug-related SAEs with concomitant medications received

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws IC and actively objects to collection of further data.

9.2.4 Long-term (survival) Follow-up period

After completing the Active Follow-up period, subjects will enter the Long-term Follow-up period. Subjects will be contacted approximately every 12 weeks by phone, and additional contacts may be required for overall (survival assessment). The following information will be collected:

- Record subsequent antineoplastic treatments for prostate cancer with start and stop dates and reason for treatment change (PSA progression, clinical progression, radiological progression, toxicity, other)
- Record survival status
- Study drug-related SAEs with concomitant medications received

Additional contacts may be required during this period, for example, for the collection of survival information for survival sweep data collection.

As this information will be collected over the telephone from the subject, it is the clinical site's responsibility to obtain and make available the source documentation for the information collected from the subject and document it in the eCRF.

9.2.5 Unscheduled visits

In the event that there are significant abnormal safety findings or suspected disease progression, control assessments may be performed at any time during the study treatment period according to the judgment of the Investigator. The following procedures (one or more) may be performed at the unscheduled visit, if indicated:

- Physical examination including weight
- 12-lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry, urinalysis)
- Status of survival
- AEs, SAEs, current medical condition, concomitant treatments
- Chest, abdomen, and pelvic CT/MRI
- Bone scan

Other unscheduled procedures may be performed, as per Investigator's judgment.

9.3 Population characteristics

9.3.1 Demographic

Demographic characteristics of date of birth (age), sex, and race/ethnicity should be entered in the eCRFs.

9.3.2 Medical history

Medical history findings (i.e. previous diagnoses, diseases, or surgeries) meeting all criteria listed below will be collected:

- Not pertaining to the study indication
- Start before signing of the ICF
- Considered relevant to the study

Detailed instructions on the differentiation between medical history and AEs can be found in Section [9.6.1.1](#).

History of prostate cancer will be collected separately from the general medical history. This includes, but is not limited to:

- Date of diagnosis
- Staging (performed at diagnosis)
- Prostate cancer history
- Disease status at study entry
- Prior diagnostic and therapeutic procedures
- Prior prostate cancer treatments (any therapy that is ongoing should be reported as concomitant medication)

9.3.3 Other baseline characteristics

Baseline characteristics relating to disease factors include:

- QoL assessment (NCCN-FACT FPSI-17)
- ECOG PS
- Cancer pain assessment

All medications and significant non-drug therapies ongoing during the screening period (28 days prior to randomization) must be entered in the eCRF, including:

- Medication trade name and dose
- Reason for medication
- Start date and end date, or if continuing at time of study entry

9.4 Efficacy

The primary efficacy variable is OS.

Survival status will be assessed from randomization until the end of the Long-term Follow-up period. Date of death and primary cause of death will be recorded.

For definition and analysis of primary variable, and for description of secondary and exploratory variables, please refer to Section [10.3](#).

9.5 Pharmacokinetics/pharmacodynamics

9.5.1 Drug measurements

The concentrations of diastereomers ORM-16497 (BAY 1896951) and ORM-16555 (BAY 1896952), metabolite ORM-15341 (BAY 1896953) and docetaxel in plasma will be determined by a validated method, e.g. liquid chromatography-tandem mass spectrometry (LC-MS/MS). ODM-201 (BAY 1841788) concentration will be calculated as the sum of diastereomers ORM-16497 and ORM-16555 concentrations.

PK samples will be collected from all subjects to maintain blinding. However PK samples from the placebo arm will not be analyzed.

In the first 20 subjects, detailed mandatory blood sampling for PK analysis of ODM-201, ORM-16555, ORM-16497, ORM-15341, and docetaxel will be collected on the day of the first docetaxel administration which should be at least 14 days after randomization (i.e. start of study drug).

On this day blood samples will be collected at the following time points:

Pre-dose (prior to the administration of study drug and docetaxel), 20±10 minutes after start of docetaxel infusion, 1 hour (within 5 minutes before the end of infusion), 90 minutes, 2, 3, 4, 6, and 8 hours following the start of docetaxel infusion. For planned time points between 90 minutes and 8 hours, blood samples should be collected within ±15 minutes of the planned time.

Two additional samples will be taken at 2 later visits (i.e. one at each visit, at any time relative to ODM-201 administration), to provide information on variability at steady state (ODM-201 only).

It is of importance that the actual date and time of blood sampling and study drug administration as well as docetaxel administration are documented in the eCRF because PK calculations will be based on the actual sampling times relative to dosing times.

In all other randomized subjects, sparse sampling for population PK will be collected to determine the exposure (i.e. area under the curve over the dosing interval at steady state) of ODM-201, the diastereomers ORM-16555 and ORM-16497, and the metabolite ORM-15341 and to quantify its variability in the target subject population.

Two samples will be taken on the day of randomization (Visit 1), at least 30 minutes after the first dose of study drug, approximately 1 hour apart to determine absorption data.

Two additional samples will be taken at 2 later visits (i.e. one at each visit), at any time relative to ODM-201 and docetaxel administration, to provide information on variability at steady state. Drug intake and sampling times have to be accurately documented.

The instructions for sample collection, processing, storage and shipment will be detailed in a separate document (e.g., sample handling sheet or lab manual).

9.5.2 Pharmacokinetic evaluation

Non-compartment approach PK analysis

In the first 20 subjects:

Pharmacokinetic parameters will be calculated using a non-compartment approach according to the current Bayer guidelines, using Win-Nonlin software. Details of the used version will be given in the clinical study report.

Based on the plasma concentration-time data, the following PK parameters are planned to be estimated for ODM-201, ORM-16555, ORM-16497, ORM-15341, and docetaxel.

Main parameter:

ODM-201, ORM-16555, ORM-16497, ORM-15341:

AUC(0-12) ¹⁾

1) : Using the predose plasma sample also as 12 hour sample

Docetaxel:

AUC(0-t_{last})

Additional parameters:

ODM-201, ORM-16555, ORM-16497, ORM-15341:

AUC(0-8), AUC(0-t_{last}), C_{max}, t_{max}, t_{last}

Docetaxel:

C_{max}, t_{max}, t_{last}

To ensure the blinding of the PK-evaluator during preliminary PK evaluation for DMC review, concentration data with masked subject IDs will be provided by Bioanalytics. The evaluation results will then be transferred to an unblinded statistician at the DMC.

Population PK evaluation

In all other randomized subjects:

Individual concentration-time data of ODM-201, ORM-16555, ORM-16497, and ORM-15341 will be provided in a clinical study report appendix. The concentration data from this study, which might be augmented with PK data from other studies, will be analyzed to evaluate the variability of the PK of ODM-201, of the respective diastereomers ORM-16555 and ORM-16497, and of the metabolite ORM-15341 in this phase III population. The possible effect of relevant covariates on the PK of these compounds might also be evaluated. In addition, exploratory exposure-response analyses of efficacy, PD or safety parameters, e.g. OS, PSA response rate, etc., are planned. A population PK approach will be applied for these evaluations, which will be described in detail in a separate Modeling & Simulation (M&S) Plan and the results will be reported separately in the M&S Report.

9.6 Safety

9.6.1 Adverse events

9.6.1.1 Definitions

Definition of adverse event

In a clinical study, an AE is any untoward medical occurrence (i.e. any unfavorable and unintended sign [including abnormal laboratory findings], symptom, or disease) in a subject or clinical investigation subject after providing written IC for participation in the study. Therefore, an AE may or may not be temporally or causally associated with the use of a medicinal (investigational) product.

A surgical procedure that was planned prior to the start of the study by any physician treating the subject should not be recorded as an AE (however, the condition for which the surgery is required may be an AE).

In the following differentiation between medical history and AEs, the term “condition” may include abnormal physical examination findings, symptoms, diseases, laboratory, or ECG results.

- Conditions that started before signing of IC and for which no symptoms or treatment are present until signing of IC will be recorded as medical history (e.g. seasonal allergy without acute complaints).
- Conditions that started before signing of IC and for which symptoms or treatment are present after signing of IC, at unchanged intensity, will be recorded as medical history (e.g. allergic pollinosis).
- Conditions that started or deteriorated after signing of the ICF including those recorded as medical history will be documented as AEs. This includes intercurrent illnesses.

Definition of a serious adverse event

An SAE is classified as any untoward medical occurrence that, at any dose, meets any of the following criteria:

- a. Results in death
- b. Is life threatening

The term ‘life threatening’ in the definition refers to an event during which the subject was at risk of death at the time of the event. It does not refer to an event that, hypothetically, might have caused death if it were more severe.

- c. Requires in-subject hospitalization or prolongation of existing hospitalization

A hospitalization or prolongation of hospitalization will not be regarded as an SAE if at least one of the following exceptions is met:

- The admission results in a hospital stay of less than 12 hours

- The admission is pre-planned
(i.e. elective or scheduled surgery arranged prior to the start of the study)
- The admission is not associated with an AE
(e.g. social hospitalization for purposes of respite care)

However, it should be noted that invasive treatment during any hospitalization may fulfill the criterion of ‘medically important’ and as such may be reportable as an SAE, dependent on clinical judgment. In addition, where local regulatory authorities specifically require a more stringent definition, the local regulation takes precedence.

d. Results in persistent or significant disability/incapacity

Disability means a substantial disruption of a person’s ability to conduct normal life functions.

- e. Is a congenital anomaly/birth defect
- f. Is another medically important serious event as judged by the Investigator

9.6.1.2 Classifications for adverse event assessment

All AEs will be assessed and documented by the Investigator according to the categories detailed below.

9.6.1.2.1 Seriousness

For each AE, the seriousness must be determined according to the criteria given in Section [9.6.1.1](#).

9.6.1.2.2 Intensity

The severity (or intensity) of AEs should be documented using NCI-CTCAE v 4.03. If no exact matching code is available in NCI-CTCAE v 4.03, the following guide should be used:

- CTC 1 = Mild AE: Transient in nature and generally not interfering with normal activities
- CTC 2 = Moderate AE: Sufficiently discomforting to interfere with normal activities
- CTC 3 = Severe AE: Prevents normal activities
- CTC 4 = Life threatening (subject was at risk of death at the time of the event)
- CTC 5 = Results in death (fatal)

9.6.1.2.3 Causal relationship

The assessment of the causal relationship between an AE and the administration of treatment is a clinical decision based on all available information at the time of the completion of the eCRF. The causal relationship should be assessed to ODM-201/placebo as well as the background docetaxel treatment independently.

The assessment is based on the question whether there was a “reasonable causal relationship” to ODM-201/placebo or docetaxel in question.

Possible answers are “yes” or “no.”

An assessment of “no” should include either:

1. The existence of a clear alternative explanation (e.g. mechanical bleeding at surgical site)
or
2. A lack of plausibility (e.g. the subject is struck by an automobile without evidence that ODM-201/placebo or docetaxel caused disorientation that may have led to the event, or cancer develops a few days after the first administration of a ODM-201/placebo or docetaxel.

An assessment of “yes” indicates that there is a reasonable suspicion that the AE is associated with the use of ODM-201/placebo or docetaxel.

Important factors to be considered in assessing the relationship of the AE to ODM-201/placebo or docetaxel include:

- The temporal sequence relating the event to drug administration. The event should occur after the drug is given. The length of time from drug exposure to event should be evaluated in the clinical context of the event.
- Recovery on drug discontinuation (dechallenge) and recurrence on drug re-introduction (rechallenge). The subject’s response after dechallenge or the subject’s response after rechallenge should be considered in the view of the usual clinical course of the event in question.
- Underlying, concomitant, intercurrent diseases: Each event should be evaluated in the context of the natural history and course of the disease being treated and any other disease the subject may have.
- Concomitant medication or treatment: The other drugs the subject is taking or the treatment the subject receives should be examined to determine whether any of these may be suspected as a cause for the event in question.
- Known response pattern for this class of drug: Clinical/preclinical.
- Exposure to physical and/or mental stresses: The exposure to stress might induce adverse changes in the recipient and provide a logical and better explanation for the event.
- The pharmacology and PK of the study treatment: The PK properties (absorption, distribution, metabolism and excretion) of the study treatment, coupled with the individual subject’s PD should be considered.
- The assessment is not possible.

Causal relationship to protocol-required procedure(s)]

The assessment of a possible causal relationship between the AE and protocol-required procedure(s) is based on the question whether there was a “reasonable causal relationship” to protocol-required procedure(s).

Possible answers are “yes” or “no”

9.6.1.2.4 Action taken with study treatment

Any action on study treatment to resolve the AE is to be documented using the categories listed below:

- Drug withdrawn
- Drug interrupted
- Dose reduced
- Dose not changed
- Not applicable
- Unknown

9.6.1.2.5 Other specific treatment(s) of adverse events

- None
- Remedial drug therapy
- Other

9.6.1.2.6 Outcome

The outcome of the AE is to be documented using the following categories:

- Recovered/resolved
- Recovering/resolving
- Recovered/resolved with sequelae
- Not recovered/not resolved
- Fatal
- Unknown

9.6.1.3 Assessments and documentation of adverse events

All AEs occurring from the time the subject has signed the ICF until 30 (+7) days after the last dose of study drug must be entered in the subject's eCRF. All treatment-emergent AEs must be followed until resolution or stabilization, wherever feasible.

The type of information that should be assessed and recorded by the Investigator for each AE is listed in Section [9.6.1.2](#).

“Death” should not be recorded as an AE on the AE page. Instead, “death” is the outcome of underlying AE(s).

For all SAEs, the Sponsor has to carry out a separate assessment for expectedness, seriousness, and causal relationship to study drug.

A laboratory test abnormality that is considered to have a clinically relevant effect on the subject (e.g. causing the subject to withdraw from the study, requiring treatment or causing apparent clinical manifestations, or judged relevant by the Investigator) should be reported as an AE.

Documentation of an AE must be supported by an entry in the subject's medical file.

Each event should be described in detail, along with start and stop dates, severity, relationship to investigational product, action taken, and outcome.

9.6.1.4 Reporting of serious adverse events

The definition of SAEs is given in Section 9.6.1.1. Each SAE must be followed up until resolution or stabilization by submission of updated reports to the designated recipient.

Investigator's notification of the Sponsor

All investigators will be thoroughly instructed and trained on all relevant aspects of the Investigator's reporting obligations for SAEs. This information, including all relevant contact details, is summarized in the Investigator site file. This information will be updated as needed.

All SAEs occurring during the observation period defined in Section 9.6.1.1 must immediately (within 24 hours of the Investigator's awareness) be reported to the recipient detailed in the Investigator's file. An SAE form must also be completed within 24 hours of the Investigator awareness and forwarded to the designated recipient. Each SAE must be followed up until resolution or stabilization by submission of updated reports to the designated recipient.

Serious AEs that occur 30 (+7) days after the last dose of study drug during Active Follow-up or Long-term Follow-up should be reported on the SAE form, if the Investigator feels that there is a reasonable possibility for the event to have been caused by the study subject's participation in the study. In such cases, the SAEs will be processed by the Sponsor according to all applicable regulations.

If disease progression leads to signs and symptoms that meet the criteria for seriousness (e.g. hospitalization), the associated signs and symptoms may be reported as an SAE, but the underlying cause (that is, "recurrent disease" or "metastatic disease") should not be reported as an SAE. In this case, recurrent or metastatic disease should be mentioned on the SAE form as an "alternative explanation."

An isolated laboratory abnormality that is assigned a severity rating of Grade 4, as defined in NCI-CTCAE v 4.03, is not reportable as an SAE unless the Investigator assesses that the event meets standard International Conference on Harmonization criteria for an SAE. Similarly, a baseline laboratory abnormality that is part of the disease profile should not be reported as an SAE when assigned a severity rating of Grade 4, as defined in NCI-CTCAE v 4.03, specifically when this abnormality is allowed (or not excluded) by the protocol inclusion/exclusion criteria (see Sections 6.1 and 6.2). If an Investigator is in doubt about the applicable reporting obligations, he/she should consult with the study monitor for the Sponsor. The NCI-CTCAE version 4.03 Grade 4 laboratory abnormalities will be documented and will be reviewed on a regular basis.

Notification of the IECs / IRBs

Notification of the IECs/IRBs about all relevant events (e.g. SAEs or SUSARs) will be performed by the Sponsor/delegate and/or by the Investigator according to all applicable regulations.

Notification of the authorities

The Sponsor will expedite all suspected unexpected SAEs and SUSARs as well as other safety issues requiring expedited reporting to the relevant authorities within applicable timelines.

Sponsor's notification of the investigational site

The Sponsor/delegate will inform all investigational sites about reported relevant events (e.g. SUSARs) according to all applicable regulations.

9.6.1.5 Expected adverse events

For this study, the applicable reference document is the most current version of the IB.

Overview listings of frequent events that have occurred so far in the clinical development are shown in the current IB. If relevant new safety information is identified, the information will be integrated into an update of the IB and distributed to all participating sites.

The expectedness of AEs will be determined by the Sponsor according to the applicable reference document and according to all local regulations.

9.6.1.6 Adverse events of special safety interest

None.

9.6.2 Pregnancies

Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the study treatment and for 3 months after the end of the treatment with ODM-201, and 6 months after treatment with docetaxel (inclusion criterion).

Pregnancies inadvertently fathered by study subjects during the study should be reported and followed up using the Bayer Pregnancy Monitoring Forms, if permissible by local legislation.

For a pregnancy in the partner of a male study subject, all efforts will be made to obtain similar information on course and outcome, subject to the partner's consent.

For all reports, the forms provided are to be used. The report should be submitted within the same timelines as an SAE, although a pregnancy per se is not considered an SAE.

9.6.2.1 Laboratory safety assessments

The PSA and testosterone values are to be assessed by a central laboratory during the study period. Blood samples to test alkaline phosphatase for stratification purpose will be analyzed

in central laboratory and thereafter by local laboratory. Other laboratory assessments are to be performed by local laboratories.

Instruction on sample collection, handling, labeling, storage, and shipment will be provided in the laboratory manual.

The following laboratory safety assessments will be performed during the screening period, study treatment period and at the end-of-study treatment visit.

Hematology:

- Hemoglobin
- Hematocrit
- Erythrocyte count
- Leukocyte count
- Differential count (lymphocytes, monocytes, eosinophils, neutrophils, basophils)
- Mean corpuscular hemoglobin
- Mean corpuscular volume
- Mean corpuscular hemoglobin concentration
- Thrombocytes

Chemistry:

- Albumin
- Alanine aminotransferase (ALT)
- Aspartate aminotransferase (AST)
- Alkaline phosphatase
 - Note: Blood samples to test alkaline phosphatase for stratification purpose will be analyzed in central laboratory and thereafter by local laboratory
- Bilirubin total (+ direct and indirect bilirubin if total bilirubin is $>1.0 \times \text{ULN}$)
- Calcium
- Blood urea nitrogen
- Creatinine
- Serum glucose
- Total protein
- Lactate dehydrogenase
- Potassium

- Sodium
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), using a LC-MS/MS method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT treatment. In addition, SHBG and albumin concentrations will be determined to calculate the concentration of non-protein-bound (free) testosterone in these subjects.
- PSA (total)
- Chromogranin A (only at Screening)
- Prothrombin time, expressed as international normalized ratio, in subjects under vitamin K antagonist therapy (local assessment)

Urinalysis:

- Glucose
- Protein
- Erythrocytes
- Leukocytes

In the event of implausible results, the laboratory may measure additional parameters to assess the quality of the sample (e.g. clotted or hemolyzed) and to verify the results. The results from such additional analyses may neither be included in the clinical database of this study nor evaluated further. If the results are relevant, the Investigator will be informed to determine follow-up activities outside of this protocol.

9.6.2.2 Physical examination

Physical examination will be performed and weight will be measured at all visits (except at Day 1 Visit if already performed at Screening). Height will be recorded at Screening only.

Abnormal physical examination findings are recorded either as medical history or as AEs (see Section [9.6.1.1](#)).

9.6.2.3 12-lead ECGs

All 12-lead ECGs will be recorded in a supine position after at least 10 minutes rest.

9.6.2.4 Vital signs

All vital sign assessments will be recorded in a supine position after at least 10 minutes rest.

9.7 Other procedures and variables

Other exploratory variables in this study are pain intensity and pain interference scores from the BPI-SF, medical resource use, and assessment of biomarkers.

In addition to calculating a pain score based on the worst pain question (i.e., worst pain subscale [WPS] score) on the BPI-SF (Appendix 16.4), the subject's responses on the BPI-SF can be used to calculate a pain score intensity and a pain interference score (29).

9.7.1 Biomarker investigations

Biomarker status will be correlated with clinical outcome to explore whether any biological targets appear to define subject populations that are particularly sensitive or resistant to ODM-201. Details of the biomarker analyses will be described in a separate statistical analysis plan (SAP) and the results of these analyses will be provided as a separate biomarker report.

Tumor biomarker studies will be performed to investigate the mechanisms by which ODM-201 may work in subjects with mHSPC and will use tumor biomarkers circulating in plasma or, if available, biopsies either collected prior to study entry or at disease progression.

Tumor biomarkers may be categorized as “non-genetic” (involving protein) or “genetic” (involving ribonucleic acid [RNA] or deoxyribonucleic acid [DNA]). In the current study, analyses of both non-genetic and genetic biomarkers are planned. The analysis of tumor genetic markers by tumor DNA sequencing and/or gene expression studies with tissue samples taken from the primary tumor and metastases will support identification of biomarkers contributing to clinical benefit to therapy. Utmost care will be taken to protect subject identity during biomarker analyses.

Biomarkers related to drug safety, efficacy, or mechanism of action will be analyzed from whole blood and involve targeted genetic testing. Targeted genetic testing will focus on single gene polymorphisms of genes involving, but not limited to, hepatotoxicity, metabolism, or targets involved in drug activity. Identified gene variants aim to predict the best individuals to treat, either selecting for toxicity avoidance or for positive benefit.

In addition to the biomarkers listed above, other biomarkers deemed relevant to gain further knowledge about the pathomechanism of the disease or about the drug (i.e. mode of action related effect or safety of the drug) may be measured, based on newly emerging data from other ongoing studies and / or literature data.

Details on the collection, processing, storage and shipment of biomarker samples will be provided in separate documents (e.g. sample handling sheets or lab manual).

9.7.1.1 Plasma-based biomarker analysis

Subjects will be asked to provide blood samples for biomarker analyses, which will be obtained during regular study visits in which blood will be drawn for other scheduled but unrelated laboratory tests.

Genetic biomarkers from plasma:

These samples are intended to isolate plasma to study tumor markers circulating in blood such as circulating tumor DNA (ctDNA) or circulating tumor RNA. Candidates of genes for the evaluation from ctDNA are AR, PIK3CA, and phosphatase and tensin homolog (PTEN). The types of analyses may comprise the identification of mutations or splice variants.

Furthermore, these plasma samples will serve as source for non-coding microRNA, as possible markers describing treatment response.

Non-genetic biomarkers from plasma:

Plasma samples will also be used to quantify the circulating levels of various proteins of interest, to attempt to identify a protein signature that correlates with drug response.

Plasma samples for the analyses of tumor biomarkers (genetic and non-genetic biomarkers) will be prepared from blood samples obtained at the following time points:

- Screening visit and on Day 1, Visit 1 (pre-dose) once eligibility criteria have been confirmed
- Visit 2 and subsequent Visits
- End of Treatment Visit

On treatment days, blood will be drawn for preparation of plasma samples prior to drug administration, if possible.

In addition to the proteins and genes listed above, other biomarkers deemed relevant to this study will be measured.

Data from the biomarker analyses may be correlated with measures of clinical efficacy or toxicity obtained in this study.

9.7.1.2 Circulating tumor cells

CTCs may serve as source of tumor metastases circulating in blood and may reflect the tumor phenotype or genotype. Enumeration of CTCs may serve as description of tumor burden.

Circulating tumor cells will be analyzed in detail from whole blood samples collected at various time points to assess the effect of ODM-201 in decreasing numbers of CTCs as well as to characterize the phenotype and genotype of CTCs such as presence of AR mutations, AR splice variants, and other genomic markers.

A whole blood sample for CTC analysis will be obtained at Visit 1 (once eligibility criteria have been confirmed), Visit 2 and EOT.

9.7.1.3 Tumor-based biomarker analysis

From subjects for which tumor tissue specimens (e.g. from biopsy) are available, tumor specific biomarker analyses are intended to be used for the following purposes: (1) to evaluate mutations, copy numbers or gene rearrangements in known oncogenes such as AR, TMPRSS, cMYC, PI3K, and CYP17 extracted from tumor tissue specimens; (2) to analyze the gene expression of genes of interest such as AR, cMYC, PTEN, PI3K, and AR-regulated genes in RNA extracted from tumor tissue specimens; (3) to evaluate expression of non-coding tumor relevant RNAs such as microRNAs.

Provided sufficient tissue quality and quantity, tumor tissue specimens are also intended to be used to quantify the expression of proteins of interest, to attempt to identify a protein signature that correlates with drug response.

As per Investigator's discretion, and if technically feasible, biopsy at the time of disease recurrence is highly encouraged to be taken. This sample would allow generation of information about the identification of tumor biomarkers playing a role in therapy resistance for which the subject may benefit for further guidance. Note, however, that this is not considered to be mandatory by this study protocol.

9.7.1.4 Genetic biomarker analysis from whole blood

Biomarkers related to drug safety, efficacy, or mechanism of action may be analyzed from whole blood and involve targeted pharmacogenetic testing. These studies will be conducted aiming to find gene variants associated with responses to ODM-201 treatment and to predict the best individuals to treat, either selecting for toxicity avoidance or for positive benefit.

Targeted pharmacogenetic testing will focus on single gene polymorphisms (SNPs) on genes involved but not limited to hepatotoxicity, metabolism, or targets involved in drug activity. Gene variants identified aim.

Unless precluded by local guidelines, subjects will be asked to sign a separate pharmacogenetic ICF. A whole blood sample for pharmacogenetic testing will be obtained at Visit 1 (once eligibility criteria have been confirmed) from subjects who have signed a separate pharmacogenetic ICF. From these blood samples, DNA will be isolated which will be used for the analysis SNP candidates such as CYP17.

9.7.2 Resource utilization

Information on healthcare resource use (e.g. hospitalization visits, physician visits etc.) that is associated with the management of AEs as well as non-protocol-driven subject monitoring will be collected by e-CRF that is completed by the physician at each visit.

9.7.3 Subject-reported outcomes

For this study, health-related QoL will be measured using the NCCN-FACT FPSI-17 questionnaire.

The NCCN-FACT FPSI-17 is a validated instrument that was developed to assess symptoms of prostate cancer, symptoms of treatment of prostate cancer, and health related QoL of prostate cancer subjects. The instrument was developed in accordance with recent Food and Drug Administration guidance for development of instruments for ePRO. The instrument contains 17 items, each of which utilizes a Likert scale with 5 possible responses (30). The 10 items reflect disease-related physical symptoms of disease and the responses on the items are to be summed to calculate a disease-related physical symptom subscale score. One item represents emotional symptoms of disease and the response to that item is used to calculate a disease-related emotional symptom subscale score. Four items represent treatment-related symptoms and the responses to these items are summed to calculate a treatment side effect subscale score. Finally, 2 items represent functional well-being and responses to those items are summed to calculate a functions / well-being subscale score. The NCCN-FACT FPSI-17 will be self-administered by the subject via ePRO device. Some subjects may skip an item altogether if they feel that it is not applicable to them. The questionnaire is to be completed at baseline either within 7 days before randomization or at Visit 1 (Day 1), at all treatment visits

(at Visit 2 and subsequent visits), at the End of Treatment (EOT), and approximately every 12 weeks during Active Follow-up.

Brief Pain Inventory – Short form is a validated instrument totaling 15 items to assess subject-reported levels of pain that has a precedent of being used in clinical trials with the goal of obtaining label claims based on the results captured with this instrument. A reduced set of 11 items is used in this trial. This reduced questionnaire is self-administered and designed to evaluate the intensity of, and the impairment caused by, pain. All BPI-SF items are scored using rating scales. Four items measure pain intensity (pain now, average pain, worst pain, and least pain) using 0 (“no pain”) to 10 (“pain as bad as you can imagine”) numeric rating scales, and 7 items measure the level of interference with function caused by pain (general activity, mood, walking ability, normal work, relations with other people, sleep, and enjoyment of life) using 0 (no interference) to 10 (complete interference) rating scales.

The items are aggregated into 2 dimensions: (1) Pain severity index, using the mean of the 4 items on the pain intensity, and (2) Function interference index, using the mean of the 7 pain interference items. All 4 severity items must be completed for aggregating the pain severity index. The function interference index is scored as the mean of the item scores multiplied by 7, given that more than 50% or 4 of 7, of the items have been completed.

The questionnaire is to be completed via ePRO at Baseline (Screening or Visit 1), at treatment visits (at Visit 2 and subsequent visits), at EOT, and approximately every 12 weeks during Active Follow-up.

For each questionnaire, a subject-reported outcome (PRO) information sheet must be completed by the study nurse/Investigator at each visit at which the questionnaires are scheduled to be administered regardless of whether or not the questionnaires were completed, in order to document information on questionnaire assessment such as completion status, date of administration, and reasons for non-completed questionnaires, missing assessments, etc.

9.8 Appropriateness of procedures / measurements

All efficacy and safety parameters, as well as the methods to measure them, are standard variables /methods in clinical studies and/or clinical practice. They are widely used and generally recognized as reliable, accurate, and relevant.

The NCCN-FACT FPSI-17 is a validated instrument that was developed to assess symptoms of prostate cancer, symptoms of treatment of prostate cancer, and health related QoL of prostate cancer subjects (28).

Brief Pain Inventory – Short form is validated to assess subject-reported levels of pain that has a precedent of being used in clinical trials with the goal of obtaining label claims based on the results captured with this instrument.

The NCI-CTCAE v 4.03 is a list of AE terms harmonized with the Medical Dictionary for Regulatory Activities (MedDRA) Terminology (version 12.0) at the AE term level and associated with a severity grading scale that is widely used within the oncology research community as the standard for documentation and analysis of AEs occurring in cancer research. This tool is useful in defining the maximum tolerated dose of study drugs, in making

recommendations concerning modification of dose and dose schedule of study drugs, and in comparing safety profiles of therapeutic interventions.

10. Statistical methods and determination of sample size

10.1 General considerations

Statistical analysis will be performed using SAS; the version used will be specified in the SAP.

10.2 Analysis sets

Full analysis set (FAS): All randomized subjects. The FAS will be used in the analysis of all efficacy endpoints. Subjects will be included in FAS according to the treatment to which they are randomized.

Safety analysis set: All randomized subjects who have received at least 1 dose of study drug. This safety population will be used in the analyses of all safety endpoints. Subjects will be included in the analyses according to the treatment they actually received.

Pharmacokinetic Analysis Set (PKS): The first 20 subjects, for whom mandatory dense PK sampling will be done, will be included in the Pharmacokinetic Analysis Set, as long as they have received at least 3 days of uninterrupted study drug treatment as well as one cycle of docetaxel and have at least one post dose PK measurement.

10.3 Variables and planned statistical analyses

The complete list of variables to be analyzed for this study will be provided in the SAP.

10.3.1 Primary efficacy variable

The primary efficacy variable is OS, defined as the time (in days) from date of randomization until death from any cause.

Analysis of OS

All randomized subjects (FAS) will be included in the primary analysis of OS, the primary efficacy endpoint. The analysis will be performed when approximately 509 deaths are observed. The primary analysis of OS will be a stratified log-rank test with the same stratification factors as used for randomization.

The null hypothesis that there is no difference in OS between treatment arms, which is equivalent to a hazard ratio (HR) of 1, will be tested against the alternative hypothesis that the HR of ODM-201 over placebo is below 1.

The HR (ODM-201 group/placebo) for OS and its 95% confidence interval will be calculated using the Cox model, stratified by the same factors as stated above. Kaplan-Meier (KM) estimates for OS will be presented for each treatment group. The KM estimates at time points such as 3 months, 6 months, etc., together with corresponding 95% confidence intervals (CIs)

and the differences of these estimates between the ODM-201 group and the placebo group will be presented.

The overall 1-sided type I error rate for the analysis of OS is 0.025. Two formal interim analyses of OS for futility and efficacy are planned. A DMC will oversee these interim analyses. For further details on these interim analyses see Section 10.5.

10.3.2 Secondary efficacy variables

The secondary efficacy variables include:

- **Time to castration-resistant prostate cancer**, defined as the time to PSA progression (Section 10.3.4) with serum testosterone being at castrate level <0.50 ng/mL, or the time to progression by soft tissue lesions (as described in Section 9.2.2.3) or time to progression by bone lesions (as described in Section 9.2.2.3), whatever comes first.
- **Time to initiation of subsequent antineoplastic therapy**, defined as the time from randomization to initiation of first subsequent antineoplastic therapy for prostate cancer.
- **Symptomatic skeletal event free survival**, defined as the time from randomization to the first occurrence of SSE or death from any cause, whichever comes first. An SSE is defined as EBRT to relieve skeletal symptoms, or new symptomatic pathologic bone fracture, or occurrence of spinal cord compression or tumor-related orthopedic surgical intervention, whichever comes first.
- **Time to first symptomatic SSE**, defined as the time from randomization to the first occurrence of SSE. An SSE is defined as EBRT to relieve skeletal symptoms, or new symptomatic pathologic bone fracture, or occurrence of spinal cord compression or tumor-related orthopedic surgical intervention, whichever comes first.
- **Time to initiation of opioid use for ≥ 7 consecutive days**, defined as the time from date of randomization to the date of first opiate use for ≥ 7 consecutive days. Time to opiate use will be determined by analgesic use captured via eCRF (recorded analgesics, see schedule of assessments).
- **Time to pain progression is defined as** the interval from randomization to the first date a subject experiences a pain progression. Pain will be assessed with the BPI-SF questionnaire (Appendix 16.4) during the visit, prior to any procedures or examination by physician. Subjects who have not experienced pain progression at the time of analysis will be censored on the last date the subject was known to have not progressed. Subjects with no on study assessment or no baseline assessment will be censored at the date of randomization.

For **asymptomatic subjects** (WPS 0 at baseline), pain progression is defined as an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) score from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart

OR

initiation of short- or long-acting opioid use for pain.

For **symptomatic subjects** (WPS >0 at baseline), pain progression is defined as an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) score from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart and a WPS of ≥ 4

OR

initiation of short- or long-acting opioid use for pain.

- **Time to worsening of physical symptoms of disease** is defined as the interval from randomization to the first date a subject experiences an increase in physical symptoms based on the NCCN-FACT-FPSI-17 questionnaire. Symptoms and QoL will be assessed with the NCCN-FACT-FPSI-17 questionnaire (Appendix 16.3) during the visit, prior to any procedures or examination by physician. An increase in physical symptoms of disease is defined as 3-point decrease (a lower score = higher symptom burden) from baseline in the physical symptoms of disease subscale (FPSI-DRS-P subscale of the NCCN-FACT-FPSI-17 questionnaire) observed at 2 consecutive evaluations ≥ 4 weeks apart.

Secondary efficacy variables analysis

The secondary efficacy variables will be analyzed for the FAS population unless otherwise specified in SAP. Time-to-event endpoints will be analyzed using stratified log-rank test with randomization stratification factors. Hazard ratio and 95% CI will be provided using the Cox model stratified by the same factors as stated above. Detailed analysis methods and the plan for type 1 error control for secondary endpoints will be specified in SAP.

10.3.3 Safety variables

The safety variables include:

- AEs until the end-of-study treatment visit
- SAEs until the end-of-study treatment visit
- Study drug-related SAEs until the end of Long-term Follow-up
- Vital signs: BP and heart rate (HR)
- 12-lead electrocardiogram (ECG)
- Physical examination
- Laboratory safety assessments (hematology, clinical chemistry, urinalysis)

Safety analysis

AEs will be coded using MedDRA. Descriptive summary tables will be presented on all safety parameters by treatment group. The safety assessment period is from the first dose of study drug until 30 days (+7 days) after the last treatment with ODM-201/placebo. Subjects will be monitored for AEs using the NCI-CTCAE v 4.03.

The AEs and safety laboratory parameters will be presented with their worst NCI-CTCAE v 4.03 grade (Section 16.2).

10.3.4 Other exploratory variables

- **Rate of absolute PSA response at 6 and 12 months**

Absolute PSA response is defined as blood PSA level <0.2 ng/mL, confirmed by a second subsequent PSA value <0.2 ng/mL 3 or more weeks later. Rate of absolute PSA response is defined as the number of subjects with absolute PSA response, divided by the total number of subjects evaluable for absolute PSA response. Rate of absolute PSA response will be evaluated on subject data up to 6 months and 12 months after randomization.

- **Rate of relative PSA response at 3, 6 and 12 months**

Relative PSA response is defined as a $\geq 30\%$ reduction of the blood PSA level compared to the screening value, confirmed by a second subsequent PSA value with a $\geq 30\%$ reduction from Screening 3 or more weeks later. Rate of relative PSA response is defined as the number of subjects with relative PSA response, divided by the total number of subjects evaluable for relative PSA response. Rate of relative PSA response will be evaluated on subject data up to 3 months, 6 months, and 12 months after randomization.

- **Time to PSA progression**

Prostate-specific antigen progression is defined as a $\geq 25\%$ increase above the nadir (lowest Screening or baseline) value, which is confirmed by a second value 3 or more weeks later, and an increase in absolute value of ≥ 2 ng/mL above nadir.

The time to PSA progression is defined as the time (in days) from the date of randomization to the date of first PSA progression. Subjects without PSA progression as of database cut-off, whether or not surviving, will be censored at the last total PSA lab assessment date.

- **Changes in symptom burden and QoL based on the NCCN-FACT FPSI-17 questionnaire (see Section 10.3.8).**
- **Medical resource use (see Section 9.7.2)**
- **Pharmacokinetics (see Section 9.5.2)**
- **Biomarker evaluations (see Section 9.7.1)**

The details of the analysis for the exploratory variables to be analyzed for this study will be provided in the SAP.

Missing data / drop outs

Every effort should be made to retain subjects who discontinue the treatment period for any reason. These subjects are to be encouraged to remain on the study for follow-up of primary, secondary, and exploratory endpoints.

The method used for imputation of missing data will be described in the SAP (see Section [11.4](#)).

10.3.5 Pharmacokinetics

Pharmacokinetics in the first 20 subjects

The concentration-time courses of all substances will be tabulated separated by treatment. The following statistics will be calculated for each of the sampling points: arithmetic mean, standard deviation (SD), and coefficient of variation (CV); geometric mean, geometric SD (re-transformed SD of the logarithms), and CV; minimum, median, maximum value, and the number of measurements.

Means at any time will only be calculated if at least 2/3 of the individual data were measured and were above the lower limit of quantification (LLOQ). For the calculation of the mean value, a data point below LLOQ will be substituted by one-half of this limit. In tables showing mean values, where values below LLOQ are included in the calculation of mean values, these means will be marked.

Individual and geometric mean concentration versus time curves of all substances (using the actual sampling times for individual plots and the planned sampling times for mean plots) will be plotted by treatment using both a linear and semilogarithmic scale.

Pharmacokinetic characteristics ($t_{\max}$, t_{last} excluded) will be summarized by the statistics mentioned above. The $t_{\max}$, t_{last} will be described utilizing minimum, maximum, and median as well as frequency counts.

Pharmacokinetics in all subjects

Individual concentration-time data of ODM-201, ORM-16555, ORM-16497, and ORM-15341 will be provided in a clinical study report appendix. All other parameters will be described in detail in a separate M&S report.

10.3.6 Biomarker analyses

Results from exploratory biomarker studies will be reported in a separate biomarker report.

10.3.7 Medical Resource use

Medical resource use is described in Section [9.7.2](#).

10.3.8 Subject-reported outcomes

Subject-reported outcome procedures are described in Section [9.7.3](#).

10.3.9 Baseline and demographic characteristics

Baseline and demographic characteristics will be summarized by descriptive statistics by treatment group ODM-201 for the FAS population.

In addition, exploratory exposure-response analyses of efficacy, PD, or safety parameters, e.g. OS, PSA response rate, etc., are planned.

10.4 Determination of sample size

The sample size of the study is based on the primary endpoint of OS. The study is designed to have 90% power to detect a 33% increase in median time of OS with ODM-201 compared to placebo (from 60 to 80 months, corresponding to a HR of 0.75) with a 1-sided alpha of 0.025. The OS data will be considered as mature when approximately 509 deaths are observed.

With the additional assumptions that subjects will enroll at a rate of 50 subjects per month, exponential distributions of the OS event times, median time of OS in the control group of 60 months, 5% dropout rate of subjects, and a 6-months enrollment ramp-up period, it follows that approximately 1,300 subjects must be enrolled to observe 509 deaths after approximately 70 months.

10.5 Planned interim analyses

Two formal interim analyses of OS are planned. The first interim analysis for futility will be performed after approximately 153 deaths are observed. The second interim analysis for both futility and efficacy will be performed after approximately 357 deaths are observed. The DMC (see Section 13.1.1) will oversee these interim analyses. Detailed analysis methods and stopping rules will be specified in the SAP and the DMC charter.

For the futility interim and final analyses together, a one-sided overall beta of 0.1 will be used. Stopping boundaries will be calculated with an O'Brien-Fleming alpha-spending function based on the actual number of events observed up to the cut-off date. The interim futility analyses are planned when approximately 153 deaths and approximately 357 events are observed (information fraction=0.3 and 0.7). The critical boundaries for the futility analyses will be calculated separately, in order to not interfere with the type 1 error of the efficacy analyses.

For the efficacy interim and final analysis together, a one-sided overall alpha of 0.025 will be used. An early stopping boundary will be calculated with an O'Brien-Fleming alpha-spending function based on the actual number of events observed up to the cut-off date. The interim efficacy analysis is planned when approximately 357 deaths are observed (information fraction=0.7).

A DMC (see Section 13.1.1) will be instituted for independent review of ongoing data from this study in accordance with a separate DMC charter. The DMC will operate independently of the Sponsor and Investigators.

The DMC will review safety data as per a separate DMC charter.

11. Data handling and quality assurance

11.1 Data recording

The data collection tool for this study will be an eCRF; a validated electronic data capture system called RAVE. Subject data necessary for analysis and reporting will be entered/transmitted into a validated database or data system (CIE/TOSCA; SAS).

Data required according to this protocol will be recorded by investigational site personnel via data entry into the internet based electronic data capture (EDC) software system RAVE, which Bayer has licensed from Medidata Solutions Worldwide. RAVE has been validated by Medidata Solutions Worldwide and Bayer for use in its clinical studies. RAVE allows for the application of software logic to set up data entry screens and data checks to ensure the completeness and accuracy of the data entered by the site personnel. Bayer extensively applies the logic to ensure data are complete and reflect the clinical data requirements of the study. Data queries resulting from the application of the software logic are resolved by the site personnel. The data are stored at a secure host facility maintained by Medidata Solutions Worldwide and transferred on a periodic basis to Bayer's internal computer system via a secure Virtual Private Network.

All access to the RAVE system is through a password-protected security system that is part of the RAVE software. All internal Bayer and external Investigator site personnel seeking access must go through a thorough RAVE training process before they are granted access to RAVE for use in Bayer's clinical studies. Training records are maintained.

All personnel with access to the RAVE system are supported by a Service Desk staffed with trained personnel to answer questions and ensure access is maintained such that data entry can proceed in a timely manner.

The RAVE system contains a system-generated audit trail that captures any changes made to a data field, including who made the change, why the change was made and the date and time it was made. This information is available both at the Investigator's site and at Bayer. Data entries made in the RAVE EDC screens are supported by source documents maintained for all subjects enrolled in this study.

Source documentation

It is the expectation of the Sponsor that key data entered into the CRF has source documentation available at the site. Study-specific data not needed for the subject's routine medical care (QoL-related data: NCCN-FACT FPSI-17 and BPI-SF questionnaires) may be entered directly into the validated database or data system, without availability of corresponding source documentation.

The site must implement processes to ensure availability of all required source documentation. A source document checklist (not part of this protocol) will be used at the site to identify the source data for key data points collected and the monitor will work with the site to complete this.

Data recorded from screening failures

Data of 'only screened subjects' will be recorded at least as source data, as far as the reason for the premature discontinuation is identifiable. At a minimum, the following data should be recorded in the eCRF:

- Demographic information (subject number; year of birth / age; sex; if applicable, race/ethnicity)
- Relevant Inclusion and/or Exclusion criteria
- Date of IC
- Reason for premature discontinuation
- Date of last visit.

These data will be transferred to the respective database.

For screening failures with an SAE, the following data should be collected in the CRF in addition to the data specified above:

- All information related to the SAE such as:
 - Concomitant medication
 - Medical history
 - Other information needed for SAE complementary page

11.2 Monitoring

In accordance with applicable regulations, GCP, and Sponsor's/CRO's procedures, monitors will contact the site prior to the start of the study to review with the site staff the protocol, study requirements, and their responsibilities to satisfy regulatory, ethical, and Sponsor's requirements. When reviewing data collection procedures, the discussion will also include identification and documentation of source data items.

The Sponsor/designee will monitor the site activity to verify that the:

- Data are authentic, accurate, and complete.
Supporting data may be requested (example: blood glucose readings to support a diagnosis of diabetes).
- Safety and rights of subjects are being protected.
- Study is conducted in accordance with the currently approved protocol (including study treatment being used in accordance with the protocol).
- Any other study agreements, GCP, and all applicable regulatory requirements are met.

The Investigator and the head of the medical institution (where applicable) agrees to allow the monitor direct access to all relevant documents.

11.3 Data processing

Data will be collected as described in Section 11.1. Clinical data management will be performed in accordance with applicable Sponsor's/CRO's standards and data cleaning procedures. This is applicable for data recorded on CRF as well as for data from other sources (e.g. IXRS, laboratory, ECG, ePRO, adjudication committees).

For data coding (e.g. AEs, medication), internationally recognized and accepted dictionaries will be used.

After its initial release for biometrical analysis, the clinical database is planned to be re-opened for the inclusion of the following additional data: PK data, antibody data.

11.4 Missing data

The analysis of the primary endpoint is based on the intent to treat population. Subjects who discontinue the treatment prematurely or leave the study prematurely will not be replaced.

Handling of missing data, which in this context are events not observed for the time-to-event endpoints (primary variable and secondary variables), is implicitly handled in the statistical methods applied (log-rank test, Cox model) by censoring these subjects. Further aspects of handling of missing data will be discussed in the SAP.

11.5 Audit and inspection

To ensure compliance with GCP and regulatory requirements, a member of the Sponsor's (or a designated CRO's) quality assurance unit may arrange to conduct an audit to assess the performance of the study at the study site and of the study documents originating there. The Investigator/institution will be informed of the audit outcome.

In addition, inspections by regulatory health authority representatives and IEC(s)/IRB(s) are possible. The Investigator should notify the Sponsor immediately of any such inspection.

The Investigator/institution agrees to allow the auditor or inspector direct access to all relevant documents and allocate his/her time and the time of his/her staff to the auditor/inspector to discuss findings and any issues. Audits and inspections may occur at any time during or after completion of the study.

11.6 Archiving

Essential documents shall be archived safely and securely in such a way that ensures that they are readily available upon authorities' request.

Subject (hospital) files will be archived according to local regulations and in accordance with the maximum period of time permitted by the hospital, institution, or private practice. Where the archiving procedures do not meet the minimum timelines required by the Sponsor, alternative arrangements must be made to ensure the availability of the source documents for the required period.

The Investigator/institution notifies the Sponsor if the archival arrangements change (e.g. relocation or transfer of ownership).

The Investigator site file is not to be destroyed without the Sponsor's approval.

The contract with the Investigator/institution will contain all regulations relevant for the study center.

12. Premature termination of the study

The Sponsor has the right to close this study (or, if applicable, individual segments thereof [e.g. treatment arms; dose steps; centers]) at any time, which may be due, but not limited to, the following reasons:

- If risk-benefit ratio becomes unacceptable owing to, for example,
 - Safety findings from this study (e.g. SAEs)
 - Results of any interim analysis
 - Results of parallel clinical studies
 - Results of parallel animal studies
(e.g. toxicity, teratogenicity, carcinogenicity, or reproduction toxicity).
- If the study conduct (e.g. recruitment rate; dropout rate; data quality; protocol compliance) does not suggest a proper completion of the trial within a reasonable time frame.

The Investigator has the right to close his/her center at any time.

For any of the above closures, the following applies:

- Closures should occur only after consultation between involved parties. Final decision on the closure must be in writing.
- All affected institutions (e.g. IEC[s]/IRB[s]; competent authority[ies]; study center; head of study center) must be informed as applicable according to local law.
- All study materials (except documentation that has to remain stored at site) must be returned to the Sponsor. The Investigator will retain all other documents until notification is given by the Sponsor for destruction.
- In the event of a partial study closure, ongoing subjects, including those in post-study follow-up, must be taken care of in an ethical manner.

Details for an individual subject's withdrawal can be found in Section [6.4.1](#).

13. Ethical and legal aspects

13.1 Investigator(s) and other study personnel

All other study personnel not included in this section are identified in a separate personnel list (not part of this clinical study protocol) as appropriate. This list will be updated as needed; an abbreviated version with personnel relevant for the centers will be available in each center's Investigator site file.

Whenever the term ‘Investigator’ is noted in the protocol text, it may refer to either the PI at the site, or an appropriately qualified, trained and delegated individual of the investigational site.

The PI of each center must sign the protocol signature page and must receive all required external approvals (e.g. health authority, ethics committee, Sponsor) before subject recruitment may start at the respective center. Likewise, all amendments to the protocol must be signed by the PI and must have received all required external approvals before coming into effect at the respective center.

A complete list of all participating centers and their investigators, as well as all required signature documents, will be maintained in the Sponsor’s study file.

The global Sponsor of this study is identified on the title page of this protocol. If required by local law, local co-sponsors will be nominated; they will be identified on the respective country-specific signature pages.

13.1.1 Independent data monitoring committee

A DMC will be instituted to ensure ongoing safety of study subjects with respect to a risk/benefit assessment during periodic data review meetings, review results from planned interim analyses and provide a formal recommendation for continuation/termination of the study and monitor study conduct to ensure the overall integrity of trial is maintained. Recommendation for trial continuation will be guided by safety evaluations at all safety data reviews. The committee will include at least 3 independent oncologists and one independent statistician. Data review meetings will be held periodically as per separate DMC charter. Enrollment in the study will continue throughout the scheduled meetings of the DMC. Decisions on trial termination, protocol amendment, on hold of subject recruitment, or cessation of subject recruitment based on risk/benefit assessments will be made after recommendations from the DMC have been assessed by the Sponsor. Safety review meetings will be held in accordance with the DMC charter. Decisions on study termination or cessation of subject recruitment based on safety or OS outcomes will be made after recommendations from the DMC have been assessed by Bayer AG.

The DMC will also review the safety assessment, including PK, which will be performed after the first 20 subjects are randomized and have received at least 1 cycle of docetaxel. Blinding of Investigators, subjects and the Sponsor will be maintained during this assessment based on the preclinical mechanisms and known safety profiles for the ODM-201 and docetaxel (see Section 3.3), the toxicities from the combination are expected to be manageable. Therefore, enrollment will continue during this 20-subject safety review unless unexpected or excessive toxicities related to ODM-201 in addition to docetaxel plus ADT compared to the treatment of docetaxel plus ADT alone are reported, in which case subject enrollment will be placed on hold until completion of the DMC safety review. Further safety reviews to ensure subjects’ safety may be implemented for different subject populations following the first 20 subjects due to lack of experience in treatment of ODM-201 in addition to docetaxel plus ADT. The details of the evaluations will be described in the DMC charter.

13.2 Funding and financial disclosure

Funding

This study will be funded by its Sponsor.

Financial disclosure

Each Investigator (including principal and/or any sub-investigators) who is directly involved in the treatment or evaluation of research subjects has to provide a financial disclosure according to all applicable legal requirements. All relevant documentation will be filed in the trial master file.

13.3 Ethical and legal conduct of the study

The procedures set out in this protocol, pertaining to the conduct, evaluation, and documentation of this study, are designed to ensure that the Sponsor and Investigator abide by GCP guidelines and the guiding principles detailed in the Declaration of Helsinki. The study will also be carried out in keeping with applicable local law(s) and regulation(s).

Documented approval from appropriate IEC(s)/IRBs will be obtained for all participating centers/countries before start of the study, according to GCP, local laws, regulations, and organizations. When necessary, an extension, amendment or renewal of the IEC/IRB approval must be obtained and also forwarded to the Sponsor. The responsible unit (e.g. IEC/IRB, head of the study center/medical institution) must supply to the Sponsor, upon request, a list of the IEC/IRB members involved in the vote and a statement to confirm that the IEC/IRB is organized and operates according to GCP and applicable laws and regulations.

Strict adherence to all specifications laid down in this protocol is required for all aspects of study conduct; the Investigator may not modify or alter the procedures described in this protocol.

Modifications to the study protocol will not be implemented by either the Sponsor or the Investigator without agreement by both parties. However, the Investigator or the Sponsor may implement a deviation from, or a change of, the protocol to eliminate an immediate hazard(s) to the trial subjects without prior IEC/IRB/Sponsor approval/favorable opinion. As soon as possible, the implemented deviation or change, the reasons for it and if appropriate the proposed protocol amendment should be submitted to the IEC/IRB/head of medical institution/Sponsor. Any deviations from the protocol must be explained and documented by the Investigator.

Details on discontinuation of the entire study or parts thereof can be found in Section [12](#).

13.4 Subject information and consent

All relevant information on the study will be summarized in an integrated subject information sheet and ICF provided by the Sponsor or the study center. A sample subject information and ICF is provided as a document separate to this protocol.

Based on this subject information sheet, the Investigator or designee will explain all relevant aspects of the study to each subject/legal representative or proxy consentor (if the subject is under legal protection) prior to his/her entry into the study (i.e. before any examinations and procedures associated with the selection for the study are performed or any study-specific data is recorded on study-specific forms).

The Investigator will also mention that written approval of the IRB/IEC has been obtained.

Each subject/legal representative or proxy consentor will be informed about the following aspects of premature withdrawal:

- Each subject has the right to withdraw from the study at any time without any disadvantage and without having to provide reasons for this decision.
- The subject's consent covers end-of-study examinations as specified in the visit description described in Section 9.2 to be conducted after withdrawal of consent.
- The subject's data that have been collected until the time of withdrawal will be retained and statistically analyzed in accordance with the SAP.
- Subject-specific data on the basis of material obtained before withdrawal may be generated after withdrawal (e.g. image reading, analysis of biological specimen such as blood, urine or tissues); these data would also be retained and statistically analyzed in accordance with the SAP. The subject has the right to object to the generation and processing of this post-withdrawal data. For this, he/she needs to sign a corresponding declaration of objection; alternatively, the subject's oral objection may be documented in the subject's source data.

If at any time during the study the subject would like to withdraw consent, the Investigator must discuss with the subject the active and Long-term Follow-up period parts of the study. Withdrawal of consent should be documented in the subject's medical file.

Each subject/legal representative or proxy consentor will have ample time and opportunity to ask questions.

Only if the subject/legal representative or proxy consentor voluntarily agrees to sign the ICF and has done so, may he/she enter the study. Additionally, the Investigator will personally sign and date the form. The subject/legal representative or proxy consentor will receive a copy of the signed and dated form.

The signed IC statement is to remain in the Investigator site file or, if locally required, in the subject's note/file of the medical institution.

In the event that IC is obtained on the date that baseline study procedures are performed, the study record or subject's clinical record must clearly show that IC was obtained prior to these procedures.

If the subject is not capable of providing a signature, a verbal statement of consent can also be given in the presence of an impartial witness (independent of the Sponsor and the Investigator). This is to be documented by a signature from the informing physician as well as by a signature from the witness.

The ICF and any other written information provided to subjects/legal representatives or proxy consenters will be revised whenever important new information becomes available that may be relevant to the subject's consent, or there is an amendment to the protocol that necessitates a change to the content of the subject information and/or the written ICF. The Investigator will inform the subject/legal representative or proxy consenters of changes in a timely manner and will ask the subject to confirm his/her participation in the study by signing the revised ICF. Any revised written ICF and written information must receive the IEC/IRB's approval/favorable opinion in advance of use.

13.5 Publication policy and use of data

The Sponsor has made the information regarding the study protocol publicly available on the internet at www.clinicaltrials.gov.

All data and results and all intellectual property rights in the data and results derived from the study will be the property of the Sponsor who may utilize them in various ways, such as for submission to government regulatory authorities or disclosure to other investigators.

Regarding public disclosure of study results, the Sponsor will fulfill its obligations according to all applicable laws and regulations. The Sponsor is interested in the publication of the results of every study it performs.

The Sponsor recognizes the right of the Investigator to publish the results upon completion of the study. However, the Investigator, while free to utilize study data derived from his/her center for scientific purposes, must obtain written consent of the Sponsor on the intended publication manuscript before its submission. To this end, the Investigator must send a draft of the publication manuscript to the Sponsor within a time period specified in the contract. The Sponsor will review the manuscript promptly and will discuss its content with the Investigator to reach a mutually agreeable final manuscript.

13.6 Compensation for health damage of subjects / insurance

The Sponsor maintains clinical trial insurance coverage for this study in accordance with the laws and regulations of the country in which the study is performed.

13.7 Confidentiality

All records identifying the subject will be kept confidential and, to the extent permitted by the applicable laws and/or regulations, will not be made publicly available.

Subject names will not be supplied to the Sponsor. Only the subject number will be recorded in the CRF, and if the subject name appears on any other document (e.g. pathologist report), it must be obliterated before a copy of the document is supplied to the Sponsor. Study findings stored on a computer will be stored in accordance with local data protection laws. As part of the IC process, the subjects will be informed in writing that representatives of the Sponsor, IEC/IRB, or regulatory authorities may inspect their medical records to verify the information collected, and that all personal information made available for inspection will be handled in strictest confidence and in accordance with local data protection laws.

If the results of the study are published, the subject's identity will remain confidential.

The Investigator will maintain a list to enable subjects to be identified.

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15. Protocol amendments

Not applicable.

16. Appendices

16.1 Eastern Cooperative Oncology Group performance status

Grade	Description
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work)
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Death

16.2 National Cancer Institute-Common Terminology Criteria for Adverse Events

This study will utilize the NCI-CTC for Adverse Events Version 4.03 for toxicity and serious AEs reporting. A copy of the CTC Version 4.03 can be downloaded in PDF form from <http://evs.nci.nih.gov/ftp1/CTCAE/About.html>.

All appropriate treatment areas should have access to a copy of the NCI-CTC Version 4.03.

16.3 NCCN-FACT FPSI-17

Below is a list of statements that other people with your illness have said are important.
Please circle or mark one number per line to indicate your response as it applies to the past 7 days.

			Not at all	A little bit	Some what	Quite a bit	Very much
D R S- P	GP1	I have a lack of energy.....	0	1	2	3	4
	GP4	I have pain	0	1	2	3	4
	P7	I have difficulty urinating	0	1	2	3	4
	C2	I am losing weight	0	1	2	3	4
	BP1	I have bone pain.....	0	1	2	3	4
	HI7	I feel fatigued.....	0	1	2	3	4
	NCCN3	I have weakness in my legs	0	1	2	3	4
D R S- E	P3	My pain keeps me from doing things I want to do	0	1	2	3	4
	C6	I have a good appetite.....	0	1	2	3	4
	GF5	I am sleeping well.....	0	1	2	3	4
	GE6	I worry that my condition will get worse .	0	1	2	3	4
	GP2	I have nausea	0	1	2	3	4
	P6	I have trouble moving my bowels	0	1	2	3	4
	GS7	I am satisfied with my sex life.....	0	1	2	3	4
T S E	GP5	I am bothered by side effects of treatment	0	1	2	3	4
	GF3	I am able to enjoy life	0	1	2	3	4
	GF7	I am content with the quality of my life right now.....	0	1	2	3	4

DRS-P=Disease-Related Symptoms Subscale – Physical
DRS-E=Disease-Related Symptoms Subscale – Emotional
TSE=Treatment Side Effects Subscale
FWB=Function and Well-Being Subscale
English (Universal), Copyright 2001

03 March 2010

STUDY ID # _____ DO NOT WRITE ABOVE THIS LINE HOSPITAL # _____

Brief Pain Inventory (Short Form)

Date: ____/____/____ Time: _____

Name: _____

____ Last _____ First _____ Middle Initial _____

Items 1 and 2 removed
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3. Please rate your pain by circling the one number that best describes your pain at its worst in the last 24 hours.

0	1	2	3	4	5	6	7	8	9	10
No Pain										Pain as bad as you can imagine

4. Please rate your pain by circling the one number that best describes your pain at its least in the last 24 hours.

0	1	2	3	4	5	6	7	8	9	10
No Pain										Pain as bad as you can imagine

5. Please rate your pain by circling the one number that best describes your pain on the average.

0	1	2	3	4	5	6	7	8	9	10
No Pain										Pain as bad as you can imagine

6. Please rate your pain by circling the one number that tells how much pain you have right now.

0	1	2	3	4	5	6	7	8	9	10
No Pain										Pain as bad as you can imagine

STUDY ID #: _____ DO NOT WRITE ABOVE THIS LINE HOSPITAL #: _____
_____/_____/_____
Last First Middle Initial

Items 7 and 8 removed
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9. Circle the one number that describes how, during the past 24 hours, pain has interfered with your:

A. General Activity

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

B. Mood

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

C. Walking Ability

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

D. Normal Work (includes both work outside the home and housework)

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

E. Relations with other people

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

F. Sleep

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

G. Enjoyment of life

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

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Cover page of the integrated protocol

A randomized, double-blind, placebo-controlled Phase III study of darolutamide (ODM-201) versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer

This protocol version is an integration of the following documents / sections:

- **Original protocol**, Version 1.0, dated 25 MAY 2016
- **Amendment 2** (global amendment described in Section 15.1), forming integrated protocol, Version 2.0, dated 04 OCT 2016
- **Amendment 5** (global amendment described in Section 15.2), forming integrated protocol, Version 3.0, dated 12 FEB 2018
- **Amendment 6** (global amendment described in Section 15.3), forming integrated protocol, Version 4.0, dated 10 DEC 2019
- **Amendment 7** (global amendment described in Section 15.4), forming integrated protocol, Version 5.0, dated 26 MAY 2020

Amendments not included in the consecutive numbering of amendments are local amendments not forming part of this integrated global protocol.

- **Amendment 1**, dated 20 SEP 2016
(local amendment valid for China only)
- **Amendment 3**, dated 04 NOV 2016
(local amendment valid for UK only)
- **Amendment 4**, dated 31 JAN 2017
(local amendment valid for Japan only)

1. Title page

A randomized, double-blind, placebo-controlled Phase III study of darolutamide (ODM-201) versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer

Darolutamide in addition to standard androgen deprivation therapy and docetaxel in metastatic hormone-sensitive prostate cancer

Acronym: **ARASENS**

Test drug: BAY 1841788 / darolutamide / ODM-201

Study purpose: Efficacy and Safety

Clinical study phase: III Date: 26 MAY 2020

EudraCT: no: 2015-002590-38 Version no.: 5.0

Sponsor's study no.: 17777

Sponsor: a) Non-US: Bayer AG, [REDACTED]
b) US territory: Bayer HealthCare Pharmaceuticals Inc.,
[REDACTED]

Sponsor's medical expert: [REDACTED]

This document is electronically signed. The eSignature is archived in the trial master file (TMF) and can be provided upon request.

The study will be conducted in compliance with the protocol, ICH-GCP and any applicable regulatory requirements.

Confidential

The information provided in this document is strictly confidential and is intended solely for the guidance of the clinical investigation. Reproduction or disclosure of this document – whether in part or in full – to parties not associated with the clinical investigation, or its use for any other purpose, without the prior written consent of the sponsor is not permitted.

Throughout this document, symbols indicating proprietary names (®, TM) may not be displayed. Hence, the appearance of product names without these symbols does not imply that these names are not protected.

Signature of the sponsor's medically responsible person

This document is electronically signed. The eSignature is archived in the trial master file (TMF) and can be provided upon request.

The signatory agrees to the content of the final integrated clinical study protocol as presented.

Name: [REDACTED]

Role: [REDACTED]

Date:

Signature:

Signature of principal investigator

The signatory agrees to the content of the final integrated clinical study protocol as presented.

Name:

Affiliation:

Date:

Signature:

Signed copies of this signature page are stored in the sponsor's study file and in the respective center's Investigator site file.

2. Synopsis

Title	A randomized, double-blind, placebo-controlled Phase III study of darolutamide (ODM-201) versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer
Short title	Darolutamide in addition to standard androgen deprivation therapy (ADT) and docetaxel in metastatic hormone-sensitive prostate cancer
Acronym	ARASENS
Clinical study phase	III
Study objective(s)	The primary objective of this study is: - To demonstrate the superiority in overall survival (OS) of darolutamide in addition to standard ADT and docetaxel over placebo in addition to standard ADT and docetaxel The secondary objectives of this study are to evaluate: - Time to castration-resistant prostate cancer - Time to initiation of subsequent antineoplastic therapy - Symptomatic skeletal event free survival (SSE-FS) - Time to first symptomatic skeletal event (SSE) - Time to initiation of opioid use for ≥ 7 consecutive days - Time to pain progression - Time to worsening of physical symptoms of disease based on functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire (NCCN-FACT FPSI-17) - Safety The exploratory objectives of this study include: - Quality of life - Medical resource use - Prostate-specific antigen (PSA) assessments - Pharmacokinetics and exposure-response analysis - Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease
Test drug	Darolutamide (BAY 1841788)

Name of active ingredient	Darolutamide (BAY 1841788)
Dose(s)	Darolutamide 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg.
Route of administration	Oral
Duration of treatment	Until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.
Reference drug(s)	Matching placebo
Name of active ingredient	Not applicable
Dose(s)	The dosing of placebo is the same as for darolutamide (BAY 1841788), please see above.
Route of administration	Oral
Duration of treatment	Until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.
Background treatment	ADT (luteinizing hormone–releasing hormone [LHRH] agonist/antagonist or orchiectomy) and 6 cycles of docetaxel.
Indication	Metastatic hormone–sensitive prostate cancer (mHSPC)
Diagnosis and main criteria for inclusion /exclusion	Inclusion criteria • Written informed consent • Males ≥ 18 years of age • Histologically or cytologically confirmed adenocarcinoma of prostate • Metastatic disease documented either by a positive bone scan, or for soft tissue or visceral metastases, either by contrast–enhanced abdominal/pelvic/chest computed tomography (CT) or magnetic resonance imaging (MRI) scan assessed by Investigator and confirmed by central radiology review. Metastatic disease is defined as either malignant lesions in bone scan or measurable lymph nodes above the aortic bifurcation or soft tissue/visceral lesions according to Response Evaluation Criteria In Solid Tumors (RECIST) 1.1. Lymph nodes are measurable if the short axis diameter is ≥ 15 mm, soft tissue/visceral lesions are measurable if the long axis diameter is ≥ 10 mm. Subjects with regional lymph node metastases only (N1, below the aortic bifurcation) will not be eligible for the study. Only subjects with non–regional lymph node metastases (M1a) and/or bone metastases (M1b) and/or other sites of metastases with or without

bone disease (M1c) will be eligible.

- Subjects must be candidates for ADT and docetaxel therapy per Investigator's judgment
- Started ADT (LHRH agonist/antagonist or orchiectomy) with or without first generation anti-androgen, but no longer than 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks, prior to randomization is recommended. First generation anti-androgen has to be stopped prior to randomization.
- An Eastern Cooperative Oncology Group performance status of 0 or 1
- Blood counts at Screening: hemoglobin ≥ 9.0 g/dL, absolute neutrophil count $\geq 1.5 \times 10^9/L$, platelet count $\geq 100 \times 10^9/L$ (subject must not have received any growth factor within 4 weeks or a blood transfusion within 7 days of the hematology laboratory sample obtained at Screening)
- Screening values of serum alanine aminotransferase and/or aspartate transaminase $\leq 1.5 \times$ upper limit of normal (ULN), total bilirubin $\leq$ ULN, creatinine $\leq 2.0 \times$ ULN
- Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the treatment with darolutamide/placebo and for 3 months after the end of the treatment with darolutamide/placebo and 6 months after treatment with docetaxel.

Exclusion criteria

- Prior treatment with:
 - LHRH agonist/antagonists started more than 12 weeks before randomization
 - Second-generation androgen receptor (AR) inhibitors such as enzalutamide, ARN-509, darolutamide, other investigational AR inhibitors
 - Cytochrome P 17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer
 - Chemotherapy or immunotherapy for prostate cancer prior to randomization
- Treatment with radiotherapy (external beam radiation therapy, brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization
- Known hypersensitivity to any of the study drugs, study drug classes, or excipients in the formulation of the study drugs
- Contraindication to both CT and MRI contrast agent
- Had any of the following within 6 months before randomization: stroke, myocardial infarction, severe/unstable angina pectoris,

	coronary/peripheral artery bypass graft, congestive heart failure (New York Heart Association Class III or IV) • Uncontrolled hypertension as indicated by a resting systolic blood pressure (BP) ≥ 160 mmHg or diastolic BP ≥ 100 mmHg despite medical management • Had a prior malignancy. Adequately treated basal cell or squamous cell carcinoma of skin or superficial bladder cancer that has not spread behind the connective tissue layer (i.e., pTis, pTa, and pT1) is allowed, as well as any other cancer for which treatment has been completed ≥ 5 years before randomization and from which the subject has been disease-free • A gastrointestinal disorder or procedure which is expected to interfere significantly with absorption of study drug • An active viral hepatitis, known human immunodeficiency virus infection with detectable viral load, or chronic liver disease with a need for treatment • Previous (within 28 days before the start of study drug or 5 half-lives of the investigational treatment of the previous study, whichever is longer) or concomitant participation in another clinical study with investigational medicinal product(s) • Any other serious or unstable illness, or medical, social, or psychological condition, that could jeopardize the safety of the subject and/or his/her compliance with study procedures, or may interfere with the subject's participation in the study or evaluation of the study results • Inability to swallow oral medications • Close affiliation with the investigational site (e.g., a close relative of the Investigator, dependent person [e.g., employee or student of the investigational site]) • Previous assignment to treatment in this study
Study design	Randomized, double-blind, placebo-controlled, multicenter phase III study. Approximately 1,300 subjects will be randomized (1:1 ratio) to receive one of the following study drugs: • Darolutamide 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg • Placebo matching darolutamide tablets in appearance, twice daily with food All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended. Six cycles of docetaxel will be administered after randomization. Docetaxel can be administered in combination with

	prednisone/prednisolone at the discretion of the Investigator. Subjects will be stratified at randomization as follows: • Extent of disease ○ Non-regional lymph nodes metastases only ○ Bone metastases with or without lymph node metastases ○ Visceral metastases with or without lymph node metastases with or without bone metastases • Alkaline Phosphatase (ALP) ○ ALP<ULN ○ ALP≥ULN Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.
Methodology	The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up. Screening period: All trial-related procedures and evaluations will only be performed after the subject has agreed to participate and has signed the informed consent form. The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met. Once eligibility is confirmed and documented, eligible subjects will be randomized in a ratio of 1:1 to treatment with darolutamide or placebo. Treatment period: Study treatment will be provided for all subjects bid until symptomatic progressive disease, change of antineoplastic therapy, unacceptable toxicity, until subject withdraws consent, withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, and non-compliance. Subjects will be evaluated every 12 weeks for castration-resistant prostate cancer, initiation of subsequent antineoplastic therapy, SSEs, opioid use for ≥7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, adverse events (AEs) and serious adverse events (SAEs), Quality of life (QoL) and PSA assessment. Active Follow-up period: After treatment discontinuation, subjects will enter the Active Follow-up period, which includes: • End of Treatment Visit An End of Treatment (EOT) Visit will be conducted 30 (+7) days after the last dose of study drug. The following assessments will be performed at the EOT Visit: QoL, pain assessment, analgesic consumption, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical

	progression, radiological progression, toxicity, or other), SSEs, and all AEs and SAEs regardless of causality. Active Follow-up visits During the Active Follow-up period, the following assessments will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical progression, radiological progression, toxicity, or other), SSEs, and study drug-related SAEs. After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data. Long-term (survival) Follow-up period: After Active Follow-up, subjects will continue to be contacted approximately every 12 weeks (by phone) to capture all antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs and survival status. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study. Survival sweep: For every formal analysis of OS, survival data will be collected through additional survival sweeps. All subjects considered alive at the database cut-off date and prior to any subsequent additional analysis will be contacted for survival status. After primary analysis of the study: For subjects who are ongoing with darolutamide treatment, the sponsor will continue to provide darolutamide via a separate program at least until darolutamide is approved and reimbursed in a specific country.
Type of control	Placebo
Data Monitoring Committee	Yes
Number of subjects	Approximately 1,300 subjects in total are to be randomized
Primary variable(s)	Overall survival, defined as the time (in days) from date of randomization until death from any cause.
Time point/frame of measurement for primary variable(s)	The primary analysis will be performed when approximately 509 deaths occur in the 2 treatment arms combined. The expected study duration for 509 deaths is approximately 70 months.
Plan for statistical analysis	All randomized subjects will be included in the analysis of overall survival (OS), the primary efficacy endpoint.

	The primary efficacy endpoint of OS will be analyzed using a one-sided stratified log-rank test with an overall alpha of 0.025. Analysis will be according to treatment groups as randomized, with stratification factors extent of disease and alkaline phosphatase level. The hazard ratio (HR) (darolutamide/placebo) and its 95% confidence interval will be calculated using the Cox model, stratified by the same randomization factors. Kaplan-Meier estimates and plots will be presented for each treatment group. The secondary efficacy time-to-event endpoints will be analyzed using stratified log-rank test with randomization stratification factors. HR and 95% confidence intervals will be provided using Cox model. Interim analysis for futility at approximately 30% of total planned events will be conducted.
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List of abbreviations

5-HT	5-hydroxytryptamine
ADT	Androgen deprivation therapy
AE	Adverse event
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
AR	Androgen receptor
AST	Aspartate aminotransferase
AUC	The area under the concentration–time curve
AUC(0– t_{last})	The area under the concentration–time curve from time zero to the last sample with the quantifiable concentration
AUC(0–8)	The area under the concentration–time curve from time zero to 8 h
AUC(0–12)	The area under the concentration–time curve from time zero to 12 h
AUC(0–24)	The area under the concentration–time curve from time zero to 24 h
AUC(0–72)	The area under the concentration–time curve from time zero to 72 h
AV	Atrioventricular
bid	Twice daily
BCRP	Breast cancer resistance protein
BP	Blood pressure
BPI–SF	Brief Pain Inventory – Short Form
CAB	Complete androgen blockade
CI	Confidence interval
C_{max}	Maximum concentration
CNS	Central nervous system
CRF	Case report form
CRO	Contract research organization
CRPC	Castration–resistant prostate cancer
CT	Computed tomography
CTC	Circulating tumor cell
ctDNA	Circulating tumor DNA
CV	Coefficient of variation
CYP	Cytochrome P
dL	Deciliter
DMC	Data Monitoring Committee
DNA	Deoxyribonucleic acid
EBRT	External beam radiation therapy
ECG	Electrocardiogram
ECOG PS	Eastern Cooperative Oncology Group Performance Status

eCRF	Electronic case report form
EDC	Electronic data capture
EOT	End of treatment
ePRO	Electronic patient reported outcome
ESMO	European Society of Medical Oncology
EU	European Union
FAS	Full analysis set
FPSI-DRS-P	FACT/NCCN FPSI-17 disease-related symptoms – physical
g	Gram
GABA _A	Type-A γ -aminobutyric acid
GCP	Good Clinical Practice
GI	Gastrointestinal
GMP	Good Manufacturing Practice
HDPE	High density polyethylene
HR	Hazard ratio, heart rate
IB	Investigator's brochure
IC	Informed consent
IC ₅₀	Half maximal inhibitory concentration
ICF	Informed consent form
IEC	Independent Ethics Committee
INN	International nonproprietary name
IRB	Institutional Review Board
IV	Intravenous
IXRS	Interactive Voice/Web Response System
KM	Kaplan–Meier
LC–MS/MS	Liquid chromatography–tandem mass spectrometry
LHRH	Luteinizing hormone releasing hormone
LLOQ	Lower limit of quantification
M&S	Modeling & Simulation
mCRPC	Metastatic castration-resistant prostate cancer
mCSPC	Metastatic castration-sensitive prostate cancer
mHSPC	Metastatic hormone-sensitive prostate cancer
MedDRA	Medical Dictionary for Regulatory Activities
mg	Milligram
mL	Milliliter
mmHg	Millimeter of mercury
MRI	Magnetic resonance imaging
NCCN	National Comprehensive Cancer Network
NCCN–FACT FPSI-17	Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire

NCI–CTCAE v 4.03	National Cancer Institute–Common Terminology Criteria for Adverse Events; version 4.03
ng	Nanogram
nM	Nanomolar
OS	Overall survival
P–gp	P–glycoprotein
PCWG3	Prostate Cancer Clinical Trials Working Group
PD	Pharmacodynamic
PI	Principal investigator
PI3K	Phosphatidylinositol 3–kinase
PIK3CA	Phosphatidylinositol–4,5–bisphosphate 3–kinase, catalytic subunit alpha
PK	Pharmacokinetics
PKS	Pharmacokinetic Analysis Set
PRO	Subject–reported outcomes
PSA	Prostate–specific antigen
PTEN	Phosphatase and tensin homolog
QoL	Quality of life
RECIST	Response Evaluation Criteria In Solid Tumors
RNA	Ribonucleic acid
SAE	Serious adverse event
SAP	Statistical analysis plan
SAS	Statistical Analysis Software
SD	Standard deviation
SHBG	Sex hormone–binding globulin
SNP	Single gene polymorphism
SSE	Symptomatic skeletal event
SSE–FS	Symptomatic skeletal event free survival
SUSAR	Suspected unexpected serious adverse reaction
t_{last}	Time to the last quantifiable concentration
t_{max}	Time to the maximum plasma concentration
TMF	Trial master file
ULN	Upper limit of normal
UVB	Ultraviolet B
VCaP	Vertebral cancer of the prostate
WPS	Worst pain subscale

Definitions of terms

With the approval of the INN darolutamide, the Orion drug nomenclature (ODM-201) was replaced with darolutamide throughout the protocol. In addition, the Orion codes for darolutamide diastereomers and metabolite were replaced with trivial names in the entire document as shown in the table below.

	Orion name	INN/trivial name
Drug substance and drug product	ODM-201	Darolutamide (INN)
Diastereomer	ORM-16497	(<i>S,R</i>)-darolutamide
Diastereomer	ORM-16555	(<i>S,S</i>)-darolutamide
Metabolite	ORM-15341	Keto-darolutamide

3. Introduction

3.1 Background

Prostate cancer is the second most common cancer and the fifth leading cause of cancer death among men worldwide (1). It is the most common non-cutaneous cancer and the second leading cause of cancer-related deaths in men in Europe (2) and the United States (3).

At the time of initial diagnosis, most men are found to have localized disease and can be treated with curative surgery or radiotherapy. However, a proportion of subjects will recur either locally or in distant sites. In addition, approximately 5% of new diagnoses present with de novo hormone-sensitive metastatic disease. The most common site of metastases is bone. However, soft tissue and visceral metastases can also occur (4). Upon development of metastatic disease, treatment is palliative, with the goal of prolonging quantity and maintaining quality of life (QoL) (5; 6; 7).

The mainstay of therapy for newly diagnosed metastatic hormone-sensitive prostate cancer (mHSPC), also known as metastatic castration-sensitive prostate cancer (mCSPC), remains androgen deprivation therapy (ADT), as recommended by the prostate cancer guidelines from the American Society of Clinical Oncology, the National Comprehensive Cancer Network (NCCN), and the European Association of Urology (8; 9; 10). Androgen deprivation therapy can be accomplished either by surgical orchiectomy (castration) or medical castration (using either a luteinizing hormone releasing hormone [LHRH] agonist or an LHRH antagonist). When LHRH therapy is initiated, a transient rise in luteinizing hormone can cause a surge in serum testosterone, which may stimulate prostate cancer growth. The flare phenomenon can be effectively prevented with the addition of an anti-androgen therapy to ADT, a complete androgen blockade (CAB), which blocks the effect of the increased serum testosterone (8; 9; 10).

Over the years, efforts have been made to improve upon ADT by modified timing (early versus deferred) and schedule (intermittent versus continuous) or by adding adjunctive agents, but the optimal management of this disease state is still not defined. Combined androgen blockade using both an LHRH agonist and an anti-androgen has been extensively studied as upfront therapy in mHSPC (6; 7; 11). Several meta-analyses have shown a modest 2% to 5% improvement in 5-year survival with CAB over castration, although with increased toxicity. One of the largest meta-analyses included the individual level data of 8,275 subjects from 27 trials. Although there was an overall trend toward improved 5-year survival with CAB over castration, this did not reach statistical significance ($P=0.11$) (12; 13; 14). According to NCCN prostate cancer guidelines 2014, CAB provides modest to no benefit over castration alone in metastatic subjects.

While ADT demonstrates antitumor activity in mHSPC with prolonged disease control, unfortunately resistance ultimately occurs and subjects die of castration-resistant prostate cancer (CRPC). Approximately 10% to 20% of prostate cancer subjects develop CRPC within 5 years, with a median time of 12–24 months (15), and have a poor median survival expectancy of 9 to 30 months (16). In the setting of metastatic castration-resistant prostate cancer (mCRPC), new therapeutic agents with different mechanisms of action have recently

become available and have significantly improved clinical outcomes, including chemotherapeutic agents (cabazitaxel), androgen–receptor targeted agents (abiraterone and enzalutamide), immunotherapies (sipuleucel–T), and radiopharmaceuticals (radium–223) (17; 18; 19; 20; 21; 22; 23). Conversely, the overall treatment paradigm of mHSPC has remained substantially stable and has not been strongly challenged until more recently, when the randomized controlled CHAARTED trial examined the addition of chemotherapy to standard ADT for metastatic hormone–sensitive disease, thus bringing chemotherapy to an arena historically dominated by hormone therapy only. The CHAARTED study demonstrated that combining 6 cycles of docetaxel chemotherapy with standard ADT was associated with a statistically significant overall survival (OS) benefit in metastatic hormone–sensitive disease compared with ADT alone, particularly in subjects with high volume disease (24; 25). Recent data from another ongoing study in Europe, the Medical Research Council STAMPEDE trial, confirm a significant OS benefit for docetaxel combined with ADT compared with ADT alone in subjects with high risk locally advanced or newly diagnosed metastatic disease (26; 27).

Chemo–hormonal therapy is currently recommended by the NCCN as first–line therapy for high volume mHSPC (11) and is recommended by ESMO Clinical guidelines as first–line treatment of metastatic, hormone–naïve disease in men fit enough for chemotherapy (33).

3.2 Overview of darolutamide (BAY 1841788)

3.2.1 Pharmacology

Darolutamide is a novel non–steroidal androgen receptor (AR) inhibitor which binds with high selectivity and binding affinity (9 nM) to AR when compared to known second–generation anti–androgens. Darolutamide is a 1:1 mixture of 2 pharmacologically active diastereomers: (*S,R*)-darolutamide and (*S,S*)-darolutamide. Diastereomers as well as the main circulating metabolite from darolutamide, keto–darolutamide, show no major differences in pharmacological activity and function as antagonists in AR–overexpressing cells.

In the vertebral cancer of the prostate (VCaP) cell line (derived from a bone metastasis of a subject with CRPC) with endogenous AR gene amplification and AR overexpression, darolutamide suppressed dose dependently androgen–induced VCaP cell proliferation more efficiently than bicalutamide.

W741L (tryptophan to leucine) mutation of AR has been shown to have a role in the mechanism of bicalutamide resistance. In this mutation, darolutamide functioned as an antagonist, whereas bicalutamide was a pure agonist, as previously shown.

In addition, darolutamide inhibits testosterone–mediated nuclear translocation of AR in AR overexpressing cells. Darolutamide and keto–darolutamide were found to decrease 0.3 nM testosterone–induced nuclear translocation at concentrations ≥ 100 nM and ≥ 300 nM, respectively; whereas bicalutamide failed to block testosterone–induced AR nuclear translocation at all tested concentrations.

In the immature rat assay, darolutamide showed anti–androgenic activity, by dose dependently and significantly antagonizing testosterone–induced growth of

androgen-sensitive tissues in ventral prostate and seminal vesicles after 6-day oral dosing, with maximal inhibition at 100 mg/kg/day.

In the castration-resistant VCaP xenograft model, darolutamide showed significant anti-tumor activity. The VCaP tumors grew significantly more slowly in mice treated with 50 mg/kg of darolutamide orally once daily for 37 days compared with untreated castrated mice, whereas tumor growth was fully inhibited after orally twice daily (bid) dosing at 50 mg/kg.

In the orthotopic prostate cancer model, where VCaP cells were inoculated into prostate of nude intact male mice, inhibition of tumor growth and reduced increase in serum prostate-specific antigen (PSA) level were observed in mice treated with 50 mg/kg of darolutamide orally twice daily for 3 weeks compared with vehicle controls. Darolutamide did not have any effect on serum testosterone levels.

3.2.2 Nonclinical pharmacokinetics and metabolism

Darolutamide is rapidly absorbed in mice and rats from preclinical formulations. Dog studies show that fasted and fed states differ in darolutamide absorption with fed states showing higher total exposure than fasted states. Upon repeated dosing, no accumulation was observed. Systemic exposure to darolutamide increased less than dose proportionally, suggesting solubility-limited absorption at the high dose levels.

Darolutamide consists of 1:1 ratio of diastereomers (*S,R*)-darolutamide and (*S,S*)-darolutamide, which are able to interconvert and exhibit different ratios in animals. In rat plasma the diastereomer ratio remains about 1:1, but mice and dogs show higher exposure to (*S,R*)-darolutamide (i.e. (*S,S*)-darolutamide is eliminated more rapidly than (*S,R*)-darolutamide in these species). (*S,S*)-darolutamide is the main human diastereomer (about 85%).

The main circulating metabolite keto-darolutamide is formed by oxidation of a secondary alcohol to form a ketone. The diastereomers (*S,R*)-darolutamide and (*S,S*)-darolutamide are able to interconvert via keto-darolutamide. In rats and dogs, the metabolite/parent darolutamide ratio is close to unity, while in mice and humans the ratio is around 2 to 3. No human-specific disproportionate metabolites have been found so far.

Systemic clearance of darolutamide in nonclinical species is low (0.2 L/h/kg in dogs) which is in agreement with the results obtained *in vitro* in hepatocyte incubations. Terminal volume of distribution of darolutamide in dogs is 1.8 L/kg.

Based on *in vitro* data, darolutamide is a substrate and inhibitor of P-glycoprotein (P-gp) and Breast Cancer Resistance Protein (BCRP). According to additional *in vitro* data, darolutamide is also an inhibitor of the uptake transporters OATP1B1 and OATP1B3. Interactions with P-gp or BCRP are expected *in vivo* based on these *in vitro* data, and also on preliminary clinical study results (please refer to Section 3.2.4).

In general, the potential for darolutamide to inhibit CYP reaction pathways *in vitro* in human liver microsomes is low. The lowest IC₅₀ of 30 µM was observed for CYP2C9. For its pharmacologically active metabolite keto-darolutamide, a K_i value of 27 µM was determined. Applying a mechanistic static model, no inhibition potential towards CYP2C9 *in vivo* is expected.

The CYP450 induction potential of darolutamide and keto-darolutamide was evaluated *in vitro* in primary cultures of human hepatocytes from three individual donors. Based on these data, a risk to observe weak to moderate CYP3A4 induction by darolutamide in the clinic cannot be ruled out.

In vitro darolutamide has a free fraction of 4.5% to 8% in the plasma of human, rat, and mouse. The free fraction in dog plasma is dependent on darolutamide concentration (range: 0.3% to 11%). The major binding protein is serum albumin. There are no differences in plasma protein binding between the diastereomers. The main circulating metabolite keto-darolutamide has a free fraction <1% in most species.

Tissue distribution studies in rodents suggest that darolutamide and its metabolite keto-darolutamide have very limited access to brain (brain/plasma ratio <0.05). Prostate and heart have clearly higher tissue/plasma ratios of darolutamide. The kidney/plasma ratio of darolutamide is around 1 and the liver/plasma ratio is around 3. The results from a whole body autoradiography study in rats confirm that the ¹⁴C–darolutamide–related radioactivity concentrations in tissues (except in the liver) are lower than those in plasma.

In vivo mass balance study in rats suggests that about two-thirds of darolutamide–related radioactivity is excreted to feces and the rest to urine. Maximally, 45% of darolutamide is excreted unchanged. The similar excretion pattern after intravenous (IV) and oral administration supports complete oral absorption at the studied low dose (11 mg/kg). Excretion is almost complete by 48 hours post-dose.

3.2.3 Safety pharmacology and toxicology

The repeat-dose toxicity studies with darolutamide have been conducted up to 26 weeks in the rat and up to 39 weeks in the dog. Darolutamide has been well tolerated up to the highest dose levels in both species. The no observable adverse effect level was >2 x 500 mg/kg/day in the rat and >2 x 200 mg/kg/day in the dog. All the observed effects were directly related to the anti-androgenic mechanism of action of darolutamide and no off-target toxicity could be identified in either species. The effects were also shown to be reversible after discontinuation of administration of darolutamide. Increase of exposure to darolutamide was less than dose-proportional both in rats and dogs, and no accumulation was observed. The exposure multiples of darolutamide at the highest dose level in the rat and dog were approximately up to 3-fold and 1.8-fold, respectively, compared with AUC(0–24) at the 700-mg bid dose level in subjects with mCRPC in the clinical phase I/II ARADES study (17829 / 3104001). The corresponding exposure multiples of the metabolite keto-darolutamide in the rat and dog were approximately up to 1.5-fold and 0.7-fold, respectively.

No stand-alone developmental and reproductive toxicity studies have been conducted with darolutamide. Based on the known pharmacologic effects of anti-androgens (11; 12; 13; 14), decreased fertility in males and developmental toxicity would be expected. Also it has to be taken into account that the subjects in this clinical study use gonadotropin releasing hormone agonist or antagonist treatment or have bilateral orchiectomy, which also affects fertility.

Based on the results from the completed *in vitro* and *in vivo* genotoxicity studies, darolutamide is not considered to represent any significant genotoxic risk for man. Although

both darolutamide and its metabolite, keto-darolutamide, absorb light in the UVB wavelength range of 290 to 320 nm with the absorption maximum at 290 nm, an *in vitro* 3T3 neutral red uptake phototoxicity study demonstrated that darolutamide is not phototoxic.

Cardiovascular safety evaluation of darolutamide investigated within the frame of systemic toxicity studies in dogs, did not reveal any significant cardiovascular effects up to the highest dose levels tested after oral dosing. The mean maximum concentrations ($C_{\max}$) of darolutamide in dog plasma at the highest dose levels in the 28-day, 13-week and 39-week toxicity studies were 9,990, 6,030 to 8,350 and 12070–12440 ng/mL, respectively.

In an anesthetized dog model using bolus IV administration, darolutamide and its main human diastereomer (*S,S*)-darolutamide, at observed mean $C_{\max}$ values of >9,300 ng/mL and >4,000 mg/mL in plasma, respectively, induced vasodilatation, especially at the peripheral level, leading to decreased arterial blood pressure (BP). This is not regarded to be of biological relevance as this was observed at free peak plasma concentrations of ≥ 3 -fold higher than the plasma free peak concentrations observed at 700 mg bid dose level in the Phase I/II clinical study in mCRPC subjects.

In the study with (*S,S*)-darolutamide, 1 dog out of 4 displayed a reversible complete atrioventricular (AV) block at the highest bolus dose. At this dose, the mean $C_{\max}$ of (*S,S*)-darolutamide in dog plasma was 27,200 ng/mL, indicating that (*S,S*)-darolutamide may affect AV conduction at very high systemic exposure levels. The mechanism and etiology of vasodilatation and AV block are not known, but they might be multifactorial and at least partly related to a synergistic effect of the anesthetic agents used and secondary pharmacodynamic (PD) effects of darolutamide and its main human diastereomer (*S,S*)-darolutamide. Based on *in vitro* human ether-a-go-go-related gene and calcium channel inhibition and *in vivo* dog studies, the risk for QT interval prolongation appears low.

Functional *in vitro* studies evaluating secondary PD effects showed that darolutamide inhibits 5-hydroxytryptamine (5-HT) uptake and type-A γ -aminobutyric acid (GABA_A) receptors. However, due to the low brain/plasma ratio, central nervous system (CNS) effects via 5-HT uptake and GABA_A inhibition are not expected. *In vivo* safety pharmacology evaluations confirmed that darolutamide has no remarkable effects on CNS or respiratory function. At high local doses/concentrations in the gastrointestinal (GI) tract, darolutamide may affect gastric emptying and intestinal transit.

3.2.4 Drug–drug interactions

The information on darolutamide drug-drug interactions is provided in the latest available version of the Investigator's Brochure (IB) for darolutamide.

Docetaxel will be administered after randomization according to standard treatment practice. Docetaxel is mainly metabolized by CYP3A4. Thus, inducers or inhibitors of CYP3A4 may alter the PK of docetaxel. Darolutamide has no inhibitory but may have some inducing effect on CYP3A4 substrates. Docetaxel is also a substrate of P-gp and OATP1B3. An effect of darolutamide on the PK of docetaxel could, therefore, not be excluded. The potential impact of darolutamide on docetaxel PK, and the respective impact on safety was investigated on the day of the first docetaxel administration in the first 20 subjects who were randomized and

have received at least 1 cycle of docetaxel. The data did not reveal any significant impact of darolutamide on the PK of docetaxel.

3.2.5 Clinical studies with darolutamide

As of 02 OCT 2015, 173 subjects with mCRPC have been treated with darolutamide.

Darolutamide has been studied in 7 clinical trials: a first-in-man study with cohort expansion (Study 17829 / 3104001, ARADES), a long-term safety extension study for the first-in-man study (Study 18035 / 3104002, ARADES-EXT), and a bioavailability study (Study 17830 / 3104003, ARAFOR). Additionally, there is a phase III efficacy and safety study of darolutamide in men with high risk non-metastatic ongoing CRPC (Study 17712 / 3104007, ARAMIS), a Phase I mass balance and bioavailability study in healthy male subjects (Study 17831 / 3104005, ARIADME), a phase I study in Japanese subjects with mCRPC (Study 17719) and a phase I drug-drug interaction study in healthy male and female subjects (Study 17723).

Darolutamide treatment has been well tolerated up to the highest pre-specified dose of 900 mg bid. The maximum tolerated dose was not reached, and no dose-limiting toxicities were observed in the phase I dose-escalation. Escalation of doses in the phase I study was discontinued because a plateau in bioavailability of darolutamide was observed.

Following oral administration of darolutamide at doses of 100, 200, 300, 500, 700, and 900 mg bid, both diastereomers, (*S,R*)-darolutamide and (*S,S*)-darolutamide, were slowly absorbed after administration with a light breakfast, the median t_{max} was 2–5 h.

AUC and C_{max} of darolutamide, (*S,R*)-darolutamide, and (*S,S*)-darolutamide increased in a nearly dose-proportional manner at doses 100–700 mg bid. At the 900 mg bid dose level, there was no further increase of C_{max} or AUC observed compared to 700 mg dose bid. The concentration of the active metabolite keto-darolutamide was 1.6–2.4 times higher than of the parent compound darolutamide. This ratio was dose-independent. After repeated oral administration of darolutamide, steady-state conditions seemed to be reached after 7 days at the latest. The diastereomer ratio of (*S,R*)-darolutamide to (*S,S*)-darolutamide was about 1:6 under steady-state conditions. The terminal half-life of darolutamide was in the range of 11.5–16.1 hours after single oral administration of a tablet formulation.

A food effect was observed when darolutamide was administered after a standardized high-fat, high-calorie breakfast. The exposure and C_{max} were about 50% lower when darolutamide was administered under fasting conditions.

Considering the 164 mCRPC patients as of 02 OCT 2015 treated with darolutamide in the Studies 17829 / 3104001, ARADES, 17830 / 3104003, ARAFOR, and 18035 / 3104002, ARADES-EXT, the integrated safety data showed 93.3% of subjects developed at least 1 adverse event (AE) and Grade 3 or higher AEs were reported in 28% of subjects. The most common AEs have been fatigue in 26.8% of subjects, back pain (22%). The most common AEs considered related to study treatment by the Investigator have been fatigue (9.8%) and hot flush (4.3%). The majority of AEs (91%) were classified as Grade 1 to 2. No dose-related trends were noted for the occurrence of any AEs.

Forty-five subjects (27.4%) have experienced a total of 125 serious adverse events (SAEs) from the start of study treatment and 7 subjects (4%) have died. None of these events was considered related to the study treatment by the Investigator or by the Sponsor.

Eleven subjects (6.7%) have withdrawn from the studies due to an AE. Only 1 of these AEs (fatigue, Grade 2) was considered related to the study treatment by the Investigator.

No events of seizure have been reported during darolutamide treatment period.

Seven patients (7/164) experienced falls or accidental falls. Two out of those seven patients who had a fall or accidental fall experienced SAEs. All SAEs were resolved/improved after treatment. Nine patients (9/164) experienced non-pathological fractures. Four patients experienced SAEs. All serious events of fractures were secondary to a fall, an accidental wound, or syncope. Among the other five patients who experienced non-serious fractures, only one patient had a fall. None of those events were assessed as related to darolutamide or resulted in permanent discontinuation from the study due to the events.

No safety concerns were raised from the ongoing Studies. In Japanese Phase I study, one SAE of grade 3 nausea was reported as related to darolutamide by Investigator, however, due to preexisting confounding factors, the sponsor assessed the event as not related with darolutamide treatment.

Efficacy results show that darolutamide has substantial antitumor activity in subjects with metastatic CRPC for time to PSA progression and for 12-week changes from baseline in serum PSA, soft tissue and bone lesions, and circulating tumor cell (CTC) count. Anti-tumor effects were observed at all dose levels. Anti-tumor activity was better in chemotherapy-naïve CYP17i-naïve subjects compared to post-chemotherapy CYP17i-naïve subjects and post-CYP17i subjects. The higher doses seemed to have a better response than the lower doses, particularly in chemotherapy-naïve CYP17i-naïve subjects.

Further details can be found in the latest available version of the IB, which contains comprehensive information on the study drug.

3.3 Rationale of the study

Management of advanced prostate cancer has evolved considerably in recent years and new therapeutic agents with diverse mechanisms of action have dramatically changed the paradigm of treatment. However, the vast majority of advances have been made in subjects with metastatic castration-resistant disease. Fewer advances have been made in the earlier, hormone-responsive stage of metastatic disease.

ADT remains the mainstay of treatment in metastatic hormone-sensitive prostate cancer. However, despite the demonstrated antitumor activity of ADT in mHSPC, subjects ultimately progress and die of CRPC (5; 6; 7).

The variations upon ADT timing and schedule have not resulted in substantial changes in clinical outcome. Similarly, combined androgen blockade using both an LHRH agonist and an anti-androgen has shown a modest survival benefit over castration, with higher toxicity and costs (12; 13; 14).

The treatment scenario of mHSPC has only recently been challenged, when the randomized controlled CHAARTED trial showed improved survival with upfront chemotherapy (docetaxel) combined with ADT compared to standard ADT in men with metastatic hormone-sensitive prostate cancer, with a 17-month median OS benefit in those with high volume disease (25). Recent data from the STAMPEDE trial confirm the role of upfront chemo-hormonal therapy in mHSPC, showing a significant improvement in OS for subjects with metastatic disease treated with docetaxel in addition to ADT (26; 27).

Docetaxel combined with ADT is, therefore, anticipated to become a new standard for subjects with mHSPC.

It is clear that while ADT is the backbone of treatment, and upfront chemotherapy combined with ADT considerably improves survival, hormone-sensitive disease remains an area of active investigations. Novel treatment approaches are needed to improve disease control and survival and delay castration resistance.

Darolutamide is a novel non-steroidal AR inhibitor lacking a significant agonist action on AR, unlike first generation anti-androgens. It has high binding affinity to AR and a distinctive ability to decrease testosterone-mediated nuclear localization of AR in AR overexpressing cells. In phase I-II clinical studies, darolutamide has demonstrated a very favorable safety profile, with no dose-limiting toxicities observed, and substantial antitumor activity in mCRPC, with effect on both serum PSA and soft tissue and bone lesions. A phase III trial is ongoing to compare darolutamide versus placebo in men with non-metastatic CRPC.

The excellent tolerability and the clinical activity demonstrated so far in phase I/II in mCRPC provide the rationale to explore the efficacy of darolutamide in the earlier stages of metastatic hormone-sensitive prostate cancer, where novel therapeutic strategies are needed to improve the outcome of subjects before castration resistance occurs.

This randomized phase III study aims to demonstrate that the addition of darolutamide to ADT and docetaxel chemotherapy significantly prolongs OS over placebo in mHSPC subjects.

Based on *in vitro* transporter inhibition and *in vitro* CYP3A4 induction results with darolutamide, a small effect on docetaxel exposure during concomitant treatment with darolutamide cannot be excluded. It is, therefore, planned to investigate the potential impact of darolutamide on docetaxel PK and the respective impact on safety on the day of the first docetaxel administration in the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel (see Section 13.1.1). Based on the mechanisms described above as well as known safety profiles from darolutamide and docetaxel, the toxicities from the

combination therapy are expected to be well managed. Therefore, the study will continue subject accrual during this PK and safety review period.

3.4 Benefit–risk assessment

Darolutamide treatment has been well tolerated in clinical studies in mCRPC subjects. There is no important risk identified. Fall and non–pathological fractures are currently considered important potential risks. No dose–limiting toxicities were observed in the phase I dose escalation. The AE profile observed reflects a profile that is expected in a population of subjects with advanced prostate cancer. Most AEs were Grade 1 to 2 and assessed by the investigators as not related to the study treatment. No dose–related trends were noted in the AE profile. Antitumor activity was seen at all dose levels as evaluated by PSA, CTC counts, and soft and bone lesion imaging. Subjects who were naïve to treatment with chemotherapy or CYP17i have responded best to darolutamide treatment.

With its potent anti–androgenic profile, there is a strong rationale to study darolutamide in subjects with mHSPC to delay disease progression and death. The available safety and efficacy data suggest that subjects who participate in this trial are not placed at undue risk.

As presented in the previous section, ADT and docetaxel have demonstrated significant survival benefits with the tolerable toxicities and is anticipated to become a new standard for subjects with mHSPC.

Overall, the benefit/risk assessment for darolutamide in addition to standard ADT and docetaxel therapy for mHSPC subjects is positive.

4. Study objectives

The primary objective of this study is:

- To demonstrate the superiority in OS of darolutamide in addition to standard ADT and docetaxel over placebo in addition to standard ADT and docetaxel

The secondary objectives of this study are to evaluate:

- Time to castration–resistant prostate cancer
- Time to initiation of subsequent antineoplastic therapy
- Symptomatic skeletal event free survival (SSE–FS)
- Time to first symptomatic skeletal event (SSE)
- Time to initiation of opioid use for ≥ 7 consecutive days
- Time to pain progression
- Time to worsening of physical symptoms of disease based on NCCN–FACT FPSI–17
- Safety

The exploratory objectives of this study include:

- Quality of life
- Medical resource use
- Prostate-specific antigen (PSA) assessments
- Pharmacokinetics and exposure-response analysis
- Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease

5. Study design

Design overview

This is an international, randomized, double-blind, placebo-controlled phase III study of darolutamide in subjects with mHSPC.

The estimated sample size is approximately 1,300 subjects (see Section 10.4).

The start of the study period is defined by signing of the informed consent form (ICF). After an up to 28-day Screening period, subjects meeting the eligibility criteria (i.e. all of the inclusion criteria and none of the exclusion criteria [see Sections 6.1 and 6.2]) will be randomly assigned in a 1:1 ratio to treatment with one of the study drugs (darolutamide or placebo):

- Darolutamide 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg
- Placebo matching darolutamide tablets in appearance, twice daily with food

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended.

Six cycles of docetaxel will be administered after randomization.

Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator.

For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

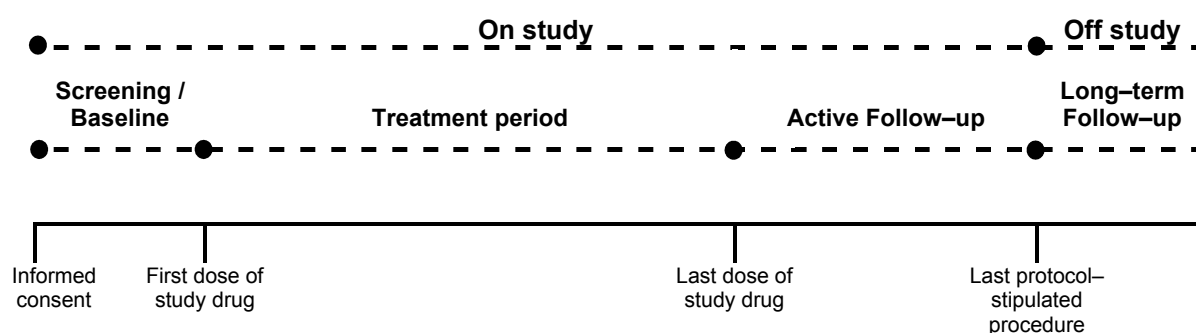
Subjects will be stratified at randomization as follows:

- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases
 - Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase (ALP)
 - $ALP < \text{upper limit of normal (ULN)}$
 - $ALP \geq \text{ULN}$

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

A schematic of the study periods is presented in [Figure 5–1](#).

Figure 5–1 Study periods schematic



The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up.

Screening period:

All trial-related procedures and evaluations will only be performed after the subject has agreed to participate and has signed the ICF. The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met.

Once eligibility is confirmed and documented, eligible subjects will be randomized in a ratio of 1:1 to treatment with darolutamide or placebo.

Treatment period:

The start of the treatment period is defined by the first administration of study drug. Treatment will be provided for all subjects twice daily until disease progression (symptomatic progressive disease, change of antineoplastic therapy), unacceptable toxicity, consent withdrawal or withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.

Subjects will be evaluated every 12 weeks for CRPC, initiation of subsequent antineoplastic therapy, SSEs, opioid use for ≥ 7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, AEs and SAEs, QoL and PSA assessment.

Active Follow-up period:

The Active Follow-up period is the interval from the end-of-study drug intake to the end of all protocol-specified post-treatment interventions.

It includes:

- End of Treatment Visit (EOT)

An EOT Visit will be conducted 30 (+7) days after the last dose of study drug. The following assessment will be performed: QoL, pain assessment, analgesic consumption, subsequent antineoplastic treatments for prostate cancer, which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, other), SSEs, and all AEs and SAEs regardless of causality.

- Active Follow-up visits

During the Active Follow-up period, the following assessment will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent antineoplastic treatments for prostate cancer, which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, other), SSEs, and study drug-related SAEs with concomitant medications received. After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data.

Long-term Follow-up period:

After Active Follow-up, subjects will continue to be contacted approximately every 12 weeks (by phone) to capture all antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs with concomitant medications received, and survival status. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study.

Survival sweep:

For every formal analysis of OS, survival data will be collected through additional survival sweeps. All subjects considered alive at the database cut-off date and prior to any subsequent additional analysis will be contacted for survival status.

End-of-study

The end of the study as a whole will be reached as soon as the last visit of the last subject has been reached in all centers in all participating countries (European Union [EU] and non-EU).

This study will end when all subjects on darolutamide treatment have transitioned into a separate program to continue receiving darolutamide (see Section 8.2) or have discontinued the study for any other reason (e.g. death, lost to follow-up, consent withdrawal with no further data collection), and all subjects on placebo have discontinued treatment.

Until transition to the separate program, subjects on darolutamide treatment will continue to follow all the protocol required procedures and visits in the current protocol. Once the separate program is available for subjects to move to, study treatment in the current study will no longer be available.

Primary completion

The first analysis of OS will be performed as an interim futility analysis when approximately 153 deaths have occurred. In case the study continues after the futility analysis, a final efficacy analysis of OS will be performed when approximately 509 deaths have occurred and will be considered the primary completion. The database cut-off date for this analysis will be the date of primary completion.

6. Study population

6.1 Inclusion criteria

Subjects must meet the following criteria at the time of Screening:

1. Written informed consent.
2. Males ≥ 18 years of age.
3. Histologically or cytologically confirmed adenocarcinoma of prostate.
4. Metastatic disease documented either by a positive bone scan, or for soft tissue or visceral metastases, either by contrast-enhanced abdominal/pelvic/chest computed tomography (CT) or magnetic resonance imaging (MRI) scan assessed by Investigator and confirmed by central radiology review. Metastatic disease is defined as either malignant lesions in bone scan or measurable lymph nodes above the aortic bifurcation or soft tissue/visceral lesions according to Response Evaluation Criteria In Solid Tumors (RECIST) 1.1 (31). Lymph nodes are measurable if the short axis diameter is ≥ 15 mm, soft tissue/visceral lesions are measurable if the long axis diameter is ≥ 10 mm.
 - Subjects with regional lymph node metastases only (N1, below the aortic bifurcation) will not be eligible for the study. Only subjects with non-regional lymph node metastases (M1a) and/or bone metastases (M1b) and/or other sites of metastases with or without bone disease (M1c) will be eligible.
5. Subjects must be candidates for ADT and docetaxel therapy, per Investigator's judgment.
6. Started ADT (LHRH agonist/antagonist or orchiectomy) with or without first generation anti-androgen, but no longer than 12 weeks before randomization. For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended. First generation anti-androgen has to be stopped prior to randomization.
7. An Eastern Cooperative Oncology Group performance status (ECOG PS) of 0 or 1.
8. Blood counts at Screening: hemoglobin ≥ 9.0 g/dL, absolute neutrophil count $\geq 1.5 \times 10^9$ /L, platelet count $\geq 100 \times 10^9$ /L (subject must not have received any growth factor within 4 weeks or a blood transfusion within 7 days of the hematology laboratory sample obtained at Screening).
9. Screening values of serum alanine aminotransferase (ALT) and/or aspartate transaminase (AST) $\leq 1.5 \times$ ULN, total bilirubin $\leq$ ULN, creatinine $\leq 2.0 \times$ ULN.

10. Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the treatment with darolutamide/placebo and for 3 months after treatment with darolutamide/placebo and 6 months after treatment with docetaxel.

6.2 Exclusion criteria

Subjects who meet any of the following criteria at the time of Screening will be excluded:

1. Prior treatment with:
 - LHRH agonist/antagonists started more than 12 weeks before randomization
 - Second-generation AR inhibitors such as enzalutamide, ARN-509, darolutamide, other investigational AR inhibitors
 - CYP17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer
 - Chemotherapy or immunotherapy for prostate cancer prior to randomization
2. Treatment with radiotherapy (external beam radiation therapy [EBRT], brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization.
3. Known hypersensitivity to any of the study drugs, study drug classes, or excipients in the formulation of the study drugs.
4. Contraindication to both CT and MRI contrast agent.
5. Had any of the following within 6 months before randomization: stroke, myocardial infarction, severe/unstable angina pectoris, coronary/peripheral artery bypass graft, congestive heart failure (New York Heart Association Class III or IV).
6. Uncontrolled hypertension as indicated by a resting systolic BP ≥ 160 mmHg or diastolic BP ≥ 100 mmHg despite medical management.
7. Had a prior malignancy. Adequately treated basal cell or squamous cell carcinoma of skin or superficial bladder cancer that has not spread behind the connective tissue layer (i.e., pTis, pTa, and pT1) is allowed, as well as any other cancer for which treatment has been completed ≥ 5 years before randomization and from which the subject has been disease-free.
8. A GI disorder or procedure which is expected to interfere significantly with absorption of study drug.
9. An active viral hepatitis, known human immunodeficiency virus infection with detectable viral load, or chronic liver disease with a need of treatment.
10. Previous (within 28 days before the start of study drug or 5 half-lives of the investigational treatment of the previous study, whichever is longer) or concomitant participation in another clinical study with investigational medicinal product(s).

11. Any other serious or unstable illness, or medical, social, or psychological condition, that could jeopardize the safety of the subject and/or his/her compliance with study procedures, or may interfere with the subject's participation in the study or evaluation of the study results.
12. Inability to swallow oral medications.
13. Close affiliation with the investigational site (e.g., a close relative of the Investigator, dependent person [e.g., employee or student of the investigational site]).
14. Previous assignment to treatment in this study.

For prohibited concomitant therapy, please refer to Section [8.1.3](#).

6.3 Justification of selection criteria

The inclusion/exclusion criteria used in this study, including concomitant conditions and medications as outlined above and in Section [8.1](#) are based upon knowledge of the known toxicity profile of the agent under investigation and the understanding of the concomitant conditions associated with prostate cancer. To ensure that subjects are in the earliest stages of mHSPC, before castration resistance occurs, ADT initiation within 12 weeks before randomization was selected as an inclusion criterion.

The selection criteria were chosen to ensure, to the best of the Sponsor's ability, the safety of the subjects who may participate in this study.

6.4 Withdrawal of subjects from study

6.4.1 Withdrawal

Withdrawal criteria

General procedures

In all cases, the reason for withdrawal must be recorded in the case report form (CRF) and in the subject's medical records.

The subject may object to the generation and processing of post-withdrawal data as specified in Section [13.4](#).

Details for the premature termination of the study as a whole (or components thereof) are provided in Section [12](#).

Withdrawal from the treatment with study drug

Subjects must be withdrawn from the treatment with study drug if any of the following occurs:

- If, in the Investigator's opinion, continuation of the study drug would be harmful to the subject's well-being
- Disease progression (symptomatic progressive disease, change of antineoplastic therapy)
- Unacceptable toxicity
- Any subject requiring study drug interruption >28 consecutive days
- Any subject requiring dosing of the study drug below 300 mg bid
- Occurrence of Grade 3 or higher treatment-related AE while the subject is on 300 mg bid

Note: Study drug discontinuation (i.e., discontinuation during the treatment period) does not constitute withdrawal from the study. Every effort should be made to retain subjects who discontinue the treatment period for any reason. These subjects are to be encouraged to remain on the study for follow-up of primary, secondary, and exploratory endpoints (i.e., continue in the Active Follow-up period or Long-term Follow-up period). Subjects are expected to participate in follow-up unless they explicitly object. Withdrawal of consent should be documented in the subject's medical file. For guidance on docetaxel background treatment, see Section 7.4.3.

Withdrawal from study

Subjects must be withdrawn from the study if any of the following occurs:

- At their own request or at the request of their legally acceptable representative. At any time during the study and without giving reasons, a subject may decline to participate further. The subject will not suffer any disadvantage as a result.
- At the specific request of the Sponsor and in liaison with the Investigator (e.g. obvious non-compliance, safety concerns)
- Sponsor terminates the study

Depending on the time point of withdrawal, a withdrawn subject is referred to as either "screening failure" or "dropout" as specified below:

Screening failure

A subject who, for any reason (e.g. failure to satisfy the selection criteria), terminates the study before the time point used for the definition of "dropout" (see below) is regarded a "screening failure". The reason will be documented in the source documentation and the electronic case report form (eCRF) and the subject will discontinue participation in study with no further follow-up required.

Re-screening of a subject who did not fulfill the eligibility criteria is allowed within 2 weeks from the prior screen failure date. If a subject is re-screened, a new screening number will be allocated and a link will be made between the screening numbers of the same subject.

Re-screening of screen failed subjects may only be allowed once after discussion with the Bayer-designated medical representative or Sponsor and after approval by the Sponsor. Sponsor approval of re-screening for a screen failed subject must be documented.

The subjects who need to repeat the screening procedures will be re-consented.

In any case, the Investigator has to ensure that the repeated screening procedures do not expose the subject to an unjustifiable health risk.

Dropout

A subject who discontinues study participation prematurely for any reason is defined as a “dropout” if the subject has already been randomized.

6.4.2 Replacement

A subject withdrawn from the treatment with study drug will not be replaced.

6.5 Subject identification

The subject number is a 9-digit number consisting of:

Digits 1 to 2 = Country code

Digits 3 to 5 = Center number within the country

(Digits 1 to 5 = Trial unit)

Digits 6 to 9 = Current subject number within the center

7. Treatment(s)

7.1 Treatments to be administered

Study drugs

The following treatment groups are defined for this study:

- Darolutamide 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg
- Placebo matching darolutamide tablets in appearance, twice daily orally with food

Background treatment

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy.

Six cycles of docetaxel will be administered after randomization. Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator. The first cycle of docetaxel should be administered within 6 weeks after start of study drug. For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug). Docetaxel should be prepared at the study centers in accordance with the product information and/or local standards and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy following within one hour after the administration of study drug. Administration of study drug in these 20 subjects should come before docetaxel administration.

7.2 Identity of study treatment

7.2.1 Darolutamide and placebo

All study drugs will be labeled according to the requirements of local law and legislation. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request.

For all study drugs, a system of numbering in accordance with all requirements of Good Manufacturing Practice (GMP) will be used, ensuring that each dose of study drug can be traced back to the respective bulk batch of the ingredients. Lists linking all numbering levels will be maintained by the Sponsor's clinical supplies Quality Assurance group.

A complete record of batch numbers and expiry dates of all study drugs as well as the labels will be maintained in the Sponsor's study file.

Darolutamide and matching placebo tablets will be provided by Bayer AG. The items in [Table 7–1](#) describe the elements used to identify each study drug.

Table 7–1 Study drug identifiers

Name	Darolutamide	Placebo
Formulation	Film coated tablets for oral administration	Film coated tablets for oral administration
Appearance	Blue oval shaped tablets embossed with 'OR-300'	Blue oval shaped tablets embossed with 'OR-300'
Galencial form	Film coated tablet	Film coated tablet
Composition	Drug substance: 300 mg BAY1841788 Excipients: Lactose monohydrate Calcium hydrogen phosphate anhydrous Croscarmellose sodium Povidone Magnesium stearate Laquer blue	Excipients: Lactose monohydrate Cellulose microcrystalline Calcium hydrogen phosphate anhydrous Croscarmellose sodium Povidone Magnesium stearate Laquer blue
Strength	300 mg	Not applicable
Bayer number	BAY 1841788	Not applicable
Packaging	White plastic bottles of HDPE each containing 140 tablets	White plastic bottles of HDPE each containing 140 tablets

Abbreviations: HDPE = high density polyethylene.

7.2.2 Background docetaxel treatment

Docetaxel is commercially available. The participating investigators are required to consult the product information for docetaxel.

All treatment provided by the Sponsor will be labeled according to the requirements of local law and legislation. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request. For all study treatment provided by the Sponsor, a system of numbering in accordance with all requirements of GMP will be used, ensuring that each dose of study treatment can be traced back to the respective bulk ware of the ingredients. Lists linking all numbering levels will be maintained by the Sponsor's clinical supplies Quality Assurance group. A complete record of batch numbers and expiry dates as well as the labels of all study treatments provided by the Sponsor will be maintained in the Sponsor study file.

7.3 Treatment assignment

At the end of the screening period, all subjects meeting all of the inclusion criteria and none of the exclusion criteria will be randomly assigned on a 1:1 basis in a blinded fashion to treatment with darolutamide or matching placebo plus ADT and docetaxel. In addition, randomization will be stratified by:

- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases
 - Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase
 - $ALP < ULN$
 - $ALP \geq ULN$

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

A computer-generated randomization list will be prepared by the Bayer Randomization Manager. The randomization number will be used to link the subject to a treatment group and to a medication package.

The Investigator (or designated associate) will obtain the randomization number and the medication package numbers assigned to a subject by accessing the Interactive Voice/Web Response System (IXRS) after confirming that the subject meets all the inclusion criteria and does not meet any of the exclusion criteria.

An IXRS description and handbook will be provided to each investigational site.

7.4 Dosage and administration of study treatment

7.4.1 Dose modifications of study drug

The study drug will be administered as oral 300 mg tablets. The dose of study drug to be administered is 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg. Placebo matching darolutamide tablets in appearance will be administered twice daily with food.

The dose or dosing schedule of the study drug may be modified following the occurrence of clinically significant AEs. Doses of study drug may be delayed or reduced in case of clinically significant toxicities that are considered by the Investigator to be related to study drug. Toxicities will be graded using the National Cancer Institute–Common Terminology Criteria for Adverse Events (NCI–CTCAE) v 4.03. All dose modifications regardless of relatedness should be recorded on the eCRF. If a subject experiences several study drug–related toxicities with different grading, the recommendation of the worst grading should be used.

If a dose of study drug is delayed, dose can be taken up to 6 hours later with food to make up for the missed one.

7.4.1.1 Dose interruption

The maximum time for a dose interruption period is 28 consecutive days. Any subject requiring treatment interruption >28 consecutive days must be withdrawn from treatment with study drug.

7.4.1.2 Dose reduction

If considered necessary for subject's safety, the dose of study drug may be reduced to 300 mg bid. The medical monitor must be notified of any darolutamide dose reduction.

Dosing of the study drug below 300 mg bid is not allowed. If a Grade 3 or higher treatment-related AE occurs while the subject is on 300 mg bid, the subject must be withdrawn from the treatment with study drug.

7.4.1.3 General recommendations for dose modifications of study drug

A subject who experiences a treatment-related Grade 3 or 4 AE should interrupt study drug until the AE improves to Grade 2 or less. Treatment with study drug is then to be restarted at 300 mg bid.

Additional details are provided in [Table 7-2](#).

Table 7-2 Study drug dose modifications

Severity grade (NCI-CTCAE v 4.03)	Dose modifications	Study treatment withdrawal
Grade 0-2	Treat on time. Per Investigator's decision to interrupt or reduce study drug ^{a,b}	—
Grade 3 or 4	Delay until ≤ Grade 2 ^a When the severity is Grade ≤2, restart at a reduced dose of 300 mg bid ^{b,c}	If the dosing of the study drug is temporally or permanently reduced to 300 mg bid and a Grade 3 or higher treatment-related AE occurs while the subject is on a dose of 300 mg bid, the subject must be withdrawn from treatment with study drug

Excludes clinically nonsignificant and asymptomatic laboratory abnormalities.

Abbreviations: AE = adverse event; bid = twice daily; NCI-CTCAE v 4.03 = National Cancer Institute–Common Terminology Criteria for Adverse Events version 4.03.

^a If there is no recovery after 28 consecutive days, treatment with study drug should be permanently discontinued.

^b When AE returns to baseline or is resolved, dose escalation to 600 mg bid may be considered at the discretion of the Investigator.

^c If, following a dose reduction, dose is escalated to 600 mg bid and a second treatment-related AE with a severity of Grade 3 or higher occurs, a permanent dose reduction to 300 mg bid is required. A third occurrence of a Grade 3 or higher treatment-related AE requires permanent discontinuation of treatment with study drug.

7.4.2 Dosage and administration for background treatment

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy.

The docetaxel dose to be administered is 75 mg/m² Day 1 as 1 hour IV infusion every 21 days, in line with the summary of products characteristics. At Investigator's discretion, longer infusions are allowed for medical reasons. The cycle should be repeated every 3 weeks for 6 cycles. Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator. To prevent hypersensitivity reactions and fluid retention, the recommended pre-medication regimen is oral dexamethasone 8 mg, 12 hours, 3 hours and 1 hour before the docetaxel infusion. Anti-emetic regimens are recommended as per local clinical practice.

7.4.3 Dose modifications of docetaxel background treatment

Dose adjustments of docetaxel should be made based on the specific types of toxicities observed, graded using NCI-CTCAE v4.03.

Docetaxel should be administered when the neutrophil count is $\geq 1,500$ cells/mm³. In order to monitor the occurrence of neutropenia, which may be severe and result in infection, frequent blood cell counts should be performed on all subjects receiving docetaxel.

Docetaxel should not be given to subjects with bilirubin >ULN, or to subjects with AST and/or ALT >1.5 x ULN. Bilirubin, AST and ALT values should be obtained prior to each cycle of docetaxel. In subjects who experienced either febrile neutropenia, neutrophil count <500 cells/mm³ for more than one week, severe or cumulative cutaneous reactions or severe peripheral neuropathy during docetaxel therapy, the dose of docetaxel should be reduced from 75 to 60 mg/m². If the subject continues to experience these reactions at 60 mg/m², the docetaxel treatment should be discontinued. Should treatment with docetaxel be discontinued for any reason before completion of 6 cycles, the treatment with study drug may be continued.

7.5 Blinding

Subjects will be randomized to receive darolutamide or matching placebo in a double-blind fashion such that neither the Investigator, nor the Sponsor, nor the subject will know which agent is being administered. The randomization number will be assigned through the IXRS based on information supplied by the Investigator at the time of randomization.

Darolutamide and placebo will be identical in appearance in order to preserve blinding. To maintain the blind, study drugs (either darolutamide or matching placebo) will be packaged in a drug pack labeled with a unique drug pack number. The study drug pack number will be assigned to the subject through the IXRS.

The DMC will review in an unblinded manner the safety data and PK of the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel. However, the investigators, subjects and the Sponsor will remain blinded.

Bioanalytic analysis of darolutamide and docetaxel PK samples will be performed by an unblinded bioanalyst at an external laboratory.

SUSAR unblinding

In the event of a Suspected Unexpected Serious Adverse Reaction (SUSAR) (see Section 9.6.1.4) related to the blinded treatment, the subject's treatment code will usually be broken before reporting to health authorities and central ethics committees in an unblinded fashion. In compliance with the European Commission's Detailed Guidance on the collection and verification of presentation of AE/reaction reports arising from clinical trials on medicinal products for human use, investigators and persons responsible for the ongoing conduct of the study will not usually receive unblinded copies of SUSAR reports, unless unblinded information is judged necessary for safety reasons.

Emergency unblinding by the Investigator

Investigators may only unblind subjects under emergency unblinding rules. If a subject is unblinded by the Investigator, they must discontinue study drugs.

Investigators should note that emergency unblinding is reserved for emergency situations where lack of knowledge about the actual study drug treatment interferes with appropriate emergency management.

The occurrence of an SAE should not routinely precipitate the immediate unblinding of the label. Progressive disease itself does not routinely constitute an emergency situation, therefore unblinding the label only for subsequent treatment decision is not allowed.

If emergency unblinding is necessary for the treatment of a subject for an SAE, the treatment with study drug can be unblinded via the IXRS system (refer to the IXRS manual for instructions). The participating site has unrestricted and immediate access to break the treatment code in IXRS. Should the blinded code be broken for a subject, the medical monitor or designee should be contacted by the principal investigator (PI) within 1 working day of unblinding to discuss the rationale for the premature unblinding.

7.6 Drug logistics and accountability

All study drugs will be stored at the investigational site in accordance with Good Clinical Practice (GCP) and GMP requirements and the instructions given by the clinical supplies department of the Sponsor (or its affiliate/contract research organization [CRO]), and will be inaccessible to unauthorized personnel. Special storage conditions and a complete record of batch numbers and expiry dates can be found in the Sponsor's study file; the site-relevant elements of this information will be available in the Investigator site file. On the day of receipt, the responsible site personnel will confirm receipt of study drug via IXRS. The personnel will use the study drug only within the framework of this clinical study and in accordance with this protocol. Receipt, distribution, return, and destruction (if any) of the study drug must be properly documented according to the Sponsor's agreed and specified procedures.

Written instructions on medication destruction will be made available to affected parties as applicable.

The Investigator or designated associate(s) are responsible for dispensing the study drugs (darolutamide or placebo).

The Investigator is responsible for control of the distribution and use of the study drug. Bottles must be returned to the Investigator with all unused medication. Throughout the study, all unused study drug must be accounted for. This information will be recorded in the drug dispensing log.

Study subjects should be instructed to return all unused tablets to the study site at their next study visit for drug accountability and proper disposal. Study subjects must be instructed not to dispose of unused tablets on their own. Drug accountability should be done at the time bottles are returned by the subject. Any discrepancies should be reported immediately upon discovery.

7.7 Treatment compliance

Any discrepancies between actual and expected amount of returned study medication must be discussed with the subject at the time of the visit, and any explanation must be documented in the source records.

8. Non–study therapy

8.1 Prior and concomitant therapy

All subjects must receive an ADT of the Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy started ≤ 12 weeks before randomization (if combined with a first-generation anti-androgen, such as bicalutamide, flutamide, nilutamide, or cyproterone acetate, it must be stopped before randomization). For subjects receiving LHRH agonists, treatment in combination with a first generation anti-androgen for at least 4 weeks prior to randomization is recommended.

Six cycles of docetaxel will be administered after randomization.

Docetaxel can be administered in combination with prednisone/prednisolone at the discretion of the Investigator.

8.1.1 Prior therapy

All medications taken in the first 28 days prior to randomization (including start/stop dates, dose frequency, route of administration, and indication) must be recorded in the subject's source documentation, as well as entered in the appropriate pages of the eCRF. Prior local treatment for prostate cancer (surgery and/or radiation) before the time of informed consent (IC) is allowed and should be recorded in the medical record and in the appropriate eCRF. Prior treatment with LHRH agonist/antagonists started more than 12 weeks before randomization, second-generation AR inhibitors, other investigational AR inhibitors, CYP17 enzyme inhibitor such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer is not allowed. Prior chemotherapy or immunotherapy for prostate cancer is not allowed. Treatment with radiotherapy (EBRT, brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization is not allowed. Blood transfusions are not allowed within 7 days and growth factors are not allowed within 4 weeks of the hematology laboratory sample obtained at Screening. For a detailed list of prohibited medications and treatments, see Sections [6.2](#) and [8.1.3](#).

8.1.2 Permissible concomitant medications and treatments

All concomitant treatments from the time of IC until the end-of-study treatment visit must be recorded on the CRFs. Once the subject has been withdrawn from the treatment with study drug, follow-up treatments will be recorded if used to treat new study drug-related SAEs or unresolved related AEs or if being used as a systemic antineoplastic therapy for prostate cancer.

Analgesic use will be captured via eCRF. Physicians are to record which opioid and non-opioid medications were used since the last visit, and physicians are to record exact daily doses of each analgesic consumed in the last 24 hours prior to the visit (Analgesic 24 hour consumption log –eCRF). Note that analgesic medication should be recorded via eCRF for each visit that pain is assessed via the Brief Pain Inventory short form (BPI-SF).

Palliative radiation therapy or surgical intervention as needed are allowed during study treatment. Treatment with bisphosphonates and denosumab is allowed.

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy. Switching ADT to an LHRH antagonist is permitted during study treatment.

For docetaxel pre-medication please refer to Section 7.4.2. Supportive care in case of toxicity related to docetaxel including use of biologic response modifiers such as granulocyte colony-stimulating factor or granulocyte-macrophage colony-stimulating factor, should be applied according to standard practice.

Based on results from a drug-drug interaction study (Study 17726), repeated co-administration of strong CYP3A4 inducers (see listing of drugs in Appendix 16.5) with darolutamide is expected to reduce darolutamide plasma concentrations. Therefore, concomitant intake of strong CYP3A4 inducers should be avoided. It is strongly recommended to use alternative treatments. Concomitant short term use is allowed.

Administration of 600 mg darolutamide bid over 4 days prior to administration of a single dose of 5 mg rosuvastatin, a BCRP, OATP1B1 and OATP1B3 substrate, together with food resulted in a 5.2-fold increase in mean exposure [AUC(0–24)] and a 4.9-fold increase in C_{max} of rosuvastatin in Study 17723. This indicates that co-administration of darolutamide can also increase the plasma concentrations of other concomitant BCRP, OATP1B1 and OATP1B3 substrates (e.g. methotrexate, sulfasalazine, fluvastatin, atorvastatin). Therefore, the patients should be closely monitored for signs and symptoms of increased exposure to BCRP, OATP1B1 or OATP1B3 substrates. Dose modification of these substrates should be considered based on the prescriber information or such compounds should be avoided.

8.1.3 Prohibited concomitant medications and treatments

Concomitant treatment with another systemic antineoplastic therapy or another investigational medicinal product is prohibited with the exception of ADT throughout the study and 6 cycles of docetaxel after randomization.

Initiation of the following medications during the study treatment period is prohibited:

- Any investigational medicinal product
- Radiopharmaceuticals
- Immunotherapy (e.g. sipuleucel-T)
- Cytotoxic chemotherapy other than docetaxel for 6 cycles after randomization
- Enzalutamide, ARN-509, bicalutamide, flutamide, nilutamide
- Abiraterone acetate, TAK-700, or other CYP17 inhibitors
- Systemic ketoconazole as antineoplastic treatment for prostate cancer
- ADT switch to LHRH agonist
- Another systemic antineoplastic therapy may be initiated no sooner than 7 days after the last dose of study drug

For prohibited prior therapy, please refer to Section [6.2](#).

8.2 Post-study therapy

After treatment discontinuation, subjects will enter the Active Follow-up period and then the Long-term (Survival) Follow-up period. Therapy after discontinuation of study treatment will be collected as follows:

During Active Follow-up (includes EOT Visit and Active Follow-up visits):

- EOT Visit: collection of medications related to pain and QoL, and collection of medications for study drug-related SAEs.
- Active Follow-up Visit: collection of all subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change PSA progression, clinical progression, radiological progression, toxicity, or other); collection of medications related to pain and QoL, and collection of medications for study drug-related SAEs.

During Long-term (Survival) Follow-up period:

- Long-term Follow-up visits: collection of subsequent antineoplastic treatments for prostate cancer which should be provided with start and stop dates and reason for change (PSA progression, clinical progression, radiological progression, toxicity, or other), and collection of medications for study drug-related SAEs.

After primary analysis of the study:

- For subjects who are ongoing with darolutamide treatment, the sponsor will continue to provide darolutamide via a separate program at least until darolutamide is approved and reimbursed in a specific country.
- Subjects receiving placebo will discontinue the treatment and complete the study. Further treatment is at the discretion of the Investigator.
- Subjects in the Active or Long-term follow-up will complete the study.

9. Procedures and variables

9.1 Tabular schedule of evaluations

All procedures, including efficacy and safety measurements obtained during the course of the study, are summarized in the schedule of assessments ([Table 9-1](#)).

Table 9–1 Schedule of assessments

Procedures	Screening Period		Treatment Period ^a		Follow-up Period		
			Visit 1	Visit 2 and subsequent visits	Active Follow-up Period EOT Visit	Active Follow-up Visits	Long-term Follow-up Period
	Within 28 days before randomization	Within 7 days before randomization	Day 1 (+3 days)	Week 12 (±7 days) and every 12 weeks	30 days (+7 days) ^u after last dose of study drug	Approximately every 12 weeks from discontinuation of treatment up to 1 year	Approximately every 12 weeks after Active Follow-up, until death, lost to follow-up, consent withdrawal, or end-of-study
Informed consent	X						
Pharmacogenetic consent (whole blood only)	X						
Demography	X						
Medical history	X						
Prostate cancer history ^b	X						
Eligibility criteria	X	X					
Medical resource use			X	X	X	X	
Symptom/QoL questionnaire NCCN–FACT–FPI–17 ^c		X	X ^d	X	X	X	
Pain questionnaire BPI–SF ^{c,e}		X	X ^d	X	X	X	
Analgesic 24 hour consumption log – eCRF (physician recording analgesic pain meds consumed over last 24 hours) ^{c,e}		X	X ^d	X	X	X	
Randomization			X ^s				
Physical examination including weight and height ^f	X		X ^g	X	X		
12-lead ECG		X	X ^g	X	X		
Vital signs (BP, HR)		X	X ^g	X	X		
Laboratory safety assessments (hematology, general chemistry and urinalysis) ^h		X		X	X		

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Procedures	Screening Period		Treatment Period ^a		Follow-up Period		
			Visit 1	Visit 2 and subsequent visits	Active Follow-up Period		Long-term Follow-up Period
					EOT Visit	Active Follow-up Visits	
	Within 28 days before randomization	Within 7 days before randomization	Day 1 (+3 days)	Week 12 (±7 days) and every 12 weeks	30 days (+7 days) ^u after last dose of study drug	Approximately every 12 weeks from discontinuation of treatment up to 1 year	Approximately every 12 weeks after Active Follow-up, until death, lost to follow-up, consent withdrawal, or end-of-study
Serum PSA ^h	X			X	X		
Testosterone	X ^h			X ^h	X ^h		
Alkaline Phosphatase for screening ^h	X						
PK samples ⁱ			X	X			
Plasma collection for genetic biomarkers		X	X	X	X		
Plasma collection for non-genetic biomarkers		X	X	X	X		
Whole blood for CTC analyses			X	X ^j	X		
Whole blood collection for pharmacogenetic analysis – only if pharmacogenetic consent is signed			X	(X) ^t			
Collection of tumor tissue (e.g. biopsy)	X ^k				X ^k		
ECOG PS	X		X	X	X		
CT/MRI ^l	X ^m			(X) ⁿ			
Bone scan ^l	X ^o			(X) ^p			
First SSE				X	X	X	
SSE-FS				X	X	X	
Post-study antineoplastic therapy ^q						X	X
Survival status ^v						X	X
AEs/SAEs	X	X	X	X	X	X ^r	X ^r
Concomitant treatments	X	X	X	X	X	X ^r	X ^r
Study drug dispensing			X	X			

AE=adverse events, BP=blood pressure, BPI-SF=Brief Pain Inventory – Short Form; CT=computed tomography, CTC=circulating tumor cells, ECG=electrocardiogram, ECOG PS=Eastern Cooperative Oncology Group Performance Status, eCRF=electronic case report form, HR=heart rate, MRI=magnetic resonance imaging, NCCN-FACT FPSI-17=Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17-item questionnaire; PK=pharmacokinetics, PSA=prostate-specific antigen, QoL=quality of life, SAE=serious adverse event, SSE=symptomatic skeletal event, SSE-FS=symptomatic skeletal event free survival

^a Subjects will receive 6 cycles of docetaxel every 21 days after check of the routine hematology and biochemistry blood tests (see Section 9.6.3.1). The first cycle of docetaxel should be administered within 6 weeks after start of study drug. For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

^b Including treatments and procedures.

^c Questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

^d If not collected at the Screening visit.

^e Analgesic consumption and pain assessment questionnaires must be collected on the same day.

^f Height to be measured at Screening only.

^g If not previously done.

^h PSA and testosterone values are to be assessed by a central laboratory during the study period. In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), and at the EOT Visit using a liquid chromatography–tandem mass spectrometry method for the analysis of total testosterone. The non–protein–bound (free) testosterone will be analyzed and also calculated with the sex hormone–binding globulin and albumin concentrations, both of which will also be determined. Blood samples to test alkaline phosphatase for stratification purpose will be analyzed in a central laboratory and thereafter by a local laboratory. Other laboratory assessments are to be performed by local laboratories. Specific laboratory assessments are detailed in Section 9.6.3.1.

ⁱ In the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, several blood samples will be collected on the day of first docetaxel administration, which should be at least 14 days after start of study drug. On this day, blood samples will be collected at the following time points: Pre–Dose (prior to the administration of darolutamide and docetaxel), 20±10 minutes after start of docetaxel infusion, 1 hour (within 5 minutes before the end of the infusion), 90 minutes, 2, 3, 4, 6, and 8 hours following the start of docetaxel infusion. For planned time points between 90 minutes and 8 hours, blood samples should be collected within ±15 minutes of the planned time. For all other randomized subjects, 2 blood samples for PK will be collected at least one hour apart on the first day of study drug administration independent of docetaxel administration. The first sample should be taken at least 30 minutes after the first study drug dose. In all randomized subjects, two additional blood samples will be collected at 2 subsequent docetaxel cycles (one sample per docetaxel cycle), at any time relative to study drug administration and after docetaxel administration on the same day. For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, the first additional blood sample must be taken from docetaxel cycle 2 onwards. Please refer to Section 9.5.1 for further guidance.

^j To be performed at Visit 2 only.

^k Collection of tumor tissue at Screening is only if available. Biopsy at disease progression is optional and if available.

^l Chest, abdomen and pelvic CT or MRI and bone scan should be performed at the end of docetaxel treatment (within 30 days from last cycle of docetaxel) and yearly thereafter.

^m Contrast–enhanced chest, abdomen, and pelvic CT or MRIs performed within 42 days prior to start of treatment with study drug are acceptable as screening scans if the standard acquisition procedure according to the Imaging Manual was followed.

ⁿ Chest, abdomen and pelvic CT or MRI can be performed at any time in case of PSA progression, symptomatic progressive disease, or change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment; (X) indicates optional assessment.

^o Bone scan performed within 42 days prior to start of treatment with study drug are acceptable as screening scan if the standard acquisition procedure according to the Imaging Manual was followed.

^p Bone scan can be performed at any time in case of PSA progression, symptomatic progressive disease, change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment; (X) indicates optional assessment.

^q Please note that documentation of post–study antineoplastic therapy must include the start and stop dates of each treatment and reason for treatment changes (PSA progression, clinical progression, radiological progression, toxicity, other).

^r Only study drug–related SAEs with concomitant medication received.

^s Randomization must occur within 28 days from ICF signature.

^t If missed at Visit 1.

^u For further details, see Section 9.2.3.

^v For every formal analysis of OS, survival data will be collected through additional survival sweeps. All subjects considered alive at the database cut-off date and prior to any subsequent additional analysis will be contacted for survival status.

9.2 Visit description

Subject study visits are to occur every 12 weeks (± 7 days) starting from randomization. The EOT Visit should occur within a minimum of 30 days (+7 days) after last dose of study drug. The details of these visits are listed in [Table 9–1](#).

9.2.1 Screening period

A prospective study subject will receive both written and verbal information about the study, and will have an opportunity to ask questions and should have sufficient time to decide whether or not to participate in the study. The original signed and dated ICF must be retained in the Investigator's study file and a copy must be provided to the subject.

All subjects for whom the IC procedure is completed successfully will be assigned a subject number.

Subject screening must occur within 28 days prior to randomization. Screening procedures can be performed on separate occasions within the allowed 28 day timeframe. Re-testing is allowed during the 28-day window only if there is a medical or logistical reason. In case screening procedures need to be repeated outside the allowed 28 day timeframe of the screening period, subject should be considered a screen failure. It is estimated that it will take 7 to 14 days to obtain results from the safety laboratory assessments, testosterone, and PSA.

Each subject's eligibility will be confirmed by the Bayer-designated medical representative's written agreement. The Investigator will complete a paper eligibility form confirming the inclusion and exclusion criteria, this will be forwarded to the Bayer-designated medical representative for review and eligibility confirmation which will take into account the results of the central image review for eligibility. The subject will be identified only by their study-specific subject number.

A subject who, for any reason (e.g. failure to satisfy the selection criteria), terminates the study at any time prior to randomization is regarded a "screening failure".

Please refer to [Section 6.4.1](#) for guidance on re-screening procedures.

The following procedures will be performed during the Screening period:

Within **28** days before randomization:

- Signed and dated ICF
- Informed consent for pharmacogenetic consent (whole blood only)
- Demographic data (date of birth, age, racial group)
- Medical history and current medical condition
- Prostate cancer history including prior treatments and procedures

- Assess eligibility criteria
 - Eligibility must be confirmed by written agreement of the Bayer–designated medical representative
- ECOG PS
- Physical examination including weight and height
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), and at the EOT Visit using a liquid chromatography-tandem mass spectrometry (LC–MS/MS) method for the analysis of total testosterone. The non–protein–bound (free) testosterone will be analyzed and calculated with the sex hormone–binding globulin (SHBG) and albumin concentrations, both of which will also be determined
- Alkaline phosphatase for screening (stratification)
 - Note for alkaline phosphatase test during Screening: blood samples to test alkaline phosphatase for stratification purpose will be analyzed in central laboratory and during treatment period by local laboratory
- Contrast–enhanced chest, abdomen, and pelvic CT or MRI. CT/MRI performed within 42 days prior to start of treatment with study drug are acceptable as screening scans if fulfilling the requirements of the Imaging Manual.

Radiologic tumor assessments will include, chest, abdomen and pelvis, and will be evaluated locally at the study site and by the central review. The first radiological assessment must be comprised of an IV (and oral, if indicated, per Imaging Manual) contrast enhanced CT/MRI scans of chest, abdomen and pelvis. Scans done up to 42 days prior to first dose can be used as baseline scans fulfilling the requirements in the Imaging Manual. Contrast–enhanced MRI shall be performed instead of CT when local regulations or allergy to CT contrast media do not permit the use of CT as requested per protocol schedule. Copies of scans should be forwarded to the central imaging facility for independent assessment. Subjects with a contraindication to both, CT and MRI contrast agent, should not be enrolled in this study (see exclusion criterion 4).
- Bone scan. Bone scan performed within 42 days prior to start of treatment with study drug is acceptable as screening scan if performed according to the Imaging Manual. Copies of scans should be forwarded to the central imaging facility for independent assessment
- AEs/SAEs and concomitant treatments, including opioid use for pain
- Optional collection of archival tissue – if available– for tissue biomarker analyses unless precluded by local guidelines, e.g. Independent Ethics Committee (IEC)/ Institutional Review Board (IRB) or regulatory authorities

Within 7 days before randomization:

- Functional assessment of cancer therapy/National Comprehensive Cancer Network prostate cancer symptom index (NCCN–FACT FPSI–17) questionnaire (collected via electronic patient reported outcome [ePRO] device)
- Pain questionnaire – BPI–SF (collected via ePRO device)
- Analgesic 24–hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject). Should be collected on the same day as pain assessment questionnaires
- A 12–lead ECG
- Vital signs (BP and HR)
- Laboratory safety assessments (hematology, chemistry and urinalysis) (see Section 9.6.3.1)
- A blood (plasma) sample for tumor–related biomarker analysis (genetic and non–genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities if eligibility has been confirmed (see Section 13)
- AEs/SAEs
- Concomitant treatments, including opioid use for pain
- Confirmation of eligibility

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study–related procedure or other clinical activities.

At the end of the screening period, the subject will be randomized.

9.2.2 Study treatment period

9.2.2.1 Day 1 (Visit 1)

Randomization must occur within 28 days from ICF signature. The subject will arrive at the study center on the morning of Day 1 (+3 days). The first dose of study drug will be administered with breakfast at the study center. The following procedures will be performed:

- Medical resource use
- Blood draw for 2 PK samples, which are collected at least 1 hour apart, and the first sample should be taken at least 30 minutes after the first darolutamide dose

In the first 20 subjects who were randomized and have received at least one cycle of docetaxel, detailed mandatory blood sampling for PK analysis of darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, keto-darolutamide, and docetaxel will be collected on the day of the first docetaxel administration which should be at least 14 days after randomization (i.e. start of study drug; see Section 9.5).

- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities (see Section 13)
- Whole blood draw for CTC
- Whole blood draw for pharmacogenomics analysis if PI/ICF Pharmacogenetic consent is signed unless precluded by local guidelines, e.g., IEC/IRB or regulatory authorities (see Section 13)
- ECOG PS
- Current medical condition, AEs, SAEs, and concomitant treatments
- Study drug will be dispensed for 12 weeks

If not performed at Screening:

- Functional assessment of cancer therapy/National Comprehensive Cancer Network prostate cancer symptom index (NCCN-FACT-FPSI-17) questionnaire (collected via ePRO device)
- Pain questionnaire – BPI-SF (collected via ePRO device)
- Analgesic 24-hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Physical examination including weight
- A 12-lead ECG
- Vital signs (BP and HR)

Subjects will continue taking study drug at home throughout the study treatment period.

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

Note for docetaxel administration: Subjects will receive 6 cycles of docetaxel every 21 days after check of the routine hematology and biochemistry blood tests (see Section 9.6.3.1). The first cycle of docetaxel should be administered within 6 weeks after start of study drug. For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization (i.e. start of study drug).

9.2.2.2 Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days)

The following procedures will be performed:

- Medical resource use
- Symptom / QoL questionnaire NCCN-FACT-FPSI-17

- Pain questionnaire – BPI–SF
- Analgesic 24–hour consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Physical examination including weight
- A 12–lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry and urinalysis) (see Section 9.6.3.1)
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), and at the EOT Visit using a LC–MS/MS method for the analysis of total testosterone. The non–protein–bound (free) testosterone will be analyzed and calculated with the SHBG and albumin concentrations, both of which will also be determined
- At 2 visits, an additional blood draw for PK analysis (see Section 9.5)
- A blood (plasma) sample for tumor–related biomarker analysis (genetic and non–genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities (see Section 13)
- Collect whole blood sample for CTC (at Visit 2 only)
- ECOG PS
- Chest, abdomen and pelvic CT/MRI: (please refer to Section 9.2.2.3)
- Bone scan: (please refer to Section 9.2.2.3)
- Date of first SSE (if applicable)
- Current medical condition, AEs, SAEs, and concomitant treatments
- Study drug will be dispensed for 12 weeks
- If missed at Visit 1, whole blood collection for pharmacogenetic analysis – only if pharmacogenetic consent is signed

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study–related procedure or other clinical activities.

The treatment period will end at each site when all subjects at the site who are ongoing on darolutamide treatment have transitioned into the separate program to continue receiving darolutamide (see Section 8.2) or have discontinued from the study for any other reason (e.g. lost to follow-up, consent withdrawal), and all subjects on placebo have discontinued the treatment.

9.2.2.3 Requirement for chest, abdomen and pelvic CT/MRI and bone scan during the study treatment period

Contrast-enhanced chest, abdomen and pelvic CT or MRI for the assessment of soft tissue/visceral lesions and bone scan for the assessment of bone lesions should be performed at the end of docetaxel treatment (within 30 days from last cycle of docetaxel) and yearly thereafter.

In addition, contrast-enhanced chest, abdomen and pelvic CT or MRI and bone scan can be performed at any time in case of PSA progression, symptomatic progressive disease, change of antineoplastic therapy or if considered as appropriate in the Investigator's judgment. (Section 9.2.5).

If a subject develops contraindication to CT contrast media during the course of the study after the baseline imaging, contrast enhanced MRI shall be performed. In rare cases where the subject develops contraindication to both CT and MRI contrast media, the post-baseline imaging examination could continue without contrast media.

The method chosen at the baseline should be the same throughout the study. The same technique (e.g., slice thickness, field of view) should be used for all scans during the study treatment period. Preferably all scans should be interpreted by the same investigator/radiologist during the study whenever possible.

Radiological progression by soft tissue/visceral lesions is determined according to RECIST criteria, version 1.1 (31), based on MRI / CT scans of the chest, abdomen, and pelvis performed by the Investigator, as recommended by Prostate Cancer Clinical Trials Working Group (PCWG3) (32).

Bone lesions will be recorded separately from soft tissue/visceral lesions. Radiological progression by bone lesions is determined according to PCWG3 criteria based on whole body ^{99m}Tc methylene diphosphonate bone scans performed by the Investigator.

Further guidance for acquisition of images and assessment rules will be provided in the Imaging Manual.

9.2.3 Active Follow-up period

End-of-study treatment visit

Subjects will have an EOT Visit 30 days (+7 days) after the last dose of study drug. Another systemic antineoplastic therapy may be initiated no sooner than 7 days after the last dose of study drug.

After primary analysis of the study:

- Subjects in the darolutamide arm, who decide to continue darolutamide treatment in a separate program, will have their EOT Visit for the current study at the same time as their first treatment visit in the separate program.
- Subjects in the darolutamide arm, who then do not decide to continue treatment, will have their EOT visit 30 days (+7 days) after the last dose.
- Subjects in the placebo arm will discontinue treatment and will have their EOT Visit at their earliest convenience, and at the latest 30 days (+7 days) after the last dose.

The following procedures will be performed during the EOT Visit:

- Medical resource use
- Symptom / QoL questionnaire NCCN-FACT-FPSI-17
- Pain questionnaire – BPI-SF
- Analgesic consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
- Physical examination including weight
- A 12-lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry, and urinalysis) (see Section 9.6.3.1)
- PSA
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at screening and repeated every 6 months (Visit 3, Visit 5, etc.), and at the EOT Visit using a LC-MS/MS method for the analysis of total testosterone. The non-protein-bound (free) testosterone will be analyzed and calculated with the SHBG and albumin concentrations, both of which will also be determined
- A blood (plasma) sample for tumor-related biomarker analysis (genetic and non-genetic) unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities
- Whole blood draw sample for CTC analysis

- Optional collection of biopsy tissue taken at disease progression, if available, for tissue biomarker analyses unless precluded by local guidelines, e.g. IEC/IRB or regulatory authorities
- ECOG PS
- Date of first SSE (if applicable)
- Current medical condition, AEs, SAEs and concomitant treatments

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

Active Follow-up Visits

After treatment discontinuation, subjects will begin the Active Follow-up period. Active Follow-up visits occur as standard of care clinic visits approximately every 12 weeks for up to 1 year to collect the following:

- Medical resource use
- Review handheld device or tablet and ensure subjects complete the following:
 - Symptom / QoL questionnaire NCCN-FACT-FPSI-17
 - Pain questionnaire – BPI-SF
 - Analgesic consumption log –eCRF (Physician to record whether or not the subject has initiated opioid therapy since last visit and to record analgesic pain meds consumed by the subject over last 24 hours). Should be collected on the same day as pain assessment questionnaires
 - To facilitate assessment of analgesic use, subjects should be requested to bring all pain medication consumed to each visit
- Record SSEs if applicable
- Subsequent antineoplastic treatments for prostate cancer which should be provided with start and stop dates and reason for treatment change (PSA progression, clinical progression, radiological progression, toxicity, other)
- Record survival status
- Study drug-related SAEs with concomitant medications received

Note: For all visits where questionnaires are administered, the questionnaires should be administered at the start of the visit prior to any study-related procedure or other clinical activities.

After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws IC and actively objects to collection of further data.

After primary analysis of the study, all subjects in the Active Follow-up will complete the study.

9.2.4 Long-term (survival) Follow-up period

After completing the Active Follow-up period, subjects will enter the Long-term Follow-up period. Subjects will be contacted approximately every 12 weeks by phone, and additional contacts may be required for overall (survival assessment). The following information will be collected:

- Record subsequent antineoplastic treatments for prostate cancer with start and stop dates and reason for treatment change (PSA progression, clinical progression, radiological progression, toxicity, other)
- Record survival status
- Study drug-related SAEs with concomitant medications received

As this information will be collected over the telephone from the subject, it is the clinical site's responsibility to obtain and make available the source documentation for the information collected from the subject and document it in the eCRF.

After primary analysis of the study, all subjects in the Long-term Follow-up will complete the study.

9.2.5 Unscheduled visits

In the event that there are significant abnormal safety findings or suspected disease progression, control assessments may be performed at any time during the study treatment period according to the judgment of the Investigator. The following procedures (one or more) may be performed at the unscheduled visit, if indicated:

- Physical examination including weight
- 12-lead ECG
- Vital signs (BP, HR)
- Laboratory safety assessments (hematology, chemistry, urinalysis)
- Status of survival
- AEs, SAEs, current medical condition, concomitant treatments
- Chest, abdomen, and pelvic CT/MRI
- Bone scan

Other unscheduled procedures may be performed, as per Investigator's judgment.

The laboratory tests routinely obtained before each cycle of docetaxel must be reported in the unscheduled hematology and biochemistry eCRF.

For every formal analysis of OS, survival data will be collected through additional survival sweeps. All subjects considered alive at the database cut-off date and prior to any subsequent additional analysis will be contacted for survival status.

9.3 Population characteristics

9.3.1 Demographic

Demographic characteristics of date of birth (age), sex, and race/ethnicity should be entered in the eCRFs.

9.3.2 Medical history

Medical history findings (i.e. previous diagnoses, diseases, or surgeries) meeting all criteria listed below will be collected:

- Not pertaining to the study indication
- Start before signing of the ICF
- Considered relevant to the study

Detailed instructions on the differentiation between medical history and AEs can be found in Section [9.6.1.1](#).

History of prostate cancer will be collected separately from the general medical history. This includes, but is not limited to:

- Date of diagnosis
- Staging (performed at diagnosis)
- Prostate cancer history
- Disease status at study entry
- Prior diagnostic and therapeutic procedures
- Prior prostate cancer treatments (any therapy that is ongoing should be reported as concomitant medication)

9.3.3 Other baseline characteristics

Baseline characteristics relating to disease factors include:

- QoL assessment (NCCN–FACT FPSI–17)
- ECOG PS
- Cancer pain assessment

All medications and significant non–drug therapies ongoing during the screening period (28 days prior to randomization) must be entered in the eCRF, including:

- Medication trade name and dose
- Reason for medication
- Start date and end date, or if continuing at time of study entry

9.4 Efficacy

The primary efficacy variable is OS.

Survival status will be assessed from randomization until the end of the Long-term Follow-up period. Date of death and primary cause of death will be recorded.

For definition and analysis of primary variable, and for description of secondary and exploratory variables, please refer to Section [10.3](#).

9.5 Pharmacokinetics/pharmacodynamics

9.5.1 Drug measurements

The concentrations of diastereomers (*S,R*)-darolutamide (BAY 1896951) and (*S,S*)-darolutamide (BAY 1896952), metabolite keto-darolutamide (BAY 1896953) and docetaxel in plasma will be determined by a validated method, e.g. liquid chromatography–tandem mass spectrometry (LC–MS/MS). Darolutamide (BAY 1841788) concentration will be calculated as the sum of diastereomers (*S,R*)-darolutamide and (*S,S*)-darolutamide concentrations.

PK samples will be collected from all subjects to maintain blinding. However PK samples from the placebo arm will not be analyzed.

In the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, detailed mandatory blood sampling for PK analysis of darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, keto-darolutamide, and docetaxel will be collected on the day of the first docetaxel administration which should be at least 14 days after randomization (i.e. start of study drug).

On this day blood samples will be collected at the following time points:

Pre-dose (prior to the administration of study drug and docetaxel), 20±10 minutes after start of docetaxel infusion, 1 hour (within 5 minutes before the end of infusion), 90 minutes, 2, 3, 4, 6, and 8 hours following the start of docetaxel infusion. For planned time points between 90 minutes and 8 hours, blood samples should be collected within ±15 minutes of the planned time.

Two additional samples will be taken at 2 subsequent docetaxel cycles (i.e. one sample per docetaxel cycle, at any time relative to study drug administration and after docetaxel administration on the same day), from docetaxel cycle 2 onwards. These samples will be used to provide information on variability at steady state for darolutamide and to explore any potential effect of darolutamide on the PK of docetaxel.

It is of importance that the actual date and time of blood sampling and study drug administration as well as docetaxel administration are documented in the eCRF because PK calculations will be based on the actual sampling times relative to dosing times.

In all other randomized subjects, sparse sampling for population PK will be collected to determine the exposure (i.e. area under the curve over the dosing interval at steady state) of darolutamide, the diastereomers (*S,S*)-darolutamide and (*S,R*)-darolutamide, the metabolite keto-darolutamide, and docetaxel and to quantify its variability in the target subject population.

Two samples will be taken on the day of first study drug intake (Visit 1 Day 1), at least 30 minutes after the first dose of study drug, approximately 1 hour apart to determine darolutamide absorption data.

Two additional samples will be taken at 2 docetaxel cycles (i.e. one sample per docetaxel cycle from the first cycle onwards, at any time relative to study drug administration and after docetaxel administration on the same day). These samples will be used to provide information on variability at steady state for darolutamide and to explore any potential effect of darolutamide on the PK of docetaxel. The first additional sample in this subset of subjects can be taken on the day of first docetaxel cycle, as long as the subject has already received at least one dose of study drug on the day before.

Drug intake and sampling times have to be accurately documented.

The instructions for sample collection, processing, storage and shipment will be detailed in a separate document (e.g., sample handling sheet or lab manual).

9.5.2 Pharmacokinetic evaluation

Non-compartment approach PK analysis

In the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel:

Pharmacokinetic parameters will be calculated using a non-compartment approach according to the current Bayer guidelines, using Win-Nonlin software. Details of the used version will be given in the clinical study report.

Based on the plasma concentration–time data, the following PK parameters are planned to be estimated for darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, keto-darolutamide, and docetaxel.

Main parameter:

Darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, keto-darolutamide:

AUC(0–12) ¹⁾

1) : Using the predose plasma sample also as 12 hour sample

Docetaxel:

AUC(0–t_{last})

Additional parameters:

Darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, keto-darolutamide:

AUC(0–8), AUC(0– t_{last}), C_{max} , t_{max} , t_{last}

Docetaxel:

C_{max} , t_{max} , t_{last}

To ensure the blinding of the PK–evaluator during preliminary PK evaluation for DMC review, concentration data with masked subject IDs will be provided by Bioanalytics. The evaluation results will then be transferred to an unblinded statistician at the DMC.

Population PK evaluation

In all other randomized subjects:

Individual concentration–time data of darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, and keto-darolutamide will be provided in a clinical study report appendix. The concentration data from this study, which might be augmented with PK data from other studies, will be analyzed to evaluate the variability of the PK of darolutamide, of the respective diastereomers (*S,S*)-darolutamide and (*S,R*)-darolutamide, and of the metabolite keto-darolutamide in this phase III population. The possible effect of relevant covariates on the PK of these compounds might also be evaluated. In addition, exploratory exposure–response analyses of efficacy, PD or safety parameters, e.g. OS, PSA response rate, etc., are planned. In addition, the docetaxel exposure in the two treatment groups will be compared. A population PK approach will be applied for these evaluations, which will be described in detail in a separate Modeling & Simulation (M&S) Plan and the results will be reported separately in the M&S Report.

9.6 Safety

9.6.1 Adverse events

9.6.1.1 Definitions

Definition of adverse event

In a clinical study, an AE is any untoward medical occurrence (i.e. any unfavorable and unintended sign [including abnormal laboratory findings], symptom, or disease) in a subject or clinical investigation subject after providing written IC for participation in the study. Therefore, an AE may or may not be temporally or causally associated with the use of a medicinal (investigational) product.

A surgical procedure that was planned prior to the start of the study by any physician treating the subject should not be recorded as an AE (however, the condition for which the surgery is required may be an AE).

In the following differentiation between medical history and AEs, the term “condition” may include abnormal physical examination findings, symptoms, diseases, laboratory, or ECG results.

- Conditions that started before signing of IC and for which no symptoms or treatment are present until signing of IC will be recorded as medical history (e.g. seasonal allergy without acute complaints).
- Conditions that started before signing of IC and for which symptoms or treatment are present after signing of IC, at unchanged intensity, will be recorded as medical history (e.g. allergic pollinosis).
- Conditions that started or deteriorated after signing of the ICF including those recorded as medical history will be documented as AEs. This includes intercurrent illnesses.

Definition of a serious adverse event

An SAE is classified as any untoward medical occurrence that, at any dose, meets any of the following criteria:

- a. Results in death
- b. Is life threatening

The term ‘life threatening’ in the definition refers to an event during which the subject was at risk of death at the time of the event. It does not refer to an event that, hypothetically, might have caused death if it were more severe.

- c. Requires in–subject hospitalization or prolongation of existing hospitalization

A hospitalization or prolongation of hospitalization will not be regarded as an SAE if at least one of the following exceptions is met:

- The admission results in a hospital stay of less than 12 hours
- The admission is pre–planned
(i.e. elective or scheduled surgery arranged prior to the start of the study)
- The admission is not associated with an AE
(e.g. social hospitalization for purposes of respite care)

However, it should be noted that invasive treatment during any hospitalization may fulfill the criterion of ‘medically important’ and as such may be reportable as an SAE, dependent on clinical judgment. In addition, where local regulatory authorities specifically require a more stringent definition, the local regulation takes precedence.

- d. Results in persistent or significant disability/incapacity

Disability means a substantial disruption of a person’s ability to conduct normal life functions.

- e. Is a congenital anomaly/birth defect
- f. Is another medically important serious event as judged by the Investigator

9.6.1.2 Classifications for adverse event assessment

All AEs will be assessed and documented by the Investigator according to the categories detailed below.

9.6.1.2.1 Seriousness

For each AE, the seriousness must be determined according to the criteria given in Section [9.6.1.1](#).

9.6.1.2.2 Intensity

The severity (or intensity) of AEs should be documented using NCI–CTCAE v 4.03. If no exact matching code is available in NCI–CTCAE v 4.03, the following guide should be used:

- CTC 1 = Mild AE: Transient in nature and generally not interfering with normal activities
- CTC 2 = Moderate AE: Sufficiently discomforting to interfere with normal activities
- CTC 3 = Severe AE: Prevents normal activities
- CTC 4 = Life threatening (subject was at risk of death at the time of the event)
- CTC 5 = Results in death (fatal)

9.6.1.2.3 Causal relationship

The assessment of the causal relationship between an AE and the administration of treatment is a clinical decision based on all available information at the time of the completion of the eCRF. The causal relationship should be assessed to darolutamide/placebo as well as the background docetaxel treatment independently.

The assessment is based on the question whether there was a “reasonable causal relationship” to darolutamide/placebo or docetaxel in question.

Possible answers are “yes” or “no.”

An assessment of “no” should include either:

1. The existence of a clear alternative explanation (e.g. mechanical bleeding at surgical site)
or
2. A lack of plausibility (e.g. the subject is struck by an automobile without evidence that darolutamide/placebo or docetaxel caused disorientation that may have led to the event, or cancer develops a few days after the first administration of a darolutamide/placebo or docetaxel.

An assessment of “yes” indicates that there is a reasonable suspicion that the AE is associated with the use of darolutamide/placebo or docetaxel.

Important factors to be considered in assessing the relationship of the AE to darolutamide/placebo or docetaxel include:

- The temporal sequence relating the event to drug administration. The event should occur after the drug is given. The length of time from drug exposure to event should be evaluated in the clinical context of the event.
- Recovery on drug discontinuation (dechallenge) and recurrence on drug re-introduction (rechallenge). The subject's response after dechallenge or the subject's response after rechallenge should be considered in the view of the usual clinical course of the event in question.
- Underlying, concomitant, intercurrent diseases: Each event should be evaluated in the context of the natural history and course of the disease being treated and any other disease the subject may have.
- Concomitant medication or treatment: The other drugs the subject is taking or the treatment the subject receives should be examined to determine whether any of these may be suspected as a cause for the event in question.
- Known response pattern for this class of drug: Clinical/preclinical.
- Exposure to physical and/or mental stresses: The exposure to stress might induce adverse changes in the recipient and provide a logical and better explanation for the event.
- The pharmacology and PK of the study treatment: The PK properties (absorption, distribution, metabolism and excretion) of the study treatment, coupled with the individual subject's PD should be considered.
- The assessment is not possible.

Causal relationship to protocol-required procedure(s)]

The assessment of a possible causal relationship between the AE and protocol-required procedure(s) is based on the question whether there was a "reasonable causal relationship" to protocol-required procedure(s).

Possible answers are "yes" or "no"

9.6.1.2.4 Action taken with study treatment

Any action on study treatment to resolve the AE is to be documented using the categories listed below:

- Drug withdrawn
- Drug interrupted
- Dose reduced
- Dose not changed
- Not applicable
- Unknown

9.6.1.2.5 Other specific treatment(s) of adverse events

- None

- Remedial drug therapy
- Other

9.6.1.2.6 Outcome

The outcome of the AE is to be documented using the following categories:

- Recovered/resolved
- Recovering/resolving
- Recovered/resolved with sequelae
- Not recovered/not resolved
- Fatal
- Unknown

9.6.1.3 Assessments and documentation of adverse events

All AEs occurring from the time the subject has signed the ICF until 30 (+7) days after the last dose of study drug must be entered in the subject's eCRF. All treatment-emergent AEs must be followed until resolution or stabilization, wherever feasible.

The type of information that should be assessed and recorded by the Investigator for each AE is listed in Section [9.6.1.2](#).

Disease progression per se should not be regarded as an AE. Instead, the associated signs and symptoms should be recorded as AEs.

“Death” should not be recorded as an AE on the AE page. Instead, “death” is the outcome of underlying AE(s).

For all SAEs, the Sponsor has to carry out a separate assessment for expectedness, seriousness, and causal relationship to study drug.

A laboratory test abnormality that is considered to have a clinically relevant effect on the subject (e.g. causing the subject to withdraw from the study, requiring treatment or causing apparent clinical manifestations, or judged relevant by the Investigator) should be reported as an AE.

Documentation of an AE must be supported by an entry in the subject's medical file.

Each event should be described in detail, along with start and stop dates, severity, relationship to investigational product, action taken, and outcome.

9.6.1.4 Reporting of serious adverse events

The definition of SAEs is given in Section [9.6.1.1](#). Each SAE must be followed up until resolution or stabilization by submission of updated reports to the designated recipient.

Investigator's notification of the Sponsor

All investigators will be thoroughly instructed and trained on all relevant aspects of the Investigator's reporting obligations for SAEs. This information, including all relevant contact details, is summarized in the Investigator site file. This information will be updated as needed.

All SAEs occurring during the observation period defined in Section 9.6.1.1 must immediately (within 24 hours of the Investigator's awareness) be reported to the recipient detailed in the Investigator's file. An SAE form must also be completed within 24 hours of the Investigator awareness and forwarded to the designated recipient. Each SAE must be followed up until resolution or stabilization by submission of updated reports to the designated recipient.

Serious AEs that occur 30 (+7) days after the last dose of study drug during Active Follow-up or Long-term Follow-up should be reported on the SAE form, if the Investigator feels that there is a reasonable possibility for the event to have been caused by the study subject's participation in the study. In such cases, the SAEs will be processed by the Sponsor according to all applicable regulations.

If disease progression leads to signs and symptoms that meet the criteria for seriousness (e.g. hospitalization), the associated signs and symptoms may be reported as an SAE, but the underlying cause (that is, "recurrent disease" or "metastatic disease") should not be reported as an SAE. In this case, recurrent or metastatic disease should be mentioned on the SAE form as an "alternative explanation."

An isolated laboratory abnormality that is assigned a severity rating of Grade 4, as defined in NCI-CTCAE v 4.03, is not reportable as an SAE unless the Investigator assesses that the event meets standard International Conference on Harmonization criteria for an SAE. Similarly, a baseline laboratory abnormality that is part of the disease profile should not be reported as an SAE when assigned a severity rating of Grade 4, as defined in NCI-CTCAE v 4.03, specifically when this abnormality is allowed (or not excluded) by the protocol inclusion/exclusion criteria (see Sections 6.1 and 6.2). If an Investigator is in doubt about the applicable reporting obligations, he/she should consult with the study monitor for the Sponsor. The NCI-CTCAE version 4.03 Grade 4 laboratory abnormalities will be documented and will be reviewed on a regular basis.

Notification of the IECs / IRBs

Notification of the IECs/IRBs about all relevant events (e.g. SAEs or SUSARs) will be performed by the Sponsor/delegate and/or by the Investigator according to all applicable regulations.

Notification of the authorities

The Sponsor will expedite all suspected unexpected SAEs and SUSARs as well as other safety issues requiring expedited reporting to the relevant authorities within applicable timelines.

Sponsor's notification of the investigational site

The Sponsor/delegate will inform all investigational sites about reported relevant events (e.g. SUSARs) according to all applicable regulations.

9.6.1.5 Expected adverse events

For this study, the applicable reference document is the most current version of the IB.

Overview listings of frequent events that have occurred so far in the clinical development are shown in the current IB. If relevant new safety information is identified, the information will be integrated into an update of the IB and distributed to all participating sites.

The expectedness of AEs will be determined by the Sponsor according to the applicable reference document and according to all local regulations.

9.6.1.6 Adverse events of special safety interest

None.

9.6.2 Pregnancies

Sexually active male subjects must agree to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential to use a method of effective birth control, during the study treatment and for 3 months after the end of the treatment with darolutamide, and 6 months after treatment with docetaxel (inclusion criterion).

Pregnancies inadvertently fathered by study subjects during the study should be reported and followed up using the Bayer Pregnancy Monitoring Forms, if permissible by local legislation.

For a pregnancy in the partner of a male study subject, all efforts will be made to obtain similar information on course and outcome, subject to the partner's consent.

For all reports, the forms provided are to be used. The report should be submitted within the same timelines as an SAE, although a pregnancy per se is not considered an SAE.

9.6.3 Further safety

9.6.3.1 Laboratory safety assessments

The PSA and testosterone values are to be assessed by a central laboratory during the study period. Blood samples to test alkaline phosphatase for stratification purpose will be analyzed in central laboratory and thereafter by local laboratory. Other laboratory assessments are to be performed by local laboratories.

Instruction on sample collection, handling, labeling, storage, and shipment will be provided in the laboratory manual.

The following laboratory safety assessments will be performed during the screening period, study treatment period and at the end-of-study treatment visit.

At screening, all the following assessments are to be taken within 7 days prior to randomization, but PSA.

Hematology:

- Hemoglobin
- Hematocrit
- Erythrocyte count
- Leukocyte count
- Differential count (lymphocytes, monocytes, eosinophils, neutrophils, basophils)
- Mean corpuscular hemoglobin
- Mean corpuscular volume
- Mean corpuscular hemoglobin concentration
- Thrombocytes

Chemistry:

- Albumin
- Alanine aminotransferase (ALT)
- Aspartate aminotransferase (AST)
- Alkaline phosphatase
 - Note: Blood samples to test alkaline phosphatase for stratification purpose will be analyzed in central laboratory and thereafter by local laboratory
- Bilirubin total (+ direct and indirect bilirubin if total bilirubin is $>1.0 \times \text{ULN}$)
- Calcium
- Blood urea nitrogen
- Creatinine
- Serum glucose
- Total protein
- Lactate dehydrogenase
- Potassium
- Sodium
- Testosterone
 - Note for testosterone: In a subset of approximately 300 subjects, additional determination of total and free testosterone will be performed at Screening and repeated every 6 months (Visit 3, Visit 5, etc.), and at the EOT Visit using a LC-MS/MS method for the analysis of total testosterone. This method allows precise quantification of low testosterone levels in subjects during ADT

treatment. The non-protein-bound (free) testosterone will be analyzed and calculated with the sex hormone-binding globulin (SHBG) and albumin concentrations, both of which will also be determined

- PSA (total) (at screening, blood sample for PSA is to be taken within 28 days prior to randomization)
- Chromogranin A (only at Screening)
- Prothrombin time, expressed as international normalized ratio, in subjects under vitamin K antagonist therapy (local assessment)

Urinalysis:

- Glucose
- Protein
- Erythrocytes
- Leukocytes

Safety assessments for docetaxel: Hematology (for ANC check), ALT, AST, bilirubin total (+ direct and indirect bilirubin if total bilirubin is $>1.0 \times \text{ULN}$) are mandatory safety assessments to be checked before each cycle of docetaxel. Further laboratory safety assessments may also be checked according to Investigator's judgment.

In the event of implausible results, the laboratory may measure additional parameters to assess the quality of the sample (e.g. clotted or hemolyzed) and to verify the results. The results from such additional analyses may neither be included in the clinical database of this study nor evaluated further. If the results are relevant, the Investigator will be informed to determine follow-up activities outside of this protocol.

9.6.3.2 Physical examination

Physical examination will be performed and weight will be measured at all visits (except at Day 1 Visit if already performed at Screening). Height will be recorded at Screening only.

Abnormal physical examination findings are recorded either as medical history or as AEs (see Section 9.6.1.1).

9.6.3.3 12-lead ECGs

All 12-lead ECGs will be recorded in a supine position after at least 10 minutes rest.

9.6.3.4 Vital signs

All vital sign assessments will be recorded in a supine position after at least 10 minutes rest.

9.7 Other procedures and variables

Other exploratory variables in this study are pain intensity and pain interference scores from the BPI-SF, medical resource use, and assessment of biomarkers.

In addition to calculating a pain score based on the worst pain question (i.e., worst pain subscale [WPS] score) on the BPI-SF (Appendix 16.4), the subject's responses on the BPI-SF can be used to calculate a pain score intensity and a pain interference score (29).

9.7.1 Biomarker investigations

Biomarker status will be correlated with clinical outcome to explore whether any biological targets appear to define subject populations that are particularly sensitive or resistant to darolutamide. Details of the biomarker analyses may be described in a separate statistical analysis plan (SAP) and the results of these analyses may be provided as a separate biomarker report.

Tumor biomarker studies will be performed to investigate the mechanisms by which darolutamide may work in subjects with mHSPC and will use tumor biomarkers circulating in plasma or, if available, biopsies either collected prior to study entry or at disease progression.

Tumor biomarkers may be categorized as “non-genetic” (involving protein) or “genetic” (involving ribonucleic acid [RNA] or deoxyribonucleic acid [DNA]). In the current study, analyses of both non-genetic and genetic biomarkers are planned. The analysis of tumor genetic markers by tumor DNA sequencing and/or gene expression studies with tissue samples taken from the primary tumor and metastases will support identification of biomarkers contributing to clinical benefit to therapy. Utmost care will be taken to protect subject identity during biomarker analyses.

Biomarkers related to drug safety, efficacy, or mechanism of action will be analyzed from whole blood and involve targeted genetic testing. Targeted genetic testing will focus on single gene polymorphisms of genes involving, but not limited to, hepatotoxicity, metabolism, or targets involved in drug activity. Identified gene variants aim to predict the best individuals to treat, either selecting for toxicity avoidance or for positive benefit.

In addition to the biomarkers listed above, other biomarkers deemed relevant to gain further knowledge about the pathomechanism of the disease or about the drug (i.e. mode of action related effect or safety of the drug) may be measured, based on newly emerging data from other ongoing studies and / or literature data.

Details on the collection, processing, storage and shipment of biomarker samples will be provided in separate documents (e.g. sample handling sheets or lab manual).

9.7.1.1 Plasma-based biomarker analysis

Subjects will be asked to provide blood samples for biomarker analyses, which will be obtained during regular study visits in which blood will be drawn for other scheduled but unrelated laboratory tests.

Genetic biomarkers from plasma:

These samples are intended to isolate plasma to study tumor markers circulating in blood such as circulating tumor DNA (ctDNA) or circulating tumor RNA. Candidates of genes for the evaluation from ctDNA are AR, PIK3CA, and phosphatase and tensin homolog (PTEN). The types of analyses may comprise the identification of mutations or splice variants.

Furthermore, these plasma samples will serve as source for non-coding microRNA, as possible markers describing treatment response.

Non-genetic biomarkers from plasma:

Plasma samples will also be used to quantify the circulating levels of various proteins of interest, to attempt to identify a protein signature that correlates with drug response.

Plasma samples for the analyses of tumor biomarkers (genetic and non-genetic biomarkers) will be prepared from blood samples obtained at the following time points:

- Screening visit and on Day 1, Visit 1 (pre-dose) once eligibility criteria have been confirmed
- Visit 2 and subsequent Visits
- End of Treatment Visit

On treatment days, blood will be drawn for preparation of plasma samples prior to drug administration, if possible.

In addition to the proteins and genes listed above, other biomarkers deemed relevant to this study will be measured.

Data from the biomarker analyses may be correlated with measures of clinical efficacy or toxicity obtained in this study.

9.7.1.2 Circulating tumor cells

CTCs may serve as source of tumor metastases circulating in blood and may reflect the tumor phenotype or genotype. Enumeration of CTCs may serve as description of tumor burden.

Circulating tumor cells will be analyzed in detail from whole blood samples collected at various time points to assess the effect of darolutamide in decreasing numbers of CTCs as well as to characterize the phenotype and genotype of CTCs such as presence of AR mutations, AR splice variants, and other genomic markers.

A whole blood sample for CTC analysis will be obtained at Visit 1 (once eligibility criteria have been confirmed), Visit 2 and EOT.

9.7.1.3 Tumor-based biomarker analysis

From subjects for which tumor tissue specimens (e.g. from biopsy) are available, tumor specific biomarker analyses are intended to be used for the following purposes: (1) to evaluate mutations, copy numbers or gene rearrangements in known oncogenes such as AR, TMPRSS, cMYC, PI3K, and CYP17 extracted from tumor tissue specimens; (2) to analyze the gene expression of genes of interest such as AR, cMYC, PTEN, PI3K, and AR-regulated genes in RNA extracted from tumor tissue specimens; (3) to evaluate expression of non-coding tumor relevant RNAs such as microRNAs.

Provided sufficient tissue quality and quantity, tumor tissue specimens are also intended to be used to quantify the expression of proteins of interest, to attempt to identify a protein signature that correlates with drug response.

As per Investigator's discretion, and if technically feasible, biopsy at the time of disease recurrence is highly encouraged to be taken. This sample would allow generation of information about the identification of tumor biomarkers playing a role in therapy resistance for which the subject may benefit for further guidance. Note, however, that this is not considered to be mandatory by this study protocol.

9.7.1.4 Genetic biomarker analysis from whole blood

Biomarkers related to drug safety, efficacy, or mechanism of action may be analyzed from whole blood and involve targeted pharmacogenetic testing. These studies will be conducted aiming to find gene variants associated with responses to darolutamide treatment and to predict the best individuals to treat, either selecting for toxicity avoidance or for positive benefit.

Targeted pharmacogenetic testing will focus on single gene polymorphisms (SNPs) on genes involved but not limited to hepatotoxicity, metabolism, or targets involved in drug activity.

Unless precluded by local guidelines, subjects will be asked to sign a separate pharmacogenetic ICF. A whole blood sample for pharmacogenetic testing will be obtained preferably at Visit 1 (once eligibility criteria have been confirmed) or alternatively at any other visit during the treatment period, regardless of the time of study drug intake. Samples can only be collected from subjects who have signed a separate pharmacogenetic ICF. From these blood samples, DNA will be isolated which will be used for the analysis of SNP candidates such as CYP17.

9.7.2 Medical resource use

Information on medical resource use (e.g. hospitalization visits, physician visits etc.) that is associated with the management of AEs as well as non-protocol-driven subject monitoring will be collected by e-CRF that is completed by the physician at each visit.

9.7.3 Subject-reported outcomes

For this study, health-related QoL will be measured using the NCCN-FACT FPSI-17 questionnaire.

The NCCN-FACT FPSI-17 is a validated instrument that was developed to assess symptoms of prostate cancer, symptoms of treatment of prostate cancer, and health related QoL of prostate cancer subjects. The instrument was developed in accordance with recent Food and Drug Administration guidance for development of instruments for ePRO. The instrument contains 17 items, each of which utilizes a Likert scale with 5 possible responses (30). The 10 items reflect disease-related physical symptoms of disease and the responses on the items are to be summed to calculate a disease-related physical symptom subscale score. One item represents emotional symptoms of disease and the response to that item is used to calculate a disease-related emotional symptom subscale score. Four items represent treatment-related symptoms and the responses to these items are summed to calculate a treatment side effect subscale score. Finally, 2 items represent functional well-being and responses to those items are summed to calculate a functions / well-being subscale score. The NCCN-FACT FPSI-17 will be self-administered by the subject via ePRO device. Some subjects may skip an item

altogether if they feel that it is not applicable to them. The questionnaire is to be completed at baseline either within 7 days before randomization or at Visit 1 (Day 1), at all treatment visits (at Visit 2 and subsequent visits), at the End of Treatment (EOT), and approximately every 12 weeks during Active Follow-up.

Brief Pain Inventory – Short form is a validated instrument totaling 15 items to assess subject-reported levels of pain that has a precedent of being used in clinical trials with the goal of obtaining label claims based on the results captured with this instrument. A reduced set of 11 items is used in this trial. This reduced questionnaire is self-administered and designed to evaluate the intensity of, and the impairment caused by, pain. All BPI-SF items are scored using rating scales. Four items measure pain intensity (pain now, average pain, worst pain, and least pain) using 0 (“no pain”) to 10 (“pain as bad as you can imagine”) numeric rating scales, and 7 items measure the level of interference with function caused by pain (general activity, mood, walking ability, normal work, relations with other people, sleep, and enjoyment of life) using 0 (no interference) to 10 (complete interference) rating scales.

The items are aggregated into 2 dimensions: (1) Pain severity index, using the mean of the 4 items on the pain intensity, and (2) Function interference index, using the mean of the 7 pain interference items. All 4 severity items must be completed for aggregating the pain severity index. The function interference index is scored as the mean of the item scores multiplied by 7, given that more than 50% or 4 of 7, of the items have been completed.

The questionnaire is to be completed via ePRO at Baseline (Screening or Visit 1), at treatment visits (at Visit 2 and subsequent visits), at EOT, and approximately every 12 weeks during Active Follow-up.

For each questionnaire, a subject-reported outcome (PRO) information sheet must be completed by the study nurse/Investigator at each visit at which the questionnaires are scheduled to be administered regardless of whether or not the questionnaires were completed, in order to document information on questionnaire assessment such as completion status, date of administration, and reasons for non-completed questionnaires, missing assessments, etc.

9.8 Appropriateness of procedures / measurements

All efficacy and safety parameters, as well as the methods to measure them, are standard variables /methods in clinical studies and/or clinical practice. They are widely used and generally recognized as reliable, accurate, and relevant.

The NCCN-FACT FPSI-17 is a validated instrument that was developed to assess symptoms of prostate cancer, symptoms of treatment of prostate cancer, and health related QoL of prostate cancer subjects (28).

Brief Pain Inventory – Short form is validated to assess subject-reported levels of pain that has a precedent of being used in clinical trials with the goal of obtaining label claims based on the results captured with this instrument.

The NCI-CTCAE v 4.03 is a list of AE terms harmonized with the Medical Dictionary for Regulatory Activities (MedDRA) Terminology at the AE term level and associated with a severity grading scale that is widely used within the oncology research community as the

standard for documentation and analysis of AEs occurring in cancer research. This tool is useful in defining the maximum tolerated dose of study drugs, in making recommendations concerning modification of dose and dose schedule of study drugs, and in comparing safety profiles of therapeutic interventions.

10. Statistical methods and determination of sample size

10.1 General considerations

Statistical analysis will be performed using SAS; the version used will be specified in the SAP.

10.2 Analysis sets

Full analysis set (FAS): All randomized subjects. The FAS will be used in the analysis of all efficacy endpoints. Subjects will be included in FAS according to the treatment to which they are randomized.

Safety analysis set (SAF): All randomized subjects who have received at least 1 dose of study drug. This safety population will be used in the analyses of all safety endpoints. Subjects will be included in the analyses according to the treatment they actually received.

Pharmacokinetic Analysis Set (PKS): The first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom mandatory dense PK sampling will be done, will be included in the Pharmacokinetic Analysis Set, as long as they have received at least 3 days of uninterrupted study drug treatment as well as one cycle of docetaxel and have at least one post dose PK measurement.

10.3 Variables and planned statistical analyses

The complete list of variables to be analyzed for this study will be provided in the SAP.

10.3.1 Primary efficacy variable

The primary efficacy variable is OS, defined as the time (in days) from date of randomization until death from any cause.

Analysis of OS

All randomized subjects (FAS) will be included in the primary analysis of OS, the primary efficacy endpoint. The analysis will be performed when approximately 509 deaths are observed. The primary analysis of OS will be a stratified log-rank test with the same stratification factors as used for randomization.

The null hypothesis that there is no difference in OS between treatment arms, which is equivalent to a hazard ratio (HR) of 1, will be tested against the alternative hypothesis that the HR of darolutamide over placebo is below 1.

The HR (darolutamide group/placebo) for OS and its 95% confidence interval will be calculated using the Cox model, stratified by the same factors as stated above. Kaplan–Meier

(KM) estimates for OS will be presented for each treatment group. The KM estimates at time points such as 3 months, 6 months, etc., together with corresponding 95% confidence intervals (CIs) and the differences of these estimates between the darolutamide group and the placebo group will be presented.

The overall one-sided type I error rate for the analysis of OS is 0.025. One formal interim analysis of OS for futility was planned. A DMC was overseeing the interim analysis. For further details on the interim analysis, see Section 10.5.

10.3.2 Secondary efficacy variables

The secondary efficacy variables include:

- **Time to castration-resistant prostate cancer**, defined as the time to PSA progression (Section 10.3.4) with serum testosterone being at castrate level <0.50 ng/mL, or the time to progression by soft tissue/visceral lesions (as described in Section 9.2.2.3) or time to progression by bone lesions (as described in Section 9.2.2.3), whatever comes first.
- **Time to initiation of subsequent antineoplastic therapy**, defined as the time from randomization to initiation of first subsequent antineoplastic therapy for prostate cancer.
- **Symptomatic skeletal event free survival**, defined as the time from randomization to the first occurrence of SSE or death from any cause, whichever comes first. An SSE is defined as EBRT to relieve skeletal symptoms, or new symptomatic pathologic bone fracture, or occurrence of spinal cord compression or tumor-related orthopedic surgical intervention, whichever comes first.
- **Time to first SSE**, defined as the time from randomization to the first occurrence of SSE. An SSE is defined as EBRT to relieve skeletal symptoms, or new symptomatic pathologic bone fracture, or occurrence of spinal cord compression or tumor-related orthopedic surgical intervention, whichever comes first.
- **Time to initiation of opioid use for ≥ 7 consecutive days**, defined as the time from date of randomization to the date of first opiate use for ≥ 7 consecutive days. Time to opiate use will be determined by opioid use captured via eCRF (recorded opioid, see schedule of assessments).
- **Time to pain progression is defined as** the interval from randomization to the first date a subject experiences a pain progression. Pain will be assessed with the BPI-SF questionnaire (Appendix 16.4) during the visit, prior to any procedures or examination by physician. Subjects who have not experienced pain progression at the time of analysis will be censored on the last date the subject was known to have not progressed. Subjects with no on study assessment or no baseline assessment will be censored at the date of randomization.

For **asymptomatic subjects** (WPS 0 at baseline), pain progression is defined as an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) score from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart

OR

initiation of short- or long-acting opioid use for pain.

For **symptomatic subjects** (WPS >0 at baseline), pain progression is defined as an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) score from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart and a WPS of ≥ 4

OR

initiation of short- or long-acting opioid use for pain.

- **Time to worsening of physical symptoms of disease** is defined as the interval from randomization to the first date a subject experiences an increase in physical symptoms based on the NCCN-FACT-FPSI-17 questionnaire. Symptoms and QoL will be assessed with the NCCN-FACT-FPSI-17 questionnaire (Appendix 16.3) during the visit, prior to any procedures or examination by physician. An increase in physical symptoms of disease is defined as 3-point decrease (a lower score = higher symptom burden) from baseline in the physical symptoms of disease subscale (FPSI-DRS-P subscale of the NCCN-FACT-FPSI-17 questionnaire) observed at 2 consecutive evaluations ≥ 4 weeks apart.

Secondary efficacy variables analysis

The secondary efficacy variables will be analyzed for the FAS population unless otherwise specified in the SAP. Time-to-event endpoints will be analyzed using stratified log-rank test with randomization stratification factors. Hazard ratio and 95% CI will be provided using the Cox model stratified by the same factors as stated above. Detailed analysis methods and the plan for type 1 error control for secondary endpoints, including ranking of secondary endpoints, will be specified in the SAP.

10.3.3 Safety variables

The safety variables include:

- AEs until the end-of-study treatment visit
- SAEs until the end-of-study treatment visit
- Study drug-related SAEs until the end of Long-term Follow-up
- Vital signs: BP and heart rate (HR)
- 12-lead electrocardiogram (ECG)
- Physical examination
- Laboratory safety assessments (hematology, clinical chemistry, urinalysis)

Safety analysis

AEs will be coded using MedDRA. Descriptive summary tables will be presented on all safety parameters by treatment group. The safety assessment period is from the first dose of study drug until 30 days (+7 days) after the last treatment with darolutamide/placebo. Subjects will be monitored for AEs using the NCI–CTCAE v 4.03.

The AEs and safety laboratory parameters will be presented with their worst NCI–CTCAE v 4.03 grade (Section 16.2).

10.3.4 Other exploratory variables

- **Rate of absolute PSA response at 6 and 12 months**

Absolute PSA response is defined as blood PSA level <0.2 ng/mL, confirmed by a second subsequent PSA value <0.2 ng/mL 3 or more weeks later. Rate of absolute PSA response is defined as the number of subjects with absolute PSA response, divided by the total number of subjects evaluable for absolute PSA response. Rate of absolute PSA response will be evaluated on subject data up to 6 months and 12 months after randomization.

- **Rate of relative PSA response at 3, 6 and 12 months**

Relative PSA response is defined as a $\geq 30\%$ reduction of the blood PSA level compared to the screening value, confirmed by a second subsequent PSA value with a $\geq 30\%$ reduction from Screening 3 or more weeks later. Rate of relative PSA response is defined as the number of subjects with relative PSA response, divided by the total number of subjects evaluable for relative PSA response. Rate of relative PSA response will be evaluated on subject data up to 3 months, 6 months, and 12 months after randomization.

- **Time to PSA progression**

Prostate-specific antigen progression is defined as a $\geq 25\%$ increase above the nadir (lowest Screening or baseline) value, which is confirmed by a second value 3 or more weeks later, and an increase in absolute value of ≥ 2 ng/mL above nadir, at least 12 weeks from baseline.

The time to PSA progression is defined as the time (in days) from the date of randomization to the date of first PSA progression. Subjects without PSA progression as of database cut-off, whether or not surviving, will be censored at the last total PSA lab assessment date.

- **Changes in symptom burden and QoL based on the NCCN–FACT FPSI–17 questionnaire** (see Section 10.3.8)
- **Medical resource use** (see Section 9.7.2)
- **Pharmacokinetics** (see Section 9.5.2)
- **Biomarker evaluations** (see Section 9.7.1)

The details of the analysis for the exploratory variables to be analyzed for this study will be provided in the SAP.

Missing data / drop outs

Every effort should be made to retain subjects who discontinue the treatment period for any reason. These subjects are to be encouraged to remain on the study for follow-up of primary, secondary, and exploratory endpoints.

The method used for imputation of missing data will be described in the SAP (see Section 11.4).

10.3.5 Pharmacokinetics

Pharmacokinetics in the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel

The concentration–time courses of all substances will be tabulated separated by treatment. The following statistics will be calculated for each of the sampling points: arithmetic mean, standard deviation (SD), and coefficient of variation (CV); geometric mean, geometric SD (re-transformed SD of the logarithms), and CV; minimum, median, maximum value, and the number of measurements.

Means at any time will only be calculated if at least 2/3 of the individual data were measured and were above the lower limit of quantification (LLOQ). For the calculation of the mean value, a data point below LLOQ will be substituted by one-half of this limit. In tables showing mean values, where values below LLOQ are included in the calculation of mean values, these means will be marked.

Individual and geometric mean concentration versus time curves of all substances (using the actual sampling times for individual plots and the planned sampling times for mean plots) will be plotted by treatment using both a linear and semilogarithmic scale.

Pharmacokinetic characteristics ($t_{\max}$, t_{last} excluded) will be summarized by the statistics mentioned above. The $t_{\max}$, t_{last} will be described utilizing minimum, maximum, and median as well as frequency counts.

Pharmacokinetics in all subjects

Individual concentration–time data of darolutamide, (*S,S*)-darolutamide, (*S,R*)-darolutamide, and keto-darolutamide will be provided in a clinical study report appendix. All other parameters will be described in detail in a separate M&S report.

10.3.6 Biomarker analyses

Results from exploratory biomarker studies will be reported in a separate biomarker report.

10.3.7 Medical Resource use

Medical resource use is described in Section [9.7.2](#).

10.3.8 Subject-reported outcomes

Subject-reported outcome procedures are described in Section [9.7.3](#).

10.3.9 Baseline and demographic characteristics

Baseline and demographic characteristics will be summarized by descriptive statistics by treatment group darolutamide for the FAS population.

In addition, exploratory exposure-response analyses of efficacy, PD, or safety parameters, e.g. OS, PSA response rate, etc., are planned.

10.4 Determination of sample size

The sample size of the study is based on the primary endpoint of OS. The study is designed to have 90% power to detect a 33% increase in median time of OS with darolutamide compared to placebo (from 60 to 80 months, corresponding to a HR of 0.75) with a 1-sided alpha of 0.025. The OS data will be considered as mature when approximately 509 deaths are observed.

With the additional assumptions that subjects will enroll at a rate of 50 subjects per month, exponential distributions of the OS event times, median time of OS in the control group of 60 months, 5% dropout rate of subjects, and a 6-months enrollment ramp-up period, it follows that approximately 1,300 subjects must be enrolled to observe 509 deaths after approximately 70 months.

10.5 Planned interim analyses

One formal interim analysis of OS was planned. This interim analysis for futility was to be performed after approximately 153 deaths were observed. The DMC (see Section [13.1.1](#)) was overseeing the interim analysis. Detailed analysis methods and stopping rules will be specified in the SAP and the DMC charter.

For the futility interim and final analyses together, a one-sided overall beta of 0.1 will be used. Stopping boundaries will be calculated with an O'Brien-Fleming beta-spending function based on the actual number of events observed up to the cut-off date. The interim futility analysis was planned to be performed when approximately 153 deaths were observed (information fraction=0.3).

A DMC (see Section [13.1.1](#)) will be instituted for independent review of ongoing data from this study in accordance with a separate DMC charter. The DMC will operate independently of the Sponsor and Investigators.

The DMC will review safety data as per a separate DMC charter.

11. Data handling and quality assurance

11.1 Data recording

The data collection tool for this study will be an eCRF; a validated electronic data capture system called RAVE. Subject data necessary for analysis and reporting will be entered/transmitted into a validated database or data system (CIE/TOSCA; SAS).

Data required according to this protocol will be recorded by investigational site personnel via data entry into the internet based electronic data capture (EDC) software system RAVE, which Bayer has licensed from Medidata Solutions Worldwide. RAVE has been validated by Medidata Solutions Worldwide and Bayer for use in its clinical studies. RAVE allows for the application of software logic to set up data entry screens and data checks to ensure the completeness and accuracy of the data entered by the site personnel. Bayer extensively applies the logic to ensure data are complete and reflect the clinical data requirements of the study. Data queries resulting from the application of the software logic are resolved by the site personnel. The data are stored at a secure host facility maintained by Medidata Solutions Worldwide and transferred on a periodic basis to Bayer's internal computer system via a secure Virtual Private Network.

All access to the RAVE system is through a password-protected security system that is part of the RAVE software. All internal Bayer and external Investigator site personnel seeking access must go through a thorough RAVE training process before they are granted access to RAVE for use in Bayer's clinical studies. Training records are maintained.

All personnel with access to the RAVE system are supported by a Service Desk staffed with trained personnel to answer questions and ensure access is maintained such that data entry can proceed in a timely manner.

The RAVE system contains a system-generated audit trail that captures any changes made to a data field, including who made the change, why the change was made and the date and time it was made. This information is available both at the Investigator's site and at Bayer. Data entries made in the RAVE EDC screens are supported by source documents maintained for all subjects enrolled in this study.

Source documentation

It is the expectation of the Sponsor that key data entered into the CRF has source documentation available at the site. Study-specific data not needed for the subject's routine medical care (QoL-related data: NCCN-FACT FPSI-17 and BPI-SF questionnaires) may be entered directly into the validated database or data system, without availability of corresponding source documentation.

The site must implement processes to ensure availability of all required source documentation. A source document checklist will be used at the site to identify the source data for key data points collected and the monitor will work with the site to complete this.

Data recorded from screening failures

Data of 'only screened subjects' will be recorded at least as source data, as far as the reason for the premature discontinuation is identifiable. At a minimum, the following data should be recorded in the eCRF:

- Demographic information (subject number; year of birth / age; sex; if applicable, race/ethnicity)
- Relevant Inclusion and/or Exclusion criteria
- Date of IC
- Reason for premature discontinuation
- Date of last visit.

These data will be transferred to the respective database.

For screening failures with an SAE, the following data should be collected in the CRF in addition to the data specified above:

- All information related to the SAE such as:
 - Concomitant medication
 - Medical history
 - Other information needed for SAE complementary page

11.2 Monitoring

In accordance with applicable regulations, GCP, and Sponsor's/CRO's procedures, monitors will contact the site prior to the start of the study to review with the site staff the protocol, study requirements, and their responsibilities to satisfy regulatory, ethical, and Sponsor's requirements. When reviewing data collection procedures, the discussion will also include identification and documentation of source data items.

The Sponsor/designee will monitor the site activity to verify that the:

- Data are authentic, accurate, and complete.
Supporting data may be requested (example: blood glucose readings to support a diagnosis of diabetes).
- Safety and rights of subjects are being protected.
- Study is conducted in accordance with the currently approved protocol (including study treatment being used in accordance with the protocol).
- Any other study agreements, GCP, and all applicable regulatory requirements are met.

The Investigator and the head of the medical institution (where applicable) agrees to allow the monitor direct access to all relevant documents.

11.3 Data processing

Data will be collected as described in Section 11.1. Clinical data management will be performed in accordance with applicable Sponsor's/CRO's standards and data cleaning procedures. This is applicable for data recorded on CRF as well as for data from other sources (e.g. IXRS, laboratory, ECG, ePRO, adjudication committees).

For data coding (e.g. AEs, medication), internationally recognized and accepted dictionaries will be used.

After its initial release for biometrical analysis, the clinical database is planned to be re-opened for the inclusion of the following additional data: PK data, antibody data.

11.4 Missing data

The analysis of the primary endpoint is based on the intent to treat population. Subjects who discontinue the treatment prematurely or leave the study prematurely will not be replaced.

Handling of missing data, which in this context are events not observed for the time-to-event endpoints (primary variable and secondary variables), is implicitly handled in the statistical methods applied (log-rank test, Cox model) by censoring these subjects. Further aspects of handling of missing data will be discussed in the SAP.

11.5 Audit and inspection

To ensure compliance with GCP and regulatory requirements, a member of the Sponsor's (or a designated CRO's) quality assurance unit may arrange to conduct an audit to assess the performance of the study at the study site and of the study documents originating there. The Investigator/institution will be informed of the audit outcome.

In addition, inspections by regulatory health authority representatives and IEC(s)/IRB(s) are possible. The Investigator should notify the Sponsor immediately of any such inspection.

The Investigator/institution agrees to allow the auditor or inspector direct access to all relevant documents and allocate his/her time and the time of his/her staff to the auditor/inspector to discuss findings and any issues. Audits and inspections may occur at any time during or after completion of the study.

11.6 Archiving

Essential documents shall be archived safely and securely in such a way that ensures that they are readily available upon authorities' request.

Subject (hospital) files will be archived according to local regulations and in accordance with the maximum period of time permitted by the hospital, institution, or private practice. Where the archiving procedures do not meet the minimum timelines required by the Sponsor, alternative arrangements must be made to ensure the availability of the source documents for the required period.

The Investigator/institution notifies the Sponsor if the archival arrangements change (e.g. relocation or transfer of ownership).

The Investigator site file is not to be destroyed without the Sponsor's approval.

The contract with the Investigator/institution will contain all regulations relevant for the study center.

12. Premature termination of the study

The Sponsor has the right to close this study (or, if applicable, individual segments thereof [e.g. treatment arms; dose steps; centers]) at any time, which may be due, but not limited to, the following reasons:

- If risk–benefit ratio becomes unacceptable owing to, for example,
 - Safety findings from this study (e.g. SAEs)
 - Results of any interim analysis
 - Results of parallel clinical studies
 - Results of parallel animal studies
(e.g. toxicity, teratogenicity, carcinogenicity, or reproduction toxicity).
- If the study conduct (e.g. recruitment rate; dropout rate; data quality; protocol compliance) does not suggest a proper completion of the trial within a reasonable time frame.

The Investigator has the right to close his/her center at any time.

For any of the above closures, the following applies:

- Closures should occur only after consultation between involved parties. Final decision on the closure must be in writing.
- All affected institutions (e.g. IEC[s]/IRB[s]; competent authority[ies]; study center; head of study center) must be informed as applicable according to local law.
- All study materials (except documentation that has to remain stored at site) must be returned to the Sponsor. The Investigator will retain all other documents until notification is given by the Sponsor for destruction.
- In the event of a partial study closure, ongoing subjects, including those in post–study follow–up, must be taken care of in an ethical manner.

Details for an individual subject's withdrawal can be found in Section [6.4.1](#).

13. Ethical and legal aspects

13.1 Investigator(s) and other study personnel

Sponsor's medical expert

Name:

Address:

Telephone no.:

Co-ordinating investigators

Name:

Address:

Name:

Address:

All other study personnel not included in this section are identified in a separate personnel list (not part of this clinical study protocol) as appropriate. This list will be updated as needed; an abbreviated version with personnel relevant for the centers will be available in each center's Investigator site file.

Whenever the term 'Investigator' is noted in the protocol text, it may refer to either the PI at the site, or an appropriately qualified, trained and delegated individual of the investigational site.

The PI of each center must sign the protocol signature page and must receive all required external approvals (e.g. health authority, ethics committee, Sponsor) before subject recruitment may start at the respective center. Likewise, all amendments to the protocol must be signed by the PI and must have received all required external approvals before coming into effect at the respective center.

A complete list of all participating centers and their investigators, as well as all required signature documents, will be maintained in the Sponsor's study file.

The global Sponsor of this study is identified on the title page of this protocol. If required by local law, local co-sponsors will be nominated; they will be identified on the respective country-specific signature pages.

13.1.1 Independent data monitoring committee

A DMC will be instituted to ensure ongoing safety of study subjects with respect to a risk/benefit assessment during periodic data review meetings, review results from the planned interim analysis and provide a formal recommendation for continuation/termination of the study and monitor study conduct to ensure the overall integrity of trial is maintained. Recommendation for trial continuation will be guided by safety evaluations at all safety data reviews. The committee will include at least 3 independent oncologists and one independent statistician. Data review meetings will be held periodically as per separate DMC charter. Enrollment in the study will continue throughout the scheduled meetings of the DMC. Decisions on trial termination, protocol amendment, on hold of subject recruitment, or cessation of subject recruitment based on risk/benefit assessments will be made after recommendations from the DMC have been assessed by the Sponsor. Safety review meetings will be held in accordance with the DMC charter. Decisions on study termination or cessation of subject recruitment based on safety or OS outcomes will be made after recommendations from the DMC have been assessed by Bayer AG.

The DMC will also review the safety assessment, including PK, which will be performed after the first 20 subjects were randomized and have received at least 1 cycle of docetaxel. Blinding of Investigators, subjects and the Sponsor will be maintained during this assessment based on the preclinical mechanisms and known safety profiles for the darolutamide and docetaxel (see Section 3.3), the toxicities from the combination are expected to be manageable. Therefore, enrollment will continue during this 20–subject safety review unless unexpected or excessive toxicities related to darolutamide in addition to docetaxel plus ADT compared to the treatment of docetaxel plus ADT alone are reported, in which case subject enrollment will be placed on hold until completion of the DMC safety review. Further safety reviews to ensure subjects' safety may be implemented for different subject populations following the first 20 subjects who were randomized and have received at least one cycle of docetaxel, due to lack of experience in treatment of darolutamide in addition to docetaxel plus ADT. The details of the evaluations will be described in the DMC charter.

13.2 Funding and financial disclosure

Funding

This study will be funded by its Sponsor.

Financial disclosure

Each Investigator (including principal and/or any sub–investigators) who is directly involved in the treatment or evaluation of research subjects has to provide a financial disclosure according to all applicable legal requirements. All relevant documentation will be filed in the trial master file.

13.3 Ethical and legal conduct of the study

The procedures set out in this protocol, pertaining to the conduct, evaluation, and documentation of this study, are designed to ensure that the Sponsor and Investigator abide by GCP guidelines and the guiding principles detailed in the Declaration of Helsinki. The study will also be carried out in keeping with applicable local law(s) and regulation(s).

Documented approval from appropriate IEC(s)/IRBs will be obtained for all participating centers/countries before start of the study, according to GCP, local laws, regulations, and organizations. When necessary, an extension, amendment or renewal of the IEC/IRB approval must be obtained and also forwarded to the Sponsor. The responsible unit (e.g. IEC/IRB, head of the study center/medical institution) must supply to the Sponsor, upon request, a list of the IEC/IRB members involved in the vote and a statement to confirm that the IEC/IRB is organized and operates according to GCP and applicable laws and regulations.

Strict adherence to all specifications laid down in this protocol is required for all aspects of study conduct; the Investigator may not modify or alter the procedures described in this protocol.

Modifications to the study protocol will not be implemented by either the Sponsor or the Investigator without agreement by both parties. However, the Investigator or the Sponsor may implement a deviation from, or a change of, the protocol to eliminate an immediate hazard(s) to the trial subjects without prior IEC/IRB/Sponsor approval/favorable opinion. As soon as possible, the implemented deviation or change, the reasons for it and if appropriate the proposed protocol amendment should be submitted to the IEC/IRB/head of medical institution/Sponsor. Any deviations from the protocol must be explained and documented by the Investigator.

Details on discontinuation of the entire study or parts thereof can be found in [Section 12](#).

13.4 Subject information and consent

All relevant information on the study will be summarized in an integrated subject information sheet and ICF provided by the Sponsor or the study center. A sample subject information and ICF is provided as a document separate to this protocol.

Based on this subject information sheet, the Investigator or designee will explain all relevant aspects of the study to each subject/legal representative or proxy consentor (if the subject is under legal protection) prior to his/her entry into the study (i.e. before any examinations and procedures associated with the selection for the study are performed or any study-specific data is recorded on study-specific forms).

The Investigator will also mention that written approval of the IRB/IEC has been obtained.

Each subject/legal representative or proxy consentor will be informed about the following aspects of premature withdrawal:

- Each subject has the right to withdraw from the study at any time without any disadvantage and without having to provide reasons for this decision.

- The subject's consent covers end-of-study examinations as specified in the visit description described in Section 9.2 to be conducted after withdrawal of consent.
- The subject's data that have been collected until the time of withdrawal will be retained and statistically analyzed in accordance with the SAP.
- Subject-specific data on the basis of material obtained before withdrawal may be generated after withdrawal (e.g. image reading, analysis of biological specimen such as blood, urine or tissues); these data would also be retained and statistically analyzed in accordance with the SAP. The subject has the right to object to the generation and processing of this post-withdrawal data. For this, he/she needs to sign a corresponding declaration of objection; alternatively, the subject's oral objection may be documented in the subject's source data.

If at any time during the study the subject would like to withdraw consent, the Investigator must discuss with the subject the active and Long-term Follow-up period parts of the study. Withdrawal of consent should be documented in the subject's medical file.

Each subject/legal representative or proxy consentor will have ample time and opportunity to ask questions.

Only if the subject/legal representative or proxy consentor voluntarily agrees to sign the ICF and has done so, may he/she enter the study. Additionally, the Investigator will personally sign and date the form. The subject/legal representative or proxy consentor will receive a copy of the signed and dated form.

The signed IC statement is to remain in the Investigator site file or, if locally required, in the subject's note/file of the medical institution.

In the event that IC is obtained on the date that baseline study procedures are performed, the study record or subject's clinical record must clearly show that IC was obtained prior to these procedures.

If the subject is not capable of providing a signature, a verbal statement of consent can also be given in the presence of an impartial witness (independent of the Sponsor and the Investigator). This is to be documented by a signature from the informing physician as well as by a signature from the witness.

The ICF and any other written information provided to subjects/legal representatives or proxy consenters will be revised whenever important new information becomes available that may be relevant to the subject's consent, or there is an amendment to the protocol that necessitates a change to the content of the subject information and/or the written ICF. The Investigator will inform the subject/legal representative or proxy consentor of changes in a timely manner and will ask the subject to confirm his/her participation in the study by signing the revised ICF. Any revised written ICF and written information must receive the IEC/IRB's approval/favorable opinion in advance of use.

13.5 Publication policy and use of data

The Sponsor has made the information regarding the study protocol publicly available on the internet at www.clinicaltrials.gov.

All data and results and all intellectual property rights in the data and results derived from the study will be the property of the Sponsor who may utilize them in various ways, such as for submission to government regulatory authorities or disclosure to other investigators.

Regarding public disclosure of study results, the Sponsor will fulfill its obligations according to all applicable laws and regulations. The Sponsor is interested in the publication of the results of every study it performs.

The Sponsor recognizes the right of the Investigator to publish the results upon completion of the study. However, the Investigator, while free to utilize study data derived from his/her center for scientific purposes, must obtain written consent of the Sponsor on the intended publication manuscript before its submission. To this end, the Investigator must send a draft of the publication manuscript to the Sponsor within a time period specified in the contract. The Sponsor will review the manuscript promptly and will discuss its content with the Investigator to reach a mutually agreeable final manuscript.

13.6 Compensation for health damage of subjects / insurance

The Sponsor maintains clinical trial insurance coverage for this study in accordance with the laws and regulations of the country in which the study is performed.

13.7 Confidentiality

All records identifying the subject will be kept confidential and, to the extent permitted by the applicable laws and/or regulations, will not be made publicly available.

Subject names will not be supplied to the Sponsor. Only the subject number will be recorded in the CRF, and if the subject name appears on any other document (e.g. pathologist report), it must be obliterated before a copy of the document is supplied to the Sponsor. Study findings stored on a computer will be stored in accordance with local data protection laws. As part of the IC process, the subjects will be informed in writing that representatives of the Sponsor, IEC/IRB, or regulatory authorities may inspect their medical records to verify the information collected, and that all personal information made available for inspection will be handled in strictest confidence and in accordance with local data protection laws.

If the results of the study are published, the subject's identity will remain confidential.

The Investigator will maintain a list to enable subjects to be identified.

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15. Protocol amendments

15.1 Amendment 2

Amendment 2 is a global amendment dated 04 OCT 2016.

15.1.1 Overview of changes to the study

15.1.1.1 Modification 1: New drug–drug interaction data available

It was clarified that interaction between ODM–201 and P–gp or BCRP are expected *in vivo* based on *in vitro* data and preliminary clinical study results. Text in the protocol was revised, to mention the preliminary results of a drug–drug interaction clinical study, and to provide the Investigators with suggestions how to manage relevant concomitant medications.

Rationale: based on newly available information from a drug–drug interaction clinical study and following IB revision, the protocol text has been revised and aligned.

Sections affected by this modification: [3.2.2](#) Nonclinical pharmacokinetics and metabolism, [3.2.4](#) Drug–drug interactions, [3.2.5](#) Clinical studies with ODM–201, [8.1.2](#) Permissible concomitant medications and treatments.

15.1.1.2 Modification 2: Clarification of PK analysis

Wording defining the patient population for the detailed PK analysis was revised and clarified throughout the protocol. Wording about docetaxel analysis for the sparse PK sampling in all randomized subjects was also revised and clarified throughout the protocol.

Rationale: Modification was done to clarify that the patients participating to the detailed PK analysis must have received at least one cycle of docetaxel, and to clarify the timing of the sparse PK sampling as well as the additional analysis of docetaxel in all the randomized subjects.

Sections affected by this modification: [3.2.4](#) Drug–drug interactions, [3.3](#) Rationale of the study, [5](#) Study design, [7.1](#) Treatments to be administered, [7.5](#) Blinding, [9.1](#) Tabular schedule of evaluations, [9.2.2.1](#) Day 1 (Visit 1), [9.5.1](#) Drug measurements, [9.5.2](#) Pharmacokinetic evaluation, [10.2](#) Analysis sets, [10.3.5](#) Pharmacokinetics, [13.1.1](#) Independent data monitoring committee.

15.1.1.3 Modification 3: Addition of non-protein-bound (free) testosterone analysis

Wording throughout the protocol was revised to add the non-protein-bound (free) testosterone analysis in a subset of 300 subjects.

Rationale: Low testosterone levels are expected in subjects during ADT treatment, therefore the sex hormone-binding globulin and albumin concentrations will be determined to calculate the non-protein-bound (free) testosterone in a subset of approximately 300 patients.

Non-protein-bound (free) testosterone analysis results will be compared to the results from the calculation. In case non-protein-bound (free) testosterone is not measurable in all the subjects of that subset of approximately 300 patients, the calculation will be used.

Sections affected by this modification: [9.1](#) Tabular schedule of evaluations, [9.2.1](#) Screening period, [9.2.2.2](#) Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days), [9.2.3](#) Active Follow-up period, [9.6.3.1](#) Laboratory safety assessments

15.1.1.4 Modification 4: Administrative changes

Sponsor's study medical expert was updated due to a change in sponsor's personnel. The contact information was also added to Section [13.1](#) as per template. The contact details of Co-ordinating investigators were also added.

Sections affected by this modification: Title page, [13.1](#) Investigator(s) and other study personnel.

15.1.1.5 Modification 5: Other clarifications and corrections

- Scheme of study periods was updated: "treatment with study drug" was changed to "treatment period" to clarify that also docetaxel is given during treatment period.
- Subheading "Further safety" was added as per template, content was not changed.

Sections affected by this modification: [5](#) Study design, [9.6.3](#) Further safety

15.1.2 Changes to the protocol text

Changes to the protocol text done in Amendment 2 are provided in Section 15.1.2 of Amendment 2.

15.2 Amendment 5

Amendment 5 is a global amendment dated 12 FEB 2018.

15.2.1 Overview of changes to the study

15.2.1.1 Modification 1: New drug-drug interaction data available

Text was modified to include the results of drug-drug interaction clinical study, and to provide the Investigators with guidance how to manage relevant concomitant medications: Concomitant treatment with strong and moderate CYP3A4 inducers should be avoided. The list of strong and moderate CYP3A4 inducers was also included as an appendix.

Rationale: Based on clinical study results (Study 17726), strong CYP3A4 and P-gp inhibitors and strong CYP3A4 inducers affect the PK of darolutamide. Results indicate that co-administration of darolutamide with a strong CYP3A4 or P-gp inhibitor does not result in a clinically relevant increase of darolutamide plasma concentrations. Results also indicate that repeated co-administration of strong or moderate CYP3A4 inducers with darolutamide is expected to reduce darolutamide plasma concentrations. Therefore, concomitant intake of strong or moderate CYP3A4 inducers should be avoided.

Sections affected by this modification: List of abbreviations, [3.2.4](#) Drug-drug interactions, [8.1.2](#) Permissible concomitant medications and treatments, [16.5](#) List of strong and moderate CYP3A4 inducers (*new appendix added*)

15.2.1.2 Modification 2: Modification of the dosing language

The wording for the study drug dose to be administered was modified and harmonized throughout the protocol: 600 mg (2 tablets of 300 mg) twice daily with food, equivalent to a total daily dose of 1200 mg.

A guidance text in the event of a missed dose was added: if a dose of study drug is delayed, dose can be taken up to 6 hours to make up for the missed one.

Rationale: The change was done to align darolutamide dosing wording across the development program. In addition, instructions for handling a missed dose were missing from the protocol.

Sections affected by this modification: [2](#) Synopsis, [5](#) Study design, [7.1](#) Treatments to be administered, [7.4.1](#) Dose modifications of study drug

15.2.1.3 Modification 3: Clarification of docetaxel dosage and administration

The possibility to administer docetaxel as per local practice (if different from the regimen described in the summary of product characteristics) was removed from the protocol, and also the guidance to refer to the local prescribing information was deleted. Docetaxel should be administered at 75 mg/m² as 1 hour IV infusion every 3 weeks for 6 cycles. Infusions longer than 1 hour are acceptable for medical reasons at Investigator's discretion. It was also clarified that the first cycle of docetaxel should be administered within 6 weeks after start of study drug instead of 6 weeks after randomization. Rationale for pre-medication instructions was also clarified.

Rationale: The change was done to clarify that pre-medication, docetaxel dose and infusion schedule must be in accordance with the label and that docetaxel can only be administered after the patient has started study drug.

Sections affected by this modification: [7.1](#) Treatments to be administered, [7.4.2](#) Dosage and administration for background treatment, [7.4.3](#) Dose modifications of docetaxel background treatment [9.1](#) Tabular schedule of evaluations, [9.2.2.1](#) Day 1 (Visit 1)

15.2.1.4 Modification 4: Guidance on laboratory tests before each docetaxel cycle

The list of laboratory assessments that should be performed before each docetaxel cycle was added to the protocol. It was also clarified that docetaxel can only be infused after routine hematology and biochemistry laboratory tests have been obtained and that the results must be reported in the unscheduled hematology and biochemistry eCRF.

Rationale: The modification was done to give further guidance on laboratory safety assessments and to be in line with docetaxel label requirements.

Sections affected by this modification: [7.4.3](#) Dose modifications of docetaxel background treatment, [9.1](#) Tabular schedule of evaluations, [9.2.2.1](#) Day 1 (Visit 1), [9.2.5](#) Unscheduled visits, [9.6.3.1](#) Laboratory safety assessments

15.2.1.5 Modification 5: Clarification related to soft tissue/visceral lesions

It was clarified that the evaluation of soft tissue and visceral lesions is done with the same radiological methods and assessed by RECIST criteria.

Rationale: The change was done for clarity and for consistency throughout the protocol.

Sections affected by this modification: [9.2.2.3](#) Requirement for chest, abdomen and pelvic CT/MRI and bone scan during the study treatment period, [10.3.2](#) Secondary efficacy variables

15.2.1.6 Modification 6: Time windows for randomization and Visit 1

Protocol was modified to state that randomization must occur within 28 days from ICF signature and that the patient will be randomized at the end of the screening period. In addition, the time window for Day 1 (Visit 1) procedures was adjusted to clarify that procedures scheduled for Visit 1 can occur +3 days after randomization instead of ± 3 days.

The modification was done for clarification.

Sections affected by this modification: [9.1](#) Tabular schedule of evaluations, [9.2](#) Visit description, [9.2.1](#) Screening period, [9.2.2.1](#) Day 1 (Visit 1), [9.5.1](#) Drug measurements

15.2.1.7 Modification 7: Switch of ADT to LHRH agonist prohibited

ADT switch to LHRH agonist was added to the list of prohibited concomitant medications and treatments. It was also clarified that ADT switch to an antagonist is allowed during study treatment.

Rationale: Switch of ADT to LHRH agonist may cause flare reactions, impacting subject's safety and radiological assessments, therefore it is prohibited. Opposite a switch to an LHRH antagonist is permitted.

Sections affected by this modification: [8.1.2](#) Permissible concomitant medications and treatments, [8.1.3](#) Prohibited concomitant medications and treatments

15.2.1.8 Modification 8: Collection of whole blood sample for pharmacogenetics test allowed at other visits if missed at Visit 1

A whole blood sample for pharmacogenetic testing will be collected preferably at Visit 1 from subjects who have signed pharmacogenetic consent. However, the protocol was modified to allow the collection at any other visit during the treatment period if missed at Visit 1

Rationale: To ensure completeness of sample collection for pharmacogenetics, more flexibility was added to the timing of collecting the respective blood samples.

Sections affected by this modification: [9.1](#) Tabular schedule of evaluations, [9.2.2.2](#) Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days), [9.7.1.4](#) Genetic biomarker analysis from whole blood

15.2.1.9 Modification 9: Unblinding in non-emergency situations not permitted

It was clarified that progressive disease does not routinely constitute an emergency situation, therefore unblinding the label only for subsequent treatment decision is not allowed.

Rationale: The modification was done to clarify that unblinding in non-emergency situations is not permitted.

Sections affected by this modification: [7.5](#) Blinding

15.2.1.10 Modification 10: Clarification of PK sampling

It was clarified that two additional blood samples for sparse PK analysis can be taken from docetaxel cycle 1 onwards from patients who do not belong to the dense PK subset.

Rationale: To provide clear instructions to the sites regarding collection of blood samples for sparse PK analysis.

Sections affected by this modification: [9.1](#) Tabular schedule of evaluations, [9.5.1](#) Drug measurements

15.2.1.11 Modification 11: Clarification of laboratory safety assessments

It was clarified in the protocol text that PSA at screening can be taken within 28 days before randomization.

Rationale: To provide clear instructions about timing of the safety laboratory assessments in the screening period.

Section affected by this modification: [9.6.3.1](#) Laboratory safety assessments

15.2.1.12 Modification 12: Nomenclature change

With the approval of the INN darolutamide, the Orion drug nomenclature (ODM-201) was replaced with darolutamide throughout the protocol. In addition, the Orion codes for darolutamide diastereomers and metabolite were replaced with trivial names in the entire document as shown in the table under “definitions of terms”.

Sections affected by this modification: Definition of terms (*new section added*), the entire document (amended sections are not indicated in the protocol body)

15.2.1.13 Modification 13: Personnel change

The sponsor’s medically responsible person was changed as a result of personnel change.

Section affected by this modification: Signature of the sponsor’s medically responsible person

15.2.1.14 Modification 14: Other clarifications and corrections

- A typo was corrected in Section 9.7.1.4 (incomplete sentence).
- The abbreviation for AUC(0-x) was corrected in the list of abbreviations.
- The word “approval” was changed to “agreement” in Section 9.2.1 to clarify that the Bayer–designated medical representative provides written agreement to each subject’s eligibility. The modification was done to harmonize the language with other study documents.
- The wording related to medical resource use was harmonized (“resource utilization” was changed to “medical resource use”) for consistency.
- The statement “not part of this protocol” was removed from the language related to source document checklist to avoid confusion.

Section affected by this modification: List of abbreviations, 9.2.1 Screening period, 9.2.3 Active Follow-up period, 9.7.1.4 Genetic biomarker analysis from whole blood, 9.7.2 Medical resource use, 11.1 Data recording

15.2.2 Changes to the protocol text

Changes to the protocol text done in Amendment 5 are provided in a separate track changes version.

15.3 Amendment 6

Amendment 6 is a global amendment dated 10 DEC 2019.

15.3.1 Overview of changes to the study

Section(s)	Description of change	Rationale
2 Synopsis 8.2 Post-study therapy 9.1 Tabular schedule of evaluations 9.2.2.2 Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days) 9.2.3 Active Follow-up period 9.2.4 Long-term (survival) Follow-up period	The option to continue darolutamide treatment in a separate program was added for those subjects who are ongoing on darolutamide treatment at the time of the primary analysis of the current study. Subjects receiving placebo will discontinue the treatment and complete the study.	To enable subjects who are receiving darolutamide treatment in the current study to continue to receive darolutamide treatment when the study is completed. Subjects in the placebo arm have received standard of care with 6 cycles docetaxel after randomization. Upon disease progression, it will be at the Investigator's discretion to decide on further treatment in the best interest of the patient.
2 Synopsis 5 Study design 9.1 Tabular schedule of evaluations 9.2.5 Unscheduled visits	Additional survival sweeps were added to be performed in all subjects considered alive at the database cut-off date for the second interim analysis and prior to any subsequent additional analysis.	To ensure that survival data is current.
3.2.4 Drug-drug interactions	Detailed information on darolutamide drug-drug interactions were removed and a reference to the latest available version of darolutamide IB was provided instead. The section was updated to reflect that the data from the present study did not reveal any significant impact of darolutamide on the PK of docetaxel.	The most up to date data on drug-drug interactions can be found from the darolutamide IB. The information on the effect of darolutamide on the PK of docetaxel was updated based on the availability of blinded results and the DMC assessment.
8.1.2 Permissible concomitant medications and treatments 16.5 List of strong CYP3A4 inducers	a) The guidance to use strong inhibitors of P-gp and BCRP with caution was removed. b) The guidance to use medicinal products that are sensitive substrates for P-gp or BCRP with caution when co-administered with darolutamide was removed.	a) The results of study 17726 with itraconazole, a strong CYP3A4, P-gp and BCRP inhibitor, did not indicate a clinically relevant impact on the PK of darolutamide. b) Based on the results of study 18860 using dabigatran etexilate as a sensitive P-gp substrate, there is no effect of darolutamide on P-gp

	c) The guidance to monitor subjects receiving concomitant drugs which are mainly metabolized via CYP3A4 enzyme for signs of decreased efficacy was removed. d) The recommendation to avoid concomitant use of moderate CYP3A4 inducers was removed (while strong CYP3A4 inducers should still be avoided). e) The guidance to closely monitor patients for signs and symptoms of increased exposure was updated to include also OATP1B1 and OATP1B3 substrates.	transporter. The effect of darolutamide on BCRP is described in detail based on study 17723 with rosuvastatin. c) Based on the results of study 18860 using midazolam as a sensitive CYP3A4 substrate, darolutamide has only a minor inducing effect on CYP3A4 which is not considered to be clinically relevant. d) Recent results have established the darolutamide exposure-response relationship which showed a flat relationship (maximum PSA response) over the entire exposure range of 600 mg bid and also at exposures below that range. Thus, even with a darolutamide exposure decrease of 50% (driven by a CYP3A4 inducer), there would still be a strong effect on PSA. Therefore, the modification allows the use of moderate CYP3A4 inducers. e) New <i>in vitro</i> data on OATP1B1 and OATP1B3 indicated that the effect of darolutamide on such substrates may be clinically relevant.
9.6.1.3 Assessments and documentation of adverse events	The guidance on AE reporting was modified to clarify that disease progression should not be reported as an AE. Instead, the associated signs and symptoms should be recorded as AEs.	The change was done for clarification.
9.1 Tabular schedule of evaluations 9.2.1 Screening period 9.2.2.2 Visit 2 and subsequent visits (every 12 weeks $\pm$ 7 days) 9.2.3 Active Follow-up period 9.6.3.1 Laboratory safety assessments	In a subset of subjects, additional determination of total and free testosterone was added to be performed also at the EOT Visit	Emerging scientific data suggest that investigating free testosterone at the time of disease progression (which likely will lead to stopping study treatment) will be worth analyzing in relation to the efficacy endpoints (34)
9.8 Appropriateness of procedures /	Clarifications and corrections	Modifications were done for clarification

measurements 10.3.2 Secondary efficacy variables 10.3.4 Other exploratory variables	were done: • MedDRA version number removed • “Analgesic use” changed to “opioid use” • Definition of “time to PSA progression” clarified • Typos corrected	and consistency.
Title page Signature of the sponsor’s medically responsible person 13.1 Investigator and other study personnel	Sponsor’s medical expert and medically responsible person were changed.	Administrative changes.
15.1.2 Changes to the protocol text	The detailed old vs. new text comparison for protocol amendment 2 was replaced with a reference to the respective protocol amendment.	To improve readability, and to reduce complexity of the protocol.

15.3.2 Changes to the protocol text

Changes to the protocol text done in Amendment 6 are provided in a separate track changes version.

15.4 Amendment 7

Amendment 7 is a global amendment dated 26 MAY 2020.

15.4.1 Overview of changes to the study

Section(s)	Description of change	Rationale
1 Title page	New e-signature text added.	New e-signature process implemented at Bayer.
2 Synopsis 5 Study design 9.1 Tabular schedule of evaluations (footnote v) 9.2.5 Unscheduled visits 10.3.1 Primary efficacy variable 10.5 Planned interim analyses 13.1.1 Independent data monitoring committee	Planned second interim analysis will not be performed.	Due to the implications of the COVID-19 pandemic on the conduct of study procedures and data collection at the study sites, the risk for not achieving the needed quality of data for a formal analysis is considered to be too high. Therefore, the second interim analysis, which was expected to be conducted in the second half of 2020, will not be performed. As a consequence, there is no splitting of alpha for efficacy between the now dropped second interim analysis and the final analysis. This change does not affect the regular DMC for safety.
9.7.1 Biomarker investigations	Clarification added for biomarker analysis and reporting.	To allow for inclusion of biomarker analysis in the main study SAP and to be consistent with other parts of the protocol.
10.3.2 Secondary efficacy variables	Added text regarding ranking of secondary endpoints.	To clarify that type 1 error control will include the ranking of secondary endpoints.
10.5 Planned interim analyses	Stopping boundaries for the futility interim and final analyses together will use a beta-spending function based on the actual number of events.	Due to removal of interim analysis 2, the sentence regarding alpha-spending has been removed. Added a statement about beta-spending for clarification.

15.4.2 Changes to the protocol text

Changes to the protocol text done in Amendment 7 are provided in a separate track changes version.

16. Appendices

16.1 Eastern Cooperative Oncology Group performance status

Grade	Description
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work)
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Death

16.2 National Cancer Institute–Common Terminology Criteria for Adverse Events

This study will utilize the NCI–CTC for Adverse Events Version 4.03 for toxicity and serious AEs reporting. A copy of the CTC Version 4.03 can be downloaded in PDF form from <http://evs.nci.nih.gov/ftp1/CTCAE/About.html>.

All appropriate treatment areas should have access to a copy of the NCI–CTC Version 4.03.

16.3 NCCN-FACT FPSI-17

Below is a list of statements that other people with your illness have said are important.
Please circle or mark one number per line to indicate your response as it applies to the past 7 days.

		Not at all	A little bit	Some what	Quite a bit	Very much	
D R S — P	GP1	I have a lack of energy.....	0	1	2	3	4
	GP4	I have pain	0	1	2	3	4
	P7	I have difficulty urinating	0	1	2	3	4
	C2	I am losing weight	0	1	2	3	4
	BP1	I have bone pain.....	0	1	2	3	4
	HI7	I feel fatigued.....	0	1	2	3	4
	NCCN3	I have weakness in my legs	0	1	2	3	4
D R S — E	P3	My pain keeps me from doing things I want to do	0	1	2	3	4
	C6	I have a good appetite.....	0	1	2	3	4
	GF5	I am sleeping well.....	0	1	2	3	4
	GE6	I worry that my condition will get worse .	0	1	2	3	4
	GP2	I have nausea	0	1	2	3	4
	P6	I have trouble moving my bowels	0	1	2	3	4
	GS7	I am satisfied with my sex life.....	0	1	2	3	4
T S E	GP5	I am bothered by side effects of treatment	0	1	2	3	4
	GF3	I am able to enjoy life	0	1	2	3	4
	GF7	I am content with the quality of my life right now.....	0	1	2	3	4

DRS-P=Disease-Related Symptoms Subscale – Physical
DRS-E=Disease-Related Symptoms Subscale – Emotional
TSE=Treatment Side Effects Subscale
FWB=Function and Well-Being Subscale
English (Universal), Copyright 2001

03 March 2010

16.4 BPI-SF

STUDY ID #	DO NOT WRITE ABOVE THIS LINE		HOSPITAL #
Brief Pain Inventory (Short Form)			
Date: / /			Time:
Name:	Last	First	Middle Initial

**Items 1 and 2 removed
with permission of
copyright owner**

3. Please rate your pain by circling the one number that best describes your pain at its <u>worst</u> in the last 24 hours.
0 1 2 3 4 5 6 7 8 9 10 No Pain Pain as bad as you can imagine
4. Please rate your pain by circling the one number that best describes your pain at its <u>least</u> in the last 24 hours.
0 1 2 3 4 5 6 7 8 9 10 No Pain Pain as bad as you can imagine
5. Please rate your pain by circling the one number that best describes your pain on the <u>average</u> .
0 1 2 3 4 5 6 7 8 9 10 No Pain Pain as bad as you can imagine
6. Please rate your pain by circling the one number that tells how much pain you have <u>right now</u> .
0 1 2 3 4 5 6 7 8 9 10 No Pain Pain as bad as you can imagine

Page 1 of 2

STUDY ID #: _____ DO NOT WRITE ABOVE THIS LINE HOSPITAL #: _____
_____/_____/_____
Last First Middle Initial

Items 7 and 8 removed
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9. Circle the one number that describes how, during the past 24 hours, pain has interfered with your:

A. General Activity

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

B. Mood

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

C. Walking Ability

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

D. Normal Work (includes both work outside the home and housework)

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

E. Relations with other people

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

F. Sleep

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

G. Enjoyment of life

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

16.5 List of strong CYP3A4 inducers

Strong

Ajmaline phenobarbital

Carbamazepine

Enzalutamide

Fosphenytoin

Hypericum perforatum (St. John's Wort)

Lumacaftor

Methylphenobarbital

Mitotane

Phenobarbital

Phenytoin

Primidone

Propranolol phenobarbital

Rifabutin

Rifampicin

Rifamycin

Rifapentin

A randomized, double-blind, placebo-controlled Phase III study of ODM-201 versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer

[ODM-201 in addition to standard androgen deprivation therapy and docetaxel in metastatic hormone-sensitive prostate cancer]

Acronym: ARASENS

Bayer study drug BAY 1841788 / ODM-201

Study purpose: Efficacy and Safety

Clinical study phase: III **Date:** 07 APR 2017

Study No.: 17777 **Version:** 1.0

Author: [REDACTED]

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List of Abbreviations

ADT	Androgen deprivation therapy
AE	Adverse event
AEE	Actual extent of exposure
ALP	Alkaline phosphatase
ATC	Anatomical-Therapeutic-Chemical
bid	Twice daily
BPI-SF	Brief Pain Inventory – Short Form
CI	Confidence interval
CT	Computed tomography
CV	Coefficient of variation
DMC	Data Monitoring Committee
EBRT	External beam radiation therapy
ECG	Electrocardiogram
ECOG PS	Eastern Cooperative Oncology Group Performance Status
EOT	End of treatment
FAS	Full analysis set
FPSI-DRS-P	FACT/NCCN FPSI-17 disease-related symptoms - physical
GCP	Good Clinical Practice
HR	Hazard ratio
ICF	Informed consent form
KM	Kaplan-Meier
LHRH	Luteinizing hormone releasing hormone
LKAD	Last known alive date
LLOQ	Lower limit of quantification
M&S	Modeling & Simulation
MedDRA	Medical Dictionary for Regulatory Activities
mg	Milligram
mHSPC	Metastatic hormone-sensitive prostate cancer
mL	Milliliter
MRI	Magnetic resonance imaging
NCCN	National Comprehensive Cancer Network
NCCN-FACT FPSI-17	Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire
NCI-CTCAE v4.03	National Cancer Institute-Common Terminology Criteria for Adverse Events; version 4.03
ng	Nanogram
OEE	Overall extent of exposure
OS	Overall survival
PCWG3	Prostate Cancer Clinical Trials Working Group
PK	Pharmacokinetics
PKS	Pharmacokinetic Analysis Set
PRO	Subject-reported outcomes
PSA	Prostate-specific antigen
PT	Preferred Term
QoL	Quality of life
RECIST	Response Evaluation Criteria In Solid Tumors
SAE	Serious adverse event
SAF	Safety analysis set
SAP	Statistical Analysis Plan
SAS	Statistical Analysis Software
SD	Standard deviation
SOC	System Organ Class
SSE	Symptomatic skeletal event

SSE-FS	Symptomatic skeletal event free survival
TEAE	Treatment emergent adverse events
TESAE	Treatment emergent serious adverse events
t_{last}	Time to the last quantifiable concentration
t_{max}	Time to the maximum plasma concentration
ULN	Upper limit of normal
WHO	World Health Organization
WHO-DD	WHO Drug Dictionary
WPS	Worst pain subscale

1. Introduction

The original protocol, Version 1.0, is dated 25 MAY 2016. This statistical analysis plan (SAP) v1.0, describes the analyses to be included in the clinical study report for study17777. It is based on the integrated clinical study protocol Version 2.0, dated 04 OCT 2016.

2. Study Objectives

The primary objective of this study is:

- To demonstrate the superiority in overall survival (OS) of ODM-201 in addition to standard androgen deprivation therapy (ADT) and docetaxel over placebo in addition to standard ADT and docetaxel

The secondary objectives of this study are to evaluate:

- Time to castration-resistant prostate cancer
- Time to initiation of subsequent antineoplastic therapy
- Symptomatic skeletal event free survival (SSE-FS)
- Time to first symptomatic skeletal event (SSE)
- Time to initiation of opioid use for ≥ 7 consecutive days
- Time to pain progression
- Time to worsening of physical symptoms of disease based on functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire (NCCN-FACT FPSI-17)
- Safety

The exploratory objectives of this study include the evaluation of:

- Quality of life (QoL), measured using the NCCN-FACT FPSI-17 questionnaire
- Medical resource use
- Prostate-specific antigen (PSA) assessments
- PK and exposure-response analysis
- Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease

3. Study Design

Design overview

This is an international, randomized, double-blind, placebo-controlled, multicenter phase III study of ODM-201 in subjects with metastatic hormone-sensitive prostate cancer (mHSPC). Approximately 1,300 subjects will be randomized (1:1 ratio) to receive one of the following study drugs:

- ODM-201, 600 mg (2 x 300-mg tablets) bid orally with food
- Placebo matching ODM-201 tablets in appearance, bid orally with food

All subjects must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. Six cycles of docetaxel will be administered after randomization.

For the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization.

Subjects will be stratified at randomization as follows:

- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases
 - Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase (ALP)
 - ALP < upper limit of normal (ULN)
 - ALP $\geq$ ULN

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

A schematic of the study periods is presented in [Figure 3–1](#).

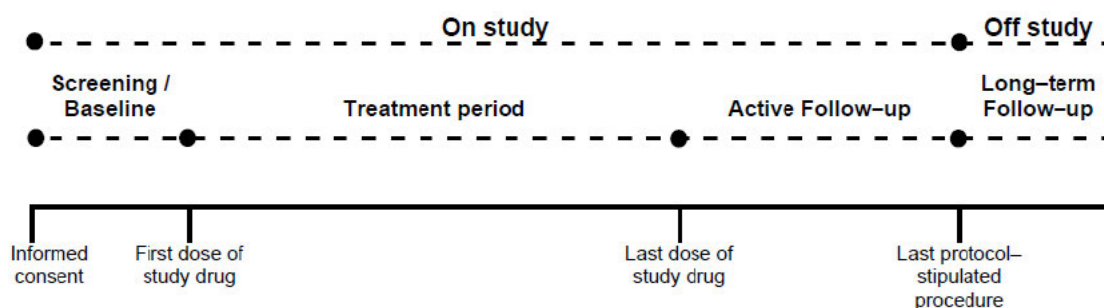


Figure 3–1 Study periods schematic

The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up.

Screening period:

All trial-related procedures and evaluations will only be performed after the subject has agreed to participate and has signed the informed consent form (ICF).

The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met.

Once eligibility is confirmed and documented, eligible subjects will be randomized in a ratio of 1:1 to treatment with ODM-201 or placebo.

Treatment period:

The start of the treatment period is defined by the first administration of study drug. Study treatment will be provided for all subjects to be taken twice daily until disease progression (symptomatic progressive disease, change of antineoplastic therapy), unacceptable toxicity, consent withdrawal or withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.

Subjects will be evaluated every 12 weeks for castration-resistant prostate cancer, initiation of subsequent antineoplastic therapy, SSEs, opioid use for ≥ 7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, adverse events (AEs) and serious adverse events (SAEs), QoL and PSA assessment.

Active Follow-up period:

The Active Follow-up period is the interval from the end-of-study drug intake to the end of all protocol-specified post-treatment interventions. It includes:

- End of Treatment Visit

An End of Treatment (EOT) Visit will be conducted 30 (+7) days after the last dose of study drug. The following assessments will be performed at the EOT Visit: QoL, pain assessment, analgesic consumption, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical progression, radiological progression, toxicity, or other), SSEs, and all AEs and SAEs regardless of causality.

- Active Follow-up visits

During the Active Follow-up period, the following assessments will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change: PSA progression, clinical progression, radiological progression, toxicity, or other), SSEs, and study drug-related SAEs. After approximately 1 year of Active Follow-up, subjects will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of treatment period for up to 1 year or until the subject can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data.

Long-term (survival) Follow-up period:

After Active Follow-up, subjects will continue to be contacted approximately every 12 weeks (by phone) to capture all antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs and survival status. Additional contacts may be required during this period, for example, for the collection of survival information for survival sweep data collection. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study.

4. General Statistical Considerations**4.1 General Principles**

Statistical analyses will be conducted by or under the supervision of the sponsor's Study Statistician, except for the analysis of biomarker data, which will be performed by or under the direction of the sponsor's Genomics and Biomarker Statistical Expert. Statistical analyses will be performed by using the software package Statistical Analysis System (SAS) release 9.2 or higher (SAS Institute Inc., Cary, NC, USA).

All variables will be analyzed by descriptive statistical methods by treatment group and overall, unless otherwise specified. The number of data available and missing data, mean, standard deviation (SD), minimum, quartiles, median, and maximum will be calculated for continuous variables. Frequency tables will be generated for categorical data.

Definition of efficacy and safety endpoints, analyses strategies, structure of analyses datasets and layout of analyses data displays are following Bayer standards as documented in the Bayer standard system: Oncology Therapeutic Area Standards, and Global Medical Standards, respectively. Where the given ordering reflects the priority of the different standards, meaning specifications of the latter ones have to be followed only if not specified in standards mentioned before. Study-specific specifications may be included in addition to the project standards, if needed.

4.2 Handling of Dropouts

A subject who discontinues study participation prematurely for any reason is defined as a "dropout" if the subject has already been randomized. Subjects who drop out will not be replaced.

4.3 Handling of Missing Data

In order to achieve the goal of a well conducted clinical trial according to Good Clinical Practice (GCP), every effort should be made to collect all data. However, despite best efforts, it may be inevitable that missing or incomplete data are reported. All missing or partial data will be presented in the patient data listing as they are recorded on the Case Report Form. Except as noted, missing data will not be estimated or carried forward in any statistical analysis.

When appropriate, the following rules will be implemented so as not to exclude patients from statistical analyses due to missing or incomplete data:

Efficacy Variables

Death Date

Every effort should be made to resolve incomplete or missing dates during the course of the study (i.e. edit checks, data cleaning/monitoring etc.). However, in rare circumstances, missing parts of either date of last contact or the date of death may occur where an imputation algorithm has to be defined. In general the following rule should be followed: Missing month or year is not acceptable. To impute a value for day, available data from other sources should be utilized.

If a patient died and the date of death is completely or partially missing, but there is an AE with the outcome as “Death”, the date of death can be replaced by the end date of the AE.

Other Time-to-event Endpoints

The time-to-event endpoints include time to castration-resistant prostate cancer, time to initiation of subsequent antineoplastic therapy, SSE-FS, time to first symptomatic SSE, time to initiation of opiate use for ≥ 7 consecutive days, time to pain progression, and time to worsening of physical symptoms of disease. The following rule in [Table 4-1](#) to impute the missing event assessment date will be used.

Table 4-1 Imputation rules for time-to-event endpoints assessment date

Scenario: Partially Missing	A. If partial date in the specific time-to-event endpoint data panel has day and month missing, then January 01 will be assigned to the missing fields. B. If partial date in the specific time-to-event endpoint data panel has missing day only, then the first day of the month will be assigned to the missing day. C. If imputed date from A or B above is before the available date of adjacent previous assessment, then the date of adjacent previous assessment will assigned to overwrite the imputed date from A or B. D. If imputed date from A or B or C above is before the [reference date], then [reference date] will assigned to overwrite the imputed date from A or B or C. E. If imputed date from A or B or C above is after the cut-off date, then the cut-off date will assigned to overwrite the imputed date from A or B or C or D.
Scenario: Completely Missing	No imputation need.

Questionnaire

In this study, a reduced set of 11 items of Brief Pain Inventory – Short form (BPI-SF) is used. The items are aggregated into two dimensions: pain severity index and function interference index. All 4

severity items must be completed for aggregating the pain severity index. The function interference index is scored as the mean of the item scores multiplied by 7, given that more than 50% or 4 of 7, of the items have been completed.

For NCCN-FACT FPSI-17, total scores and subscales scores (disease-related physical symptoms, disease-related emotional symptom, treatment side-effects, function/well-being) will be assessed. Where there are missing items, subscale scores can be prorated if >50% of items on subscale are completed. The total score is then calculated as the sum of the un-weighted subscale scores.

4.4 Interim Analyses and Data Monitoring

4.4.1 Interim Analyses

Two formal interim analyses of OS are planned. The first interim analysis is for futility and the second interim analysis is for both futility and efficacy. The DMC (Data Monitoring Committee; see [4.4.2 Independent Data Monitoring Committee](#)) will oversee these interim analyses.

For the futility interim and final analyses together, a one-sided overall beta of 0.1 will be used. Stopping boundaries will be calculated with an O'Brien-Fleming alpha-spending function based on the actual number of events observed up to the cut-off date. The interim futility analyses are planned when approximately 153 deaths and approximately 357 deaths are observed (information fraction=0.3 and 0.7). The critical boundaries for the futility analyses will be calculated separately from the efficacy boundaries, in order to not interfere with the type 1 error of the efficacy analyses.

For the efficacy interim and final analyses together, a one-sided overall alpha of 0.025 will be used. An early stopping boundary will be calculated with an O'Brien-Fleming alpha-spending function based on the actual number of events observed up to the cut-off date. The interim efficacy analysis is planned when approximately 357 deaths are observed (information fraction=0.7).

The following scenario in [Table 4-2](#) serves as an example.

Table 4-2 Example of stopping boundaries for interim analyses

Analysis Time	# Events	Stopping Boundaries Hazard Ratio Scale (ODM-201/placebo) (% Improvement) / Z Score		Cumulative Alpha Spent	Cumulative Beta Spent
		Crossing Lower Bound (efficacy)	Crossing Upper Bound (futility - lack of efficacy)		
1 st Interim	■	-	1.166 (-14.2%) / 0.952	-	0.003
2 nd Interim	■	0.773 (29.4%) / -2.438	0.886 (12.9%) / -1.140	0.007	0.049
Final	■	0.838 (19.3%) / -2.000	0.838 (19.3%) / -2.0	0.025	0.100

Software EAST 6.4 was used for this design

At the first interim analysis, if the estimated hazard ratio (ODM-201 over placebo) is greater than or equal to 1.166, corresponding to greater than or equal to approximately 14.2% decrease in median OS compared to placebo, one would consider stopping the trial for lack of efficacy (i.e. futility). Otherwise, if the estimated hazard ratio is less than 1.166, the trial would continue to the second interim analysis, providing there were no other safety considerations to warrant discontinuation of subject treatment and follow-up.

At the time of the second interim analysis, the amount of alpha spent would be 0.007. If the estimated hazard ratio (ODM-201 over placebo) is less than or equal to 0.773, this corresponds to a greater than or equal to 29.4% improvement in OS compared with placebo (i.e., median =77.6 months for the ODM-201 group versus 60 months assumed for the placebo group), one would consider stopping the trial and conclude that ODM-201 prolongs OS compared to placebo.

On the other hand, if the estimated hazard ratio is greater than or equal to 0.886, corresponding to less than or equal to approximately 12.9% increase in median OS compared to placebo, one would consider stopping the trial for lack of efficacy (i.e. futility). Otherwise, if the estimated hazard ratio falls between 0.773 and 0.886, the trial would continue to the final analysis.

At the final analysis, if the estimated hazard ratio (ODM-201 over placebo) is less than or equal to 0.838, this corresponds to greater than or equal to 19.3% improvement in median OS compared to placebo (i.e., median=71.6 months for the ODM-201 group versus 60 months assumed for the placebo group), one would conclude that ODM-201 is efficacious in prolonging OS.

4.4.2 Independent Data Monitoring Committee

A DMC will be instituted to ensure ongoing safety of study subjects with respect to a risk/benefit assessment during periodic data review meetings, review results from planned interim analyses and provide a formal recommendation for continuation/termination of the study and monitor study conduct to ensure the overall integrity of trial is maintained. The DMC will operate independently of the Sponsor and Investigators.

4.5 Data Rules

Baseline

Baseline is defined as the last measurements performed prior to the first study drug administration in Screening or Visit 1, unless otherwise specified.

Repeated measures

If there are repeated measurements per time point (e.g. laboratory values, vital signs, etc.), the following rules will be used (unless otherwise specified):

- Before the start of the study drug administration (i.e., for screening and baseline value), the latest measurement at scheduled visits will be used. Unscheduled visits will be used, if there are no measurements at scheduled visits. If the latter is the case, the last unscheduled visit will be used.

- In case of repeated measurements at any post screening time point, the first measurement at scheduled visits will be used. Unscheduled visits will be used, if there are no measurements at scheduled visits. If the latter is the case, the first unscheduled visit will be used.

5. Analysis Sets

5.1 Assignment of analysis sets

Full analysis set (FAS)

All randomized subjects represent the FAS. The FAS will be used in the analysis of all efficacy endpoints. Subjects will be included in FAS according to the treatment to which they are randomized.

Safety analysis set (SAF)

All randomized subjects who have received at least 1 dose of study drug represent the SAF. This safety population will be used in the analyses of all safety endpoints. Subjects will be included in the analyses according to the treatment they actually received.

Pharmacokinetic Analysis Set (PKS)

The first 20 subjects who were randomized and have received at least 1 cycle of docetaxel, and for whom mandatory dense PK sampling will be done, will be included in the PKS, as long as they have received at least 3 days of uninterrupted study drug treatment as well as one cycle of docetaxel and have at least one post dose PK measurement.

The SAF will be used for the analyses of the safety variables. The pharmacokinetic data will be analyzed in the PKS. The FAS will be used for the analysis of efficacy variables and all other variables.

Patients who signed the informed consent but for any reasons don't proceed to randomization, will be considered screening failures. They will be listed separately.

6. Statistical Methodology

6.1 Population characteristics

6.1.1 Disposition

The number of patients starting, completing, and discontinuing together with the primary reason for discontinuing of the screening, treatment, active follow-up, and long-term follow-up phases will be presented overall and by treatment group (except for the screening phase). Disposition of screening will be summarized for all enrolled patients, and disposition of other phases for FAS.

In addition, the number of patients with major and minor protocol deviations will be presented for treatment groups and overall. The frequencies of each major protocol deviation will be presented by treatment group and overall.

6.1.2 Demographic and Other Baseline Characteristics

Descriptive summaries of demographics and baseline characteristics are presented by treatment group and overall for the FAS populations. Comparability of the treatment groups with respect to demographics and baseline characteristics is assessed using descriptive summaries.

The following demographic data will be summarized:

- Age at screening (years)
- Age category (< 65, ≥65 years)
- Region (Western, Asia, Rest of the World)¹
- Race and ethnicity
- Height
- Weight at baseline

The following baseline characteristics are summarized:

- Histology
- Time from initial diagnosis to the date of the first actual dose of study drug (months)
- Time from initial diagnosis of metastases to the date of the first actual dose of study drug (weeks)
- Stage of prostate cancer (TNM Classification) at initial diagnosis
- Gleason score at initial diagnosis of prostate cancer
- Status of primary tumor at study entry
- Stage of prostate cancer (TNM Classification) at study entry
- Gleason score at study entry
- Extent of metastatic disease at study entry
- PSA at baseline
- Testosterone and ALP from central lab at baseline
- Baseline Eastern Cooperative Oncology Group Performance Status (ECOG PS)
- Vital signs at baseline: blood pressure, heart rate
- Baseline in the physical symptoms of disease subscale of the NCCN-FACT FPSI-17
- Worst pain subscale of the BPI-SF

¹ Western comprises of Australia, Belgium, Canada, Finland, France, Germany, Israel, Italy, Netherlands, Spain, Sweden, United Kingdom, United States; Asia comprises of China, Japan, South Korea, Taiwan; Rest of the World comprises of Brazil, Bulgaria, Czech Republic, Poland, Russia, and Mexico.

6.1.3 Medical History and Concurrent Medical Conditions

For data coding, Medical Dictionary for Regulatory Activities (MedDRA) [version 18.0 or most recent version] will be used for medical history. Medical history findings (i.e. previous diagnoses, diseases or surgeries) not pertaining to the study indication, start before signing of the ICF and considered relevant to the study will be presented for each MedDRA Primary System Organ Class (SOC) and Preferred Term (PT) by treatment group and overall.

6.1.4 Prior and Concomitant Medication

Prior therapy is defined as all medications taken in the first 28 days prior to randomization.

All medications and significant non-drug therapies ongoing during the screening period (28 days prior to randomization) will be recorded and summarized.

Prior, concomitant and post-treatment medications will be coded by the World Health Organization Drug Dictionary (WHO-DD). Prior and concomitant medication will be summarized based on Anatomical-Therapeutic-Chemical (ATC) Class (level 1, which is the first character from the WHO-DD code) and ATC Sub-class (level 2, which is the first 3 characters from the WHO-DD code). Note that the same medication can appear more than once in the table as it can have several ATC codes.

Concomitant opioid medication and concomitant analgesic medication during the 24 hours prior to each study visit will be recorded and summarized.

6.1.5 Prior and Concomitant Anti-cancer Therapy and Procedures

All subjects must receive an ADT of the Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy started ≤ 12 weeks before randomization. Six cycles of docetaxel will be administered after randomization.

Prior, concomitant and post-study antineoplastic therapy will be summarized based on ATC Class (level 1, which is the first character from the WHO-DD code) and ATC Sub-class (level 3, which is the first 4 characters from the WHO-DD code).

The following frequency tables will be provided by treatment group and overall for FAS:

- Diagnostic and therapeutic procedures for the cancer under study (Prior / Concomitant / During follow-up)
- Radiotherapy (Prior / Concomitant / During follow-up)
- ADT (Prior / Concomitant)
- Docetaxel therapy
- Systemic anti-cancer therapies during follow-up

6.1.6 Extent of Exposure

As a general rule, trailing “0 mg” records, which are not followed by any positive amount of drug, will not be included in the calculation of any drug duration or amount. Similarly, the according trailing “drug interruptions” will not be used in statistical tables. A footnote will be included, stating that “Interruption becoming permanent study treatment discontinuation is not accounted as an interruption”.

Descriptive statistical summaries will be provided for ODM-201, ADT and docetaxel separately by treatment group in SAF population for the following variables:

- Overall time under treatment ^a
- Actual time under treatment ^b
- Actual dose per day ^c
- Total amount of dose ^d
- Percent of planned dose received ^e

a. including time off drug and dose interruptions

b. excluding time off drug and dose interruptions

c. actual dose per day = total amount of dose / number of days with intake > 0

d. total amount of dose = sum of dose received over total time under treatment

e. percent of planned dose received = total amount of dose [mg] / planned dose [mg] * 100%;
planned dose is sum of the intended initial dose according to protocol over total time under treatment.

For patients with dose reduction, interruption, or delay, the number of dose reductions, interruptions, or delays per patient and their reasons will be summarized.

The maximum time for a dose interruption period is 28 consecutive days. Any subject requiring treatment interruption >28 consecutive days must be withdrawn from treatment with study drug.

6.2 Efficacy

For time-to-event analyses the censoring mechanism is assumed to be non-informative. Patients will be handled as right-censored in time-to-event analyses.

6.2.1 Primary efficacy variable

Definition of OS

The primary efficacy variable is OS, defined as the time (in days) from date of randomization until death from any cause.

The following censoring rules in [Table 6-1](#) will be applied:

Table 6-1 Censoring rules for OS

Situation	End Date	Censored
Documented death during study	Death date	No
No documented death with no contacts after randomization	Date of randomization	Yes
No documented death at the time of analysis (database cut-off date)	Last known alive date (LKAD) or at the database cut-off date, whichever comes earlier	Yes

Last Known Alive Date

The LKAD is derived from the main data sources. The last available date across all selected data panels listed below will be picked as the LKAD by subject. Information from selected data, i.e. visit dates, exposure information, laboratory measurements, tumor assessment dates, SSE dates, demographics, survival status date, and disposition events or follow up assessments will be used to determine survival status. Within all the dates from the selected data panels, identify the latest available date as the LKAD for each patient.

For partial missing dates, impute the dates in each selected data panel; see imputation rules below in [Table 6-2](#).

Table 6-2 Imputation rules for death date derivation

Scenario: Partially Missing	A. If partial death date has only day missing, then impute the missing day as LKAD + 1 or the first day of the death month, whichever comes later. B. If month and/or year are missing, censor the patient at LKAD. (The survival status should be changed to censored on LKAD.) C. If imputed death date is after the cut-off date, then the cut-off date will be used as censoring date.
Scenario: Completely Missing	No imputation will be done.

Analysis of OS

All randomized subjects (FAS) will be included in the primary analysis of OS, the primary efficacy endpoint. The analysis will be performed when approximately 509 deaths are observed.

The primary analysis of OS will be a stratified log-rank test with the same IxRS stratification factors as used for randomization. The null hypothesis that there is no difference in OS between treatment arms, which is equivalent to a hazard ratio (HR) of 1, will be tested against the alternative hypothesis that the HR of ODM-201 over placebo is below 1. If the p-value from the one-sided log-rank test is less than 0.025 with the HR (ODM-201 group/placebo) less than 1, the null hypothesis will be rejected in favour of the alternative hypothesis.

The SAS code will be similar to the following:

```
PROC LIFETEST DATA=<DATASET>;  
TIME EFFVAL * CENSORNY;  
STRATA {strata variables};  
TEST TREATMGR;  
RUN;
```

The HR (ODM-201 group/placebo) for OS and its 95% confidence interval (CI) will be calculated using the Cox model, stratified by the same factors as stated above. The SAS code similar to the following will be used:

```
PROC PHREG DATA = <DATASET>;  
MODEL EFFVAL * CENSORNY = TREATMGR;  
STRATA {strata variables};  
RUN;
```

Kaplan-Meier (KM) estimates for the median time of OS (including 95% CI) and first and third quartiles will be presented for each treatment group. The KM estimates at time points such as 3 months, 6 months, etc., together with corresponding 95% CIs and the differences of these estimates between the ODM-201 group and the placebo group will be presented. The SAS code will be similar to the following:

```
PROC LIFETEST DATA = <DATASET>;  
TIME EFFVAL * CENSORNY;  
STRATA = TREATMGR;  
RUN;
```

The overall 1-sided type I error rate for the analysis of OS is 0.025. Two formal interim analyses of OS (the first one for futility and the second one for both futility and efficacy) are planned. A DMC will oversee these interim analyses. For further details on these interim analyses see Section 4.4.2

[Independent Data Monitoring Committee](#)

Descriptive statistics of survival follow-up time will be calculated by treatment arm and total. Follow-up time is the time to death as described for the OS endpoint for censored and uncensored patients.

Sensitivity analysis

OS will be evaluated with the unstratified log-rank test and Cox model for the FAS population.

OS will also be evaluated with stratified log-rank test and Cox model for the FAS population using stratification factors collected from the eCRF, in case there are more than 5% of patients with different values in any stratification variables between IxRS and eCRF.

Subgroup analysis

Subgroup analyses will be conducted for the primary efficacy endpoint OS based on the FAS population. Descriptive statistics and HR estimates with 95% CI will be provided at least for the subgroups listed below, provided there are a sufficient number of events in total within the subgroup across the treatment arms. Forest plots of the HRs will be generated.

- Stratification Factor: Extent of disease (non-regional lymph nodes metastases only, bone metastases with or without lymph node metastases, or visceral metastases with or without lymph node metastases or with or without bone metastases)
- Stratification Factor: ALP ($<ULN$, $\geq ULN$)
- Age group (<65 , ≥ 65)
- Ethnic / Race group (Caucasian, non-caucasian)
- Region (Western, Asia, Rest of the World)²
- Baseline PSA values ($< median$, $\geq median$)
- ECOG PS at baseline (0, 1)

6.2.2 Secondary efficacy variable

Definitions and censoring rules for secondary efficacy variables

The secondary efficacy variables include:

- **Time to castration-resistant prostate cancer**, defined as the time to the occurrence of the following events, whatever comes first:
 - PSA progression with serum testosterone being at castrate level <0.50 ng/mL.
PSA progression is defined as a $\geq 25\%$ increase above the nadir (lowest at or after baseline) value, which is confirmed by a second value 3 or more weeks later (in the next assessment) and an increase in absolute value of ≥ 2 ng/mL above nadir.
 - Radiological progression by soft tissue and visceral lesions, which is determined according to Response Evaluation Criteria In Solid Tumors (RECIST) version 1.1 (1), based on MRI / CT scans of the chest, abdomen, and pelvis performed by the Investigator, as recommended by Prostate Cancer Clinical Trials Working Group (PCWG3) (2).
 - Radiological progression by bone lesions, which is determined according to PCWG3 criteria based on whole body ^{99m}Tc methylene diphosphonate bone scans performed by the Investigator. Bone lesions will be recorded separately from soft tissue and visceral lesions.

The following censoring rules in [Table 6-3](#) will be applied:

² Western comprises of Australia, Belgium, Canada, Finland, France, Germany, Israel, Italy, Netherlands, Spain, Sweden, United Kingdom, United States; Asia comprises of China, Japan, South Korea, Taiwan; Rest of the World comprises of Brazil, Bulgaria, Czech Republic, Poland, Russia, Mexico.

Table 6-3 Censoring rules for time to castration-resistant prostate cancer

Situation	End Date	Censored
Castration-resistant prostate cancer during study	Event assessment date	No
No baseline or post-baseline event assessment	Date of randomization	Yes
Castration-resistant prostate cancer after two or more consecutive missing assessments	Date of the last assessment before the consecutive missing ones	Yes
No castration-resistant prostate cancer at the time of analysis	Date of last visit before discontinuation or randomization date, whichever is later	Yes
Prohibited ³ new anticancer therapy before castration-resistant prostate cancer and database cut-off	Date of last visit before new therapy start date or randomization date, whichever is later	Yes

- **Time to initiation of subsequent antineoplastic therapy⁴**, defined as the time from randomization to initiation of first subsequent antineoplastic therapy for prostate cancer.

The following censoring rules in [Table 6-4](#) will be applied:

Table 6-4 Censoring rules for time to initiation of subsequent antineoplastic therapy

Situation	End Date	Censored
Initiation of first subsequent antineoplastic therapy for prostate cancer during study	Start date of first subsequent antineoplastic therapy for prostate cancer	No
No subsequent antineoplastic therapy for prostate cancer at the time of analysis	Date of last visit, or randomization date, whichever comes later	Yes

- **Symptomatic skeletal event free survival**, defined as the time from randomization to the first occurrence of SSE or death from any cause, whichever comes first.

An SSE is defined as the occurrence of one of the followings, whichever comes first:

- External beam radiation therapy (EBRT) to relieve skeletal symptoms, or
- new symptomatic pathologic bone fracture, or

³ Allowed concomitant anticancer therapies are ADT and docetaxel for 6 cycles after randomization. All eCRF entered drug names will be transformed into a standard drug name.

⁴ Antineoplastic therapy is the same as “anticancer therapy”.

- occurrence of spinal cord compression, or
- tumor-related orthopaedic surgical intervention.

The following censoring rules in [Table 6-5](#) will be applied:

Table 6-5 Censoring rules for SSE-FS

Situation	End Date	Censored
SSE during study	Date of the first assessment reporting event	No
Death without prior SSE event during the study with less than two consecutive missed SSE assessments immediately prior to death (less than $2 * (12+1) = 26$ weeks between last SSE assessment prior to death)	Date of death	No
Death without prior SSE event during the study with two or more consecutive missed SSE assessments immediately prior to death	Last SSE assessment before the missing SSE assessments	Yes
Discontinued the study without documented event	Date of last SSE assessment or randomization date, whatever comes later	Yes
Prohibited new anticancer therapy before documented event and database cut-off	Date of last last SSE assessment before new anticancer therapy or randomization date, whatever comes later	Yes

- **Time to first symptomatic SSE**, defined as the time from randomization to the first occurrence of SSE. Death is not considered as event in this endpoint; otherwise the same rules as for SSE-FS apply. The following censoring rules in [Table 6-6](#) will be applied:

Table 6-6 Censoring rules for first symptomatic SSE

Situation	End Date	Censored
Symptomatic SSE during study	Date of the first assessment reporting event	No
No symptomatic SSE at the time of analysis	Last SSE assessment before the missing SSE assessments	Yes

- **Time to initiation of opioid use for ≥ 7 consecutive days**, defined as the time from randomization to the date of first opiate use for ≥ 7 consecutive days. Only data of opioid use related to prostate cancer pain will be included. Time to opiate use will be determined by analgesic use captured via electronic case report form. The following censoring rules in [Table 6-7](#) will be applied:

Table 6-7 Censoring rules for time to initiation of opioid use for ≥ 7 consecutive days

Situation	End Date	Censored
Opioid use for ≥ 7 consecutive days during study	Start date of initiation of opioid use for ≥ 7 consecutive days for prostate cancer	No
No opioid use for ≥ 7 consecutive days at the time of analysis	Date of last SSE assessment or randomization date, whatever comes later	Yes
Opioid use for ≥ 7 consecutive days at/before randomization date	Randomization date	Yes

- **Time to pain progression** is defined as the time from randomization to the first date a subject experiences a pain progression. Pain will be assessed with the BPI-SF questionnaire. Pain progression is defined as follows:
 - For asymptomatic subjects (worst pain subscale [WPS] 0 at baseline), pain progression is defined as
 - an increase of 2 or more points in the “worst pain in 24 hours” score (i.e. 2 or more point increase in WPS score) from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart, or
 - initiation of short- or long-acting opioid use for pain.
 - For symptomatic subjects (WPS > 0 at baseline), pain progression is defined as
 - an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) score from baseline observed at 2 consecutive evaluations ≥ 4 weeks apart and a WPS of ≥ 4 , or
 - initiation of short- or long-acting opioid use for pain.

The following censoring rules in [Table 6-8](#) will be applied:

Table 6-8 Censoring rules for time to pain progression

Situation	End Date	Censored
Pain progression during study	Start date of pain progression	No
No baseline or post-baseline event assessment	Date of randomization	Yes
No pain progression at the time of analysis	Date of last visit or randomization date, whatever comes later	Yes
Opioids taken for any reason within 4 weeks before randomization	Date of randomization	Yes

- **Time to worsening of physical symptoms of disease** is defined as the time from randomization to the first date a subject experiences an increase in physical symptoms based on the NCCN-FACT-FPSI-17 questionnaire. An increase in physical symptoms of disease is defined as 3-point decrease (a lower score = higher symptom burden) from baseline in the physical symptoms of disease subscale (FPSI-DRS-P subscale of the NCCN-FACT-FPSI-17 questionnaire) observed at 2 consecutive evaluations ≥ 4 weeks apart. The following censoring rules in Table 6-9 will be applied:

Table 6-9 Censoring rules for time to worsening of physical symptoms of disease

Situation	End Date	Censored
Worsening of physical symptoms of disease during study	Start date of worsening of physical symptoms of disease	No
No worsening of physical symptoms of disease at the time of analysis	Date of last visit, or randomization date, whichever comes later	Yes

Secondary efficacy variables analysis

The secondary efficacy variables will be analyzed for the FAS population unless otherwise specified. Time-to-event endpoints will be analyzed using stratified log-rank test with randomization stratification factors. Hazard ratio and 95% CI will be provided using the Cox model stratified by the same factors as stated above.

Median time, the 25th and 75th percentiles and associated 95% CI of KM estimates will be presented by treatment group as well as the number and percentage of censored observations. A KM curve will be generated for each treatment group.

6.3 Safety

The safety variables include:

- AEs until the end-of-study treatment visit
- SAEs until the end-of-study treatment visit
- Study drug-related SAEs until the end of Long-term Follow-up
- Vital signs: blood pressure and heart rate
- 12-lead electrocardiogram (ECG)
- Physical examination
- Laboratory safety assessments (hematology, clinical chemistry, urinalysis)

6.3.1 Adverse Events

AEs will be coded using MedDRA. The severity (or intensity) of AEs would be documented using the National Cancer Institute-Common Terminology Criteria for Adverse Event version 4.03 (NCI-CTCAE v 4.03). Descriptive summary tables will be presented on all safety parameters by treatment group.

The AEs will be presented with their worst NCI-CTCAE v 4.03 grade. AEs will be classified by the investigator as related or not related to study drug. Treatment emergent AE (TEAE) is defined as any event arising or worsening after the first dose of study drug until 30 days after the last treatment with ODM-201/placebo.

Summary statistics (frequency and percentage of patients, not of events) will be presented by treatment group, MedDRA SOC and PT, and NCI-CTCAE worst grade for the following:

- Incidence rate of all TEAEs
- Incidence rate of TEAEs leading to permanent discontinuation of treatment (separate tables for ODM-201 and docetaxel)
- Incidence rate of TEAEs leading to dose reduction (separate tables for ODM-201 and docetaxel)
- Incidence rate of TEAEs leading to dose interruption (separate tables for ODM-201 and docetaxel)
- Incidence rate of TEAEs for NCI-CTCAE grade 3 or 4
- Incidence rate of TEAEs for NCI-CTCAE grade 5
- Incidence rate of TEAEs with incidence of greater than 5%
- Incidence rate of TEAEs with incidence of greater than 10%
- Incidence rate of treatment emergent SAEs (TESAEs)
- Incidence rate of all drug-related TEAEs
- Incidence rate of drug-related TEAEs leading to permanent discontinuation of treatment
- Incidence rate of drug-related TEAEs leading to dose reduction
- Incidence rate of drug-related TEAEs leading to dose interruption
- Incidence rate of drug-related TEAEs for NCI-CTCAE grade 3 or 4
- Incidence rate of drug-related TEAEs for NCI-CTCAE grade 5
- Incidence rate of drug-related TEAEs with incidence of greater than 5%
- Incidence rate of drug-related TESAEs

Summaries listed above that are drug-related would be repeated for both ODM-201 and docetaxel. In addition, the incidence of AEs during screening will be tabulated.

In addition, the following listings will be provided:

- Listings of AEs and laboratory toxicities by worst NCI-CTCAE grade up to 30 days after the last treatment with ODM-201/placebo
- Listings of AEs between 31 days after the last treatment with ODM-201/placebo to EOT
- Listings of study drug-related SAEs with concomitant medication received in the follow-up periods
- Listings of patients with TEAEs leading to permanent discontinuation of treatment
- Listing of all TESAEs
- Listing of patients who died during treatment or up to 30 days after last dose of study medication
- Listing of patients who died later than 30 days after last dose of study medication

No AEs of special safety interest were defined in this study.

6.3.2 Laboratory safety assessments

The laboratory safety parameters will be presented with their worst NCI-CTCAE v 4.03 grade. Results of the laboratory safety assessments (hematology, general chemistry and urinalysis) will be summarized by visit and overall with worst grade. Summary statistics will also be presented for the change from baseline for each quantitative clinical laboratory variable, at each post-baseline visit. For these variables, the number and percentage of patients with transitions from worst grade at baseline to worst grade at post-baseline visits relative to the respective laboratory's reference ranges will be presented by study phase. In case there is more than one observation for the same latest time, the worst grade will be taken.

For tables displaying treatment emergent laboratory abnormalities, patients with specimen collection between the first dose of study drug and 30 days after the last treatment will be included.

6.3.3 Other Safety Measures

In addition, results of physical examination, vital signs, 12-lead ECG, and ECOG PS will be summarized with descriptive statistics and/or frequency tables. Vital signs and ECG data will be summarized by treatment and visit, including change from baseline where appropriate.

6.4 Other exploratory variables

• Rate of absolute PSA response at 6 and 12 months

Absolute PSA response is defined as blood PSA level <0.2 ng/mL, confirmed by a second subsequent PSA value <0.2 ng/mL 3 or more weeks later. Rate of absolute PSA response is defined as the number of subjects with absolute PSA response, divided by the total number of subjects evaluable for absolute PSA response. Rate of absolute PSA response will be evaluated on subject data up to 6 months and 12 months after randomization.

- **Rate of relative PSA response at 3, 6 and 12 months**

Relative PSA response is defined as a $\geq 30\%$ reduction of the blood PSA level compared to the screening value, confirmed by a second subsequent PSA value with a $\geq 30\%$ reduction from Screening 3 or more weeks later. Rate of relative PSA response is defined as the number of subjects with relative PSA response, divided by the total number of subjects evaluable for relative PSA response. Rate of relative PSA response will be evaluated on subject data up to 3 months, 6 months, and 12 months after randomization.

- **Time to PSA progression**

The time to PSA progression is defined as the time from the date of randomization to the date of first PSA progression. The same definition of PSA progression and censoring rules applied to time to castration-resistant prostate cancer will be applied here. Moreover, subjects without PSA progression as of database cut-off, whether or not surviving, will be censored at the last total PSA lab assessment date. Time to PSA progression will be descriptively evaluated using KM estimates for quantiles, including 95% two-sided CIs.

- **Changes in symptom burden and QoL based on the NCCN-FACT FPSI-17 questionnaire**

Descriptive statistics on observed data will be presented at each assessment time point and for change from baseline by treatment group. Summary statistics of difference of treatment groups will be provided.

- **Medical resource use**

Listings of patients with healthcare resource use will be provided.

- **Biomarker evaluations**

Results from exploratory biomarker studies will be reported in a separate biomarker report.

6.5 Pharmacokinetics

Pharmacokinetics in the first 20 subjects who were randomized and have received at least 1 cycle of docetaxel

The concentration-time courses of all substances will be tabulated separated by treatment. The following statistics will be calculated for each of the sampling points: arithmetic mean, SD, and coefficient of variation (CV); geometric mean, geometric SD (re-transformed SD of the logarithms), and CV; minimum, median, maximum value, and the number of measurements.

Means at any time will only be calculated if at least 2/3 of the individual data were measured and were above the lower limit of quantification (LLOQ). For the calculation of the mean value, a data point below LLOQ will be substituted by one-half of this limit. In tables showing mean values, where values below LLOQ are included in the calculation of mean values, these means will be marked.

Individual and geometric mean concentration versus time curves of all substances (using the actual sampling times for individual plots and the planned sampling times for mean plots) will be plotted by treatment using both a linear and semilogarithmic scale.

Pharmacokinetic characteristics ($t_{\max}$, t_{last} excluded) will be summarized by the statistics mentioned above. The $t_{\max}$, t_{last} will be described utilizing minimum, maximum, and median as well as frequency counts.

Pharmacokinetics in all subjects

Individual concentration-time data of ODM-201, ORM-16555, ORM-16497, and ORM-15341 & docetaxel will be provided in a clinical study report appendix. All other parameters will be described in detail in a separate M&S report.

6.6 Determination of sample size

The sample size of the study is based on the primary endpoint of OS. The study is designed to have 90% power to detect a 33% increase in median time of OS with ODM-201 compared to placebo (from 60 to 80 months, corresponding to a HR of 0.75) with a 1-sided alpha of 0.025. The OS data will be considered as mature when approximately 509 deaths are observed.

With the additional assumptions that subjects will enroll at a rate of 50 subjects per month, exponential distributions of the OS event times, median time of OS in the control group of 60 months, 5% dropout rate of subjects, and a 6-months enrollment ramp-up period, it follows that approximately 1,300 subjects should be enrolled to observe 509 deaths after approximately 70 months.

7. Document history and changes in the planned statistical analysis

- Approval of SAP v1.0

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Title page**A randomized, double-blind, placebo-controlled Phase III study of darolutamide (ODM-201) versus placebo in addition to standard androgen deprivation therapy and docetaxel in patients with metastatic hormone-sensitive prostate cancer****Darolutamide in addition to standard androgen deprivation therapy and docetaxel in metastatic hormone-sensitive prostate cancer****Acronym: ARASENS****Bayer study drug** BAY 1841788 / Darolutamide**Study purpose:** Efficacy and Safety**Clinical study phase:** III **Date:** 11 NOV 2021**Study No.:** 17777 **Version:** 4.1**Author:** **Confidential**

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List of Abbreviations

ADT	Androgen deprivation therapy
AE	Adverse event
AEE	Actual extent of exposure
ALP	Alkaline phosphatase
ATC	Anatomical-Therapeutic-Chemical
bid	Twice daily
BHA	Bone health agent
BPI-SF	Brief Pain Inventory – Short Form
CI	Confidence interval
CT	Computed tomography
CV	Coefficient of variation
DMC	Data Monitoring Committee
EBRT	External beam radiation therapy
ECG	Electrocardiogram
ECOG PS	Eastern Cooperative Oncology Group Performance Status
EOT	End of treatment
FAS	Full analysis set
FPSI-DRS-P	FACT/NCCN FPSI-17 disease-related symptoms - physical
GCP	Good Clinical Practice
HR	Hazard ratio
ICF	Informed consent form
KM	Kaplan-Meier
LHRH	Luteinizing hormone releasing hormone
LKAD	Last known alive date
LLOQ	Lower limit of quantification
M&S	Modeling & Simulation
MedDRA	Medical Dictionary for Regulatory Activities
mg	Milligram
mHSPC	Metastatic hormone-sensitive prostate cancer
mL	Milliliter
MRI	Magnetic resonance imaging
NCCN	National Comprehensive Cancer Network
NCCN-FACT FPSI-17	Functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire
NCI-CTCAE v4.03	National Cancer Institute-Common Terminology Criteria for Adverse Events; version 4.03
ng	Nanogram
OEE	Overall extent of exposure
OS	Overall survival
PCWG3	Prostate Cancer Clinical Trials Working Group
PK	Pharmacokinetics
PKS	Pharmacokinetic Analysis Set
PRO	Patient-reported outcomes
PSA	Prostate-specific antigen
PT	Preferred Term
QoL	Quality of life
RECIST	Response Evaluation Criteria In Solid Tumors
SAE	Serious adverse event
SAF	Safety analysis set
SAP	Statistical Analysis Plan
SD	Standard deviation
SOC	System Organ Class
SSE	Symptomatic skeletal event
SSE-FS	Symptomatic skeletal event free survival
TEAE	Treatment emergent adverse events

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TESAE	Treatment emergent serious adverse events
t_{last}	Time to the last quantifiable concentration
t_{max}	Time to the maximum plasma concentration
ULN	Upper limit of normal
WHO	World Health Organization
WHO-DD	WHO Drug Dictionary
WPS	Worst pain subscale

1. Introduction

Study 17777 (ARASENS) is an international, randomized, double-blind, placebo-controlled, Phase III efficacy and safety study of darolutamide in addition to standard androgen deprivation therapy (ADT) and docetaxel in patients with metastatic hormone-sensitive prostate cancer (mHSPC).

This statistical analysis plan (SAP) describes the pre-specified analyses and data presentations for the primary completion analyses, which will be performed when the targeted number of death events for the primary efficacy endpoint of overall survival (OS) is reached.

The SAP contains definitions of analysis sets, derived variables, and statistical methods for the analysis of efficacy and safety.

For SAP version history and changes in the planned statistical analysis, please refer to Section [7](#).

2. Study Objectives

The primary objective of this study is:

- To demonstrate the superiority in overall survival (OS) of darolutamide in addition to standard androgen deprivation therapy (ADT) and docetaxel over placebo in addition to standard ADT and docetaxel

The secondary objectives of this study are to evaluate:

- Time to castration-resistant prostate cancer
- Time to pain progression
- Symptomatic skeletal event free survival (SSE-FS)
- Time to first symptomatic skeletal event (SSE)
- Time to initiation of subsequent systemic antineoplastic therapy
- Time to worsening of disease-related physical symptoms based on functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire (NCCN-FACT FPSI-17)
- Time to initiation of opioid use for ≥ 7 consecutive days
- Safety

The secondary objective of the efficacy endpoints will be tested in the hierarchical order as given above, except for safety.

The exploratory objectives of this study include the evaluation of:

- Quality of life (QoL), measured using the NCCN-FACT FPSI-17 questionnaire
- Medical resource use
- Prostate-specific antigen (PSA) assessments
- PK and exposure-response analysis
- Evaluate biomarkers to investigate the drug (i.e. mode-of-action-related effect and / or safety) and / or the pathomechanism of the disease

3. Study Design

Design overview

This is an international, randomized, double-blind, placebo-controlled, multicenter Phase III study of darolutamide in patients with mHSPC. Approximately 1,300 patients will be randomized (1:1 ratio) to receive one of the following study drugs:

- Darolutamide, 600 mg (2 x 300 mg tablets) bid orally with food
- Placebo matching darolutamide tablets in appearance, bid orally with food

All patients must receive ADT of Investigator's choice (LHRH agonist/antagonists or orchiectomy) as standard therapy, started ≤ 12 weeks before randomization. Six cycles of docetaxel will be administered as background treatment after randomization. The first cycle of docetaxel should be administered within 6 weeks after start of study drug.

For at least the first 20 randomized patients who have received at least 1 cycle of docetaxel, and for whom a detailed mandatory PK analysis will be performed, docetaxel should be administered at least 14 days after randomization.

Patients will be stratified at randomization as follows:

- Extent of disease
 - Non-regional lymph nodes metastases only
 - Bone metastases with or without lymph node metastases
 - Visceral metastases with or without lymph node metastases or with or without bone metastases
- Alkaline Phosphatase (ALP)
 - $ALP < \text{upper limit of normal (ULN)}$
 - $ALP \geq \text{ULN}$

Note: Blood samples to measure ALP levels for stratification will be analyzed in a central laboratory.

The study will comprise 4 consecutive periods: Screening, Treatment, Active Follow-up, and Long-term (survival) Follow-up.

Screening period:

All trial-related procedures and evaluations will only be performed after the patient has agreed to participate and has signed the informed consent form (ICF).

The Screening period will consist of multiple evaluations that will take place within 28 days prior to randomization to ensure that all eligibility criteria are met.

Once eligibility is confirmed and documented, eligible patients will be randomized in a ratio of 1:1 to treatment with darolutamide or placebo.

Treatment period:

The start of the treatment period is defined by the first administration of study drug (darolutamide or placebo). Study treatment will be provided for all patients to be taken twice

daily until disease progression (symptomatic progressive disease, change of systemic antineoplastic therapy), unacceptable toxicity, consent withdrawal or withdrawal from the study at the discretion of the Investigator or his/her designated associate(s), death, or non-compliance.

Patients will be evaluated every 12 weeks for castration-resistant prostate cancer, initiation of subsequent systemic antineoplastic therapy, SSEs, opioid use for ≥ 7 consecutive days, pain progression, worsening of physical symptoms of disease based on NCCN-FACT FPSI-17, adverse events (AEs) and serious adverse events (SAEs), QoL and PSA assessment.

Active Follow-up period:

The Active Follow-up period is the interval from the end of darolutamide or placebo intake to the end of all protocol-specified post-treatment interventions.

It includes the End of Treatment Visit (EOT). An EOT Visit will be conducted 30 (+ window of 7) days after the last dose of darolutamide or placebo. The following assessments will be performed at the EOT Visit: QoL, pain assessment, analgesic consumption, subsequent systemic antineoplastic treatments for prostate cancer, SSEs, and all AEs and SAEs regardless of causality.

During the Active Follow-up period, the following assessments will be performed during standard of care clinic visits approximately every 12 weeks for up to 1 year: QoL, pain assessment, analgesic consumption, survival status, subsequent systemic antineoplastic treatments for prostate cancer (which should be provided with start and stop dates and reason for change, SSEs, and study drug-related SAEs. After approximately 1 year of Active Follow-up, patients will be transitioned to Long-term (survival) Follow-up. The Active Follow-up period extends from the discontinuation of darolutamide or placebo treatment period for up to 1 year or until the patient can no longer travel to the clinic, dies, is lost to follow-up, or withdraws informed consent and actively objects to collection of further data.

Long-term (survival) Follow-up period:

After Active Follow-up, patients will continue to be contacted approximately every 12 weeks (by phone) to capture all systemic antineoplastic treatments for prostate cancer with start and stop date and reasons for change (PSA progression, clinical progression, radiological progression, toxicity, other), study drug-related SAEs and survival status. The end of Long-term Follow-up period is death, lost to follow-up, consent withdrawal, or end-of-study.

Survival sweep:

For every analysis of OS, survival data will be collected through additional survival sweeps. All patients considered alive at the database cut-off date and prior to any subsequent additional analysis will be contacted for survival status.

Primary completion

The first analysis of OS will be performed as an interim futility analysis when approximately 153 deaths have occurred. In case the study continues after the futility analysis, a final efficacy analysis of OS will be performed when approximately 509 deaths have occurred and will be considered the primary completion. The database cut-off date for this analysis will be the date of primary completion.

4. General Statistical Considerations

4.1 General Principles

Statistical analyses will be performed by using the software package SAS[®] Version 9.2 or higher (SAS Institute Inc., Cary, NC, USA).

All variables will be analyzed by descriptive statistical methods by treatment arm and overall, unless otherwise specified. The number of data available and missing data, mean, standard deviation (SD), minimum, quartiles, median, and maximum will be calculated for continuous variables. Frequency tables will be generated for categorical variables.

4.2 Handling of Dropouts

A patient who discontinues study participation prematurely for any reason is defined as a “dropout” if the patient has already been randomized. Patients who discontinue treatment or study will not be replaced.

4.3 Handling of Missing Data

All missing or partial data will be presented in the patient data listing as they are recorded on the Case Report Form. Except as noted, missing data will not be estimated or carried forward in any statistical analysis.

When appropriate, the following rules will be implemented so as not to exclude patients from statistical analyses due to missing or incomplete data:

Overall survival and other time-to-event endpoints:

Every effort should be made to resolve incomplete or missing dates during the course of the study (i.e. edit checks, data cleaning/monitoring etc.). However, in rare circumstances, missing parts of either the censoring date or the event date may occur where an imputation algorithm has to be defined. In general the following rule should be followed: Missing month or year is not acceptable. To impute a value for day, day 15 of the month will be used for the calculation. This applies to all OS and time-to-event endpoints dates if the dates are partially missing in the selected data panels.

AE and Concomitant Medication (CM) partial or missing dates

Refer to SAP appendix 9.1 imputation rule for incomplete AE/CM dates.

Laboratory Assessments

If laboratory assessment results contain an inequality sign “<” or “>”, then the lower or higher detection limit will be used as the imputed laboratory assessment value for both safety and efficacy analyses.

ePRO Questionnaire

Two scores will be derived for Brief Pain Inventory-Short Form (BPI-SF): (1) for the pain severity score if one answer is missing then scoring will be set to missing, (2) for the pain interference score if four or more answers are missing out of the seven questions then the score will be set to missing (see scoring of BPI-SF in Appendix 9.3).

For NCCN-FACT FPSI-17, total scores and subscales scores (disease-related physical symptoms, disease-related emotional symptoms, treatment side-effects, function/well-being) will be assessed. Where there are missing items, subscale scores can be prorated if >50% of items on subscale are completed. If ≤50% of the items are answered for any domain, then the score of that domain is set to missing. The total score is set to missing if the related overall item response rate is less than or equal to 80% (see scoring of NCCN-FACT FPSI-17 in Appendix 9.4). In addition, a total score should only be calculated if all of the component subscales have valid scores.

4.4 Determination of Sample Size

The sample size of the study is based on the primary endpoint of OS. The study is designed to have 90% power to detect a 25% decrease in risk of death with darolutamide compared to placebo, corresponding to a HR of 0.75 with a one-sided test with a type I error of 0.025 (equivalent a two-sided test with a type I error 0.05). The OS data will be considered mature when approximately 509 deaths are observed.

With the additional assumptions that patients will enroll at a rate of 50 patients per month, exponential distributions of the OS event times, median time of OS in the control group of 60 months, 5% dropout rate of patients, and a 6-months enrollment ramp-up period, it follows that approximately 1,300 patients need to be randomized to observe 509 deaths after approximately 70 months.

4.5 Interim Analyses and Data Monitoring

4.5.1 Interim Analyses

For the futility interim and final analyses together, a one-sided overall beta of 0.1 will be used. Stopping boundaries will be calculated with an O'Brien-Fleming beta-spending function using the actual number of events observed up to the cut-off date. The interim futility analysis is planned when approximately 153 deaths were observed (information fraction=0.3). The critical boundary for the futility analysis will be calculated separately from the efficacy boundary, in order to not interfere with the type I error of the efficacy analysis.

The following scenario in Table 4–1 serves as an example.

Table 4–1 Example of stopping boundaries for interim analyses

Analysis Time	# Events	Stopping Boundaries Hazard Ratio Scale (darolutamide+docetaxel arm vs. placebo+docetaxel arm) (% Improvement) / Z Score		Cumulative Alpha Spent	Cumulative Beta Spent
		Crossing Lower Bound (efficacy)	Crossing Upper Bound (futility - lack of efficacy)		
1 st Interim ^a	■	-	1.166 (-14.2 %) / 0.952	-	0.003
Final	■	0.841 (19.0%) / 1.96	0.841 (19.0%) / 1.96	0.025	0.100

Software ADDPLAN neo V10.0.4 was used for this design

a: the 1st interim analysis was completed and its stopping boundaries were calculated before the decision was made to remove the 2nd interim analysis, which was originally planned but removed with version 3.0 of the SAP. Please refer to Section 7 for additional details.

If the estimated hazard ratio (darolutamide over placebo) is greater than or equal to 1.166 at the interim analysis, stopping the trial for lack of efficacy (i.e. futility) would be considered. Otherwise, i.e. if the estimated hazard ratio is less than 1.166, the trial would continue to the final analysis, providing there were no other safety considerations to warrant discontinuation of patient treatment.

4.5.2 Independent Data Monitoring Committee

A DMC will be instituted to ensure ongoing safety of study patients with respect to a risk/benefit assessment during periodic data review meetings, review results from planned interim analyses and provide a formal recommendation for continuation/termination of the study and monitor study conduct to ensure the overall integrity of study is maintained. The DMC will operate independently of the Sponsor and Investigators.

4.6 Data Rules

Baseline

Baseline is defined as the last measurements performed in Screening or Visit 1 (regardless the scheduled or the unscheduled visits), unless otherwise specified prior to the first study drug administration.

Repeated measures

If there are repeated measurements per time point on the same day (e.g. laboratory values, vital signs, etc.), the following rules will be used (unless otherwise specified):

- Before the start of the study drug administration (i.e., for screening and baseline value), the latest measurement at scheduled visits will be used. Unscheduled visits will be used if there are no measurements at scheduled visits. If the latter is the case, the last unscheduled visit will be used.
- In case of repeated measurements at any post baseline time point, the first measurement at scheduled visits will be used. Unscheduled visits will be used if there are no measurements at scheduled visits. If the latter is the case, the first unscheduled visit will be used.

Stratification variables

Since differences may occur between the values of the stratification factors entered by the investigator at the time of randomization into IxRS and those collected on the eCRF, the stratified analysis of OS will be performed using both assignments to the strata. The primary stratified analysis for OS will be based on the information collected in the IxRS whereas stratification information entered in the eCRF (considered the 'true' information) will be used for the sensitivity analysis. In rare circumstances, if eCRF data cannot be documented, then IxRS data will be used.

5. Analysis Sets

5.1 Assignment of Analysis Sets

Patients will be excluded from all below-mentioned analysis set if they are related to or associated with any critical GCP violations that result in fraudulent patient data (please refer to Section 7 for rationale). The FAS will be used for the analysis of all efficacy endpoints and all other endpoints. The SAF will be used for the analyses of all safety endpoints. The pharmacokinetic data will be analyzed in the PKS. Patients who signed the informed consent but for any reasons did not proceed to randomization will be considered screening failures. They will be listed separately.

Full analysis set (FAS)

All patients who were randomized are included in the Full Analysis Set, except for cases with critical GCP violations. Following the intent-to-treat (ITT) principle, the patients in this set will be grouped according to the planned treatment they were allocated to receive at randomization, irrespective of actual treatment.

Safety analysis set (SAF)

All randomized patients who have received at least 1 dose of darolutamide or placebo represent the SAF, except for cases with critical GCP violations. This safety population will be used in the analyses of all safety endpoints. Patients will be included in the analyses according to the treatment they actually received. Patients will be included in the darolutamide+docetaxel arm if they have received any dose of darolutamide, otherwise they will be included in the placebo+docetaxel arm if they only received placebo.

Pharmacokinetic Analysis Set (PKS)

At least the first 20 randomized patients who have received at least 1 cycle of docetaxel, and for whom mandatory dense PK sampling was done, will be included in the PKS, as long as they have received at least 3 days of uninterrupted study drug treatment as well as one cycle of docetaxel and have at least one post dose PK measurement, except for cases with critical GCP violations.

In addition, a subset of Chinese patients who were randomized and have received at least 1 dose of darolutamide, and for whom mandatory dense PK sampling was done prior to the first dose of docetaxel, will also be included in the PK analysis set.

6. Statistical Methodology

6.1 Population characteristics

6.1.1 Disposition

The number of patients enrolled and included in each population will be tabulated by region, country, and center. The number of patients starting, completing, and discontinuing together with the primary reason (including COVID-19 pandemic related reason) for discontinuing of the Screening, Treatment, Active Follow-up, and Long-term Follow-up periods will be presented overall and by treatment arm (except for the Screening period). The number of patients currently on treatment at primary completion analysis cut-off will also be summarized by treatment arm. Disposition of Screening will be summarized for all enrolled patients, and

disposition of other study periods for the FAS. Patients who enter Active Follow-up are defined as patients who discontinued study treatment not due to death or lost to follow-up and had Active Follow-up visits after the end of treatment visit. The number of patients affected by COVID-19 pandemic related study disruptions (protocol deviations, premature discontinuations, and AEs) will be summarized in a separate table.

In addition, the number of patients with important protocol deviations will be presented by treatment arm and overall. The frequencies of each important protocol deviation will be presented by treatment arm and overall. All COVID-19 pandemic related protocol deviations are considered important protocol deviations and will be presented by treatment arm and overall. Listings of important protocol deviations associated with COVID-19 pandemic will be presented as well. The number of patients with missed visits or assessments in the key assessments will be presented for each treatment arm and overall.

6.1.2 Demographic and Other Baseline Characteristics

Descriptive summaries of demographics and baseline characteristics are presented by treatment arm and overall, for the FAS. Comparability of the treatment arms with respect to demographics and baseline characteristics is assessed using descriptive summaries.

The following demographic data will be summarized:

- Age at screening (years)
- Age category (<65, 65-74, 75-84, ≥85 years)
- Geographical region (North America, Asia Pacific, Rest of the World)¹
- Race (White, Asian, Black or African American, Other, Not Reported) and ethnicity (Hispanic or Latino, Not Hispanic or Latino, Not Reported)
- Weight at baseline
- Body Mass Index (BMI) (kg/cm²) (<20, 20 to <25, 25 to <30, ≥30)
- Renal function - eGFR at baseline:
 - Normal: eGFR ≥ 90 mL/min
 - Mild impairment: 60 ≤ eGFR < 90 mL/min
 - Moderate impairment: 30 ≤ eGFR < 60 mL/min
 - Severe impairment: 15 ≤ eGFR < 30 mL/min

eGFR will be calculated according to the Modification of Diet in Renal Disease (MDRD) Formula:

$$\text{eGFR (mL/min/1.73m}^2\text{)} = 186 \times \text{SCR}^{-1.154} \times \text{age}^{-0.203} \times (0.742, \text{ if female}) \times (1.212 \text{ if blacks or African American}) \times (0.881 \text{ if Japanese}) \times (1.227 \text{ for Chinese (mainland China, Taiwan and Hongkong), where SCR= serum creatinine measured in mg/dL}$$

- Hepatic function at baseline:

¹ Geographic region: North America (Canada and United States), Asia Pacific (China, Japan, South Korea, and Taiwan) and the rest of the world (Australia, Belgium, Bulgaria, Brazil, Czech Republic, Germany, Spain, Finland, France, United Kingdom, Israel, Italy, Mexico, Netherlands, Poland, Russian Federation, and Sweden).

- Normal: Total bilirubin and AST $\leq$ upper limit of normal (ULN)
- Mild impairment: Total bilirubin $>$ ULN to $1.5 \times$ ULN or (Total bilirubin $\leq$ ULN and AST $>$ ULN)
- Moderate impairment: Total bilirubin $>$ 1.5 to $3 \times$ ULN, any AST
- Severe impairment: Total bilirubin $>$ $3 \times$ ULN, any AST.

The following baseline characteristics will be summarized:

- Extent of metastatic disease at study entry (Non-regional lymph nodes metastases only, Bone metastases with or without lymph node metastases, Visceral metastases with or without lymph node metastases or with or without bone metastases), from both IxRS and eCRF
- ALP (U/L) at baseline ($<$ ULN, $\geq$ ULN) - central laboratory, from both IxRS and eCRF
- PSA (ug/L) at baseline ($<$ median of overall population, $\geq$ median of overall population) - central laboratory
- Eastern Cooperative Oncology Group Performance Status (ECOG PS) at baseline
- Stage of prostate cancer (TNM Classification) at initial diagnosis (Stage I, IIA, IIB, III, IV (M1, M0))
The study used AJCC 7 and therefore the stage 4 M0 and M1 were not captured in the eCRF. As the CRFs captured the date of initial diagnosis and the date of initial diagnosis of metastases, the following assumptions will be used to estimate the M0 or M1 status at initial diagnosis:
 - Patients who have a time interval $>$ 3 months between initial diagnosis and initial metastases diagnosis will be assumed to be M0.
 - Patients who have a time interval of $\leq$ 3 months between initial diagnosis and initial metastases diagnosis will be assumed to be M1.
- Gleason score of prostate cancer ($<$ 8, $\geq$ 8) at initial diagnosis
- Time from initial diagnosis to the date of the first actual dose of darolutamide or placebo (months)
- Time from initial diagnosis of metastases to the date of the first actual dose of darolutamide or placebo (months)
- Time from initial diagnosis to initial diagnosis of metastases (months)
- Stage of prostate cancer (TNM Classification) at study entry
- Testosterone ($<$ 0.5, $\geq$ 0.5 ng/mL) at baseline - central laboratory
- Chromogranin A (ug/L) at baseline – local laboratory.
- The disease-related physical symptoms subscale of the NCCN-FACT FPSI-17 at baseline
- Worst pain subscale, average pain subscale of the BPI-SF (0: no pain, 1-3: mild pain, 4-7: moderate pain, 8 to 10: severe pain)

- Pain severity score of the BPI-SF

Demographic and baseline characteristics tables will be presented by geographical regions (North America, Asia Pacific, Rest of the World).

6.1.3 Medical History

For data coding, the Medical Dictionary for Regulatory Activities (MedDRA) will be used for medical history. Medical history findings (i.e. previous diagnoses, diseases or surgeries) not pertaining to the study indication, start before signing of the ICF and considered relevant to the study will be presented for each MedDRA Primary System Organ Class (SOC), and Preferred Term (PT) by treatment arm and overall for the FAS.

6.1.4 Prior and Concomitant Medication and Procedures

Prior therapy is defined as all medications taken before the start of darolutamide or placebo treatment. Concomitant therapy is defined as all medications taken between the start of darolutamide or placebo treatment and the last dose date of darolutamide or placebo.

- Prior and concomitant medications excluding ADT, systemic antineoplastic therapies captured in the anti-cancer therapy eCRF page: frequency of patients for each drug category
- Concomitant statin medications: frequency of patients
- Prior systemic antineoplastic therapy, including ADT by preferred drug name: frequency of patients for each drug category. The prior systemic antineoplastic therapy, including ADT, are selected in CM dataset using ATC codes: L02BX, L02BB, H01CC, V98, G03HB, L02AE, H01CA.
- A summary of subsequent systemic antineoplastic therapy will be presented by preferred drug name based on WHO-DD drug record number.
- A summary of life-prolonging subsequent systemic antineoplastic therapy will be presented by preferred drug name based on WHO-DD drug record number. The following medications after the study treatment period are life-prolonging subsequent therapies for prostate cancer: Abiraterone, Apalutamide, Enzalutamide, Docetaxel, Cabazitaxel, Radium 223, Sipuleucel-T, Lutetium-177.
- Hormonal therapy and orchiectomy at study entry (by LHRH agonist/antagonist only, orchiectomy only, LHRH agonist/antagonist and orchiectomy).
- Prior local treatment for prostate cancer at study entry (by prostatectomy, surgery, TURP, other, radiation, no surgical treatment and no radiation).
- Concurrent prostate cancer related-invasive procedures (e.g. procedures for urinary symptoms, surgical procedures for prostate cancer): frequency of patients by grouped term and procedure. Prostate cancer-related invasive procedure is defined as any procedure needed for alleviation of symptoms, signs or findings caused by progression of prostate cancer (e.g. catheterization of the bladder, percutaneous drainage of hydronephrosis, palliative electro resection of the prostate, etc.).

The dictionary used for coding medications was the ATC Class/Subclass and the World Health Organization Drug Dictionary (WHO-DD). For selection of statins medications, please refer to Appendix 9.5.

6.1.5 Extent of Exposure

Study drug is defined as darolutamide or placebo. Six cycles of docetaxel will be administered as background treatment after randomization and is not considered as study drug. As a general rule, trailing “0 mg” records, which are not followed by any positive amount of drug, will not be included in the calculation of any drug duration or amount. Similarly, trailing “drug interruptions” will not be used in statistical tables. Interruption becoming permanent study treatment discontinuation is not accounted as an interruption.

Descriptive statistical summaries will be provided for darolutamide or placebo and docetaxel separately by treatment arm in the SAF population for the following variables:

- Overall time under treatment (or treatment duration)
Including time off drug and dose interruptions. It will be calculated in days and presented in months as (date of the last dose of any study treatment – date of the first dose of any study treatment + 1) / 30.44.
- Actual dose per day
Actual dose per day = total amount of dose / number of days with intake > 0
- Total amount of dose
Total amount of dose = sum of dose received over total time under treatment
- Percent of planned dose received.
Percent of planned dose received = total amount of dose [mg] / planned dose [mg] * 100%; planned dose is sum of the intended initial dose according to protocol over total time under treatment.
- From 1 day after the last dose of docetaxel to the last dose of darolutamide or placebo. Including time off drug and dose interruptions. It will be calculated in days and presented in months as (date of the last dose of darolutamide or placebo – date of the last dose of docetaxel + 1) / 30.44. If the last dose of docetaxel date is after the last dose of darolutamide or placebo, then 0 day of duration will be included in the calculation.

For patients with dose reduction, interruption, or delay, the number of dose reductions, re-escalations, interruptions, or delays per patient and their reasons will be summarized. Only dose modifications after start of treatment and before end of treatment are included in the dose modification summary table. For example, dose interruption or delay documented without further dose resumption afterwards.

Extent of exposure will be described by age groups (<65, 65 - 74, 75 - 84, ≥85 years) and geographical regions (North America, Asia Pacific, Rest of the World).

6.2 Efficacy

For time-to-event analyses the censoring mechanism is assumed to be non-informative. Patients without events will be handled as right-censored in time-to-event analyses. Time to

event will be calculated in days as date of event or censoring minus date of randomization and presented in months ($=\text{days}/30.44$). If event or censoring date is on date of the randomization, then time to event = 1 (or day 1).

6.2.1 Primary Efficacy Endpoint

Definition of OS

The primary efficacy variable is OS, defined as the time from date of randomization until death from any cause.

The following censoring rules in [Table 6–1](#) will be applied:

Table 6–1 Censoring rules for OS

Situation	End Date	Censored
Documented death during study before or at data cut-off date	Death date	No
No documented death with no contacts after randomization and before or at data cut-off date	Date of randomization (Day 1)	Yes
No documented death before or at data cut-off date	Last known alive date (LKAD) or at the data cut-off date, whichever comes earlier	Yes

Last Known Alive Date

The last known alive date (LKAD) is derived from the main data sources. The last available date across all selected data panels listed below will be used as the LKAD by patient. Information from selected data, i.e. visit dates, exposure information, laboratory measurements, tumor assessment dates, SSE dates, demographics, survival status date, vital signs and disposition events or follow up assessments will be used to determine survival status.

Analysis of OS

All randomized patients (FAS) will be included in the primary analysis of OS.

The primary analysis of OS will be a stratified log-rank test with the same IxRS stratification factors as used for randomization. The null hypothesis that there is no difference in OS between treatment arms, which is equivalent to a hazard ratio (HR) of 1, will be tested against the alternative hypothesis that the HR of darolutamide over placebo is below 1. If the p-value from the one-sided log-rank test is less than 0.025 (corresponding to a two-sided log-rank test less than 0.05) with the HR (darolutamide+docetaxel arm vs. placebo+docetaxel arm) less than 1, the null hypothesis will be rejected in favor of the alternative hypothesis.

The SAS code will be similar to the following:

```
PROC LIFETEST DATA=<DATASET>;  
TIME EFFVAL * CENSORNY;  
STRATA {strata variables} / GROUP=TREATMGR test=(logrank);  
RUN;
```

The HR (darolutamide or placebo) for OS and its 95% confidence interval (CI) will be calculated using the Cox model, stratified by the same factors as were used for randomization. The SAS code similar to the following will be used:

```
PROC PHREG DATA = <DATASET>;  
MODEL EFFVAL * CENSORNY = TREATMGR;  
STRATA {strata variables};  
RUN;
```

Kaplan-Meier (KM) estimates for the median time of OS (including 95% CI) and 25% and 75% percentiles will be presented for each treatment arm. The KM estimates at time points such as 12 months, 24 months, etc., together with corresponding 95% CIs and the differences of these estimates between the darolutamide+docetaxel arm and the placebo+docetaxel arm will be presented. The SAS code will be similar to the following:

```
PROC LIFETEST DATA = <DATASET>;  
TIME EFFVAL * CENSORNY;  
STRATA TREATMGR;  
RUN;
```

Descriptive statistics of survival follow-up time will be calculated by treatment arm and total. Follow-up time is the time to LKAD or death as described for the OS endpoint for censored and uncensored patients.

Sensitivity analyses

OS will be evaluated with the unstratified log-rank test and Cox model for the FAS population (OS sensitivity analysis 1 – unstratified analysis).

OS will also be evaluated with the stratified log-rank test and Cox model for the FAS population using stratification factors collected from the eCRF, in case there are more than 5% of patients with different values in any stratification variable between IxRS and eCRF (OS sensitivity analysis 2 – eCRF-variables stratified analysis).

In addition, OS will also be evaluated with the stratified log-rank test and Cox model for the FAS population using extent of disease stratification factors collected from the central imaging review (OS sensitivity analysis 3 – central imaging review extent of disease as stratification factor).

6.2.2 Secondary Efficacy Endpoints

If the primary endpoint OS is significant, secondary efficacy endpoints will be tested with the hierarchical gatekeeping procedure. The order of secondary endpoints is as follows:

- 1) Time to castration-resistant prostate cancer
- 2) Time to pain progression
- 3) Symptomatic skeletal event free survival (SSE-FS)

- 4) Time to first symptomatic skeletal event (SSE)
- 5) Time to initiation of subsequent systemic antineoplastic therapy
- 6) Time to worsening of disease-related physical symptoms based on functional assessment of cancer therapy / National Comprehensive Cancer Network prostate cancer symptom index 17 item questionnaire (NCCN-FACT FPSI-17)
- 7) Time to initiation of opioid use for ≥ 7 consecutive days

If the primary endpoint OS is statistically significant at a 0.025 level (one-sided), the secondary endpoints will be tested using a hierarchical test procedure in the order as described above at the same nominal significance level. If the OS or a secondary endpoint is not statistically significant, the hierarchical procedure is stopped and all subsequent analyses of the secondary endpoints will be considered exploratory.

6.2.2.1 Time to castration-resistant prostate cancer

Time to castration-resistant prostate cancer is defined as the time from randomization to the time of the occurrence of the following events, whatever comes first (date of first occurrence of castration-resistant prostate cancer minus date of randomization):

- PSA progression, according to PCWG3 criteria, with serum testosterone being at castrate level <0.50 ng/mL, is defined as the date that a 25% or greater increase and an absolute increase of 2 ng/mL or more from the nadir (lowest at or after baseline) is documented, which both are confirmed by a second value obtained at least 3 weeks later, including all potential PSA values ≥ 2 ng/mL above nadir and $\geq 25\%$ increase above nadir between initial assessment date and confirmation assessment date. This definition requires serum testosterone at castrate levels <0.50 ng/mL and a first assessment date at least 12 weeks from randomization.
- Radiological progression by soft tissue and visceral lesions, is defined according to Response Evaluation Criteria In Solid Tumors (RECIST) version 1.1 (1), based on MRI / CT scans of the chest, abdomen, and pelvis performed by the Investigator, as recommended by Prostate Cancer Clinical Trials Working Group (PCWG3) (2).
- Radiological progression by bone lesions, is defined according to PCWG3 criteria based on whole body ^{99m}Tc methylene diphosphonate bone scans performed by the Investigator. Bone lesions will be recorded separately from soft tissue and visceral lesions.

The following censoring rules in Table 6–2 will be applied:

Table 6–2 Censoring rules for time to castration-resistant prostate cancer

Situation	End Date	Censored
Castration-resistant prostate cancer due to any component event during study	The earliest event date among the three components	No
No baseline or post-baseline event assessment for all three components #	Date of randomization (Day 1)	Yes
PSA progression event immediately after two or more consecutive missing assessments, without any prior radiological	Date of the last PSA assessment before the consecutive missing PSA assessments or date of last radiological	Yes

progression event and before or at data cut-off	assessment, whichever comes later	
No castration-resistant prostate cancer (or no event among the three components) before or at data cut-off date	Date of the latest date among the three components' last assessment before discontinuation or randomization date (censored at Day 1 if there is no follow-up available), whichever is later	Yes
Subsequent systemic antineoplastic therapy without any prior components event, without post PSA progression event and before or at data cut-off	Date of last radiological assessment before or on subsequent systemic antineoplastic therapy start date or the last PSA assessment date or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes
# This includes subjects with superscan at baseline for radiological bone lesion assessment		

A summary of first castration-resistant prostate cancer events will be provided. Patients with multiple events are only counted for the category in which the first event occurred. If multiple CRPC events (component events) occur on the same date for one subject, the subject is only counted into one category in the order of: radiological soft tissue/ visceral lesion progression > radiological bone progression > PSA progression.

The primary analysis of time to castration-resistant prostate cancer will be based on central PSA assessments only. The sensitivity analysis of time to castration-resistant prostate cancer will be based on both central and local PSA laboratory data.

6.2.2.2 Time to pain progression

Time to pain progression is defined as the time from randomization to the first date a patient experiences pain progression (first date of pain progression minus date of randomization). Pain will be assessed with the BPI-SF questionnaire. The main analysis is based on the ePRO device questionnaire data only. Pain progression is defined as follows:

- For asymptomatic patients (worst pain subscale [WPS] 0 at baseline), pain progression is defined as
 - an increase of 2 or more points in the “worst pain in 24 hours” score (i.e. 2 or more point increase in WPS score) from nadir (i.e. zero) observed at 2 consecutive evaluations ≥ 4 weeks apart, or
 - initiation of short- or long-acting opioid use for pain.
- For symptomatic patients (WPS >0 at baseline), pain progression is defined as
 - an increase of 2 or more points in the “worst pain in 24 hours” (i.e. 2 or more point increase in WPS score) from nadir observed at 2 consecutive evaluations ≥ 4 weeks apart and a WPS of ≥ 4 , or
 - initiation of short- or long-acting opioid use for pain.

The following censoring rules in [Table 6–3](#) will be applied:

Table 6–3 Censoring rules for time to pain progression

Situation	End Date	Censored
Pain progression during study	Start date of pain progression	No
No baseline or post-baseline event assessment	Date of randomization (censored at Day 1 if there is no follow-up available)	Yes
No pain progression before or at data cut-off date	The last BPI-SF assessment date or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes
Opioids taken for any reason within 4 weeks before or on randomization	Date of randomization (Day 1)	Yes

Sensitivity analysis for time to pain progression

- Pain progression will be evaluated with an increase of 2 or more points in the “worst pain in 24 hours” score (i.e. 2 or more point increase in WPS score) from baseline.
- Pain progression will be evaluated with an increase of 2 or more points in the “worst pain in 24 hours” score (i.e. 2 or more point increase in WPS score) from nadir after completion of docetaxel.
- Pain progression will be evaluated with an increase of 2 or more points in the “worst pain in 24 hours” score (i.e. 2 or more point increase in WPS score) from baseline after completion of docetaxel.
- The main analysis is based on the ePRO device questionnaire data only. Pain progression will be evaluated in this sensitivity analysis with source data from both the ePRO device and the paper questionnaire, based on the main analysis definition (change from nadir).

6.2.2.3 Symptomatic skeletal event free survival

Symptomatic skeletal event free survival (SSE-FS) is defined as the time from randomization to the first occurrence of an SSE or death from any cause, whichever comes first (date of first occurrence of an SSE or death minus date of randomization).

An SSE is defined as the occurrence of one of the followings, whichever comes first:

- external beam radiation therapy (EBRT) to relieve skeletal symptoms, or
- new symptomatic pathologic bone fracture, or
- occurrence of spinal cord compression, or
- tumor-related orthopaedic surgical intervention.

The following censoring rules in [Table 6–4](#) will be applied:

Table 6–4 Censoring rules for SSE-FS

Situation	End Date	Censored
SSE during study	Date of the first assessment reporting event	No
Death during study without SSE event	Date of death	No
No SSE before or at data cut-off date	Last SSE assessment before or at data cut-off	Yes
Patient lost to follow-up before or at data cut-off date	Date of last SSE assessment or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes

A summary of SSE-FS events will be provided. Patients with multiple events are only counted for the category in which the first event occurred. If multiple SSEs (component events) occur on the same date for one patient, the patient is only counted into one category in the order of: spinal cord compression > bone fracture > orthopedic surgery > EBRT.

6.2.2.4 Time to first SSE

Time to first SSE is defined as the time from randomization to the first occurrence of an SSE. Death is not considered as event in this endpoint (date of first SSE minus date of randomization); otherwise the same rules as for SSE-FS apply.

The following censoring rules in [Table 6–5](#) will be applied:

Table 6–5 Censoring rules for first SSE

Situation	End Date	Censored
SSE during study	Date of the first assessment reporting event	No
No SSE before or at data cut-off date	Last SSE assessment before or at data cut-off	Yes
Patient lost to follow-up before or at data cut-off	Date of last SSE assessment or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes

A summary of first SSE will be provided. Patients with multiple events are only counted for the category in which the first event occurred. If multiple SSEs (component events) occur on the same date for one patient, the patient is only counted into one category in the order of: spinal cord compression > bone fracture > orthopedic surgery > EBRT.

6.2.2.5 Time to initiation of subsequent systemic antineoplastic therapy

Time to initiation of subsequent systemic antineoplastic therapy² is defined as the time from randomization to initiation of the first subsequent systemic antineoplastic therapy (first subsequent systemic antineoplastic therapy date minus date of randomization).

Systemic antineoplastic therapy treatment was selected in CM dataset, which includes “SYSTEMIC ANTI-CANCER THERAPY (MEDICATION) DURING FOLLOW-UP” eCRF pages and CONCOMITANT MEDICATION” and “PRIOR AND CONCOMITANT MEDICATION” eCRF pages. Docetaxel administrated concomitantly with study drug that continued after study drug stopped will not be counted as subsequent systemic antineoplastic therapy.

From above 3 eCRF pages, subsequent systemic antineoplastic therapy is identified with criteria available in Appendix 9.2. The sponsor’s assessment of systemic antineoplastic therapy is presented in a supplementary document. The final list of systemic antineoplastic therapy will be based on the sponsor’s assessment and medical review before database lock.

The following censoring rules in Table 6–6 will be applied:

Table 6–6 Censoring rules for time to initiation of subsequent systemic antineoplastic therapy

Situation	End Date	Censored
Initiation of first subsequent systemic antineoplastic therapy during study	Start date of first subsequent systemic antineoplastic therapy	No
No subsequent systemic antineoplastic therapy before or at data cut-off date	The last know alive date or death date or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes

6.2.2.6 Time to worsening of disease-related physical symptoms

Time to worsening of disease-related physical symptoms is defined as the time from randomization to the first date a patient experiences an increase in disease-related physical symptoms based on the NCCN-FACT-FPSI-17 questionnaire (date of first worsening of disease-related physical symptoms minus date of randomization). An increase in disease-related physical symptoms is defined as 3-point decrease of FPSI-DRS-P subscale (a lower score = higher symptom burden, a score of “0” is a severely symptomatic patient and the highest possible score is an asymptomatic patient) from baseline in the disease-related physical symptoms subscale (FPSI-DRS-P subscale of the NCCN-FACT-FPSI-17 questionnaire) observed at 2 consecutive evaluations ≥ 4 weeks apart.

² Antineoplastic therapy is the same as “anti-cancer therapy” in this study.

The following censoring rules in [Table 6–7](#) will be applied:

Table 6–7 Censoring rules for time to worsening of disease-related physical symptoms

Situation	End Date	Censored
Worsening of disease-related physical symptoms during study	Start date of worsening of disease-related physical symptoms	No
No baseline or post-baseline event assessment	Date of randomization (censored at Day 1 if there is no follow-up available)	Yes
No worsening of disease-related physical symptoms before or at data cut-off date	Date of last assessment, or randomization date (censored Day 1 if there is no follow-up available), whichever comes later	Yes

The main analysis is based on the ePRO device questionnaire data only. In addition, time to worsening of disease-related physical symptoms will be evaluated as a sensitivity analysis with source data from both the ePRO device and the paper questionnaire.

6.2.2.7 Time to initiation of opioid use for ≥ 7 consecutive days

Time to initiation of opioid use for ≥ 7 consecutive days is defined as the time from randomization to the date of first opioid use for ≥ 7 consecutive days (date of first opioid use for ≥ 7 consecutive days minus date of randomization). Data of opioid use related to prostate cancer pain will be included. Opioid use for non-malignant cause will be excluded. Time to opioid use will be determined by opioid use captured via electronic case report form.

The following censoring rules in [Table 6–8](#) will be applied:

Table 6–8 Censoring rules for time to initiation of opioid use for ≥ 7 consecutive days

Situation	End Date	Censored
Opioid use for ≥ 7 consecutive days after randomization during study	Start date of initiation of opioid use for ≥ 7 consecutive days for prostate cancer	No
No opioid use for ≥ 7 consecutive days before or at data cut-off date	Date of last visit at which analgesic consumption question was collected or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later.	Yes
Opioid use for ≥ 7 consecutive days at/before randomization date	Date of randomization (Day 1)	Yes

A summary of initiation of opioid use for ≥ 7 consecutive days will be provided.

6.2.2.8 Analyses of Secondary efficacy endpoints

The secondary efficacy variables will be analyzed for the FAS population unless otherwise specified. Time-to-event endpoints will be analyzed using stratified log-rank test with

randomization stratification factors using IxRS data. Hazard ratio and 95% CI will be provided using the Cox model stratified by the same factors as were used for randomization.

Median time, the 25th and 75th percentiles and associated 95% CI of KM estimates will be presented by treatment arm as well as the number and percentage of censored observations. A KM curve will be generated for each treatment arm.

6.2.3 Exploratory Efficacy Endpoints

6.2.3.1 Rate of absolute PSA response at 6 and 12 months

Absolute PSA response is defined as baseline PSA value above the detection limit and a post baseline PSA level below 0.2 ng/mL, confirmed by a second subsequent PSA value below 0.2 ng/mL 3 or more weeks later, with all potential PSA values between initial date and confirmation date below 0.2 ng/mL.

Rate of absolute PSA response is defined as the number of patients with absolute PSA response, divided by the total number of patients randomized. Rate of absolute PSA response will be evaluated on patient data up to 6 months and 12 months after randomization.

6.2.3.2 Rate of relative PSA response at 3, 6 and 12 months

Relative 30% PSA response is defined as baseline PSA value above the detection limit and a post baseline $\geq 30\%$ reduction of the PSA level compared to the baseline value, confirmed by a second subsequent PSA value with a $\geq 30\%$ reduction from baseline 3 or more weeks later, with all potential PSA values between initial date and confirmation date showing a $\geq 30\%$ reduction from baseline. Relative 50% and 90% PSA response is defined in the same way.

Rate of relative PSA response is defined as the number of patients with relative PSA response, divided by the total number of patients randomized. Rate of relative PSA response will be evaluated on patient data up to 3 months, 6 months, and 12 months after randomization.

The comparison of absolute PSA response at 6 and 12 months between two treatment arms will be done using the Cochran-Mantel-Haenszel test stratified by the IxRS stratification factors. Same comparison will be performed for relative PSA response at 3, 6 and 12 months.

In addition, descriptive statistics and frequency distribution (no decline, $< 30\%$, $30\% - < 50\%$, $50\% - < 90\%$, $\geq 90\%$) will be provided for PSA maximum percent decline from baseline at any time on study.

6.2.3.3 Time to PSA progression

The time to PSA progression is defined as the time from the date of randomization to the date of first PSA progression with testosterone at castrate level < 0.5 ng/mL. The same definition of PSA progression with testosterone at castrate level < 0.5 ng/mL and censoring rules applied to time to castration-resistant prostate cancer in Section 6.2.2 will be applied here. Baseline PSA value is the last non-missing observation on or before the first day of study drug intake for treated patients. For not treated patients the baseline value will be the last non-missing observation on or before the randomization date. Time to PSA progression will be descriptively evaluated using KM estimates for quantiles, including 95% two-sided CIs. The following censoring rules in Table 6–9 will be applied:

Table 6–9 Censoring rules for PSA progression

Situation	End Date	Censored
PSA progression event during study	Event assessment date	No
No baseline or post-baseline event assessment	Date of randomization (censored at Day 1 if there is no follow-up available)	Yes
PSA progression event immediately after two or more consecutive missing assessments	Date of the last PSA assessment before the consecutive missing ones	Yes
No PSA progression before or at data cut-off date	Date of last PSA assessment before discontinuation or randomization date (censored at Day 1 if there is no follow-up available), whichever comes later	Yes

The primary analysis of PSA response and progression will be based on central PSA assessments only. A sensitivity analysis of PSA response and progression will be based on both central and local laboratory data.

6.2.3.4 ECOG PS

ECOG PS will be summarized with descriptive statistics and frequency tables. Changes from baseline to worst post-baseline ECOG PS score during treatment will be summarized in a shift tables by treatment arm. In addition, changes from worst post-baseline ECOG PS score during treatment to EOT score will be summarized in a shift table by treatment arm.

Baseline ECOG value is the last non-missing observation on or before the randomization date.

6.2.3.5 Quality of life

PRO data as measured by the NCCN-FACT FPSI-17 and the BPI-SF will be analyzed. Quality of Life and PRO data analyses specified in this section are based on the ePRO device questionnaire data only.

The PRO analyses will be performed for the patients in the FAS. Statistical tests will be performed with a one-sided test with type I error 0.025 (equivalent a two-sided test with type I error 0.05). The p-values are for descriptive purposes only.

Descriptive statistics on observed data will be presented for the NCCN-FACT FPSI-17 questionnaire (total score and each domain subscale score), and for BPI-SF questionnaire (pain severity and pain interference scores) at each assessment time and for change from baseline by treatment arm. Questionnaires under unscheduled visits and not planned visits per protocol will not be displayed in the descriptive tables. Analyses will be done for patients with baseline assessments.

The frequency for missing health related quality of life (HRQoL) assessment by treatment arm will be summarized.

An analysis of covariance (ANCOVA) model (a mixed linear model, with a random coefficient) for BPI-SF pain severity or interference scores in the time-adjusted Area under Curve (AUC) will be used to estimate the mean difference and treatment effect between the two treatment arms.

Least-square mean estimates, standard errors and 95% CIs will be estimated for each treatment arm and for the treatment arm difference.

Calculation of Time (AUC):

AUC will not be calculated if baseline data is missing.

The trapezoidal rule will be used to derive the AUC for a patient for BPI-SF pain severity and interference scores. The time-adjusted AUC will be calculated by dividing the AUC by the duration (in days) over the treatment period. When calculating the AUC, the exact date of completion of the questionnaires should be plotted and the duration will be calculated in days. The time adjusted AUC for BPI-SF pain severity and interference scores for an individual patient over a period of time [Ta,Tb] will be calculated as follows:

$$AUC_{a-b} = \frac{1}{2(T_b - T_a)} \sum_{i=a}^{b-1} (PRO_i + PRO_{i+1})(T_{i+1} - T_i)$$

Where PRO_i is the PRO measurement at time T_i (i=a... b).

6.2.4 Subgroup Analyses for Efficacy Endpoints

Subgroup analyses will be conducted for the primary efficacy endpoint OS, based on the FAS population. Descriptive statistics and HR estimates with 95% CI will be provided at least for the subgroups listed below, provided that at least 10 total events are observed within the subgroup across the treatment arms. HRs will be presented in forest plots. All subgroup analyses will be performed using an unstratified Cox model. Number of patients, number of events and KM estimates for the median per arm for all planned subgroups will be presented in the forest plot.

- Stratification Factor based on eCRF: Extent of disease (non-regional lymph nodes metastases only, bone metastases with or without lymph node metastases, or visceral metastases with or without lymph node metastases or with or without bone metastases)
- Stratification Factor based on eCRF: ALP at baseline (<ULN, ≥ULN)
- Age category (<65, 65-74, 75-84, ≥85 years)
- Race (White, Asian, Black or African American, Other)
- Geographical region (North America, Asia Pacific, Rest of the World)
- PSA values (<median of overall population, ≥median of overall population) at baseline
- ECOG PS at baseline (0, 1)
- Gleason score (<8, ≥8) at initial diagnosis
- Metastases at initial diagnosis (Yes: Stage IV-M1, No: Stage I, IIA, IIB, III, IV-M0)

Further important baseline cancer characteristics may also be considered.
Subgroup analyses will not be performed for the sensitivity analyses.

6.3 Safety

The safety endpoints include:

- AEs until the End of Treatment visit
- SAEs until the End of Treatment visit
- Study drug-related SAEs in Active Follow-up for up to one year
- Study drug-related SAEs until the end of Long-term Follow-up
- Laboratory safety assessments (hematology, clinical chemistry, urinalysis)
- Vital signs: blood pressure and heart rate
- 12-lead electrocardiogram (ECG)

6.3.1 Adverse Events

AEs will be coded using MedDRA 24.0 or most recent version. The severity (or intensity) of AEs will be documented using the National Cancer Institute-Common Terminology Criteria for Adverse Event version 4.03 (NCI-CTCAE v 4.03). Descriptive summary tables will be presented on all safety parameters by treatment arm.

The AEs will be presented with their worst NCI-CTCAE v 4.03 grade CTC 1 = mild, CTC 2 = moderate, CTC 3 = Severe, CTC 4 = life threatening, CTC 5 = fatal. The causal relationship is based on whether there was a reasonable causal relationship to darolutamide or placebo and docetaxel. AEs will be classified by the investigator as related or not related to darolutamide or placebo and docetaxel separately.

Treatment-emergent AE (TEAE) is defined as any event arising or worsening after the first dose of darolutamide or placebo until 30 days after the last dose of darolutamide or placebo.

An overview of TEAEs will be summarized by treatment arm in the overall treatment period. The following summaries will be created separately for study drug-related (darolutamide or placebo-related) and docetaxel-related TEAEs, by treatment arm: TEAEs leading to permanent discontinuation of study treatment, TEAEs leading to dose reduction and/or dose interruption, and drug-related TEAEs. Summary statistics (frequency and percentage of patients, not of events) will be presented by treatment arm, MedDRA SOC and PT, and NCI-CTCAE worst grade for the following:

- Incidence of all TEAEs
- Incidence of TEAEs of NCI-CTCAE grade 3, 4 or 5
- Incidence of TEAEs leading to permanent discontinuation of darolutamide or placebo and docetaxel separately
- Incidence of TEAEs leading to dose reduction
- Incidence of TEAEs leading to dose interruption
- Incidence of TEAEs with incidence of at least 5%
- Incidence of all drug-related TEAEs
- Incidence of drug-related TEAEs of NCI-CTCAE grade 3, 4 or 5
- Incidence of drug-related TEAEs leading to permanent discontinuation of darolutamide or placebo and docetaxel separately

- Incidence of drug-related TEAEs leading to dose reduction
- Incidence of drug-related TEAEs leading to dose interruption
- Incidence of drug-related TEAEs with incidence of at least 5%
- Incidence of pre-treatment AEs and post-treatment AEs (after 30 days from the last dose of study drug).
- Incidence and prevalence of most common TEAEs $\geq 10\%$ incidence in either treatment arm, analyzed within pre-specified time intervals. The same summary will be presented by geographical regions (North America, Asia Pacific, Rest of the World) as well.
- Incidence of COVID-19 related TEAEs
- The following subgroup analyses will be performed for overview of TEAEs and for TEAEs showing at least 5% point differences in incidence proportions between any of the subgroup categories .
 - Age category (<65, 65–74, 75–84, ≥ 85 years)
 - Geographical region (North America, Asia Pacific / Rest of the World)
 - Renal function – eGFR at baseline (normal, mild impairment, moderate impairment, severe impairment)
 - Hepatic function at baseline (normal, mild, moderate, severe impairment)
 - Concomitant statin use (no, yes) to be determined by concomitant medication with any medication in the Standardized Drug Grouping (SDG) ‘Statins’.
- Incidence of the TEAEs in the following will be presented by groupings and worst CTCAE grade.
 - Patients with present medical history of cardiac disorder (SOC).
 - Patients with present medical history of renal impairment (SMQ Acute renal failure; SMQ Chronic kidney disease).
 - Patients with present TEAE interstitial lung disease (not medical history, SMQ interstitial lung disease (narrow search)).
 - Patients with present TEAE neutropenia (SMQ of Haematopoietic leukopenia (narrow search)), by concomitant G-CSF use, where G-CSF is defined by ATC code: L03AA. The same summary will be presented by geographical regions (North America, Asia Pacific, Rest of the World) as well.

TEAE Groupings Considered as Special Topics

The following TEAE groupings are considered as special topics and shown in Appendix 9.6, Table 9–2:

- Bone fractures excluding pathological fractures
- Diabetes mellitus and hyperglycemia
- Fall
- Fatigue/ asthenic conditions

- Weight decreased
- Rash
- Seizure
- Hypertension
- Vasodilatation and flushing
- Mental impairment disorders
- Depressed mood disorders
- Breast disorders/gynecomastia
- Cardiac disorders
- Cerebral ischaemia
- Cerebral and intracranial hemorrhage

The following tables will be created for TEAEs of special topics:

- Incidence of TEAEs
- Incidence of TEAEs leading to permanent discontinuation of (darolutamide or placebo and docetaxel separately)
- Incidence of TEAEs leading to dose reduction
- Incidence of TEAEs leading to dose interruption
- Incidence of treatment-emergent SAEs (TESAEs)
- Incidence and prevalence of most common TEAEs of special topics $\geq 5\%$ incidence in either treatment arm, analyzed within pre-specified time intervals. The same summary will be presented by geographical regions (North America, Asia Pacific, Rest of the World) as well.

Fracture events will be described by a cumulative incidence plot of fracture. A summary of treatment-emergent fracture by bone health agent use (for selection of bone health agent see Appendix 9.7) at study entry will be provided. Association with weight decrease will be presented by histogram of fracture events by patient weight change (the weight collected closest to the start of fracture will be considered and compared to baseline weight). A listing of patients with fracture events breakdown by relevant events occurring within 7 days window: fall, accident, syncope and/or loss of consciousness, dizziness will be presented.

A listing will be generated for patients with fall treatment-emergent events with syncope and/or loss of consciousness (using the MLG Syncope). Timing of occurrence of fracture events based on first dose of study drug will be presented. A summary of time to fall and fractures events will be presented in descriptive and inferential statistics, respectively. In addition, Kaplan-Meier plots will be created to display the time to fall and time to fracture.

An overview of patients at risk for developing an AE seizure will be tabulated by displaying medical history.

To adjust for unequal lengths of study treatment (darolutamide or placebo and docetaxel) between the treatment arm, additional summaries based on event rate per 100 patient year will be performed for all TEAEs, special topics TEAEs, and all TESAEs occurring after the first

dose of darolutamide or placebo. The exposure adjusted incidence rate is calculated as the total number of patients with a given TEAE divided by the total darolutamide or placebo treatment duration of all patients in years. The rate is expressed in 100 patient years.

The treatment duration in years will be calculated as treatment duration in days divided by 365.25. The risk difference 'darolutamide+docetaxel arm – placebo+docetaxel arm' and risk ratio 'darolutamide+docetaxel arm vs. placebo+docetaxel arm' will be calculated together with 95% confidence intervals (CIs). No zero-cell correction will be applied for calculation of the risk ratio, i.e. in case of no events in the placebo+docetaxel arm the risk ratio will not be calculated. The incidence rate ratio for darolutamide+docetaxel arm vs. placebo+docetaxel arm will also be calculated.

6.3.2 Deaths and Serious Adverse events

Serious adverse events (SAEs) will be classified using the NCI-CTCAE version 4.03 and MedDRA version 24.0 or most recent version.

- Treatment-emergent SAEs
- Treatment-emergent SAEs leading to permanent discontinuation of study treatment
- Treatment-emergent SAEs leading to dose reduction
- Treatment-emergent SAEs leading to drug interruption
- Treatment-emergent drug-related SAEs
- COVID-19 related treatment-emergent SAEs
- Listing of treatment-emergent SAEs
- Listing of non-treatment-emergent SAEs

The incidence of deaths in the study: within 30 days after first dose of study drug / docetaxel, during study treatment from first to last dose of study drug / docetaxel, within 30 days after last dose of study drug/ docetaxel, later than 30 days after last dose of study drug/ docetaxel, will be summarized by each treatment arm and cause of death. All deaths before treatment start, during treatment and within 30 days after last dose of study drug, and later than 30 days after last dose of study drug, will be listed by patient with start and stop date of study medication, date of death, and cause of death.

6.3.3 Additional Primary Malignancies

Summary statistics (frequency and percentage of patients, not of events) will be presented by treatment arm, MedDRA SOC and PT for additional primary malignancies. A listing will be generated for patients with additional primary malignancies.

The additional primary malignancies are defined by: SMQ Malignant tumors, excluding: the following terms:

- PT: Prostate cancer
- PT: Prostate cancer metastatic
- PT: Prostate cancer recurrent

- PT: Prostate cancer stage 0
- PT: Prostate cancer stage I
- PT: Prostate cancer stage II
- PT: Prostate cancer stage III
- PT: Prostate cancer stage IV
- PTs from HLGT: Metastases
- PT: Cancer in remission
- LLT: Progression of pre-existing cancer

6.3.4 Laboratory Safety Assessments

Descriptive statistics (number of observations, mean, standard deviation, minimum, median and maximum values) will be presented for clinical laboratory tests (hematology, clinical biochemistry and urinalysis), their changes from baseline (including baseline value), and their percent changes from baseline by treatment arm at applicable visits.

Hematological and biochemical laboratory values will be graded based on NCI CTCAE version 4.03. CTCAE severity grading for laboratory abnormalities are based on applicable laboratory threshold values outlined in NCI CTCAE v4.03. It should be noted that in the present analysis of those laboratory parameters for which additional clinical information potentially can also influence the toxicity grade, this clinical information is in general not available and only the laboratory measurements are used for grading.

Any additional specific handling of the NCI CTCAE v4.03 toxicity grading assignments will be noted in the footnotes of the corresponding tables as applicable per the data collection in the study.

- In the event of overlapping CTCAE criteria ranges for specific lab tests, the algorithm assigns the worst grade
- If calcium type is not recorded (i.e. only “calcium” is recorded), then grading is done as if the calcium is total calcium. “Calcium corrected” is computed from total calcium and serum albumin (if ≤ 4.0 g/dl) from the same time point based on CTCAE v3.0 guidance. If serum albumin (if ≤ 4.0 g/dl) from the same time point is not available or if “calcium, unspecified” was collected then grading is done as if the calcium is “corrected calcium.”
- Results with special characters (such as “>” and “<”) will be imputed with the lower or higher detection limit (refer to Section 4.3).

The frequency of laboratory abnormalities regarding hematology, international normalized ratio (INR), clinical chemistry, and urinalysis will be tabulated by visit and treatment arm. Worst grades for hematological and biochemical toxicities will be calculated according to CTCAE, v4.03 based on laboratory measurements, and will be summarized by treatment arm and NCI CTCAE v4.03 category and worst grade.

Clinical laboratory toxicities during treatment including a period of 30 days after last dose of treatment will be considered as “treatment-emergent.”

The last non-missing value before or on the first day of study drug will be retained as “baseline” data. If several assessments are performed on the same day (without timing information) the average of the values will be considered.

Incidence tables (frequency and percentage of patients) as well as tables with change in NCI CTCAE v4.03 worst grade from baseline will be presented as following:

- Hematological and biochemical toxicity during screening (the last evaluation available before assignment to treatment is taken into account).
- Treatment-emergent hematological and biochemical toxicity.
- Treatment-emergent hematological and biochemical toxicities with incidence rate above 5% in either treatment arm.
- Change in worst grade for hematological and biochemical toxicity from baseline.

The laboratory values will be also categorized into low, normal and high according to their reference ranges.

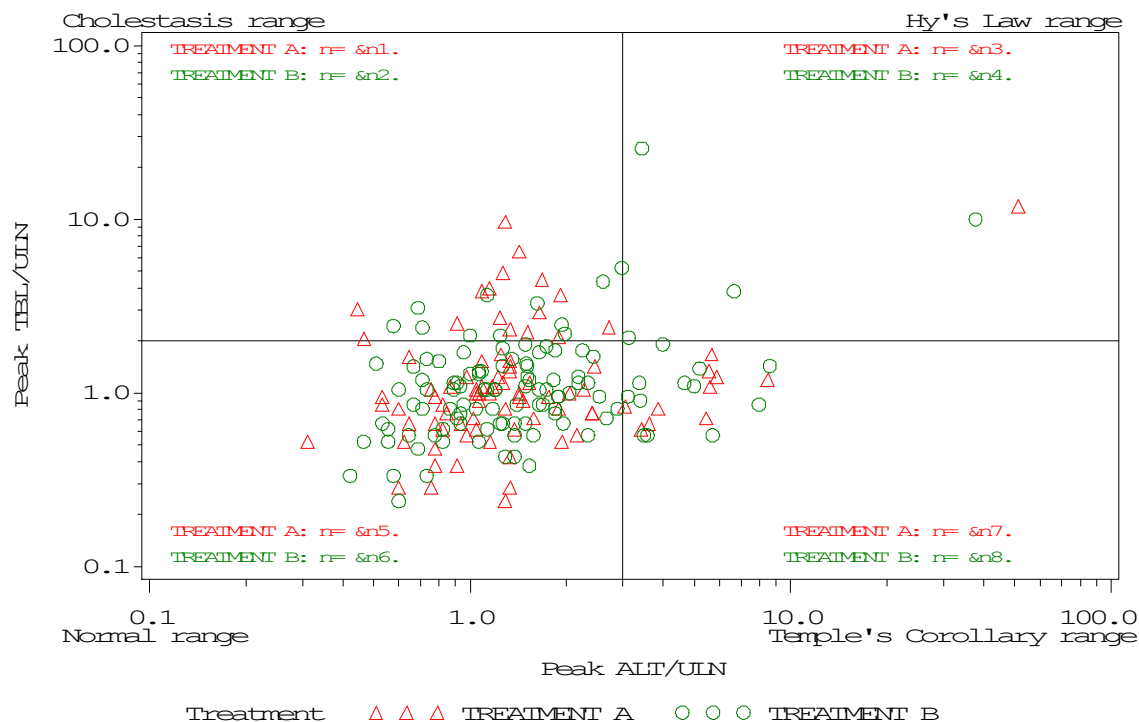
Descriptive statistics will be calculated by treatment arm and visit.

Only for ALT, AST and creatinine, descriptive statistics will be displayed for subgroups of statin users and non-statin users. Clinical laboratory tables will not be described by subgroups.

The data will also be displayed graphically using box plots and laboratory shift plots. These will show the baseline value and post baselines values up-to end of treatment.

In addition, the mean of lab values (from both scheduled and unscheduled assessments) for ALT, AST, Bilirubin, Hemoglobin, Neutrophils and Platelets will be displayed in line plots by treatment arm and time interval (1 week or less) relative to the first concomitant docetaxel treatment date, from baseline up-to Visit 6 (EOT visit will be included if the last visit number before EOT visit is < 7) for safety patients who had concomitant docetaxel with darolutamide or placebo. In the line plots, if one patient has the repeated measures within the same interval, the mean of the repeated measures for the same patient will be used in the mean calculation.

A listing will be provided for all patients possibly fulfilling Hy's Law criteria, i.e. patients with elevated AST and / or ALT > 3xULN, alkaline phosphatase < 2xULN and bilirubin ≥ 2 xULN (ref. 3). For possible Hy's Law cases relevant laboratory data will also be displayed graphically within actual patient profiles (presenting total bilirubin, ALT, AST and ALP values in terms of ULN over time) by treatment arm. Below example of Hy's law plot will be provided for peak total bilirubin vs ALT. If a patient has any total bilirubin ≥ 2 xULN then peak bilirubin will be plotted versus the maximum ALT amongst the total bilirubin ≥ 2 xULN. Otherwise, peak bilirubin will be plotted versus peak ALT.

Figure 6–1 Example for Hy's law plot

Unscheduled laboratory data will be listed (in Section 14 listings and Section 16 of the CSR), but will not be displayed in the descriptive tables.

6.3.5 Other Safety Measures

12-lead ECG

Analyses of ECG and QTc will be performed on the SAF population; no subgroup tables will be created.

Corrected QT (QTc) will be calculated using Bazett's (QTcB) and Fridericia's (QTcF) formula as below:

- Fridericia's correction: $QTc = QT/RR^{0.33}$
- Bazett's correction: $QTc = QT/RR^{0.5}$

where RR interval = 60 / Ventricular Rate

Standard 12-lead ECGs were performed at screening and at each visit in a supine position after at least 10 minutes rest.

Descriptive statistics including arithmetic mean, SD, median, minimum, and maximum will be presented for the following ECG parameters: Ventricular rate, PR, QRS, RR (derived), QT, QTcB (derived) and QTcF (derived). Parameters will be summarized for actual results and the change from baseline for the safety analysis set by treatment arm at each scheduled visit. Baseline for 12-lead ECG will be the mean of separate recordings. Similarly, post baseline values of 12-lead ECG for patients will be the mean of separate recordings.

Mean time courses +/- one standard deviation will be displayed graphically.

The last non-missing value before the first study drug intake will be considered as “baseline” data.

If several assessments are performed on the same day (without timing information) the average of the values will be considered.

In case the timing of ECG is missing at day 1 then the screening assessment will be considered as baseline data.

The number and percent of patients with absolute QTc interval prolongation or QTc increase from baseline will be defined and summarized by treatment arms. The criteria are:

QTc prolongation: QTc interval $\leq$ 450 msec, QTc interval 451-480 msec, QTc interval 481-500 msec and QTc interval $>$ 500 msec.

Increase from baseline in QTc interval: QTc interval increases from baseline $>$ 30 msec and QTc interval increases from baseline $>$ 60 msec.

The number of patients with QTc interval (max value) $\leq$ 450 msec, 451-480 msec, 481-500 msec and $>$ 500 msec data will also be displayed graphically using a bar chart.

The number and percent of patients with new clinically significant abnormalities on ECGs per the investigator’s assessment at post-baseline time points will be summarized by treatment arms.

ECG data will be presented in a listing. Values with QTc interval $\leq$ 450 msec, 451-480 msec, 481-500 msec and $>$ 500 msec, and change from baseline in QTc interval $>$ 30 msec and $>$ 60 msec will be flagged. In addition, abnormalities or clinically significant abnormalities found in the 12-lead ECGs at each scheduled time point will be provided in a listing.

Unscheduled ECG data will be listed (in Section 14 listings and Section 16 of the CSR), but will not be displayed in the summary tables.

Vital signs

For each treatment arm, vital signs (i.e. blood pressure, heart rate, weight and BMI) will be tabulated and summarized by visit for observed values and changes from baseline using descriptive statistics, as appropriate. If more than one baseline assessment is collected, the most recent one will be used. If several assessments are performed on the same day (without timing information) the average of the values will be considered.

In addition to descriptive analysis box plots will be created.

For weight parameter a graph representing mean changes in weight from baseline by visit will be created.

Outlier analyses will be conducted using the following limits:

- low systolic blood pressure: $\leq$ 90 mmHg and a decrease of $\geq$ 20 mmHg
- high systolic blood pressure: $>$ 190 mmHg and an increase of $\geq$ 20 mmHg
- low diastolic blood pressure: $\leq$ 50 mmHg and a decrease of $\geq$ 20 mmHg
- high diastolic blood pressure: $>$ 105 mmHg and an increase of $\geq$ 20 mmHg
- low heart rate: $<$ 50 bpm and a decrease of $\geq$ 15 bpm

- high heart rate: > 120 bpm and an increase of ≥ 15 bpm

The number and percentage of patients with outlying values will be tabulated by treatment arm and time interval.

Unscheduled vital signs data will be listed (in Section 14 listings and Section 16 of the CSR), but will not be displayed in the summary tables.

Concomitant statins (BCRP substrates) of interest

The number of patients reported with the below concomitant statins which are BCRP substrates will be described with frequency and percentage for the SAF population.

Concomitant statins (BCRP substrates) of interest are defined as:

Atorvastatin (ATC code C10AA05)

Fluvastatin (ATC code C10AA04)

Rosuvastatin (ATC code C10AA07)

Simvastatin (ATC code C10AA01).

More details are available in Appendix 9.8.

A listing with intervals will be created for these subjects when above statins were taken concomitantly with the study-treatment.

Interval will be created as follow: number of days when statins and study-treatment were concomitantly taken plus 14 days.

For partial dates of concomitant statins, the imputation rules in Appendix 9.1 will be applied. If the end date is completely missing it will be assumed that the concomitant medication is ongoing at study treatment end. If the start date is completely missing it is assumed that concomitant medication was taken before the start of study treatment.

The frequency of pre-defined adverse events (see Appendix 9.9) will be detailed if occurring during the interval defined above, for the subset of patients taking statins that are BCRP substrates.

6.4 Other Exploratory Variables

• Medical resource use

Summary statistics (descriptive statistics and frequency) will be presented by treatment arm for days in hospital and equivalent facility, and days in ICU (intensive care unit) facility, separately. Location/place of healthcare visit will be presented in frequency. Listings of patients with healthcare resource use will be provided.

• Biomarker evaluations

Biomarker analyses will be described in a separate biomarker analysis plan. Results from exploratory biomarker analyses will be reported in a separate biomarker report.

6.5 Pharmacokinetics

Pharmacokinetics in two sub-groups: 1) at least the first 20 randomized patients who have received at least 1 cycle of docetaxel and 2) Chinese patients in the PK substudy

The concentration-time courses of all analytes will be tabulated separated by treatment and subgroup. The following statistics will be calculated for each of the sampling points: arithmetic mean, SD, and coefficient of variation (CV); geometric mean, geometric SD (re-transformed SD of the logarithms), and CV; minimum, median, maximum value, and the number of measurements.

Means at any time will only be calculated if at least 2/3 of the individual data were measured and were above the lower limit of quantification (LLOQ). For the calculation of the mean value, a data point below LLOQ will be substituted by one-half of this limit.

In tables showing mean values, where values below LLOQ are included in the calculation of mean values, these means will be marked.

Individual and geometric mean concentration versus time curves of all analytes (using the actual sampling times for individual plots and the planned sampling times for mean plots) will be plotted by treatment using both a linear and semilogarithmic scale. For darolutamide and its diastereomers and metabolite all analytes will be plotted together.

Pharmacokinetic parameters (t_{max} , t_{last} and parameters listed under 'other parameters' excluded) will be summarized by the statistics mentioned above. The t_{max} , t_{last} will be described utilizing minimum, maximum, and median as well as frequency counts.

The diastereomer/darolutamide ratios will be calculated using the following equation:

Diastereomer/darolutamide ratios =

$AUC(0-t_{last}) (S,R)\text{-darolutamide} / AUC(0-t_{last}) \text{ darolutamide}$ and

$AUC(0-t_{last}) (S,S)\text{-darolutamide} / AUC(0-t_{last}) \text{ darolutamide}$

Metabolite keto-darolutamide to parent darolutamide $AUC(0-t_{last})$ ratios will be calculated using the following equation:

Metabolite Ratio =

$(AUC(0-t_{last}) \text{ metabolite} / MW \text{ metabolite}) / (AUC(0-t_{last}) \text{ parent} / MW \text{ parent})$

Molecular weight (MW) keto-darolutamide = 396.8 g/mol

Molecular weight (MW) darolutamide = 398.8 g/mol

Individual data points of docetaxel PK parameters C_{max} and $AUC(0-t_{last})$ for treatments of treatments of docetaxel in combination with darolutamide and docetaxel with placebo will be compared with a scatter plot.

Only dense PK sampling data from the first Visit (as well as Day 15 for the Chinese subgroup) will be used for statistical analysis. Sparse PK samples from later visits will only be listed.

Pharmacokinetics in all patients

Individual concentration-time data of darolutamide (BAY 1841788), (S,S)-darolutamide (BAY 1896952), (S,R)-darolutamide (BAY 1896951), keto-darolutamide (BAY 1896953), and docetaxel will be provided in a clinical study report appendix. The PK data will be used for population PK data analysis that will be described in detail in a separate M&S report.

7. Document History and Changes in the Planned Statistical Analysis

- Approval of SAP v1.0, dated 07 APR 2017, was based on the integrated clinical study protocol v2.0 (amendment 1), dated 04 OCT 2016.
- Approval of interim futility SAP v1.0, 11 MAR 2019 was based on SAP v1.0, dated 07 APR 2017 and the integrated clinical study protocol v2.0 (amendment 1), dated 04 OCT 2016.
- Approval of SAP v2.0, dated 03 JUL2019, was mainly an update of the efficacy censoring rules from SAP v1.0 and was based on the integrated clinical study protocol v3.0 (amendment 5), dated 12 FEB 2018
- Approval of interim futility SAP v2.0, 03 JUL 2019, was mainly an update of the efficacy censoring rules from SAP v1.0 and was based on SAP v2.0, dated 03 JUL2019 and the integrated clinical study protocol v3.0 (amendment 5), dated 12 FEB 2018.
- Approval of SAP v3.0, dated 26 MAY 2020, describes the changes of removing interim analysis 2 for futility and efficacy from the testing plan due to data collection limitations during the COVID-19 pandemic and was based on the integrated clinical study protocol v5.0 (amendment 7), dated 26 MAY 2020.
- Approval of SAP v4.0, dated 22 SEP 2021, based on the SAP v3.0, dated 26 MAY 2020, describes the changes as a result of the following:
 - Efficacy changes included applying a hierarchical gatekeeping procedure to the secondary efficacy endpoints, inclusion of a summary of ECOG PS, additional relative 50% and 90% PSA response rate and PSA maximum percent decline from baseline and
 - Safety changes included the analysis of special topics adverse events, additional primary malignancies, liver function laboratory test, 12-lead ECG, etc.
- Approval of SAP v4.1, dated 11 NOV 2021, based on the SAP v4.0, dated 22 SEP 2021, describes the changes as a result of the following:
 - Analysis set changes with an additional statement: Patients will be excluded from all FAS, SAF and PKS if they are related to or associated with any critical GCP violations that result in fraudulent patient data.

Rationale: After detection of issues with investigator fraud at one site, it was decided to exclude one affected patient from all analysis sets as these data cannot be trusted.
 - Efficacy change included a sensitivity analysis of OS with central imaging review extent of disease as stratification factor.
 - Safety changes included summary of incidence and prevalence of most common TEAEs and special topics TEAEs by geographical regions, and incidence of TEAE neutropenia by concomitant G-CSF use and geographical regions.

8. References

1. Eisenhauer EA, Therasse P, Bogaerts J, Schwartz LH, Sargent D, Ford R, et al. New response evaluation criteria in solid tumours: Revised RECIST guideline (version 1.1). Eur J Cancer. 2009;45:228-47.
2. Scher HI, Morris MJ, Stadler WM, Higano C, Basch E, Fizazi K, et al. Trial Design and Objectives for Castration-Resistant Prostate Cancer: Updated Recommendations From the Prostate Cancer Clinical Trials Working Group 3. J Clin Oncol. 2016;34(12):1402-18.
3. U.S. Department of Health and Human Services, Food and Drug Administration, Guidance for Industry Drug-Induced Liver Injury: Premarketing Clinical Evaluation, July 2009.

9. Appendix

9.1 Imputation rule for partial missing Adverse Event/ Concomitant medication dates

Table 9–1 Imputation rule for partial missing dates

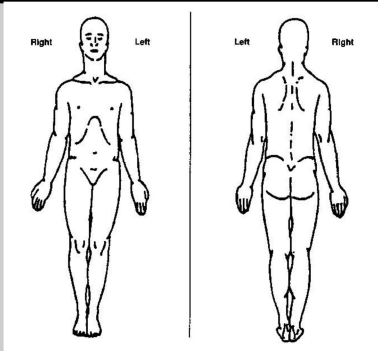
Partial Dates Imputation Rule	Impute partial AE/CM Start Date	Impute partial AE/CM Stop Date
The day missing only	IF AESTDT year and months is same as TRTSDT year and months, then impute AESTDT= TRTSDT	IF AEENDT year and months is same as last known alive date (LKAD) year and months, then impute AEENDT= LKAD
	ELSE IF AESTDT year and month is before TRTSDT year and months, then AESTDT= last date of the months	ELSE impute AEENDT= last date of the months
	ELSE IF AESTDT year and month is after TRTSDT year and months, then AESTDT= first date of the months	
Both day and months missing	IF AESTDT year is same as TRTSDT year, then impute AESTDT=TRTSDT	IF AEENDT year is same as last known alive date (LKAD) year, then impute AEENDT= LKAD
	ELSE IF AESTDT year is before TRTSDT year, then impute AESTDT=31DECYYYY	ELSE impute AEENDT=31DECYYYY
	ELSE IF AESTDT year is after TRTSDT year , then impute AESTDT=01JANYYYY	
Completely missing	No need to impute, try to query the sites by DM	No need to impute, try to query the sites by DM
Additional criteria to meet	1. AE/CM start date <= AE/CM stop date 2. The imputed dates <= last known alive date (LKAD) 3. If TRTSDT is missing, use RANDDT as reference date	

9.2 Systemic antineoplastic therapy: time to initiation of subsequent systemic antineoplastic therapy

1. Systemic antineoplastic therapy treatment was selected from eCRF page in 2 and 3 below in CM dataset
2. From “SYSTEMIC ANTI-CANCER THERAPY (MEDICATION) DURING FOLLOW-UP” eCRF pages
 - ATC code class L (antineoplastic and immunomodulating agents): L01 Antineoplastic agents, L02 endocrine therapy, L03 immunostimulants and L04 immunosuppressants.
 - ATC code class V10.
 - ATC code: G03CA, G03CB, V03AX, V, V98.
3. From “SYSTEMIC ANTI-CANCER THERAPY (MEDICATION) DURING FOLLOW-UP” eCRF pages or “CONCOMITANT MEDICATION” or “PRIOR AND CONCOMITANT MEDICATION” eCRF pages
 - If the WHO-DD drug record number is 062951 (ABIRATERONE), 900315 (ANTINEOPLASTIC AGENTS), 012711 (BICALUTAMIDE), 082361 (RADIUM RA 223 DICHLORIDE).
 - If ATC code or code class is same as any above, but the WHO-DD drug record number is 093203 (MARSDENIA TENACISSIMA STEM), 000826 (MEGESTROL ACETATE), 007269 (LEUPRORELIN ACETATE), 007321 (GOSERELIN ACETATE), 009759 (TRIPTORELIN EMBONATE), 017648 (DEGARELIX ACETATE), 010277 (UBENIMEX), 017660 (THYMALFASIN), 901166 (OTHER THERAPEUTIC PRODUCTS), then do not consider as antineoplastic therapy.

The sponsor’s assessment of antineoplastic therapy is presented in a supplementary document.

9.3 Brief Pain Inventory – Short Form (BPI-SF) questionnaire and scoring information

STUDY ID# _____	HOSPITAL # _____																						
DO NOT WRITE ABOVE THIS LINE																							
Brief Pain Inventory (Short Form)																							
Date: ____/____/____ Time: ____ Name: _____ Last First Middle Initial																							
1. Throughout our lives, most of us have had pain from time to time (such as minor headaches, sprains, and toothaches). Have you had pain other than these everyday kinds of pain today? 1. Yes 2. No																							
2. On the diagram, shade in the areas where you feel pain. Put an X on the area that hurts the most.																							
																							
3. Please rate your pain by circling the one number that best describes your pain at its worst in the last 24 hours.																							
<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%;">0</td> <td style="width: 10%;">1</td> <td style="width: 10%;">2</td> <td style="width: 10%;">3</td> <td style="width: 10%;">4</td> <td style="width: 10%;">5</td> <td style="width: 10%;">6</td> <td style="width: 10%;">7</td> <td style="width: 10%;">8</td> <td style="width: 10%;">9</td> <td style="width: 10%;">10</td> </tr> <tr> <td>No Pain</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Pain as bad as you can imagine</td> </tr> </table>		0	1	2	3	4	5	6	7	8	9	10	No Pain										Pain as bad as you can imagine
0	1	2	3	4	5	6	7	8	9	10													
No Pain										Pain as bad as you can imagine													
4. Please rate your pain by circling the one number that best describes your pain at its least in the last 24 hours.																							
<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%;">0</td> <td style="width: 10%;">1</td> <td style="width: 10%;">2</td> <td style="width: 10%;">3</td> <td style="width: 10%;">4</td> <td style="width: 10%;">5</td> <td style="width: 10%;">6</td> <td style="width: 10%;">7</td> <td style="width: 10%;">8</td> <td style="width: 10%;">9</td> <td style="width: 10%;">10</td> </tr> <tr> <td>No Pain</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Pain as bad as you can imagine</td> </tr> </table>		0	1	2	3	4	5	6	7	8	9	10	No Pain										Pain as bad as you can imagine
0	1	2	3	4	5	6	7	8	9	10													
No Pain										Pain as bad as you can imagine													
5. Please rate your pain by circling the one number that best describes your pain on the average .																							
<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%;">0</td> <td style="width: 10%;">1</td> <td style="width: 10%;">2</td> <td style="width: 10%;">3</td> <td style="width: 10%;">4</td> <td style="width: 10%;">5</td> <td style="width: 10%;">6</td> <td style="width: 10%;">7</td> <td style="width: 10%;">8</td> <td style="width: 10%;">9</td> <td style="width: 10%;">10</td> </tr> <tr> <td>No Pain</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Pain as bad as you can imagine</td> </tr> </table>		0	1	2	3	4	5	6	7	8	9	10	No Pain										Pain as bad as you can imagine
0	1	2	3	4	5	6	7	8	9	10													
No Pain										Pain as bad as you can imagine													
6. Please rate your pain by circling the one number that tells how much pain you have right now .																							
<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%;">0</td> <td style="width: 10%;">1</td> <td style="width: 10%;">2</td> <td style="width: 10%;">3</td> <td style="width: 10%;">4</td> <td style="width: 10%;">5</td> <td style="width: 10%;">6</td> <td style="width: 10%;">7</td> <td style="width: 10%;">8</td> <td style="width: 10%;">9</td> <td style="width: 10%;">10</td> </tr> <tr> <td>No Pain</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Pain as bad as you can imagine</td> </tr> </table>		0	1	2	3	4	5	6	7	8	9	10	No Pain										Pain as bad as you can imagine
0	1	2	3	4	5	6	7	8	9	10													
No Pain										Pain as bad as you can imagine													

7. What treatments or medications are you receiving for your pain?

8. In the last 24 hours, how much relief have pain treatments or medications provided? Please circle the one percentage that most shows how much relief you have received.

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
No Complete
Relief Relief

9. Circle the one number that describes how, during the past 24 hours, pain has interfered with your:

A. General Activity

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

B. Mood

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

C. Walking Ability

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

D. Normal Work (includes both work outside the home and housework)

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

E. Relations with other people

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

F. Sleep

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

G. Enjoyment of life

0 1 2 3 4 5 6 7 8 9 10
Does not Completely
Interfere Interferes

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Pain Research Group
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Scoring:

Pain Severity Score = mean of items 3-6 (pain at its worst, pain at its least, pain on the average, pain for right now)

Pain Interference Score = mean of items 9A-9G (interference of pain with: general activity, mood, walking ability, normal work, relations with other people, sleep, and enjoyment of life).

9.4 NCCN-FACT FPSI-17 questionnaire and scoring information

Below is a list of statements that other people with your illness have said are important.

Please circle or mark one number per line to indicate your response as it applies to the past 7 days.

			Not at all	A little bit	Some what	Quite a bit	Very much
D R S- P	GP1	I have a lack of energy.....	0	1	2	3	4
	GP4	I have pain	0	1	2	3	4
	P7	I have difficulty urinating	0	1	2	3	4
	C2	I am losing weight	0	1	2	3	4
	BP1	I have bone pain.....	0	1	2	3	4
	HI7	I feel fatigued.....	0	1	2	3	4
	NCCN3	I have weakness in my legs	0	1	2	3	4
D R S- E	P3	My pain keeps me from doing things I want to do	0	1	2	3	4
	C6	I have a good appetite.....	0	1	2	3	4
	GF5	I am sleeping well.....	0	1	2	3	4
	GE6	I worry that my condition will get worse .	0	1	2	3	4
	GP2	I have nausea	0	1	2	3	4
	P6	I have trouble moving my bowels	0	1	2	3	4
	GS7	I am satisfied with my sex life.....	0	1	2	3	4
T S E	GP5	I am bothered by side effects of treatment	0	1	2	3	4
	GF3	I am able to enjoy life	0	1	2	3	4
F W B	GF7	I am content with the quality of my life right now	0	1	2	3	4

DRS-P=Disease-Related Symptoms Subscale – Physical
 DRS-E=Disease-Related Symptoms Subscale – Emotional
 TSE=Treatment Side Effects Subscale
 FWB=Function and Well-Being Subscale
 English (Universal), Copyright 2001

03 March 2010

NCCN/FACT Prostate Symptom Index-17 (FPSI-17) Scoring Guidelines
(Version 2)

Instructions:

1. Record answers in "item response" column. If missing, mark with an X.
2. Perform reversals as indicated, and sum individual items to obtain a score.
3. Multiply the sum of the item scores by the number of items in the subscale, then divide by the number of items answered. This produces the symptom index score.
4. As with all FACIT questionnaires, a high score is good. Therefore, a score of “0” is a severely symptomatic patient and the highest possible score is an asymptomatic patient.

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<u>Scale</u>	<u>Item Code</u>	<u>Reverse item?</u>		<u>Item response</u>	<u>Item Score</u>
FPSI-17 Total Score range: 0-68	GP1	4	-	_____	= _____
	GP4	4	-	_____	= _____
	P7	4	-	_____	= _____
	C2	4	-	_____	= _____
	BP1	4	-	_____	= _____
	HI7	4	-	_____	= _____
	NCCN3	4	-	_____	= _____
	P3	4	-	_____	= _____
	C6	0	+	_____	= _____
	GF5	0	+	_____	= _____
	GE6	4	-	_____	= _____
	GP2	4	-	_____	= _____
	P6	4	-	_____	= _____
	GS7	0	+	_____	= _____
	GP5	4	-	_____	= _____
	GF3	0	+	_____	= _____
	GF7	0	+	_____	= _____

Sum individual item scores: _____

Multiply by 17: _____

Divide by number of items answered: _____ = FPSI-17 score

<u>Subscale</u>	<u>Item Code</u>	<u>Reverse item?</u>		<u>Item response</u>	<u>Item Score</u>
FPSI-DRS-P (Disease Related Symptoms-Physical) Score range: 0-40	GP1	4	-	_____	= _____
	GP4	4	-	_____	= _____
	P7	4	-	_____	= _____
	C2	4	-	_____	= _____
	BP1	4	-	_____	= _____
	HI7	4	-	_____	= _____
	NCCN3	4	-	_____	= _____
	P3	4	-	_____	= _____
	C6	0	+	_____	= _____
	GF5	0	+	_____	= _____

Sum individual item scores: _____

Multiply by 10: _____

Divide by number of items answered: _____ = FPSI-DRS-P score

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<u>Subscale</u>	<u>Item Code</u>	<u>Reverse item?</u>	<u>Item response</u>	<u>Item Score</u>
FPSI-DRS-E (Disease Related Symptoms-Emotional)	GE6	4	-	_____
				= _____ = FPSI-DRS-E score

Score range: 0-4

FPSI-TSE	GP2	4	-	_____	= _____
(Treatment	P6	4	-	_____	= _____
Side Effects)	GS7	0	+	_____	= _____
	GP5	4	-	_____	= _____

Score range: 0-16

Sum individual item scores: _____

Multiply by 4: _____

Divide by number of items answered: _____ = **FPSI-TSE score**

FPSI-F/WB	GF3	0	+	_____	= _____
(Function/	GF7	0	+	_____	= _____
Well-Being)					

Score range: 0-8

Sum individual item scores: _____

Multiply by 2: _____

Divide by number of items answered: _____ = **FPSI-F/WB score**

Missing data:

For NCCN-FACT FPSI-17, total scores and subscales scores (disease-related physical symptoms, disease-related emotional symptom, treatment side-effects, function/well-being) will be assessed. Where there are missing items, subscale scores can be prorated if >50% of items on subscale are completed. If ≤50% of the items are answered for any domain, then the score of that domain is set to missing. The total score is set to missing if the related overall item response rate is ≤80%. (e.g., at least 14 of 17 FACT-G items completed). This is not to be confused with individual subscale item response rate, which allows a subscale score to be prorated for missing items if >50% of items are answered. In addition, a total score should only be calculated if ALL of the component subscales have valid scores.

9.5 Statins medications

Statins will be identified using the Standardized Drug Grouping (SDG) classification ‘Statins’.

9.6 Special topic AE definition

Table 9–2 Special topic AE definitions

Grouped term	MedDRA search criteria for special topics
Bone fractures excluding pathological fractures	HLT: Fractures and dislocations NEC (without PTs: Joint dislocation, Joint dislocation pathological) HLT Limb fractures and dislocations (without PT: Radial head dislocation) HLT Pelvic fractures and dislocations HLT Skull fractures, facial bone fractures and dislocations HLT Spinal fractures and dislocations (without PT: Dislocation of vertebra) HLT Thoracic cage fractures and dislocations (without PT: Dislocation of sternum)) Exclude PT pathological fractures
Diabetes mellitus and hyperglycemia	MLG Hyperglycemia PT: Diabetes mellitus; Diabetes mellitus inadequate control; Diabetic metabolic decompensation; Type 2 diabetes mellitus; Diabetic ketoacidosis
Fall*	PT Fall PT Accident
Fatigue/asthenic conditions	MLG Decreased general strength and energy PT: Lethargy; Chronic fatigue syndrome; Malaise
Weight decreased	MLG: Weight decreased
Rash	MLG: Rash MLG: Skin erythema PT: Dermatitis
Seizure	MLG Seizures
Hypertension	MLG: Hypertension
Vasodilatation and flushing	MLG: Vasodilatation and flushing
Mental impairment disorders	HLGT: Mental impairment disorders
Depressed mood disorders	HLGT: Depressed mood disorders and disturbance
Breast disorders/gynecomastia	HLGT: Breast disorders
Cardiac disorders	HLGT Cardiac arrhythmias HLGT Coronary artery disorders HLGT Heart failures
Cerebral ischaemia	MLG: Cerebral infarction and stroke not specified as hemorrhagic or ischemic MLG: Cerebral ischemic infarction and stroke PT: Cerebral ischaemia, Transient ischaemic attack
Cerebral and intracranial hemorrhage	MLG: Cerebral and intracranial hemorrhage

MLG: MedDRA Labeling Grouping; PT: Preferred Term; HLT: High Level Term; HLGT: High Level Group Term

* Based on review of verbatims the search criteria for “Fall” was extended by the PT “Accident”.

9.7 Bone health agent (BHA)

Bone health agent were selected in CM dataset, using ATC codes:

M05BA, M05BB: Bisphosphonates
M05BX: Denosumab

9.8 Concomitant statins (BCRP substrates) of interest

Concomitant statins (BCRP substrates) of interest

The co-medication statins (BCRP substrates) were selected in CM dataset using ATC codes:

Atorvastatin - ATC code C10AA05
Fluvastatin -ATC code C10AA04
Rosuvastatin -ATC code C10AA07
Simvastatin -ATC code C10AA01

As in CM dataset, ATC code (variable CM5CL01) contains only 5 digits, the first 5 digits of the above ATCs with 7 digits will be used.

In addition, the drug name below will be selected from “standardized medication name” variable (variable CMDECOD). This list was validated per medical team based on the output from the prior selection of ATC.

CMDECOD to select:

ATORVASTATIN
ATORVASTATIN CALCIUM
FLUVASTATIN
FLUVASTATIN SODIUM
ROSUVASTATIN
ROSUVASTATIN CALCIUM
SIMVASTATIN

9.9 Pre-defined adverse events for analysis of statin (BCRP substrates)

Pre-defined adverse events of special interest were selected from AE dataset using below MedDRA PT code:

Alanine aminotransferase increased (PT 10001551)
Aspartate aminotransferase increased (PT 10003481)
Blood alkaline phosphatase increased (PT 10059570)
Blood bilirubin increased (PT 10005364)
Blood creatinine increased (PT 10005483)
Blood lactate dehydrogenase increased (PT 10005630)
Hyperbilirubinaemia (PT 10020578)
Muscular weakness (PT 10028372)
Renal failure (PT 10038435)
Renal impairment (PT 10062237)
Transaminases increased (PT 10054889).